

Mathematics Preparatory Course

Mathematics Preparatory Notes

Basic Mathematics and Pre-University Foundations

数学预科讲义：英文版



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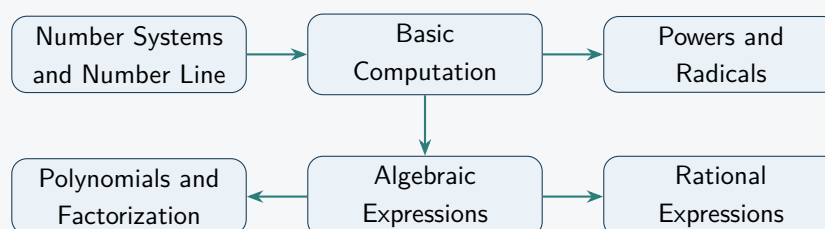
Number Systems, Operations and Algebraic Expressions

Learning Goals

After studying this chapter, you should be able to:

- ▶ recognize natural numbers, integers, rational numbers, irrational numbers and real numbers, and determine which number set a given number belongs to;
- ▶ represent numbers on the number line, compare real numbers, and understand opposite numbers and absolute value;
- ▶ carry out basic operations with integers, fractions, decimals and percentages fluently;
- ▶ use the laws of exponents correctly, and understand square roots, radicals and scientific notation;
- ▶ identify variables, constants, terms, coefficients and degrees in algebraic expressions, and perform substitution, simplification and removal of parentheses;
- ▶ carry out simple polynomial operations and master the three common multiplication formulas;
- ▶ factor polynomials by factoring out a common factor, using formulas and using the simple cross-multiplication idea;
- ▶ find the range of values for which a rational expression is meaningful, and perform reduction, common denominators and the four operations with rational expressions.

Knowledge Map



The most important thread of this chapter is: first compute numbers correctly, then understand expressions containing letters, and finally learn to simplify and factor expressions. Later equations, functions, trigonometric functions and derivatives all depend on these foundations.

Learning Route of This Chapter

This chapter moves from “numbers” to “expressions”. First, we need to know what kinds of numbers we may use, and then place numbers on the number line to compare their sizes. Next, we review the four operations, powers and radicals, because these are the basic actions in algebraic computation. Then we introduce letters and use algebraic expressions to represent quantitative relationships. Finally, we study expansion, factorization and simplification of rational expressions, preparing for solving equations in the next chapter.

When studying by yourself, you can check every step with one question: has this transformation preserved the original meaning? For example, when removing parentheses, take care of signs; when working with radicals, check the radicand; when working with rational expressions, check that the denominator is not 0.

Prerequisites and Minimum Mastery

To study this chapter, you only need the junior-middle-school foundations of integers, fractions, decimals, simple algebraic expressions and the Pythagorean theorem. The minimum goal for the first pass is not to solve difficult problems, but to avoid mistakes in fraction computation, handle negative signs clearly, substitute values correctly, expand and factor the most common expressions, and check the denominator first whenever you see a rational expression.

Common Self-Study Sticking Points in This Chapter

The three places where students most easily get stuck are negative signs, parentheses and denominators. When substituting a negative number into a power, use parentheses. When removing parentheses, distribute a preceding negative sign to every term. Before and after reducing a rational expression, keep the original restrictions on the variables.

1.1 Number Systems, Number Line and Absolute Value

We use numbers every day, but different kinds of numbers solve different kinds of problems. The expansion of number systems means continually allowing new numbers into our world of computation.

1.1.1 Extension of Number Systems (数系的扩充)

At first, people used $1, 2, 3, \dots$ to represent the number of objects. To represent “none at all”, 0 was introduced. To represent debt, decrease or the opposite direction, negative numbers were introduced. To represent division and ratios, fractions were introduced. To make equations such as $x^2 = 2$ have solutions, irrational numbers were needed.

We can also view the expansion of number systems as a series of moments when “the old numbers were no longer enough”:

Problem Encountered	Why the Old Numbers Were Not Enough	New Number Needed
There are no apples in the shop	If we only count $1, 2, 3, \dots$, we cannot express “none”	0
A debt of 5 yuan	Nonnegative numbers alone cannot express the opposite direction	-5
Half an apple	Integers cannot express one part of an equal division	$\frac{1}{2}$
The diagonal of a square with side length 1	Rational numbers cannot represent this length exactly	$\sqrt{2}$

Therefore, new numbers are not symbols added out of nowhere. They are introduced to solve problems that old numbers cannot express.

These notes use the following notation. Some of the symbols involve the language of sets. For now, you only need to recognize their inclusion relationship.

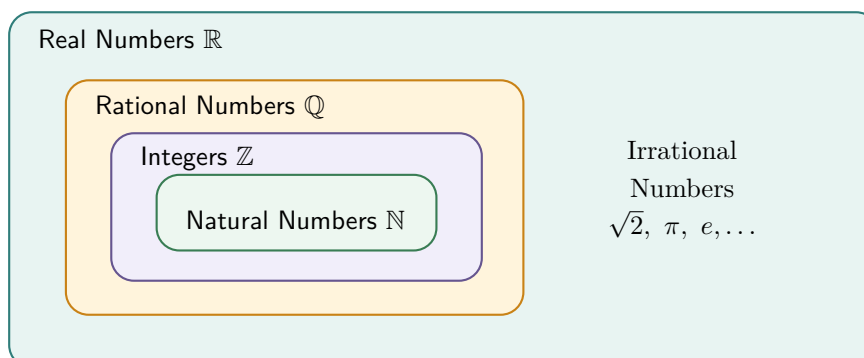
Common Number Sets (常用数集)

$\mathbb{N} = \{0, 1, 2, 3, \dots\}$	natural numbers,
$\mathbb{N}^* = \{1, 2, 3, \dots\}$	positive integers,
$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$	integers,
$\mathbb{Q} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}$	rational numbers,
$\mathbb{R} = \text{all real numbers}$	real numbers.

Every natural number is an integer, every integer is a rational number, and every rational number is a real number. Therefore,

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}.$$

In Chapter 7, real numbers will be further expanded to complex numbers, which are used to handle some equations that have no solution in the real number system. In the first pass through this chapter, you only need to master classification and operations within the real numbers.



In the figure, a smaller set completely inside a larger set means that every number in the smaller set also belongs to the larger set. For example, $7 \in \mathbb{N}$, and at the same time $7 \in \mathbb{Z}, \mathbb{Q}, \mathbb{R}$. In classification problems, only when the problem asks you to “put the number into the smallest possible number set” should you write the most precise set.

Rational and Irrational Numbers (有理数与无理数)

A number that can be written as a ratio p/q of two integers, where $q \neq 0$, is called a **rational number** (有理数). A real number that cannot be written in this form is called an **irrational number** (无理数).

Finite decimals and repeating infinite decimals are rational numbers. Non-repeating infinite decimals are irrational numbers.

Common Mistake

$\sqrt{4} = 2$ is a rational number. You cannot immediately call a number irrational just because it contains a radical sign. You must simplify first.

Example 1.1 Classifying Numbers

Place each of the following numbers into an appropriate number set, and put it into the smallest possible common number set:

- (1) -5 ;
- (2) 0 ;
- (3) $\frac{7}{3}$;
- (4) $0.\overline{27}$;
- (5) $\sqrt{49}$;
- (6) $\sqrt{5}$;
- (7) π .

Solution: First simplify the numbers that can be simplified: $\sqrt{49} = 7$, so it is an integer; $0.\overline{27}$ is an infinite repeating decimal, so it is rational.

- (1) The smallest common number set containing -5 is $\mathbb{Z}$.
- (2) The smallest common number set containing 0 is $\mathbb{N}$.
- (3) The smallest common number set containing $\frac{7}{3}$ is $\mathbb{Q}$.
- (4) The smallest common number set containing $0.\overline{27}$ is $\mathbb{Q}$.
- (5) Since $\sqrt{49} = 7$, the smallest common number set containing it is $\mathbb{N}^*$.
- (6) The smallest common number set containing $\sqrt{5}$ is $\mathbb{R}$.
- (7) The smallest common number set containing π is $\mathbb{R}$.

Here “smallest” is relative to the inclusion chain above. For example, -5 also belongs to $\mathbb{Q}, \mathbb{R}$, but the most precise common set is the integer set $\mathbb{Z}$.

1.1.2 The Number Line and the Size of Numbers (数轴与数的大小)

Real numbers correspond one-to-one with points on the number line. A number line must contain three elements: the **origin** (原点), the **positive direction** (正方向) and the **unit length** (单位长度).



On the number line, a number to the right is greater than a number to the left. Therefore, when comparing two negative numbers, the negative number farther from 0 is actually smaller.

For example, -5 is to the left of -2 , so $-5 < -2$. Many students mistakenly think that $-5 > -2$ because $5 > 2$. This confuses “size of absolute value” with “size of the number itself”.

Key Rule

For any real numbers a, b :

- ▶ if point a is to the right of point b , then $a > b$;
- ▶ if point a is to the left of point b , then $a < b$;
- ▶ positive numbers are greater than 0, negative numbers are less than 0, and every positive number is greater than every negative number.

1.1.3 Opposite Number and Absolute Value (相反数与绝对值)

Opposite Number

Two numbers that differ only in sign are opposite numbers. The opposite of a is $-a$. On the number line, they lie on opposite sides of the origin and have the same distance from the origin.

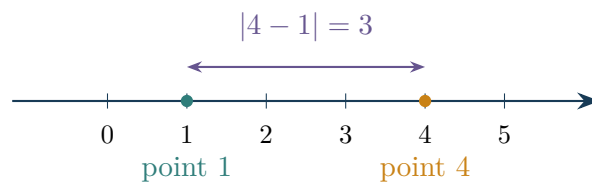
In particular, the opposite of 0 is still 0. Also note: if a itself is negative, then $-a$ is positive.

Absolute Value

The absolute value $|a|$ of a real number a represents the distance from point a to the origin:

$$|a| = \begin{cases} a, & a \geq 0, \\ -a, & a < 0. \end{cases}$$

Because distance is never negative, $|a| \geq 0$. The distance between two numbers a, b is $|a - b|$.



Similarly, $|x - 3|$ represents the distance from point x to point 3 on the number line. This interpretation will be used repeatedly when solving equations and inequalities containing absolute value.

Example 1.2 Opposite Numbers, Absolute Value and Distance (并且到原点的距离)

Let $a = -4$ and $b = 3$.

- (1) Write the opposite numbers of a and b .
- (2) Compute $|a|$, $|b|$ and $|a - b|$.
- (3) Mark a, b on the number line and explain the geometric meaning of $|a - b|$.

Solution:

- (1) The opposite of $a = -4$ is 4, and the opposite of $b = 3$ is -3 .
- (2) $|a| = |-4| = 4$, $|b| = |3| = 3$, and $|a - b| = |-4 - 3| = |-7| = 7$.
- (3) On the number line, -4 is on the left side of the origin, and 3 is on the right side of the origin. $|a - b| = 7$ represents the distance between point -4 and point 3 .

In-Class Practice 1.1.1 Basic Check

- (1) Arrange -2.1 , $-\frac{3}{2}$, 0 , $\frac{1}{4}$, 2 from least to greatest.

- (2) Simplify: $|-7|$, $-|-7|$, $|-7 + 2|$, $|3 - 8|$.

- (3) Given $|x| = 5$, mark all possible values of x on the number line.

1.2 Operations with Integers, Fractions, Decimals and Percentages

1.2.1 Four Operations (四则运算)

Addition (加法) can be understood as movement on the number line: adding a positive number moves to the right, and adding a negative number moves to the left. Subtracting a number is the same as adding its opposite.

Sign Rules

- ▶ Addition: when the signs are the same, add the absolute values and keep the sign; when the signs are different, subtract the absolute values and use the sign of the number with larger absolute value.
- ▶ Subtraction (减法) : $a - b = a + (-b)$.
- ▶ Multiplication and division: same signs give a positive result; different signs give a negative result.
- ▶ Product of several nonzero numbers: if the number of negative factors is even, the product is positive; if it is odd, the product is negative.

Any number multiplied by 0 is 0, but **division by 0 is not allowed**.

Before calculating, you can make a rough estimate. Estimation is not meant to replace calculation; it is meant to catch obvious mistakes. For example,

$$49.8 \times 2.03$$

is approximately $50 \times 2 = 100$, so the exact result should be close to 100. If you get 10 or 1000, you should immediately check the position of the decimal point.

Example 1.3 Four Operations

Compute:

$$(1) (-8) + 13 - (-4);$$

$$(2) (-3) \times 5 \div (-2);$$

$$(3) 7 - 2[3 - (-4)].$$

Solution:

$$(1) (-8) + 13 - (-4) = -8 + 13 + 4 = 5 + 4 = 9.$$

$$(2) (-3) \times 5 \div (-2) = -15 \div (-2) = \frac{15}{2}.$$

$$(3) 7 - 2[3 - (-4)] = 7 - 2(3 + 4) = 7 - 14 = -7.$$

1.2.2 Meaning and Operations of Fractions (分数的意义与运算)

A fraction a/b represents $a \div b$, where $b \neq 0$. The number a is called the numerator of the fraction, and b is called the denominator. If the numerator and denominator are both multiplied or divided by the same nonzero number, the value of the fraction does not change:

$$\frac{a}{b} = \frac{ak}{bk} \quad (k \neq 0).$$

The Four Operations with Fractions

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd},$$

$$\frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd},$$

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd},$$

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c} \quad (c \neq 0, d \neq 0).$$

Common Mistake

The following "reduction" is wrong:

$$\frac{2+6}{2+3} \neq \frac{6}{3}.$$

Reduction can only cancel a **common factor (公因式)** of the numerator and denominator. It cannot cancel one term inside an addition.

Likewise, when fractions have different denominators, you cannot add the numerators and add the denominators:

$$\frac{1}{2} + \frac{1}{3} \neq \frac{1+1}{2+3} = \frac{2}{5}.$$

The correct method is to first use a common denominator:

$$\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}.$$

Example 1.4 Mixed Fraction (引入了分数) Operations

Compute and write each result as a fraction in lowest terms:

$$(1) \frac{3}{4} - \frac{5}{6} + \frac{1}{3};$$

$$(2) \left(-\frac{2}{5}\right) \div \frac{4}{15};$$

$$(3) 1 - \frac{2}{3 - \frac{1}{2}}.$$

Solution:

$$(1) \frac{3}{4} - \frac{5}{6} + \frac{1}{3} = \frac{9}{12} - \frac{10}{12} + \frac{4}{12} = \frac{3}{12} = \frac{1}{4}.$$

$$(2) \left(-\frac{2}{5}\right) \div \frac{4}{15} = -\frac{2}{5} \cdot \frac{15}{4} = -\frac{30}{20} = -\frac{3}{2}.$$

$$(3) 1 - \frac{2}{3 - \frac{1}{2}} = 1 - \frac{2}{\frac{5}{2}} = 1 - \frac{4}{5} = \frac{1}{5}.$$

1.2.3 Decimals, Percentages and Proportions (小数、百分数与比例)

A percentage means "how many parts out of every one hundred parts". For example, $35\% = 35/100 = 0.35$. A common conversion relationship is

$$\text{decimal} \xleftrightarrow[\text{remove \% , then divide by 100}]{\text{multiply the value by 100, then add \%}} \text{percentage}.$$

If a quantity increases from A to B , its growth rate is

$$\text{growth rate} = \frac{B - A}{A} \times 100\% \quad (A > 0).$$

If the ratio of two quantities is $a : b$, it can be written as the fraction a/b . The proportion

$$\frac{a}{b} = \frac{c}{d}$$

is equivalent to $ad = bc$ when the denominators are nonzero. This is called cross multiplication.

Example 1.5 Percentages and Proportions

- (1) The original price of an item is 480 yuan. It is first discounted by 20%, and then the discounted price is increased by 10%. Find the final price, and determine whether it returns to the original price.
- (2) In a solution, the mass ratio of solute to water is 2 : 7. If the total mass of the solution is 450 grams, find the masses of solute and water.

Solution:

- (1) After the discount, the price is $480(1 - 20\%) = 480 \times 0.8 = 384$ yuan. After it is increased by 10%, the price becomes $384(1 + 10\%) = 384 \times 1.1 = 422.4$ yuan. The final price is 422.4 yuan, so it does not return to the original price.
- (2) Solute and water together make $2 + 7 = 9$ parts, and each part has mass $450 \div 9 = 50$ grams. Therefore the solute has mass $2 \times 50 = 100$ grams, and the water has mass $7 \times 50 = 350$ grams.



After a 20% discount, a 10% increase is based on 384 yuan, not on the original price 480 yuan.

1.2.4 Order of Operations (运算顺序)

Order of Mixed Operations

- (1) First compute inside parentheses, in the order of parentheses, brackets and braces.
- (2) Then compute powers and roots.
- (3) Then compute multiplication and division.
- (4) Finally compute addition and subtraction.
- (5) Operations of the same level are usually performed from left to right.

Seeing an Expression as a Tree

A complicated expression is not computed simply from left to right; it has levels. For example,

$$7 - 2[3 - (-4)]$$

first compute the innermost part $3 - (-4)$, obtaining the whole bracket; then multiply that whole bracket by 2; finally subtract the product from 7. Each time you compute, first ask: "What is the innermost, highest-priority operation right now?" This can reduce order-of-operations mistakes.

In-Class Practice 1.2.1 Computation Practice

- (1) Compute: $-18 + 7 - (-11)$.
- (2) Compute: $(-2)^4 - 3 \times (-5)$.
- (3) Compute: $\frac{5}{6} - \frac{3}{4} \div \frac{9}{8}$.
- (4) Convert 0.0625 into a fraction in lowest terms and a percentage.
- (5) 15% of a number is 42. Find the number.
- (6) The price of an item changes from 250 yuan to 290 yuan. Find the growth rate.

1.3 Powers and Radicals (幂与根式)

Exponents write repeated multiplication more compactly; radicals are inverse processes of powers. Later exponential functions, logarithmic functions and derivatives will all use the rules in this section repeatedly.

1.3.1 Exponentiation and Integer Exponents (乘方与整数指数)

For a positive integer n ,

$$a^n = \underbrace{a \cdot a \cdots a}_{n \text{ factors } a}.$$

Here a is called the **base (底数)**, n is called the **exponent (指数)**, and a^n is called a **power (幂)**.

Common Mistake

$(-2)^4$ and -2^4 are different. In the first expression the base is -2 , while the second expression means $-(2^4)$. When the base is negative, parentheses are very important.

Laws of Integer Exponents

Let m, n be integers, and assume that the expressions are meaningful. Then

$$\begin{aligned}a^m \cdot a^n &= a^{m+n}, \\ \frac{a^m}{a^n} &= a^{m-n} \quad (a \neq 0), \\ (a^m)^n &= a^{mn}, \\ (ab)^n &= a^n b^n, \\ \left(\frac{a}{b}\right)^n &= \frac{a^n}{b^n} \quad (b \neq 0), \\ a^0 &= 1 \quad (a \neq 0), \\ a^{-n} &= \frac{1}{a^n} \quad (a \neq 0, n > 0).\end{aligned}$$

These laws are not isolated facts to memorize. They come from repeated multiplication. For example,

$$a^3 \cdot a^2 = (a \cdot a \cdot a)(a \cdot a) = a^5,$$

so when powers with the same base are multiplied, the exponents are added. Similarly,

$$\frac{a^5}{a^2} = \frac{a \cdot a \cdot a \cdot a \cdot a}{a \cdot a} = a^3 \quad (a \neq 0),$$

so when powers with the same base are divided, the exponents are subtracted. Zero exponents and negative exponents are also defined so that these patterns remain consistent when exponents decrease or are subtracted.

These laws apply only to the corresponding operation structures. For example, in general, $(a + b)^n$ cannot be written as $a^n + b^n$.

Common Mistake

A negative exponent means a reciprocal; it does not mean a negative number. For example,

$$2^{-3} = \frac{1}{2^3} = \frac{1}{8},$$

not -8 . To determine the sign of a power, first look at the base and parentheses. To handle a negative exponent, first rewrite it as a reciprocal.

Example 1.6 Simplifying Exponential Expressions

Assuming all letters are nonzero, simplify:

$$(1) \frac{a^5 a^{-2}}{a^4};$$

$$(2) \frac{(2x^2 y^{-1})^3}{4xy^{-2}};$$

$$(3) \left(\frac{3a^{-2}}{b^3}\right)^{-2}.$$

The final expressions should contain no negative exponents.

Solution:

$$(1) \frac{a^5 a^{-2}}{a^4} = a^{5-2-4} = a^{-1} = \frac{1}{a}.$$

$$(2) \frac{(2x^2 y^{-1})^3}{4xy^{-2}} = \frac{8x^6 y^{-3}}{4xy^{-2}} = 2x^5 y^{-1} = \frac{2x^5}{y}.$$

$$(3) \left(\frac{3a^{-2}}{b^3} \right)^{-2} = 3^{-2} a^4 b^6 = \frac{a^4 b^6}{9}.$$

1.3.2 Square Root and Principal Square Root (平方根与算术平方根)

Square roots can be introduced from an area problem: if the area of a square is 9, its side length is 3; if the area is a , the positive side length is $\sqrt{a}$. A square root asks "which number gives a after being squared", while the principal square root specifically takes the nonnegative one.

If $x^2 = a$, then x is called a square root of a . In the real number system:

- ▶ if $a > 0$, then a has two square roots $\pm\sqrt{a}$;
- ▶ if $a = 0$, then the square root is 0;
- ▶ if $a < 0$, then there is no real square root.

$\sqrt{a}$ specifically represents the **principal square root (算术平方根)** of a , so $\sqrt{a} \geq 0$.

The easiest confusion here is between "square root" and "principal square root". For example, 9 has two square roots, 3 and -3 , because both square to 9. But $\sqrt{9}$ only represents the nonnegative one, so $\sqrt{9} = 3$, not ± 3 .

Basic Properties of Radicals

When the radicals are meaningful,

$$(\sqrt{a})^2 = a \quad (a \geq 0),$$

$$\sqrt{a^2} = |a|,$$

$$\sqrt{ab} = \sqrt{a}\sqrt{b} \quad (a \geq 0, b \geq 0),$$

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}} \quad (a \geq 0, b > 0).$$

Common Mistake

In general,

$$\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}.$$

The radical sign cannot be distributed to each term in an addition the way multiplication can.

1.3.3 Fractional Exponents (分数指数)

Fractional exponents unify radicals and powers. For a positive number a , an integer m , and a positive integer n ,

$$a^{1/n} = \sqrt[n]{a}, \quad a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m.$$

This chapter mainly discusses the case $a > 0$. Later, if the radicand contains variables, then when the root index n is even, the radicand must be nonnegative.

Why do we define $a^{1/n} = \sqrt[n]{a}$? Because it should continue to satisfy the laws of exponents:

$$\left(a^{1/n}\right)^n = a^{(1/n) \cdot n} = a.$$

That is, $a^{1/n}$ should be "the number that becomes a after being raised to the n -th power", which is exactly the meaning of an n -th root.

Example 1.7 Converting Between Radicals and Fractional Exponents

Complete the following conversions and simplifications:

(1) $\sqrt[3]{x^2}$;

(2) $a^{5/2}$;

(3) $\sqrt{72}$;

(4) $\frac{\sqrt{48}}{\sqrt{3}}$.

Solution:

(1) $\sqrt[3]{x^2} = x^{2/3}$.

(2) $a^{5/2} = a^2 \sqrt{a} \quad (a \geq 0)$.

(3) $\sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}$.

(4) $\frac{\sqrt{48}}{\sqrt{3}} = \sqrt{\frac{48}{3}} = \sqrt{16} = 4$.

1.4 Variables and Algebraic Expressions (变量与代数式)

A number represents a fixed quantity, while a letter can represent a changing quantity or a temporarily unknown quantity. Combining numbers, letters and operation symbols gives an algebraic expression.

1.4.1 Variables, Constants and Algebraic Expressions

Variables and Constants

In a problem, a quantity that can take different values is called a **variable** (变量); a quantity that remains unchanged is called a **constant** (常数). Variables are usually represented by letters such as x, y, a, b .

A letter may play different roles in different problems:

- in the equation $2x + 3 = 9$, x is the unknown to be found;

- ▶ in the function $y = 2x + 3$, x is the input variable, and y is the output variable;
- ▶ in the formula $y = kx + b$, if we study all lines, k, b can be regarded as parameters; if we study one fixed line, they are fixed constants.

When reading a problem, first determine the role of each letter, and then decide whether you should ``solve for it'', ``substitute it'', or ``treat it as a fixed but unspecified number''.

An expression made from numbers, letters and operation symbols is called an **algebraic expression (代数式)**. For example,

$$3x - 2, \quad \frac{x+1}{x-3}, \quad \sqrt{2a+b}, \quad \pi r^2$$

are all algebraic expressions.

An algebraic expression itself has no equal sign. An expression with an equal sign that asks us to find unknowns is usually an equation, which will be studied in the next chapter.

1.4.2 Terms, Coefficients and Degrees

In a polynomial, each part separated by a plus sign or minus sign is called a term. A minus sign should be taken together with the term after it.

For example, in

$$4x^3 - 2x^2 + 7x - 5,$$

the terms are $4x^3$, $-2x^2$, $7x$ and -5 . The number before the letters is called the coefficient. The sum of the exponents of all letters in a term is called the degree of a term. The largest degree among all terms in a polynomial is called the degree of the polynomial.

Term (项)	$4x^3$	$-2x^2$	$7x$	-5
Coefficient (系数)	4	-2	7	-5
Degree	3	2	1	0

When separating terms, always take the sign in front with the term. If you treat $-2x^2$ as $2x^2$, later combining like terms and evaluating by substitution will both go wrong.

Example 1.8 Identifying the Structure of an Algebraic Expression

For the polynomial

$$-3x^4y^2 + 5x^2y - 7,$$

- (1) how many terms does it have, and what are they?
- (2) what are the coefficient and degree of the first term?
- (3) what is the constant term?
- (4) what is the degree of this polynomial?

Solution: Separate terms by plus and minus signs, and keep each minus sign with the following

term.

- (1) The three terms are $-3x^4y^2$, $5x^2y$ and -7 .
- (2) The coefficient of the first term is -3 , and its degree is $4 + 2 = 6$.
- (3) The constant term is the term with no letters, so it is -7 .
- (4) The degrees of the terms are $6, 3, 0$. The largest degree is 6 , so the degree of the polynomial is 6 .

1.4.3 Evaluation by Substitution (代入求值)

When the values of letters are known, substitute those numbers into the algebraic expression. Pay special attention to the following:

- ▶ negative numbers usually need parentheses;
- ▶ fractions should be substituted as whole expressions;
- ▶ substitute first, then calculate according to the order of operations;
- ▶ after substitution, check whether the original expression is meaningful.

Standard Steps for Evaluation by Substitution

- (1) Write down the original algebraic expression.
- (2) Use parentheses to replace the corresponding letters with the given numbers.
- (3) Calculate according to the order of operations.
- (4) Check signs, parentheses and denominators.

Example 1.9 Evaluation by Substitution

Given $x = -2$ and $y = \frac{1}{2}$, find the values of the following algebraic expressions:

- (1) $x^2 - 3x + 1$;
- (2) $2xy - y^2$;
- (3) $\frac{x + y}{x - y}$.

Solution:

- (1) $x^2 - 3x + 1 = (-2)^2 - 3(-2) + 1 = 4 + 6 + 1 = 11$.
- (2) $2xy - y^2 = 2 \cdot (-2) \cdot \frac{1}{2} - \left(\frac{1}{2}\right)^2 = -2 - \frac{1}{4} = -\frac{9}{4}$.
- (3) $\frac{x + y}{x - y} = \frac{-2 + \frac{1}{2}}{-2 - \frac{1}{2}} = \frac{-\frac{3}{2}}{-\frac{5}{2}} = \frac{3}{5}$.

Parentheses When Substituting Negative Numbers

When $x = -2$,

$$x^2 = (-2)^2 = 4, \quad -x^2 = -(-2)^2 = -4.$$

The incorrect writing $x^2 = -2^2 = -4$ changes the base from x to 2 and loses the meaning of substituting the whole value.

1.4.4 Like Terms and Combining Like Terms (同类项与合并同类项)

Terms that contain the same letters, with the same exponents on the corresponding letters, are called **like terms (同类项)**. Constant terms are also like terms with other constant terms.

You can first understand like terms as "the same kind of object". $3x + 5x$ is like 3 apples plus 5 apples, so it can be combined into $8x$. But $3x + 5y$ is like apples and pears; they cannot be combined merely because their coefficients are numbers.

When combining like terms, only add or subtract the coefficients; the letters and their exponents stay unchanged:

$$ax^m + bx^m = (a + b)x^m.$$

$3x^2$ and $-5x^2$ are like terms. $3x^2$ and $3x$ are not like terms. $2ab$ and $2a^2b$ are not like terms either.

1.4.5 Removing Parentheses (去括号)

The basis for removing parentheses is the distributive property:

$$a(b + c) = ab + ac.$$

When a parenthesis is preceded by a plus sign, the signs of all terms remain unchanged after removing the parentheses. When a parenthesis is preceded by a minus sign, it can be viewed as multiplication by -1 , and every term inside the parentheses must change sign.

First look at two smallest examples:

$$-(a - b) = -a + b, \quad -2(a - b) = -2a + 2b.$$

The first expression only handles the negative sign. The second expression handles both the coefficient 2 and the negative sign. Do not write $-2(a - b)$ as $-2a - b$, because -2 must also multiply $-b$.

Common Mistake

When removing parentheses preceded by a negative sign, you must change the sign of **every term** inside the parentheses, not only the first term.

Example 1.10 Combining Like Terms and Removing Parentheses

Simplify:

$$(1) \quad 3x - 2 + 5x + 7;$$

$$(2) \ 2(3x - 1) - 3(x + 4);$$

$$(3) \ a - [2a - 3(a - 2b) + b].$$

Solution:

$$(1) \ 3x - 2 + 5x + 7 = (3x + 5x) + (-2 + 7) = 8x + 5.$$

$$(2) \ 2(3x - 1) - 3(x + 4) = 6x - 2 - 3x - 12 = 3x - 14.$$

$$(3)$$

$$\begin{aligned} a - [2a - 3(a - 2b) + b] &= a - [2a - 3a + 6b + b] \\ &= a - [-a + 7b] = 2a - 7b. \end{aligned}$$

In-Class Practice 1.4.1 Algebraic Expression Basics

(1) Identify the terms, the coefficient of each term, the constant term and the degree of $5x^3 - 2x + 9$.

(2) When $a = -1, b = 3$, find the value of $2a^2 - ab + b$.

(3) Combine like terms: $7x^2 - 3x + 5 - 2x^2 + 8x - 9$.

(4) Remove parentheses and simplify: $4(2x - 3) - 2(3x + 1)$.

(5) Remove parentheses and simplify: $3a - \{2a - [a - 2(a - b)]\}$.

(6) Determine whether $2x^2y$ and $-5xy^2$ are like terms, and explain why.

1.5 Polynomials and Factorization (多项式与因式分解)

Expansion changes multiplication into addition. Factorization rewrites an additive form as a product. These two processes go in opposite directions.

1.5.1 Polynomial Operations (多项式的基本运算)

The core of polynomial addition and subtraction is removing parentheses and combining like terms. When multiplying polynomials, use the distributive property so that every term of one factor multiplies every term of the other factor.

A table makes it clear that "every term must multiply every term":

$\times$	x	5
$2x$	$2x^2$	$10x$
-3	$-3x$	-15

Therefore, $(2x - 3)(x + 5) = 2x^2 + 10x - 3x - 15$. If one of the four boxes in the table is missing, the expansion has missed a term.

Polynomial Multiplication Checklist

- Has every term been multiplied?
- Are the negative signs correct?
- When powers with the same base are multiplied, have the exponents been added?
- Have like terms been combined at the end?

Example 1.11 Polynomial Operations

Expand and simplify:

(1) $(2x - 3)(x + 5)$;

(2) $(a + 2b)(3a - b)$;

(3) $(x - 1)(x^2 + x + 1)$.

Solution:

(1) $(2x - 3)(x + 5) = 2x^2 + 10x - 3x - 15 = 2x^2 + 7x - 15$.

(2) $(a + 2b)(3a - b) = 3a^2 - ab + 6ab - 2b^2 = 3a^2 + 5ab - 2b^2$.

(3) $(x - 1)(x^2 + x + 1) = x^3 + x^2 + x - x^2 - x - 1 = x^3 - 1$.

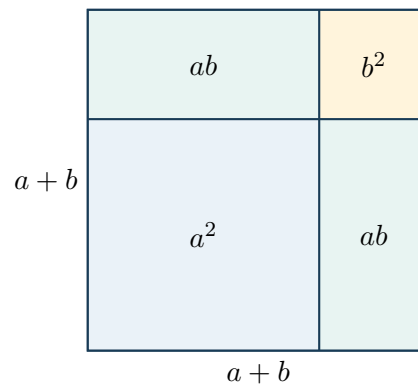
1.5.2 Common Multiplication Formulas (常用乘法公式)

The following three formulas should be mastered fluently:

Three Common Formulas

$$\begin{aligned}(a + b)^2 &= a^2 + 2ab + b^2, \\(a - b)^2 &= a^2 - 2ab + b^2, \\(a + b)(a - b) &= a^2 - b^2.\end{aligned}$$

The first two are called the perfect square formulas. The third is called the difference of squares formula. In the formulas, a, b may be numbers, letters, or complete algebraic expressions.



The diagram shows that the area of a square with side length $a + b$ can be divided into four parts:
 a^2, ab, ab, b^2 .

The difference of squares formula can also be understood by area. Remove a small square of area b^2 from a large square of area a^2 . The remaining part can be rearranged into a rectangle of length $a + b$ and width $a - b$, so

$$a^2 - b^2 = (a + b)(a - b).$$

Common Mistake

The middle term of $(a + b)^2$ is $2ab$. The most common mistake is to omit the middle term and incorrectly write $a^2 + b^2$.

1.5.3 What Factorization Is (什么是因式分解)

Factorization

Writing a polynomial as a product of several polynomials is called factoring the polynomial.

The motivation for factorization is to rewrite a complicated ``sum'' as a ``product'' that is easier to use. In the next chapter, when solving equations, if we can write

$$x^2 - 5x + 6 = 0$$

as

$$(x - 2)(x - 3) = 0,$$

then a quadratic problem can be split into two linear problems.

Factorization must produce a product form. For example, $x(x + 2)$ is a product form, while $x^2 + 2x$ is still a sum form. Factorization and expansion are inverse processes, so you can check an answer by expanding it again.

What Is Not Factorization

$$x^2 + 2x + 1 = x(x + 2) + 1$$

is not factorization, because the right side is still ``a product plus 1'', not a product of several factors as a whole. The final form of factorization must be readable as ``factor $\times$ factor $\times \dots$ ''.

1.5.4 Factoring Out a Common Factor (提取公因式)

If every term of a polynomial contains the same factor, use the distributive property in reverse to factor out the common factor:

$$ma + mb = m(a + b).$$

Usually, first factor out the greatest common divisor of the numerical coefficients, and then factor out the common letters with their lowest powers.

For example, the common factor of $12x^4y^2$ and $18x^2y^5$ can be found as follows:

	$12x^4y^2$	$18x^2y^5$
numerical part	12	18
power of x	4	2
power of y	2	5

For the numerical part, take the greatest common divisor 6. For the letters, take the lowest powers that appear in both terms, x^2y^2 . Therefore the common factor is $6x^2y^2$.

Example 1.12 Factoring Out a Common Factor (让一个因式)

Factor:

(1) $6x^2 - 9x$;

(2) $8a^3b^2 + 12a^2b^3$;

(3) $-3x^3 + 6x^2 + 9x$.

Solution:

(1) $6x^2 - 9x = 3x(2x - 3)$.

(2) $8a^3b^2 + 12a^2b^3 = 4a^2b^2(2a + 3b)$.

(3) $-3x^3 + 6x^2 + 9x = -3x(x^2 - 2x - 3) = -3x(x + 1)(x - 3)$.

1.5.5 Formula Method (公式法)

Using multiplication formulas from right to left gives the common factorization formulas:

Formula Method for Factorization

$$a^2 + 2ab + b^2 = (a + b)^2,$$

$$a^2 - 2ab + b^2 = (a - b)^2,$$

$$a^2 - b^2 = (a + b)(a - b).$$

If the terms first have a common factor, factor it out first, and then continue using formulas.

When recognizing a formula, first ask yourself: ``Who plays the role of a in the formula, and who

plays the role of b !" For example,

$$4x^2 + 12x + 9 = (2x)^2 + 2 \cdot (2x) \cdot 3 + 3^2,$$

where a is $2x$ and b is 3 , so the result is $(2x + 3)^2$.

Example 1.13 Recognizing Formula Structure

Factor:

(1) $x^2 - 16$;

(2) $4x^2 + 12x + 9$;

(3) $3x^3 - 27x$.

Factor until no further factorization is possible.

Solution:

(1) $x^2 - 16 = (x + 4)(x - 4)$.

(2) $4x^2 + 12x + 9 = (2x + 3)^2$.

(3) $3x^3 - 27x = 3x(x^2 - 9) = 3x(x + 3)(x - 3)$.

1.5.6 Simple Quadratic Trinomials (简单二次三项式)

For a quadratic trinomial with leading coefficient 1,

$$x^2 + px + q,$$

if we can find two numbers m, n satisfying

$$m + n = p, \quad mn = q,$$

then

$$x^2 + px + q = (x + m)(x + n).$$

For a general expression $ax^2 + bx + c$, you can also try this idea, and you can first look for a common factor. More systematic methods for quadratic equations will be studied in the next chapter.

When looking for m, n , first make a small table. For example, to factor $x^2 + 7x + 12$, we need two numbers whose product is 12 and whose sum is 7:

m, n	mn	$m + n$
1, 12	12	13
2, 6	12	8
3, 4	12	7

Therefore choose 3, 4.

Recommended Order for Factorization

- (1) First check whether there is a common factor.
- (2) Then check whether the expression matches the difference of squares or a perfect square formula.
- (3) Then consider the two numbers in a quadratic trinomial.
- (4) After each step, check whether further factorization is possible.
- (5) Finally expand to check.

Example 1.14 Quadratic Trinomials

Factor:

- (1) $x^2 + 7x + 12$;
- (2) $x^2 - 5x - 14$;
- (3) $2x^2 + 7x + 3$.

Solution:

- (1) $x^2 + 7x + 12 = (x + 3)(x + 4)$.
- (2) $x^2 - 5x - 14 = (x - 7)(x + 2)$.
- (3) $2x^2 + 7x + 3 = (2x + 1)(x + 3)$.

When checking, expand the right side; it should return to the original expression in each case.

In-Class Practice 1.5.1 Expansion and Factorization

- (1) Expand: $(x - 4)(x + 6)$.
 - (2) Expand: $(2a - 3b)^2$.
 - (3) Factor: $5x^2 - 10x$.
 - (4) Factor: $9a^2 - 25b^2$.
 - (5) Factor: $x^2 - 10x + 25$.
 - (6) Factor: $x^2 + x - 20$.
 - (7) Factor: $3x^3 - 12x$.
 - (8) Determine whether $x^2 + 4$ can be factored by the difference of squares formula over the real numbers, and explain why.
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1.6 Rational Expressions (分式)

Rational expressions extend fraction operations with integers to expressions containing letters. When working with rational expressions, the first step is not computation, but checking the denominator.

1.6.1 Rational Expressions and the Range of Values for Which They Are Meaningful (分式与有意义的取值范围)**Rational Expression**

An expression of the form

$$\frac{A}{B}$$

is called a rational expression, where A, B are integral expressions and B contains letters. The necessary condition for the rational expression to be meaningful is $B \neq 0$.

All variable values that make a rational expression meaningful are called the **range of values for which the rational expression is meaningful (有意义的取值范围)**. When finding this range, make sure every denominator is not 0. In Chapter 3, when studying functions, a similar input range will be formally called the domain.

Example 1.15 Finding Where a Rational Expression Is Meaningful

Find the values for which the following rational expressions are meaningful:

(1) $\frac{2x+1}{x-3}$;

(2) $\frac{x}{x^2-4}$;

(3) $\frac{x+1}{(x-2)(x^2+1)}$.

Solution: The condition for a rational expression to be meaningful is that the denominator is not 0.

(1) $x-3 \neq 0 \Rightarrow x \neq 3$.

(2) $x^2-4 = (x-2)(x+2) \neq 0 \Rightarrow x \neq 2, -2$.

(3) $(x-2)(x^2+1) \neq 0$. Over the real numbers, $x^2+1 > 0$ always holds, so we only need $x \neq 2$.

1.6.2 Basic Property of Rational Expressions and Reduction (分式的基本性质与约分)

If the numerator and denominator of a rational expression are both multiplied or divided by the same nonzero integral expression, the value of the rational expression does not change:

$$\frac{A}{B} = \frac{AC}{BC} \quad (B \neq 0, C \neq 0).$$

Before reducing, usually factor both the numerator and the denominator. What is canceled is a common **factor (因式)**, not a term inside an addition.

Common Mistake

When

$$\frac{x^2-1}{x-1}$$

is reduced to $x+1$, the original rational expression still requires $x \neq 1$. Reduction can change the appearance of an expression, but it cannot add back values that were not allowed in the original range of meaningful values.

Example 1.16 Reducing Rational Expressions

Reduce, and state the restrictions on the variables in the original rational expression:

(1) $\frac{6x^2y}{9xy^2}$;

(2) $\frac{x^2-9}{x^2-3x}$;

(3) $\frac{x^2+5x+6}{x^2-4}$.

Solution:

$$(1) \frac{6x^2y}{9xy^2} = \frac{2x}{3y}, \quad x \neq 0, y \neq 0.$$

$$(2) \frac{x^2 - 9}{x^2 - 3x} = \frac{(x-3)(x+3)}{x(x-3)} = \frac{x+3}{x}, \quad x \neq 0, 3.$$

$$(3) \frac{x^2 + 5x + 6}{x^2 - 4} = \frac{(x+2)(x+3)}{(x-2)(x+2)} = \frac{x+3}{x-2}, \quad x \neq 2, -2.$$

Variable restrictions must come from the original denominator. A restriction that disappears after reduction cannot be deleted.

1.6.3 Multiplication and Division of Rational Expressions (分式的乘除)

When multiplying rational expressions, multiply numerator by numerator and denominator by denominator. When dividing rational expressions, multiply by the reciprocal of the divisor. In actual computation, we usually factor first, reduce second, and multiply last.

Multiplication and Division of Rational Expressions

$$\begin{aligned} \frac{A}{B} \cdot \frac{C}{D} &= \frac{AC}{BD}, \\ \frac{A}{B} \div \frac{C}{D} &= \frac{A}{B} \cdot \frac{D}{C} = \frac{AD}{BC}. \end{aligned}$$

All denominators and the divisor must be nonzero.

1.6.4 Addition, Subtraction and Common Denominators (分式的加减与通分)

For rational expressions with the same denominator, add or subtract the numerators directly. For different denominators, you must first find a common denominator. The least common denominator is usually formed by factoring each denominator and taking the highest power of each distinct factor.

Addition and Subtraction of Rational Expressions

$$\begin{aligned} \frac{A}{B} + \frac{C}{B} &= \frac{A+C}{B}, \\ \frac{A}{B} + \frac{C}{D} &= \frac{AD+BC}{BD}. \end{aligned}$$

Example 1.17 The Four Operations with Rational Expressions

Simplify, and state the variable restrictions:

$$(1) \frac{2}{x} + \frac{3}{2x};$$

$$(2) \frac{1}{x-1} - \frac{2}{x+1};$$

$$(3) \frac{x^2 - 4}{x^2 - x - 2} \cdot \frac{x-2}{x+2};$$

$$(4) \frac{x}{x-1} \div \frac{x^2}{x^2-1}.$$

Solution:

$$(1) \frac{2}{x} + \frac{3}{2x} = \frac{4}{2x} + \frac{3}{2x} = \frac{7}{2x}, \quad x \neq 0.$$

$$(2) \frac{1}{x-1} - \frac{2}{x+1} = \frac{x+1-2(x-1)}{(x-1)(x+1)} = \frac{3-x}{(x-1)(x+1)}, \quad x \neq \pm 1.$$

$$(3) \frac{x^2-4}{x^2-x-2} \cdot \frac{x-2}{x+2} = \frac{(x-2)(x+2)}{(x-2)(x+1)} \cdot \frac{x-2}{x+2} = \frac{x-2}{x+1},$$

where $x \neq 2, -1, -2$.

$$(4) \frac{x}{x-1} \div \frac{x^2}{x^2-1} = \frac{x}{x-1} \cdot \frac{(x-1)(x+1)}{x^2} = \frac{x+1}{x},$$

where $x \neq 1, 0, -1$.

Standard Process for Simplifying Rational Expressions

- (1) First write all restrictions that denominators are not 0.
- (2) Factor all numerators and denominators that can be factored.
- (3) For multiplication and division, reduce first; for addition and subtraction, find common denominators first.
- (4) Complete the operation and reduce again.
- (5) Keep the variable restrictions from the original expression.

In-Class Practice 1.6.1 Rational Expression Basics

- (1) Find the values for which $\frac{1}{x+5}$ is meaningful.
 - (2) Find the values for which $\frac{x-1}{x^2-5x+6}$ is meaningful.
 - (3) Reduce: $\frac{4a^2b}{6ab^2}$.
 - (4) Reduce: $\frac{x^2-4x}{x^2-16}$.
 - (5) Compute: $\frac{3}{x} + \frac{2}{x-1}$.
 - (6) Compute: $\frac{x^2-1}{x^2+x} \cdot \frac{x}{x-1}$.
 - (7) Compute: $\frac{x}{x+2} \div \frac{x^2}{x^2-4}$.
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1.7 Chapter Summary**Main Knowledge Thread**

- (1) **Basic computation** must pay attention to values, signs, denominators and order of operations at the same time.
- (2) **Exponents and radicals** are connected operations; check conditions before using formulas.
- (3) **Algebraic expressions** use letters to represent quantities; simplification depends on the distributive property and combining like terms.
- (4) **Expansion** writes a product as a sum, while **factorization** writes a sum as a product.
- (5) For a **rational expression**, the first step is to determine that the denominator is not 0; reduction can only cancel common factors.

Examples of Common Mistakes

- ▶ When substituting a negative number into a power, use parentheses: when $x = -2$, $x^2 = (-2)^2 = 4$, not $-2^2 = -4$.
- ▶ Terms with different letters cannot be combined: $2x + 3x = 5x$, but $2x + 3y$ cannot be combined.
- ▶ Reduction can only cancel common factors: $\frac{x+1}{x}$ cannot be reduced to $1 + 1$.

Formula List for This Chapter

$$a - b = a + (-b),$$

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd},$$

$$a^m a^n = a^{m+n}, \quad (a^m)^n = a^{mn},$$

$$a^{-n} = \frac{1}{a^n}, \quad \sqrt{a^2} = |a|,$$

$$a(b + c) = ab + ac,$$

$$(a + b)^2 = a^2 + 2ab + b^2,$$

$$(a - b)^2 = a^2 - 2ab + b^2,$$

$$(a + b)(a - b) = a^2 - b^2,$$

$$\frac{A}{B} + \frac{C}{D} = \frac{AD + BC}{BD}.$$

1.8 Chapter Exercises

How to Use These Exercises

Group A checks the minimum mastery for this chapter. If you make mistakes, return to the corresponding section and reread the examples. Group B is for integrated practice before a quiz. Group C contains extension problems; during a first self-study pass, you may skip it without affecting your entry into the next chapter.

Group A: Basic Check

Difficulty: Basic

- (1) Connect the following numbers with $<$: -3 , 1 , $-\frac{5}{2}$, 0 , $\frac{1}{3}$.
- (2) Compute: $|-6| + |2 - 7| - |-3|$.
- (3) Compute: $-12 + 5 - (-8) - 9$.
- (4) Compute: $(-4) \times (-3) \div 6$.
- (5) Compute: $\frac{2}{3} - \frac{5}{6} + \frac{1}{4}$.
- (6) Compute: $1 - \frac{3}{2 - \frac{1}{2}}$.
- (7) Write 0.045 as a fraction in lowest terms and as a percentage.
- (8) An item is discounted from 800 yuan to 680 yuan. Find the percentage decrease.
- (9) Fill in the laws of exponents:
 $a^m a^n =$ _____, $\frac{a^m}{a^n} =$ _____, $(a^m)^n =$ _____, $a^{-n} =$ _____.
- (10) Simplify: $3^4 \times 3^{-2}$, $\frac{x^8}{x^3}$, $(2a^2b)^3$.
- (11) Simplify: $\sqrt{32}$, $\sqrt{75}$, $\sqrt{12a^2}$.
- (12) Write 72 500 000 and 0.00000408 in scientific notation.
- (13) When $x = -2$, $y = 3$, find the value of $2x^2 - xy + y$.
- (14) Combine like terms: $4x^2 - 3x + 7 - 9x^2 + 5x - 2$.
- (15) Remove parentheses and simplify: $3(2x - 5) - 2(x + 4)$.
- (16) Expand: $(x + 5)(x - 2)$.
- (17) Expand: $(2x - 3)^2$.
- (18) Factor: $6x^2 - 15x$.
- (19) Factor: $x^2 - 49$.

(20) Factor: $x^2 + 8x + 16$.

(21) Factor: $x^2 - 3x - 10$.

(22) Find the values for which $\frac{x+2}{x^2-9}$ is meaningful.

(23) Reduce $\frac{x^2-9}{x^2+4x+3}$, and state the variable restrictions.

Group B: Quiz Level

Difficulty: Core

(1) Given $a = -\frac{1}{2}$, $b = 3$, compute

$$\frac{2a^2 - ab}{a - b}.$$

(2) Arrange the following numbers from least to greatest:

$$-\sqrt{5}, \quad -2.1, \quad -\frac{9}{4}, \quad 0, \quad \sqrt{4}.$$

(3) Compute and simplify:

$$\left(\frac{27}{8}\right)^{-2/3} + 16^{3/4} - \sqrt{49}.$$

(4) Assuming all letters are positive, simplify:

$$\frac{(2x^2y^{-1})^3}{4x^{-1}y^{-2}}.$$

(5) Simplify:

$$2x - \{3x - [2 - (x - 4)]\}.$$

(6) The length of a rectangle is $2x + 3$, and its width is $x - 1$.

- Write algebraic expressions for its perimeter and area.
- Simplify the expressions obtained.

c. State the restrictions that x must satisfy.

(7) Expand and simplify:

$$(2x - 1)(x + 3) - (x - 2)^2.$$

(8) Factor:

$$4x^3 - 36x.$$

(9) Factor:

$$6x^2 + x - 2.$$

(10) Factor and state whether it can be factored further:

$$x^4 - 16.$$

(11) Simplify and state all variable restrictions:

$$\frac{x^2 - 4}{x^2 - x - 2} \cdot \frac{x^2 - 1}{x + 2}.$$

(12) Simplify and state all variable restrictions:

$$\frac{2}{x - 1} - \frac{x + 1}{x^2 - 1}.$$

Difficulty: Extension

Self-Check

For each item, choose ``Can do'', ``Mostly can'', or ``Not yet''.

Skill	Can do	Mostly can	Not yet
Determine which number set a number belongs to	☐	☐	☐
Compare real numbers on the number line	☐	☐	☐
Carry out mixed fraction operations correctly	☐	☐	☐
Use the laws of exponents	☐	☐	☐
Simplify radicals and handle absolute value	☐	☐	☐
Substitute into and simplify algebraic expressions	☐	☐	☐
Expand polynomials and use multiplication formulas	☐	☐	☐
Use the three basic methods of factorization	☐	☐	☐
Find where rational expressions are meaningful and perform operations	☐	☐	☐

Before Entering the Next Chapter

If you selected ``Not yet'' for fraction operations, removing parentheses, or factorization, it is recommended that you first complete and correct the corresponding Group A exercises before entering Chapter 2, Equations. These abilities will directly affect every step of solving equations.

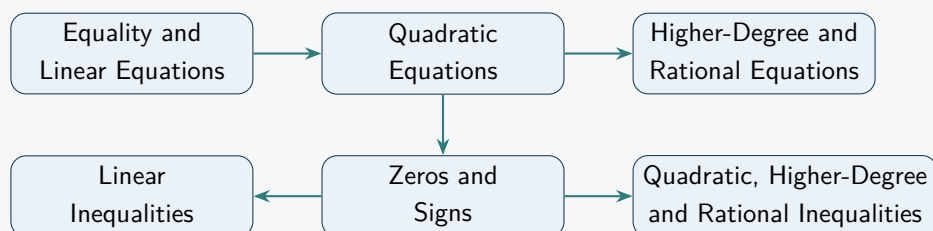
Equations and Inequalities

Learning Goals

After studying this chapter, you should be able to:

- ▶ understand the meaning of a solution of an equation, equivalent transformations, extraneous roots and lost roots;
- ▶ solve linear equations in one variable, systems of linear equations in two variables and linear equations with parameters;
- ▶ solve quadratic equations by factoring, completing the square and using the quadratic formula;
- ▶ use the discriminant to determine the number of real roots of a quadratic equation;
- ▶ handle simple higher-degree equations and rational equations by factoring or substitution;
- ▶ write the solution set of an inequality in inequality form;
- ▶ solve linear, quadratic, higher-degree and rational inequalities in one variable.

Knowledge Map



This chapter has two main threads: equations look for the points that make an expression equal to zero; inequalities determine whether an expression is positive or negative on the intervals separated by its zeros. Factorization connects these two threads.

Learning Route of This Chapter

This chapter studies two kinds of problems:

- (1) An equation asks: "Which numbers make the equality true after substitution?" When solving an equation, every step should keep the original equation and the new equation having the same solutions as much as possible.
- (2) An inequality asks: "Which numbers make the size relationship true after substitution?" For complicated inequalities, we usually first find the boundary points where the expression is 0 or undefined, and then determine signs on each interval.

When studying by yourself, pay special attention to two things: multiplying or dividing by an expression containing the unknown may change the solutions; candidate solutions obtained after clearing denominators must be checked in the original equation.

Prerequisites and Minimum Mastery

To study this chapter, you need Chapter 1 skills: removing parentheses, combining like terms, factoring, and restrictions for rational expressions. The minimum goal for the first pass is: solve linear equations without parameters, systems of equations and quadratic equations; write denominators as nonzero first in rational equations; solve already-factored inequalities using a number line or sign table.

Common Self-Study Sticking Points in This Chapter

Equations with parameters, higher-degree equations and rational inequalities belong to the second level. If you get stuck during the first reading, you may skip parameter discussions and higher-degree comprehensive problems first. Make sure that "standard equations, standard inequalities and standard checking" are fluent before returning to them.

2.1 Equality and Linear Equations in One Variable (等式与一元一次方程)

The goal of solving an equation is to isolate the unknown step by step. Each step should preserve the solution set.

2.1.1 Equations, Solutions and Equivalent Transformations (方程、解与同解变形)

An equality containing an unknown is called an **equation (方程)**. A value of the unknown that makes the two sides of the equation equal is called a **solution (解)** of the equation.

For example, $3x + 2 = 11$ is an equation, because it contains the unknown x , and the two sides are connected by an equal sign. Substituting $x = 3$, the left side is $3 \cdot 3 + 2 = 11$, exactly equal to the right side, so $x = 3$ is a solution of this equation. Substituting $x = 2$, the left side is 8, not equal to 11, so $x = 2$ is not a solution.

You can think of an equation as a balance scale: the left side and the right side were originally equal; as long as exactly the same legal operation is performed on both sides, the balance will not be broken. The collection of all solutions is called the solution set of the equation. Solving an equation

means changing a complicated equation into a simpler equation with the same solution set.



Subtract 2 from both sides and then divide both sides by 3; the balance is unchanged.

Basic Properties of Equality

If $a = b$, then:

- ▶ $a + c = b + c$;
- ▶ $ac = bc$;
- ▶ when $c \neq 0$, $\frac{a}{c} = \frac{b}{c}$.

Adding or subtracting the same expression on both sides of an equation, or multiplying or dividing both sides by the same nonzero number, gives an equivalent equation.

Common Mistake

Multiplying both sides of an equation by an expression that may equal zero may add solutions. Dividing both sides by an expression containing the unknown may lose solutions. Extra solutions are called extraneous roots, and solutions that are divided away are called lost roots. Whenever multiplication or division involves an unknown expression, you must first discuss whether that expression can be zero.

For example, the equation

$$x(x - 1) = 0$$

originally has two solutions, $x = 0$ or $x = 1$. If you directly divide both sides by x , you get $x - 1 = 0$, leaving only $x = 1$, and $x = 0$ is lost. The problem is that x may equal 0, so we cannot treat it as nonzero and divide by it without discussion.

Safety Table for Equivalent Transformations

Adding or subtracting the same expression on both sides is safe. Multiplying both sides by a known nonzero number is safe. Dividing both sides by an expression containing the unknown is risky. After squaring both sides or clearing denominators, candidate solutions must be checked.

2.1.2 Linear Equations in One Variable (一元一次方程)

An equation that contains only one unknown and whose highest degree after simplification is 1 can be written as

$$ax + b = 0.$$

When $a \neq 0$, its unique solution is $x = -b/a$.

What a Linear Equation in One Variable Looks Like

One variable means there is only one unknown. **Linear** means the highest degree of the unknown is 1. For example,

$$5x - 7 = 3, \quad 2(x - 1) = x + 4$$

are both linear equations in one variable. $x^2 - 1 = 0$ is not a linear equation because the highest degree is 2. $x + y = 3$ is not an equation in one variable because it has two unknowns.

This result comes from the properties of equality:

$$ax + b = 0 \Rightarrow ax = -b \Rightarrow x = -\frac{b}{a} \quad (a \neq 0).$$

If $a = 0$, we cannot divide by a , so equations with parameters must discuss separately whether the coefficient is zero.

Standard Steps for Solving a Linear Equation

- (1) Clear denominators.
- (2) Remove parentheses.
- (3) Move terms.
- (4) Combine like terms.
- (5) Make the coefficient of the unknown equal to 1.
- (6) Substitute back into the original equation to check.

Example 2.1 Linear Equations in One Variable

Solve:

$$(1) \quad 3(2x - 1) - 2(x + 4) = 5 - x;$$

$$(2) \quad \frac{x - 1}{3} - \frac{2x + 1}{4} = 1.$$

Solution:

(1)

$$3(2x - 1) - 2(x + 4) = 5 - x$$

$$6x - 3 - 2x - 8 = 5 - x$$

$$4x - 11 = 5 - x$$

$$5x = 16$$

$$x = \frac{16}{5}.$$

(2) Multiply both sides by 12:

$$4(x - 1) - 3(2x + 1) = 12.$$

Simplifying gives $4x - 4 - 6x - 3 = 12$, so $-2x - 7 = 12$, and $x = -\frac{19}{2}$.

When checking, substitute the obtained x back into the original equation; the two sides should be equal. Although a linear equation usually does not produce extraneous roots, forming the habit of checking directly helps with rational equations and squaring transformations later.

2.1.3 Linear Equations with Parameters (含参数的一次方程)

A parameter is a letter whose value is temporarily uncertain. For the equation $ax = b$, we cannot directly write $x = b/a$, because a may be zero.

During the first study, you may first think of a parameter as "a specific number that has not yet been told to you". For example, $2x = 6$ has a unique solution, $0x = 6$ has no solution, and $0x = 0$ has infinitely many solutions. An equation with parameters simply writes these three cases uniformly with letters.

For example, in $(m - 1)x = 2$, x is the unknown and m is the parameter. The goal is not to find one fixed number, but to answer: for which values of m is there a unique solution? When is there no solution? When are there infinitely many solutions?

Solutions of the Equation $ax = b$

Condition	Equation	Solution Situation
$a \neq 0$	$ax = b$	Unique solution $x = b/a$
$a = 0, b \neq 0$	$0x = b$	No solution
$a = 0, b = 0$	$0x = 0$	Every real number is a solution

Example 2.2 Equation with a Parameter

Discuss how many solutions the equation

$$(m - 2)x = m^2 - 4$$

has for different values of m , and write the solution when it has a unique solution.

Solution: First factor the right side:

$$m^2 - 4 = (m - 2)(m + 2).$$

When $m \neq 2$, divide both sides of the equation by $m - 2$, giving the unique solution

$$x = m + 2.$$

When $m = 2$, the original equation becomes $0 \cdot x = 0$. Every real number x is a solution, so there are infinitely many solutions.

In-Class Practice 2.1.1 Linear Equations

- (1) Solve: $5x - 7 = 3x + 9$.
 - (2) Solve: $2[3 - (x - 1)] = x + 5$.
 - (3) Solve: $\frac{2x - 1}{6} + \frac{x + 2}{4} = 1$.
 - (4) If $x = 3$ is a solution of $ax - 5 = 7$, find a .
 - (5) Discuss the solution situation of $(k + 1)x = k + 1$.
-
-
-
-
-

2.2 Systems of Linear Equations in Two Variables (二元一次方程组)

A system of equations requires the same group of unknowns to satisfy all equations at the same time. Elimination is the most basic method.

A linear equation in two variables is a linear equation containing two unknowns x, y , such as $x + y = 3$ or $3x - y = 5$. Solving a system of linear equations in two variables means finding values of x, y that satisfy two linear equations at the same time. For example, $x = 2, y = 1$ is a solution of the system

$$\begin{cases} x + y = 3 \\ 3x - y = 5 \end{cases}$$

The basic method for solving a system of linear equations in two variables is elimination.

System of Linear Equations in Two Variables

A single linear equation in two variables usually has infinitely many solutions. For example, in $x + y = 3$, $(0, 3), (1, 2), (2, 1)$ all satisfy it. Putting two or more linear equations in two variables together and requiring them to hold **at the same time** gives a system of linear equations in two variables. The solution of a system is usually written as an ordered pair (x, y) .

2.2.1 Substitution Method (代入消元法)

From one equation, express one unknown in terms of the other unknown, and then substitute it into the other equation. This changes a two-variable problem into a one-variable problem.

Example 2.3 Substitution Elimination

Solve the system:

$$\begin{cases} y = 2x - 1, \\ 3x + 2y = 12. \end{cases}$$

Solution: Substitute $y = 2x - 1$ into the second equation:

$$3x + 2(2x - 1) = 12.$$

Then $3x + 4x - 2 = 12$, so $7x = 14$, and $x = 2$. Substitute back into $y = 2x - 1$, giving $y = 3$. Therefore the solution of the system is

$$\begin{cases} x = 2, \\ y = 3. \end{cases}$$

2.2.2 Elimination Method (加减消元法)

Add or subtract the corresponding sides of two equations so that the coefficient of one unknown becomes zero.

Choosing an Elimination Method

If one unknown has already been isolated, use substitution first. If the coefficients of one unknown are the same, opposite, or easy to make the same, use addition/subtraction elimination first.

Example 2.4 Elimination

Solve the systems:

$$(1) \begin{cases} 2x + y = 6, \\ 3x - y = 4. \end{cases}$$

$$(2) \begin{cases} 3x + 2y = 7, \\ 5x - 3y = 4. \end{cases}$$

Solution:

(1) Add the two equations:

$$(2x + y) + (3x - y) = 6 + 4.$$

This gives $5x = 10$, so $x = 2$. Substituting into $2x + y = 6$, we get $y = 2$.

(2) To eliminate y , multiply the first equation by 3 and the second equation by 2:

$$\begin{cases} 9x + 6y = 21, \\ 10x - 6y = 8. \end{cases}$$

Adding them gives $19x = 29$, so $x = \frac{29}{19}$. Substitute into $3x + 2y = 7$:

$$2y = 7 - \frac{87}{19} = \frac{46}{19},$$

$$\text{so } y = \frac{23}{19}.$$

In-Class Practice 2.2.1 Systems and Applications

(1) Solve $\begin{cases} x + y = 8, \\ 2x - y = 1. \end{cases}$

(2) Solve $\begin{cases} 2x + 3y = 12, \\ 4x - y = 5. \end{cases}$

(3) The sum of two numbers is 17, and their difference is 5. Set up a system and find the two numbers.

(4) One adult ticket and two student tickets cost 110 yuan in total. Two adult tickets and three student tickets cost 190 yuan in total. Find the unit prices of the two kinds of tickets.

2.3 Quadratic Equations (一元二次方程)

The standard form of a quadratic equation in one variable is $ax^2 + bx + c = 0$, where $a \neq 0$. First convert to standard form, then choose a method.

Quadratic Equation in One Variable

A **quadratic equation in one variable** is a polynomial equation containing only one unknown, whose highest degree after simplification is 2. Its standard form is

$$ax^2 + bx + c = 0 \quad (a \neq 0).$$

Here ax^2 is the quadratic term, bx is the linear term, and c is the constant term. For example, $x^2 - 5x + 6 = 0$ and $2x^2 + x - 3 = 0$ are both quadratic equations in one variable. $x^3 - 1 = 0$ is not a quadratic equation, and $\frac{1}{x} + x = 2$ is not a polynomial quadratic equation.

When studying quadratic equations, the core question is: how do we find the x -values that make a quadratic expression equal to 0? Some problems can be factored directly, some are suitable for completing the square, and all quadratic equations can be handled uniformly by the quadratic formula.

Choosing a Method for Quadratic Equations

Method	Suitable Problems	Reminder
Factoring	Product equals zero can be seen, such as $x^2 - 5x + 6 = 0$	Do not try too long if factoring is difficult
Completing the square	You want to see a square structure, or derive the quadratic formula	Steps are longer when coefficients are complicated
Quadratic formula	Works for general quadratic equations and is most stable	Identify a, b, c accurately first
Discriminant	Determine the number of real roots first	It only determines the number; it does not replace complete solving

2.3.1 Factoring Method (因式分解法)

If $AB = 0$, then $A = 0$ or $B = 0$. Therefore, by factoring a quadratic polynomial into two linear factors, a quadratic equation can be changed into two linear equations.

This fact is called the zero product principle. For the product of two numbers to be 0, at least one factor must be 0; for example, $7 \cdot 0 = 0$ and $0 \cdot (-3) = 0$, but the product of two nonzero numbers cannot be 0.

For example,

$$x^2 - 5x + 6 = 0$$

does not directly show what x is; but if it is written as

$$(x - 2)(x - 3) = 0,$$

then it becomes "the product of two factors equals 0", so we only need to solve $x - 2 = 0$ and $x - 3 = 0$ separately.

Example 2.5 Factoring Method

Solve:

(1) $x^2 - 5x + 6 = 0$;

(2) $2x^2 + x - 3 = 0$;

(3) $4x^2 - 9 = 0$.

Solution:

$$(1) \quad x^2 - 5x + 6 = (x - 2)(x - 3) = 0, \text{ so } x = 2 \text{ or } 3.$$

$$(2) \quad 2x^2 + x - 3 = (2x + 3)(x - 1) = 0, \text{ so } x = -\frac{3}{2} \text{ or } 1.$$

$$(3) \quad 4x^2 - 9 = (2x - 3)(2x + 3) = 0, \text{ so } x = \frac{3}{2} \text{ or } -\frac{3}{2}.$$

2.3.2 Completing the Square (配方法)

Rewrite the quadratic expression into a perfect square form:

$$x^2 + px = \left(x + \frac{p}{2}\right)^2 - \left(\frac{p}{2}\right)^2.$$

The idea of completing the square is to turn a quadratic equation into "a square equals a number". Once we get

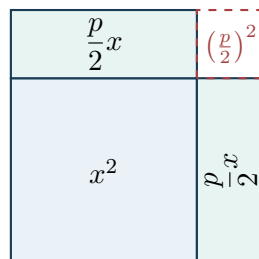
$$(x - h)^2 = k,$$

we can take square roots and obtain $x - h = \pm\sqrt{k}$. Therefore, the key to completing the square is to complete $x^2 + px$ into a full square.

First look at a smallest example:

$$x^2 + 6x = x^2 + 6x + 9 - 9 = (x + 3)^2 - 9.$$

Half of 6 is 3, so we add $3^2 = 9$. The "complete" in completing the square means adding the missing small piece and subtracting it back at the same time, so that the expression is not changed.



Add the upper-right corner to $x^2 + px$, turning it into a square of side $x + \frac{p}{2}$.

Example 2.6 Completing the Square

Solve $x^2 - 6x + 2 = 0$ by completing the square, and write $x^2 - 6x + 2$ in the form $a(x - h)^2 + k$.

Solution:

$$x^2 - 6x + 2 = (x^2 - 6x + 9) - 9 + 2 = (x - 3)^2 - 7.$$

The equation $x^2 - 6x + 2 = 0$ is equivalent to

$$(x - 3)^2 - 7 = 0,$$

that is, $(x - 3)^2 = 7$. Therefore

$$x = 3 \pm \sqrt{7}.$$

2.3.3 Quadratic Formula and Discriminant (求根公式与判别式)

The quadratic formula is a general tool for solving quadratic equations. It answers the question: for any standard form

$$ax^2 + bx + c = 0 \quad (a \neq 0),$$

can we write the roots directly using only a, b, c ?

For $ax^2 + bx + c = 0$ ($a \neq 0$), the discriminant is

$$\Delta = b^2 - 4ac.$$

The discriminant is the expression under the square root. If $\Delta > 0$, the square root is of a positive number, corresponding to two distinct real roots. If $\Delta = 0$, the square root is 0, and the two roots coincide. If $\Delta < 0$, the square root cannot be taken in the real number system, so there are no real roots.

Derivation of the Quadratic Formula

The quadratic formula is essentially completing the square for a general quadratic equation. Start from

$$ax^2 + bx + c = 0 \quad (a \neq 0).$$

Divide both sides by a :

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0.$$

Move the constant term to the right:

$$x^2 + \frac{b}{a}x = -\frac{c}{a}.$$

The left side should be completed into a perfect square. Since the coefficient of x in $x^2 + \frac{b}{a}x$ is $\frac{b}{a}$, we should add

$$\left(\frac{b}{2a}\right)^2.$$

Thus add this number to both sides:

$$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = -\frac{c}{a} + \left(\frac{b}{2a}\right)^2.$$

The left side becomes a square, and the right side is put over a common denominator:

$$\left(x + \frac{b}{2a}\right)^2 = -\frac{c}{a} + \frac{b^2}{4a^2} = \frac{-4ac + b^2}{4a^2} = \frac{b^2 - 4ac}{4a^2}.$$

If the right side is nonnegative, we can take square roots:

$$x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a}.$$

Finally move the term:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Therefore, $b^2 - 4ac$ under the square root determines whether real roots can be obtained. This is the discriminant.

Number of Real Roots (实根) and the Quadratic Formula

- $\Delta > 0$: two unequal real roots;
- $\Delta = 0$: two equal real roots;
- $\Delta < 0$: no real roots.

When $\Delta \geq 0$,

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}.$$

Common Mistake

Before using the quadratic formula, first confirm a, b, c in the standard form, especially negative signs. The denominator is the whole $2a$.

Seeing the Formula Through a Concrete Equation

Take $2x^2 - 8x + 6 = 0$ as an example. First divide by 2:

$$x^2 - 4x + 3 = 0.$$

Move the constant term and complete the square:

$$x^2 - 4x = -3, \quad x^2 - 4x + 4 = 1,$$

so

$$(x - 2)^2 = 1, \quad x = 2 \pm 1.$$

Therefore $x = 1$ or $x = 3$. If we apply the formula, $a = 2, b = -8, c = 6$, giving

$$x = \frac{8 \pm \sqrt{64 - 48}}{4} = \frac{8 \pm 4}{4},$$

and the result is still 1, 3. The formula simply writes this whole completing-the-square process at once.

Example 2.7 Discriminant and Quadratic Formula

Compute the discriminant and solve each equation:

- (1) $2x^2 + x + 1 = 0$;
- (2) $2x^2 - 8x + 8 = 0$;
- (3) $3x^2 + 5x + 1 = 0$.

Solution:

- (1) For $2x^2 + x + 1 = 0$, $\Delta = 1^2 - 4 \cdot 2 \cdot 1 = -7 < 0$, so there is no real solution.
- (2) For $2x^2 - 8x + 8 = 0$, $\Delta = (-8)^2 - 4 \cdot 2 \cdot 8 = 0$, so there are two equal real roots:

$$x = \frac{-(-8)}{2 \cdot 2} = 2.$$

- (3) For $3x^2 + 5x + 1 = 0$, $\Delta = 5^2 - 4 \cdot 3 \cdot 1 = 13$, so

$$x = \frac{-5 \pm \sqrt{13}}{6}.$$

2.3.4 Vieta's Formulas (根与系数的关系)

Vieta's formulas study this question: if we do not want to find the roots completely, can we directly know ``the sum of the two roots'' and ``the product of the two roots''? For example, the roots

of $x^2 - 5x + 6 = 0$ are 2, 3, their sum is 5, and their product is 6. This is exactly reflected in the coefficient -5 and the constant term 6.

If x_1, x_2 are the two roots of $ax^2 + bx + c = 0$, then

$$x_1 + x_2 = -\frac{b}{a}, \quad x_1x_2 = \frac{c}{a}.$$

This relationship comes from the factored form. If the two roots of a quadratic equation are x_1, x_2 , then the corresponding quadratic expression can be written as

$$a(x - x_1)(x - x_2).$$

Expanding gives

$$a(x^2 - (x_1 + x_2)x + x_1x_2) = ax^2 - a(x_1 + x_2)x + ax_1x_2.$$

Comparing coefficients with $ax^2 + bx + c$, we get

$$-a(x_1 + x_2) = b, \quad ax_1x_2 = c,$$

which is Vieta's formulas.

We can also view it in reverse: if the two roots are 2 and 3, then the corresponding equation with leading coefficient 1 is

$$(x - 2)(x - 3) = 0.$$

Expanding gives $x^2 - 5x + 6 = 0$. The sum of roots $2 + 3 = 5$ becomes the opposite of the linear coefficient, and the product $2 \cdot 3 = 6$ becomes the constant term.

Example 2.8 Computing Without Directly Finding the Roots

Let x_1, x_2 be the two roots of $2x^2 - 3x - 1 = 0$. Without directly finding the roots, compute

(1) $x_1 + x_2$;

(2) x_1x_2 ;

(3) $x_1^2 + x_2^2$;

(4) $\frac{1}{x_1} + \frac{1}{x_2}$.

Solution: By Vieta's formulas,

$$x_1 + x_2 = -\frac{-3}{2} = \frac{3}{2}, \quad x_1x_2 = \frac{-1}{2} = -\frac{1}{2}.$$

(1) $x_1 + x_2 = \frac{3}{2}$.

(2) $x_1x_2 = -\frac{1}{2}$.

(3)
$$x_1^2 + x_2^2 = (x_1 + x_2)^2 - 2x_1x_2 = \left(\frac{3}{2}\right)^2 - 2\left(-\frac{1}{2}\right) = \frac{9}{4} + 1 = \frac{13}{4}.$$

(4) Since $x_1x_2 \neq 0$,

$$\frac{1}{x_1} + \frac{1}{x_2} = \frac{x_1 + x_2}{x_1x_2} = \frac{\frac{3}{2}}{-\frac{1}{2}} = -3.$$

In-Class Practice 2.3.1 Quadratic Equations

- (1) Solve $x^2 + x - 12 = 0$ by factoring.
- (2) Solve $x^2 + 4x - 1 = 0$ by completing the square.
- (3) Solve $3x^2 - 2x - 2 = 0$ using the quadratic formula.
- (4) Determine the number of real roots of $x^2 - 4x + 5 = 0$.
- (5) The equation $x^2 - kx + 1 = 0$ in x has two unequal real roots. Find the range of k .
- (6) Given that the sum of two roots is 5 and the product is 6, write a quadratic equation with leading coefficient 1.

2.4 Higher-Degree Equations and Substitution Method

Higher-degree equations do not have one simple formula suitable for all cases. In this course, we focus on factoring, trial roots and substitution.

Higher-Degree Equation

In these notes, a **higher-degree equation** means a polynomial equation in one variable whose highest degree is greater than 2. For example,

$$x^3 - 2 = 0, \quad x^4 - 5x^2 + 4 = 0$$

are both higher-degree equations. They still ask "which x -values make the equality true after substitution", but their highest degree is greater than two, so the quadratic formula can no longer be used directly.

At the high-school preparatory stage, we do not pursue a universal method for all higher-degree equations. We only handle problems with visible structure: factor first if possible, try simple roots first if possible, and use substitution when the same whole expression appears repeatedly.

2.4.1 Factoring and Trial Roots (因式分解与试根)

The trial-root method is an introductory method for handling some higher-degree equations. Its idea is very simple: if you suspect that a number r is a solution of $P(x) = 0$, substitute $x = r$ to check. Once $P(r) = 0$, it means $x - r$ is a factor, and the degree of the equation can be reduced by one.

For example, for

$$P(x) = x^3 - 6x^2 + 11x - 6,$$

substituting $x = 1$ gives $P(1) = 0$, so $x = 1$ is a root and $x - 1$ is a factor. Then the cubic can be factored into a linear factor times a quadratic factor.

If a polynomial $P(x)$ satisfies $P(r) = 0$, then $x - r$ is a factor of $P(x)$. For a monic polynomial with integer coefficients, first try the positive and negative factors of the constant term.

Before trying roots, observe in a fixed order:

Factor out common factors? $\longrightarrow$ Group terms? $\longrightarrow$ Try small integer roots?

This avoids random guessing when there is no direction.

Trial Roots: Which Roots Should Be Guessed First?

Trial roots are not random guesses. If the integer-coefficient polynomial

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

has a rational root $\frac{p}{q}$, where p, q are coprime, then the possible root must satisfy

$$p \mid a_0, \quad q \mid a_n.$$

That is, the numerator can only be chosen from factors of the constant term a_0 , and the denominator can only be chosen from factors of the leading coefficient a_n .

In particular, if the polynomial is **monic**, meaning the leading coefficient $a_n = 1$, then any rational root can only be a positive or negative factor of the constant term. When trying roots, usually try

$$\pm 1, \pm 2, \pm 3, \dots$$

in this order, starting with numbers of smaller absolute value.

Common Mistake

If the constant term $a_0 = 0$, try $x = 0$ first. At this time every term of the polynomial has a factor x , so factor out x first, and then handle the remaining equation.

Example 2.9 Cubic Equations

Solve:

(1) $x^3 - 6x^2 + 11x - 6 = 0;$

(2) $x^3 - 2x^2 + x - 2 = 0.$

First list the integer roots that can be tried, and then complete the factorization.

Solution:

- (1) Possible integer roots are $\pm 1, \pm 2, \pm 3, \pm 6$. Substitution shows that $x = 1$ is a root, so

$$x^3 - 6x^2 + 11x - 6 = (x - 1)(x^2 - 5x + 6) = (x - 1)(x - 2)(x - 3).$$

Therefore $x = 1, 2, 3$.

- (2) Possible integer roots are $\pm 1, \pm 2$. Substitution shows that $x = 2$ is a root, so

$$x^3 - 2x^2 + x - 2 = (x - 2)(x^2 + 1).$$

Over the real numbers, $x^2 + 1 = 0$ has no solution, so the real solution is $x = 2$.

2.4.2 Substitution Method (换元法)

When the same whole expression appears repeatedly in an equation, we can set that whole expression as a new unknown.

The prerequisite for substitution is that "the same whole expression appears repeatedly". For example, in $(x^2 + x)^2 - 5(x^2 + x) + 6 = 0$, $x^2 + x$ appears twice, so it is suitable to set $u = x^2 + x$. But in $x^4 + x + 1 = 0$, there is no repeated whole expression; forcing $u = x^2$ cannot eliminate x , and instead makes the problem more confusing.

The substitution method does not change the problem. It temporarily gives a short name to a complicated whole expression. Its advantage is that it turns an equation that looks complicated into an equation we already know how to solve. For example, in

$$x^4 - 5x^2 + 4 = 0,$$

we have $x^4 = (x^2)^2$. After letting $t = x^2$, it becomes

$$t^2 - 5t + 4 = 0.$$

Four Steps of the Substitution Method

Set a new variable $\rightarrow$ solve the new equation $\rightarrow$ substitute back $\rightarrow$ check.

Example 2.10 Biquadratic Equation and Whole-Expression Substitution

Solve:

- (1) $x^4 - 5x^2 + 4 = 0$;
 (2) $(x^2 + x)^2 - 5(x^2 + x) + 6 = 0$.

Solution:

- (1) Let $t = x^2$. Then

$$t^2 - 5t + 4 = 0 = (t - 1)(t - 4).$$

Therefore $t = 1$ or $t = 4$. Substituting back gives $x^2 = 1$ or $x^2 = 4$, so $x = \pm 1, \pm 2$.

(2) Let $u = x^2 + x$. Then

$$u^2 - 5u + 6 = 0 = (u - 2)(u - 3).$$

If $u = 2$, then $x^2 + x - 2 = 0$, giving $x = 1$ or -2 . If $u = 3$, then $x^2 + x - 3 = 0$, giving

$$x = \frac{-1 \pm \sqrt{13}}{2}.$$

In-Class Practice 2.4.1 Higher-Degree Equations

(1) Solve $x^3 - 4x = 0$.

(2) Solve $x^3 + 2x^2 - x - 2 = 0$.

(3) Solve $x^4 - 10x^2 + 9 = 0$.

(4) Solve $(x - 1)^4 - 5(x - 1)^2 + 4 = 0$.

2.5 Rational Equations (分式方程)

For a rational equation, first determine where the denominators are nonzero. Any solution obtained after clearing denominators must be checked in the original equation.

Rational Equation

A **rational equation** is an equation containing rational expressions, with the unknown appearing in a denominator. For example,

$$\frac{1}{x+2} = 3x, \quad \frac{x}{x-1} = 2$$

are rational equations. $x + 2 = 5$ is not a rational equation. Although $\frac{1}{2}x = 3$ has a fractional coefficient, the unknown is not in a denominator, so it is not a rational equation in this section.

The biggest difference between rational equations and polynomial equations is: some values of x make a denominator equal to 0, and then the original equation is meaningless. Therefore the first step must be to write the restrictions.

2.5.1 Clearing Denominators (去分母法)

The purpose of clearing denominators is to turn an equation containing rational expressions into a more familiar polynomial equation. But clearing denominators is essentially multiplying both sides of the equation by an expression containing the unknown. If a candidate solution makes this expression equal to 0, the original multiplication is illegal, so checking is necessary.

For example,

$$\frac{2}{x-1} = 3$$

first requires $x \neq 1$, and then both sides are multiplied by $x - 1$, giving $2 = 3(x - 1)$. This step can turn the rational equation into a linear equation, but the final candidate solution still has to be confirmed not to equal 1.

Standard Steps for Solving a Rational Equation

- (1) Write all restrictions that denominators are nonzero.
- (2) Find the least common denominator and multiply both sides of the equation by it.
- (3) Solve the resulting polynomial equation.
- (4) Check whether candidate solutions satisfy the original restrictions and the original equation.
- (5) Write the final solution.

Example 2.11 Rational Equations

Solve:

$$(1) \frac{2}{x-1} = \frac{3}{x+1};$$

$$(2) \frac{x}{x-2} - \frac{4}{x^2-4} = 1.$$

Solution:

- (1) Require $x \neq 1, -1$. Multiply both sides by $(x-1)(x+1)$:

$$2(x+1) = 3(x-1).$$

Solving gives $2x + 2 = 3x - 3$, so $x = 5$, which satisfies the restrictions.

- (2) Require $x \neq 2, -2$. Multiply both sides by $(x-2)(x+2)$:

$$x(x+2) - 4 = (x-2)(x+2).$$

Simplifying gives $x^2 + 2x - 4 = x^2 - 4$, so $2x = 0$, $x = 0$, which satisfies the restrictions.

Example 2.12 An Extraneous Root (增根) Produced by Clearing Denominators

Solve:

$$\frac{x}{x-1} = 2 + \frac{1}{x-1}.$$

Solution: First write the restriction: the denominator $x - 1$ in the original equation cannot be 0, so

$$x \neq 1.$$

Multiply both sides by $x - 1$ to clear denominators:

$$x = 2(x - 1) + 1.$$

Simplifying gives

$$x = 2x - 1,$$

so the polynomial equation after clearing denominators gives the candidate solution

$$x = 1.$$

However, substituting $x = 1$ back into the original equation makes the denominator $x - 1 = 0$, and the original equation is meaningless. Therefore $x = 1$ is not a solution of the original equation; it is an **extraneous root** produced during clearing denominators. Hence the original equation has **no solution**.

Four-Step Table for a Rational Equation

Step	Content in This Example
Original restriction	$x - 1 \neq 0$, that is, $x \neq 1$
Equation after clearing denominators	$x = 2(x - 1) + 1$
Candidate solution	$x = 1$
Checking conclusion	$x = 1$ makes the original denominator 0, so it is rejected; the original equation has no solution

Common Mistake

The polynomial equation obtained after clearing denominators may produce an **extraneous root (增根)**. Checking is not optional; any number that makes an original denominator zero cannot be a solution.

In-Class Practice 2.5.1 Rational Equations

(1) Solve $\frac{1}{x} + \frac{1}{x+1} = 1$.

(2) Solve $\frac{x+2}{x-2} = \frac{2x}{x-1}$.

(3) Solve $\frac{3}{x^2-1} = \frac{2}{x-1} - 1$.

(4) Solve $\frac{x-1}{x} + \frac{x}{x-1} = \frac{5}{2}$.

2.6 Linear Inequalities (一次不等式)

The solution of an inequality is usually not a single number, but a set of numbers satisfying the condition. In this chapter, we describe solution sets only using inequalities themselves.

Inequality and Linear Inequality

A relation containing $>$, $<$, $\geq$, $\leq$ is called an **inequality (不等式)**. An inequality containing one unknown, whose highest degree after simplification is 1, is called a **linear inequality in one variable**, for example

$$3x - 4 > 2x + 5, \quad 5 - 2x \leq 11.$$

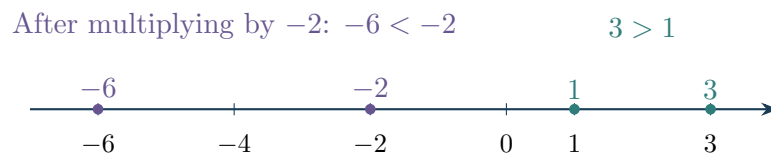
$x^2 - 1 > 0$ is not a linear inequality because the highest degree is 2. $x + y > 3$ is not an inequality in one variable because it has two unknowns.

Solving an inequality means finding all values of the unknown x that make the inequality true; this range is called the solution set of the inequality. The difference from equations is that the solution of an equation is usually one or several numbers and can be listed, while the solution set of an inequality is usually a range containing infinitely many numbers.

2.6.1 Properties of Inequalities (不等式的性质)

The idea for solving a linear inequality is similar to solving a linear equation: use addition, subtraction, multiplication and division to isolate x . But because an inequality represents a size relationship, every transformation must ensure that the size relationship remains correct.

The most important difference between inequalities and equalities is: when both sides are multiplied or divided by a negative number, the direction of the inequality reverses. For example, $3 > 1$, but after multiplying both sides by -2 , we get $-6 < -2$. This is because multiplying by a negative number reverses direction on the number line.



Transformation Rules for Inequalities

If $a > b$, then:

- ▶ $a + c > b + c$;
- ▶ when $c > 0$, $ac > bc$;
- ▶ when $c < 0$, $ac < bc$, so the **direction of the inequality changes**.

Example 2.13 Linear Inequality and System of Inequalities

Solve the following inequality and system of inequalities:

(1) $3x - 4 > 2x + 5$;

(2)
$$\begin{cases} 2x - 3 \geq 1, \\ -x \geq -6. \end{cases}$$

Solution:

(1)

$$3x - 4 > 2x + 5 \Rightarrow x > 9,$$

so the solution set is $x > 9$.

(2) In the system,

$$2x - 3 \geq 1 \Rightarrow 2x \geq 4 \Rightarrow x \geq 2,$$

$$-x \geq -6 \Rightarrow x \leq 6.$$

Notice that when both sides are multiplied or divided by a negative number, the direction of the inequality must change. The intersection is

$$2 \leq x \leq 6.$$

In-Class Practice 2.6.1 Linear Inequalities(1) Solve $5 - 2x < 11$.(2) Solve $\frac{x-1}{2} - \frac{x+2}{3} \geq 1$.(3) Solve $-2 < 3x + 1 \leq 7$.(4) Solve $\begin{cases} 3x - 4 > 5, \\ x + 1 \leq 0. \end{cases}$

2.7 Quadratic, Higher-Degree and Rational Inequalities

First find the boundary points that make the expression equal to zero, and then study the signs on the intervals cut out by those points.

The inequalities studied in this section cannot be solved only by moving terms. Their common feature is that the left side or right side contains a quadratic expression, a higher-degree polynomial or a rational expression. When solving, first move all terms to one side and convert the inequality into the form

$$F(x) > 0, \quad F(x) < 0, \quad F(x) \geq 0, \quad F(x) \leq 0,$$

and then study where $F(x)$ is positive, negative, zero or undefined.

2.7.1 Quadratic Inequalities (二次不等式)**Quadratic Inequality**

A **quadratic inequality** is an inequality containing one unknown whose highest degree after simplification is 2. For example,

$$x^2 + x - 12 > 0, \quad 3x^2 + 5x + 2 \leq 0$$

are quadratic inequalities. Their connection with quadratic equations in one variable is: first solve the corresponding quadratic equation, find the points where the quadratic expression equals 0, and then determine the signs on both sides of these points.

If a quadratic expression can be written as $a(x - x_1)(x - x_2)$, it can only change sign at its zeros. The leading coefficient a determines the sign on the rightmost interval.

Why find zeros? Because for a continuously changing product to go from positive to negative, it must pass through 0. A rational expression may also break where the denominator is 0. Therefore

zeros and denominator zeros cut the number line into intervals. Inside each interval, the sign of each factor does not change, so one test point is enough to determine the sign of the whole interval.

Sign Analysis (符号分析法)

- (1) Move one side of the inequality to 0.
- (2) Factor and find the zeros.
- (3) Mark the zeros on a number line from least to greatest.
- (4) Determine the sign on each interval.
- (5) Choose the corresponding range according to $>$, $<$, $\geq$, or $\leq$, and decide whether zeros are included.

Sign Table Template

For students just starting, it is recommended to draw a sign table every time. Write the key points in the first row, write the signs of each factor on each interval in the next rows, and write the sign of the whole product or rational expression in the last row. Do not skip steps only by saying "opens upward", especially for rational inequalities and higher-degree inequalities.

For example, for $(x + 4)(x - 3)$, the key points are $-4, 3$:

x	$(-\infty, -4)$	$(-4, 3)$	$(3, +\infty)$
$x + 4$	—	+	+
$x - 3$	—	—	+
$(x + 4)(x - 3)$	+	—	+

This table shows that the solution of $(x + 4)(x - 3) > 0$ lies in the two outer intervals.

Example 2.14 Quadratic Inequalities

Solve the following quadratic inequalities:

- (1) $x^2 + x - 12 > 0$;
- (2) $3x^2 + 5x + 2 \leq 0$;
- (3) $x^2 - 4x + 5 < 0$.

Solution:

- (1) $x^2 + x - 12 = (x + 4)(x - 3)$. The parabola opens upward, so it is greater than 0 outside the two roots. The solution set is $x < -4$ or $x > 3$.
- (2) $3x^2 + 5x + 2 = (3x + 2)(x + 1)$. The parabola opens upward, so it is less than or equal to 0 between the two roots, including endpoints. The solution set is $-1 \leq x \leq -\frac{2}{3}$.
- (3) $x^2 - 4x + 5 = (x - 2)^2 + 1$ is always greater than 0, so $x^2 - 4x + 5 < 0$ has no real solution.

The solution set is $\emptyset$.

2.7.2 Polynomial Inequalities (高次不等式)

Higher-Degree Inequality

A **higher-degree inequality** is a polynomial inequality containing one unknown whose highest degree after simplification is greater than 2. For example,

$$x(x-2)(x+1) \geq 0, \quad x^3 - 4x \leq 0$$

are higher-degree inequalities. In the first pass, the focus is not on expanding the polynomial, but on writing it as a product of factors and then determining the sign of the product.

Factor the polynomial completely. When crossing a root of odd multiplicity, the sign changes. When crossing a root of even multiplicity, the sign does not change.

The first-pass goal of this section only requires you to handle already-factored expressions such as $x(x-2)(x+1) \geq 0$. If a problem still requires complicated factorization, first review Chapter 1 factorization.

Example 2.15 Higher-Degree Inequalities

Solve:

- (1) $x(x-2)(x+1) \geq 0$;
- (2) $(x-1)^2(x+2)(x-3) < 0$.

Solution:

- (1) The zeros are $-1, 0, 2$. Determine the signs on the intervals separated by these zeros:

$$x < -1 : - \quad -1 < x < 0 : + \quad 0 < x < 2 : - \quad x > 2 : +.$$

Since the inequality asks for ≥ 0 , and zeros may be included, the solution set is

$$-1 \leq x \leq 0 \quad \text{or} \quad x \geq 2.$$

- (2) The zeros are $-2, 1, 3$. Since $(x-1)^2 \geq 0$, and at $x = 1$ it is only 0 and does not change sign, mainly look at $(x+2)(x-3)$. Requiring the product to be less than 0 gives $-2 < x < 3$, but at $x = 1$ the original expression is 0, so it must be excluded. Therefore the solution set is

$$-2 < x < 1 \quad \text{or} \quad 1 < x < 3.$$

2.7.3 Rational Inequalities (分式不等式)

Rational Inequality

A **rational inequality** is an inequality containing a rational expression, with the unknown appearing in the denominator. For example,

$$\frac{x-1}{x+2} > 0, \quad \frac{x^2-4}{x-1} \leq 0$$

are rational inequalities. Compared with a quadratic inequality, it has one additional type of key point: points where the denominator is 0. Such points are not solutions, but they divide the number line into intervals.

Zeros of the numerator may be included, but zeros of the denominator can never be included. You cannot casually multiply both sides of an inequality by a denominator containing the unknown, because the sign of the denominator is unknown.

On the number line, if a numerator zero satisfies an equality requirement, draw a solid point. A denominator zero must always be drawn as an open point, because the original expression is undefined there.

Example 2.16 Rational Inequalities

Solve:

$$(1) \frac{x-1}{x+2} > 0;$$

$$(2) \frac{x^2-4}{x-1} \leq 0.$$

Solution:

- (1) The key points are $x = 1$ and $x = -2$, and $x = -2$ cannot be taken. The rational expression is positive when numerator and denominator have the same sign:

$$x < -2 \quad \text{or} \quad x > 1.$$

Therefore the solution set is $x < -2$ or $x > 1$.

- (2) First factor:

$$\frac{x^2-4}{x-1} = \frac{(x-2)(x+2)}{x-1}, \quad x \neq 1.$$

The key points are $-2, 1, 2$. Determine signs on the intervals separated by the key points:

$$x < -2 : - \quad -2 < x < 1 : + \quad 1 < x < 2 : - \quad x > 2 : +.$$

Requiring ≤ 0 , the zeros $-2, 2$ may be included, while the denominator zero 1 cannot be included. Hence the solution set is

$$x \leq -2 \quad \text{or} \quad 1 < x \leq 2.$$

In-Class Practice 2.7.1 Comprehensive Sign Analysis

- (1) Solve $x^2 - 5x + 6 \geq 0$.
- (2) Solve $2x^2 + x - 3 < 0$.
- (3) Solve $x^3 - 4x \leq 0$.
- (4) Solve $(x - 2)^2(x + 1) > 0$.
- (5) Solve $\frac{x + 3}{x - 2} \geq 0$.
- (6) Solve $\frac{x^2 - x - 6}{x^2 - 1} < 0$.

2.8 Chapter Summary**Main Knowledge Thread**

- (1) Linear equations rely on properties of equality; systems rely on elimination.
- (2) The three basic methods for quadratic equations are factoring, completing the square and the quadratic formula.
- (3) The discriminant determines the number of real roots; Vieta's formulas can avoid directly finding roots.
- (4) Higher-degree equations commonly use factoring and substitution; rational equations must restrict denominators and check solutions.
- (5) When inequalities are multiplied or divided by negative numbers, the inequality sign changes; complicated inequalities use zeros to divide intervals and study signs.

Formula List for This Chapter

$$\Delta = b^2 - 4ac, \quad x = \frac{-b \pm \sqrt{\Delta}}{2a}, \quad x_1 + x_2 = -\frac{b}{a}, \quad x_1 x_2 = \frac{c}{a}.$$

$$AB = 0 \iff A = 0 \text{ or } B = 0.$$

2.9 Chapter Exercises**How to Use These Exercises**

Group A checks the minimum mastery for this chapter. If you make mistakes, return to the corresponding section and reread the examples. Group B is for integrated practice before a quiz. Group C contains extension problems; during a first self-study pass, you may skip it without affecting your entry into the next chapter.

Group A: Basic Check

Difficulty: Basic

- (1) Solve $4x - 7 = 2x + 9$.
- (2) Solve $\frac{x+1}{3} - \frac{x-2}{2} = 1$.
- (3) Solve the system $\begin{cases} x + 2y = 7, \\ 3x - y = 7. \end{cases}$
- (4) Solve $x^2 - 7x + 10 = 0$.
- (5) Solve $2x^2 + 3x - 1 = 0$.
- (6) Determine the number of real roots of $3x^2 - 6x + 4 = 0$.
- (7) Solve $x^4 - 13x^2 + 36 = 0$.
- (8) Solve $\frac{2}{x-1} = \frac{x}{x+1}$.
- (9) Solve $2x - 5 \leq 7$, and write the solution set.
- (10) Solve $x^2 - x - 6 > 0$.
- (11) Solve $x(x-1)(x+2) \leq 0$.
- (12) Solve $\frac{x-3}{x+1} < 0$.

Group B: Quiz Level

Difficulty: Core

- (1) Discuss the solutions of $(a - 1)x = a^2 - 1$.
- (2) If x_1, x_2 are the roots of $x^2 - 4x - 2 = 0$, find $x_1^2x_2 + x_1x_2^2$.
- (3) The equation $x^2 - kx + k = 0$ in x has two equal real roots. Find k .
- (4) Solve $x^3 - 3x^2 - 4x + 12 = 0$.
- (5) Solve $(x^2 - 2x)^2 - 5(x^2 - 2x) + 4 = 0$.
- (6) Solve $\frac{x+1}{x-2} + \frac{x-2}{x+1} = \frac{5}{2}$.
- (7) Solve $2x^2 - 2x - 4 \geq 0$.
- (8) Solve $(x+2)(x-1)^2(x-4) \leq 0$.
- (9) Solve $\frac{(x-1)(x+3)}{(x-2)(x+1)} \geq 0$.
- (10) An item first increases in price by 20%, then decreases by 20%, and its final price is 192 yuan. Find the original price.

Group C: Extension and Thinking

Difficulty: Extension

- (1) If the two roots of $x^2 - (m + 1)x + m = 0$ differ by 3, find m .
- (2) Without solving the equation, determine whether $x^2 + 2x + 3 = 0$ has a solution in the real numbers, and explain the graphical meaning.
- (3) Solve the inequality $\frac{x^2 - 5x + 6}{x^2 - 4} \leq 0$.
- (4) Find all real numbers m such that $x^2 - 2mx + m + 2 > 0$ holds for all real numbers x .
- (5) Given $x + 1/x = 4$, find $x^2 + 1/x^2$, and solve for all possible x .

Self-Check

Skill	Can do	Mostly can	Not yet
Solve linear equations and systems in a standard way	☐	☐	☐
Solve quadratic equations by three methods	☐	☐	☐
Use the discriminant to determine the number of roots	☐	☐	☐
Solve higher-degree equations by factoring or substitution	☐	☐	☐
Solve rational equations and check extraneous roots	☐	☐	☐
Write solution sets using inequalities	☐	☐	☐
Solve complicated inequalities by sign analysis	☐	☐	☐

Before Entering the Next Chapter

If you still cannot fluently determine whether an expression is meaningful, or cannot write a solution set using inequalities, review this chapter first. These abilities will be used directly in the next chapter for sets, domains and ranges.

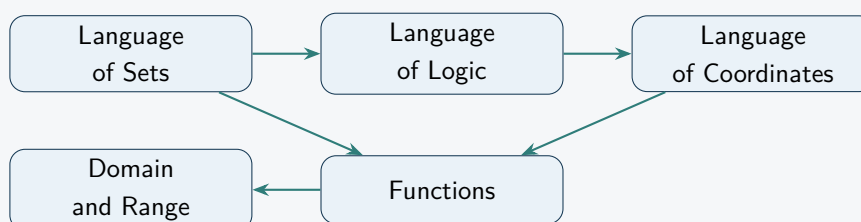
Sets, Logic and Foundations of Functions

Learning Goals

After studying this chapter, you should be able to:

- ▶ represent sets by the roster method, set-builder notation and interval notation;
- ▶ determine the relationships between elements and sets, and between sets and sets;
- ▶ compute intersections, unions and complements of sets, and understand the results using number lines or Venn diagrams;
- ▶ identify propositions, and understand ``and'', ``or'', ``not'', sufficient conditions and necessary conditions;
- ▶ plot points in the Cartesian coordinate plane, and understand quadrants, distance and midpoint;
- ▶ determine whether a correspondence is a function, and find the domain and range of a function;
- ▶ make a table from an analytic expression, plot points, and understand domain and range from a graph.

Knowledge Map



Sets describe ``which objects are put together'', logic explains ``how conditions are connected'', and coordinates and functions describe ``how variables correspond to each other''.

Learning Route of This Chapter

This chapter contains four kinds of language. They will appear together repeatedly later:

- (1) the language of sets first answers ``which objects are being studied'';
- (2) the language of logic answers ``what conditions these objects satisfy, and how the conditions are combined'';
- (3) the language of coordinates turns ordered pairs into points in the plane;
- (4) the language of functions connects inputs, outputs and graphs into one whole.

When reading this chapter, do not only memorize symbols. Whenever you see a new symbol, ask: does it solve a problem about ``range'', ``condition'', ``position'', or ``correspondence''?

Prerequisites and Minimum Mastery

To study this chapter, you need to be able to represent numbers on a number line, read simple inequalities, and plot points in a coordinate plane. The minimum goal for the first pass is: translate set conditions into ordinary language; understand intersection, union and complement using a number line or Venn diagram; judge whether a sentence is a proposition; and use ``one input has only one output'' to judge whether a correspondence is a function.

Common Self-Study Sticking Points in This Chapter

Set-builder notation, quantifiers, and sufficient and necessary conditions are relatively abstract. When studying by yourself, first translate symbols into ordinary language, and then judge truth values or ranges. Do not try to make the expression fully formal at the very beginning.

3.1 Representing Sets and Relations Between Sets

A set is a language for putting a group of objects together for study. We do not study sets in order to memorize more symbols, but in order to make the ``scope of study'' clear: which objects are under discussion, and which objects are not.

3.1.1 From Concrete Objects to Sets

Mathematical problems often first specify a range. For example, ``solve the equation over the integers'', ``find all positive real numbers'', or ``count the students in this class who joined a club''. If the range is not clear, later calculations may count too much or miss something.

When we say ``all students in Class 1'', ``positive integers less than 10'', or ``all real solutions of the equation $x^2 = 4$ '', we are describing a set. A set is not a number or a formula, but a whole made up of objects.

Whether the same object is ``inside the range under discussion'' depends on the universal set given by the problem. For example,  $-2$  is an integer, so it belongs to the range of ``integers''; but if the problem says ``discuss within the natural numbers'', then -2 is not in the range under discussion. The language of sets first helps us say clearly: exactly which objects are allowed to take part in the discussion now?

Set and Element

A whole made up of definite objects is called a **set** (集合). Each object in a set is called an **element** (元素) of the set. Sets are usually denoted by uppercase letters, and elements by lowercase letters. A set containing no elements is called the empty set, written as $\emptyset$.

If a is an element of the set A , we write $a \in A$. If a is not an element of the set A , we write $a \notin A$.

The elements of a set have three basic features:

- **definiteness**: for any object, whether it belongs to the set must be decidable;
- **distinctness**: writing the same element several times still counts as only one element;
- **unorderedness**: the order in which elements are written does not affect the set itself.

Example 3.1 Which Descriptions Can Form Sets

Decide whether the following objects can form a set, and explain the reason.

- (1) all examples that appear in Chapter 3 of these notes;
- (2) positive even numbers less than 10;
- (3) relatively large natural numbers;
- (4) all real solutions of the equation $x^2 - 5x + 6 = 0$;
- (5) the 5 students with the best mathematics scores in this class.

Solution:

- (1) Yes. The content of the notes is definite, so we can decide which examples have appeared.
- (2) Yes. They are 2, 4, 6, 8, so the objects are clear.
- (3) No. "Relatively large" has no clear standard, and different people may judge differently.
- (4) Yes. Solving the equation gives $(x - 2)(x - 3) = 0$, so the solutions form the set $\{2, 3\}$.
- (5) Usually yes, but only if the scores and ranking rules have already been determined. If there is a tie, one must also explain how ties are handled.

3.1.2 Roster Method (列举法)

The roster method means writing the elements of a set one by one inside braces, separated by commas. Finite sets are most suitable for the roster method.

The advantage of the roster method is that it is intuitive: one glance tells you which elements are in the set. Its limitation is also clear: if there are too many elements, or if the elements cannot all be listed, then it is not suitable to write all of them out.

Sometimes an infinite set can be represented with an ellipsis, for example $\{2, 4, 6, 8, \dots\}$, which represents the positive even numbers. But an ellipsis is appropriate only when the pattern is very clear. If the pattern is not unique, we should use set-builder notation to write the condition

explicitly.

Steps for the Roster Method

- (1) First read the range of objects clearly, such as ``positive integers'', ``integers'', or ``real numbers''.
- (2) Find all elements that satisfy the condition.
- (3) Write the elements inside braces $\{\}$.
- (4) Remove repeated elements. The order may be from small to large, which makes checking easier.

Example 3.2 Representing Sets by the Roster Method

Represent the following sets by the roster method:

- (1) positive odd numbers less than 10;
- (2) integers x satisfying $x^2 = 9$;
- (3) the set of letters appearing in the word **math**;
- (4) integers x satisfying $-2 < x \leq 3$.

Solution:

- (1) The positive odd numbers less than 10 are 1, 3, 5, 7, 9, so the set is $\{1, 3, 5, 7, 9\}$.
- (2) The integer solutions of $x^2 = 9$ are $x = -3$ or $x = 3$, so the set is $\{-3, 3\}$.
- (3) The letters appearing in **math** are m, a, t, h , so the set is $\{m, a, t, h\}$.
- (4) The integers greater than -2 and not greater than 3 are $-1, 0, 1, 2, 3$, so the set is $\{-1, 0, 1, 2, 3\}$.

Common Mistake

$\{1, 2, 2, 3\} = \{1, 2, 3\}$, because the same element is kept only once in a set. $\{1, 2, 3\} = \{3, 2, 1\}$, because a set does not care about order.

In-Class Practice 3.1.1 Practice: Roster Method

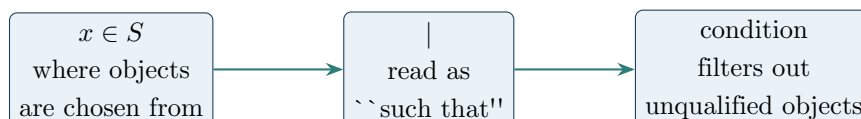
- (1) Represent the set of positive prime numbers less than 15 by the roster method.
- (2) Represent the solution set over the real numbers of $x^2 - 4 = 0$ by the roster method.
- (3) Represent the set of integers x satisfying $-3 \leq x < 2$ by the roster method.
- (4) Decide whether $\{a, b, a, c\}$ and $\{c, b, a\}$ are equal, and explain why.

3.1.3 Set-Builder Notation (描述法)

When a set has many elements, infinitely many elements, or elements that are inconvenient to list completely, we often use set-builder notation. Set-builder notation writes ``where the elements come from'' and ``what condition the elements satisfy''.

$$\{x \in S \mid x \text{ satisfies condition } A\}.$$

Here S is the range, and the vertical bar $|$ is read as ``such that'' or ``satisfying''. This set means ``the set of all elements in S that satisfy condition A ''.



When reading set-builder notation, first look before the vertical bar: it tells us where the elements are chosen from. Then look after the vertical bar: it tells us which elements fail the condition and should be filtered out. For example,

$$\{x \in \mathbb{Z} \mid -2 < x \leq 3\}$$

means ``choose from all integers the numbers satisfying $-2 < x \leq 3$ ', so the result is $\{-1, 0, 1, 2, 3\}$.

Steps for Set-Builder Notation

- (1) Choose a representative element, usually written as x .
- (2) Write clearly the range of x , such as $x \in \mathbb{N}, \mathbb{Z}, \mathbb{R}$.
- (3) Write the filtering condition after $|$.
- (4) Check whether the condition is clear enough.

Example 3.3 Converting Between Roster Method and Set-Builder Notation

Complete the following conversions:

- (1) write $\{2, 4, 6, 8, \dots\}$ in set-builder notation;
- (2) write $\{-2, -1, 0, 1, 2\}$ in set-builder notation;
- (3) write $\{x \in \mathbb{Z} \mid x^2 < 10\}$ by the roster method;
- (4) write $\{x \in \mathbb{N} \mid x \leq 5\}$ by the roster method.

Solution:

- (1) This is the set of positive even numbers, and can be written as $\{x \in \mathbb{N}^* \mid x \text{ is even}\}$.
- (2) These are the integers from -2 to 2 , and can be written as $\{x \in \mathbb{Z} \mid -2 \leq x \leq 2\}$.
- (3) The condition $x^2 < 10$ is equivalent to $-\sqrt{10} < x < \sqrt{10}$. The integer values of x are $-3, -2, -1, 0, 1, 2, 3$, so the set is $\{-3, -2, -1, 0, 1, 2, 3\}$.
- (4) In these notes we agree that $\mathbb{N} = \{0, 1, 2, \dots\}$, so $\{x \in \mathbb{N} \mid x \leq 5\} = \{0, 1, 2, 3, 4, 5\}$.

In-Class Practice 3.1.2 Practice: Set-Builder Notation

- (1) Use set-builder notation to represent the set of positive odd numbers $\{1, 3, 5, 7, \dots\}$.
- (2) Use set-builder notation to represent the set of positive multiples of 3.
- (3) Write $\{x \in \mathbb{Z} \mid -1 < x < 4\}$ by the roster method.
- (4) Write $\{x \in \mathbb{R} \mid x^2 = 1\}$ by the roster method.

3.1.4 Interval Notation (区间表示)

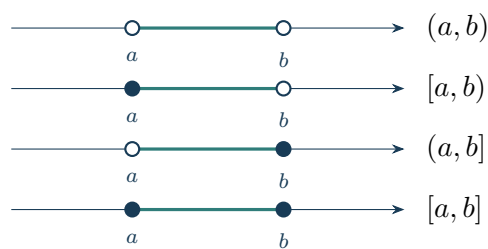
Interval notation is a special shorthand for sets of real numbers. It is suitable for representing a continuous stretch of real numbers, such as ``all real numbers greater than 2'' or ``all real numbers between -1 and 3 ''.

$$x > 2 \iff \{x \in \mathbb{R} \mid x > 2\} = (2, +\infty),$$

$$-1 < x \leq 3 \iff \{x \in \mathbb{R} \mid -1 < x \leq 3\} = (-1, 3].$$

Rules for Interval Endpoints

- Strict inequalities $<$ and $>$ correspond to parentheses, meaning that the endpoint is not included.
- Non-strict inequalities $\leq$ and $\geq$ correspond to brackets, meaning that the endpoint is included.
- $+\infty$ and $-\infty$ are not actual numbers; they only indicate directions, so parentheses must be used.
- A set consisting of several intervals is connected by the union symbol $\cup$. For example, $x \leq -1$ or $x > 2$ is written as $(-\infty, -1] \cup (2, +\infty)$.



An open point means the endpoint is not included; a solid point means the endpoint is included.

Example 3.4 Inequality Conditions and Interval Notation

Represent the following sets of real numbers using intervals:

- (1) $\{x \in \mathbb{R} \mid x \geq -2\}$;
- (2) $\{x \in \mathbb{R} \mid -3 < x < 5\}$;
- (3) $\{x \in \mathbb{R} \mid x < 0 \text{ or } x \geq 4\}$.

Solution:

- (1) $x \geq -2$ includes the endpoint -2 and extends to the right, so the set is $[-2, +\infty)$.
- (2) $-3 < x < 5$ includes neither endpoint, so the set is $(-3, 5)$.
- (3) $x < 0$ is $(-\infty, 0)$, and $x \geq 4$ is $[4, +\infty)$. When they are connected by "or", we take the union, so the set is

$$(-\infty, 0) \cup [4, +\infty).$$

In-Class Practice 3.1.3 Practice: Interval Notation

- (1) Use interval notation to represent $\{x \in \mathbb{R} \mid x < 5\}$.
 - (2) Use interval notation to represent $\{x \in \mathbb{R} \mid -2 \leq x < 4\}$.
 - (3) Use interval notation to represent $x \leq -1$ or $x > 2$.
 - (4) Rewrite $(-3, 6]$ in set-builder notation.
-
-
-
-

3.1.5 Choosing Among the Three Representations

The three representations express the same set, but their emphases are different: the roster method is intuitive, set-builder notation is concise and general, and interval notation is suitable for continuous ranges of real numbers.

For example, the same set can be translated in these ways:

$$\{0, 1, 2, 3\} = \{x \in \mathbb{Z} \mid 0 \leq x \leq 3\}.$$

If the range is changed to real numbers, then the corresponding continuous interval is $[0, 3]$, which contains many numbers not in the roster form, such as 0.5 and $\sqrt{2}$. Therefore, when writing a set, the "range" cannot be omitted.

Experience for Choosing a Set Representation

- If there are few elements and all of them can be written out, use the roster method first.
- If there are many elements or a clear pattern, use set-builder notation first.
- If the elements form a continuous stretch of real numbers, use interval notation first.
- For the same condition, first look at the range: an integer range can often be listed, while a real-number range often needs to be written as an interval.

Example 3.5 Three Representations of Sets

Let

$$A = \{x \in \mathbb{Z} \mid -2 < x \leq 3\}, \quad B = \{x \in \mathbb{R} \mid -2 < x \leq 3\}.$$

- (1) Represent A by the roster method.
- (2) Decide whether $-2, 0, 3, 4$ belong to A .
- (3) Represent B using interval notation.

- (4) Explain why A cannot be written directly as the interval $(-2, 3]$.

Solution:

- (1) The elements of A must be integers and must satisfy $-2 < x \leq 3$, so $A = \{-1, 0, 1, 2, 3\}$.
- (2) $-2 \notin A$, $0 \in A$, $3 \in A$, and $4 \notin A$.
- (3) The elements of B are all real numbers satisfying $-2 < x \leq 3$, so $B = (-2, 3]$.
- (4) The interval $(-2, 3]$ represents all real numbers satisfying the condition, including non-integers such as $0.5, \sqrt{2}$. But A contains only integers, so A cannot be written directly as $(-2, 3]$.

3.1.6 Subsets and Equal Sets (子集与相等集合)

If every element of A belongs to B , then A is a **subset** (子集) of B , written as $A \subseteq B$. If $A \subseteq B$ and $B \subseteq A$, then $A = B$.

Do Not Confuse Belongs To and Is Contained In

$a \in A$ means ``the element  $a$  is in the set  $A$ ".  $\{a\} \subseteq A$  means ``the small set containing only a is contained in A ". For example, if $A = \{1, 2, 3\}$, then $1 \in A$ and $\{1\} \subseteq A$, but we cannot write $1 \subseteq A$.

In addition, the empty set $\emptyset$ is a subset of every set. The reason is: to prove $\emptyset \subseteq A$, we need to check whether ``every element of the empty set is in A ". But the empty set has no elements, so there is no counterexample.

Example 3.6 Subsets and Equal Sets

Let

$$A = \{1, 2, 3\}, \quad B = \{3, 2, 1\}, \quad C = \{1, 2, 3, 4\}.$$

- (1) Decide whether A and B are equal.
- (2) Decide whether A is a subset of C .

Solution:

- (1) A set does not care about the order of its elements, so

$$A = \{1, 2, 3\} = \{3, 2, 1\} = B.$$

Therefore $A = B$.

- (2) Every element $1, 2, 3$ in A is in C , so

$$A \subseteq C.$$

But the element 4 in C is not in A , so $C \not\subseteq A$.

In-Class Practice 3.1.4 Set Representation and Relations

- (1) Decide which of the following can form a set: "excellent students", "students in this class taller than 170 cm", and "prime numbers less than 20".
- (2) Represent $\{x \in \mathbb{Z} \mid x^2 \leq 9\}$ by the roster method.
- (3) Represent $\{4, 8, 12, 16, \dots\}$ using set-builder notation.
- (4) Use interval notation to represent $\{x \in \mathbb{R} \mid x < -3 \text{ or } x \geq 2\}$.
- (5) Let $A = \{1, 2\}$. Write all subsets of A .
- (6) Decide: $\emptyset \subseteq \{0\}$, $0 \in \emptyset$, $\{1\} \in \{1, 2\}$.

3.2 Set Operations (集合的运算)

One set describes one range. When two sets are put together, three questions naturally arise: "belong to both", "belong to at least one", and "belong to the universal set but not to it". These are exactly intersection, union and complement.

3.2.1 Intersection, Union and Complement

Suppose the universal set U is all students in this class, A is the set of students who joined the basketball club, and B is the set of students who joined the choir. Then:

- ▶ students who "joined both the basketball club and the choir" belong to $A \cap B$;
- ▶ students who "joined the basketball club or the choir, at least one of them" belong to $A \cup B$;
- ▶ students who "are in this class but did not join the basketball club" belong to $\complement_U A$.

Therefore, set operations are not new calculation tricks. They write the everyday words "and", "or" and "not" as mathematical symbols.

Set Operations

Let A, B be subsets of a universal set U . The intersection of A and B , the union of A and B , the complement of A with respect to U , and the difference of A and B are respectively:

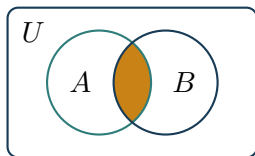
$$A \cap B = \{x \mid x \in A \text{ and } x \in B\},$$

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\},$$

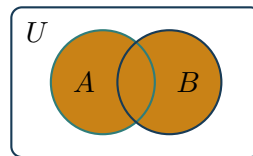
$$\complement_U A = \{x \in U \mid x \notin A\},$$

$$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}.$$

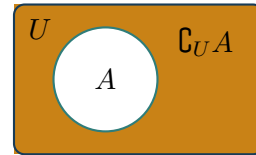
A **Venn diagram** (韦恩图) uses a large rectangle to represent the universal set U , and circles inside the rectangle to represent sets. Inside a circle means belonging to that set; outside a circle means not belonging to that set; the overlapping region of two circles means belonging to both sets. The following diagrams correspond to intersection, union, complement and difference.



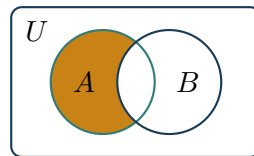
Intersection $A \cap B$



Union $A \cup B$



Complement $\complement_U A$



Difference $A \setminus B$

When reading a Venn diagram, look only at the shaded region: for an intersection, shade the overlapping part; for a union, shade all parts covered by the two circles; for a complement, shade the part of the universal set outside circle A . The difference $A \setminus B$ is read as “ A minus B ”, and represents elements that are in A but not in B . It is not ordinary subtraction; it means deleting from one range the part covered by another range.

Example 3.7 Operations on Finite Sets

Given $U = \{1, 2, 3, 4, 5, 6, 7\}$, $A = \{1, 2, 3, 4\}$, and $B = \{3, 4, 5, 6\}$, find

- (1) $A \cap B$;
- (2) $A \cup B$;
- (3) $\complement_U A$;
- (4) $A \setminus B$.

Solution:

- (1) $A \cap B = \{3, 4\}$, because 3, 4 are in both A and B .
- (2) $A \cup B = \{1, 2, 3, 4, 5, 6\}$, because the union collects all elements that appear in either set.

(3) $\complement_U A = \{5, 6, 7\}$, because these elements are in the universal set U but not in A .

(4) $A \setminus B = \{1, 2\}$, because these elements are in A but not in B .

3.2.2 Operations on Interval Sets (区间集合的运算)

Operations on intervals deal with the question: how do two ranges on a number line merge or overlap? Earlier we learned to use intervals to represent a stretch of real numbers. Now we need to solve this problem: if one condition gives a range A , and another condition gives a range B , then which part of the number line corresponds to "satisfies both", "satisfies at least one", and "does not satisfy a certain condition"?

When handling intervals, draw a number line first. The intersection takes the overlapping part, and the union takes the whole part after merging.

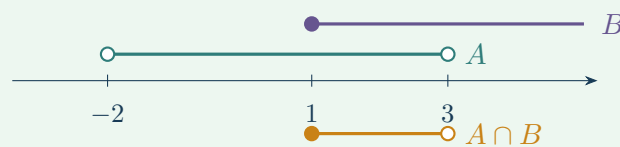
Steps for Reading Interval Operations from a Diagram

- (1) Mark the endpoints of each interval on the number line. Remember that an open point means not included, and a solid point means included.
- (2) For an intersection, keep only the part covered by all intervals.
- (3) For a union, keep every part covered by at least one interval.
- (4) Finally translate the part read from the diagram back into interval notation.

Example 3.8 Intersection and Union of Intervals

Let $A = (-2, 3)$, $B = [1, +\infty)$. Find $A \cap B$ and $A \cup B$.

Solution: First draw the two intervals on the same number line. A runs from the right side of -2 to the left side of 3 , while B starts from 1 and extends to the right.



The intersection must satisfy both conditions:

$$-2 < x < 3, \quad x \geq 1,$$

so $A \cap B = [1, 3)$. The union means numbers that belong to at least one of the sets. The two intervals connect together, starting from the right side of -2 and continuing to $+\infty$, so

$$A \cup B = (-2, +\infty).$$

In-Class Practice 3.2.1 Set Operations

- (1) $A = \{-1, 0, 1, 2\}$, $B = \{0, 2, 4\}$. Find $A \cap B$ and $A \cup B$.
- (2) $A = (-\infty, 2]$, $B = (-1, 4)$. Find $A \cap B$ and $A \cup B$.
- (3) The universal set is $\mathbb{R}$. Find the complement of $[0, 3)$.
- (4) Given $A = (-2, 3)$, $B = [a, +\infty)$, and $A \cap B = \emptyset$, find the range of a .

3.3 Elementary Logic (简单逻辑)

Mathematical proofs and problem-solving steps both rely on clear logical language. The role of logic is not to make language complicated, but to say accurately ``whether a condition holds'', ``whether a conclusion can be derived'', and ``what exactly is negated when taking a negation''.

Translation Order for Learning Logic

Do not memorize logical symbols mechanically. First translate a proposition into a complete ordinary sentence, and then look for keywords: ``and'' means both conditions must be satisfied; ``or'' means at least one condition is satisfied; ``not'' means the original proposition does not hold; ``for all'' means every object must be checked; ``there exists'' means finding one example is enough.

3.3.1 Propositions and Truth Values (命题与真假)

A declarative sentence whose truth value can be determined is called a **proposition (命题)**. A proposition has only two truth states: true or false.

To decide whether a sentence is a proposition, first ask two questions: is it a declarative sentence? In the given context, can its truth value be determined? For example, ``Please open the book'' is not a declarative sentence, so it is not a proposition. Although `` $x + 1 = 0$ '' looks like a mathematical sentence, x has no specified range or value, so by itself its truth value cannot yet be determined.

A sentence such as $x + 1 = 0$ is often called an open sentence: it contains an unspecified variable and temporarily cannot be judged true or false. Once a condition is given, such as ``when  $x = -1$ ,  $x + 1 = 0$ '', or ``there exists a real number x such that $x + 1 = 0$ '', it can become a proposition.

Example 3.9 Determining Whether a Sentence Is a Proposition

Determine whether each of the following sentences is a proposition. If it is, determine whether it is true or false.

- (1) 2 is an even number;
- (2) $3 > 5$;
- (3) please open the book;
- (4) $x + 1 = 0$;
- (5) for every real number x , $x^2 \geq 0$.

Solution:

- (1) It is a proposition, and it is true.
- (2) It is a proposition, and it is false.
- (3) It is not a proposition, because it is an imperative sentence and cannot be judged true or false.
- (4) By itself it is not a proposition, because x has no specified value. If we say ``when $x = -1$, $x + 1 = 0$ '', then it is a proposition.
- (5) It is a proposition, and it is true.

3.3.2 And, Or and Not (且、或、非)

Simple propositions can be combined to obtain new propositions. Let p, q be two propositions. The words ``and'', ``or'' and ``not'' here are close to everyday language, but in mathematics their truth values must be judged by definite rules.

In mathematics, ``or'' means ``at least one is true'', and allows both to be true. For example, `` $x = 0$ or $y = 0$ '' includes $x = 0, y \neq 0$, also includes $x \neq 0, y = 0$, and also includes $x = 0, y = 0$.

You can understand this using ``entry requirements''. If the requirement is ``bring a student ID and an identity card'', then both are required. If the requirement is ``bring a student ID or an identity card'', then at least one is enough. If we say ``not p '', it means p is not true.

Logical Connectives (逻辑联结词)

- $p \wedge q$: `` p and q '', true only when both p and q are true;
- $p \vee q$: `` p or q '', true as long as at least one of p, q is true;
- $\neg p$: ``not p '', whose truth value is opposite to that of p .

p	q	$p \wedge q$	$p \vee q$	$\neg p$
True	True	True	True	False (或假)
True	False	False	True	False
False	True	False	True	True
False	False	False	False	True

Example 3.10 Operations on Propositions

Let

p : 6 is an even number, q : 6 is a prime number.

- (1) Determine the truth value of p .
- (2) Determine the truth value of q .
- (3) Determine the truth value of $p \wedge q$.
- (4) Determine the truth value of $p \vee q$.
- (5) Determine the truth value of $\neg q$.

Solution: p is true, because 6 is divisible by 2. q is false, because $6 = 2 \times 3$, so it is not prime.

- (1) p is true.
- (2) q is false.
- (3) $p \wedge q$ is false, because "and" requires both propositions to be true.
- (4) $p \vee q$ is true, because "or" requires at least one proposition to be true.
- (5) $\neg q$ is true, because "not" reverses the truth value.

Example 3.11 Expressing Conditions Using And and Or

Rewrite the following conditions using "and" or "or".

- (1) $-1 \leq x < 3$;
- (2) $x = 0$ or $y = 0$;
- (3) x is not in $[1, 4]$.

Solution:

- (1) $-1 \leq x < 3$ means $x \geq -1$ and $x < 3$.
- (2) This is already an "or" proposition, meaning that at least one of the two equations holds.
- (3) x is not in $[1, 4]$, which is equivalent to $x < 1$ or $x > 4$.

3.3.3 Negating And and Or Propositions

Negating a proposition means writing the condition under which "the original proposition does not hold". The most important rule is: **the negation of and becomes or, and the negation of or becomes and.**

Why does the negation of "and" become "or"? Look at a concrete condition:

$$-1 \leq x < 3.$$

It means " $x \geq -1$ and $x < 3$ ". To make this condition fail, it is enough for one side to fail: either $x < -1$, or $x \geq 3$. Therefore the negation is

$$x < -1 \quad \text{or} \quad x \geq 3.$$

This is where "the negation of and becomes or" comes from.

The same happens in everyday language. The negation of "Ming is both tall and fast" is not "Ming is both not tall and not fast", but "Ming is not tall or not fast". As long as one condition fails, the original "and" proposition fails.

De Morgan's Laws (德摩根律)

$$\neg(p \wedge q) \iff (\neg p) \vee (\neg q),$$

$$\neg(p \vee q) \iff (\neg p) \wedge (\neg q).$$

That is:

- the negation of " p and q " is "not p or not q ";
- the negation of " p or q " is "not p and not q ".

Basic Table for Negating Inequalities

Original Condition	Negation
$x > a$	$x \leq a$
$x \geq a$	$x < a$
$x < a$	$x \geq a$
$x \leq a$	$x > a$
$x = a$	$x \neq a$
$x \neq a$	$x = a$

Example 3.12 Negating And and Or Propositions

Write the negation of each proposition:

- (1) $x > 2$;

$$(2) -1 \leq x < 3;$$

$$(3) x = 0 \text{ or } y = 0;$$

$$(4) a > 0 \text{ and } b > 0.$$

Solution:

(1) The negation of $x > 2$ is $x \leq 2$.

(2) $-1 \leq x < 3$ means $x \geq -1$ and $x < 3$. When negating, "and" becomes "or", so the negation is

$$x < -1 \quad \text{or} \quad x \geq 3.$$

(3) The negation of " $x = 0$ or $y = 0$ " is $x \neq 0$ and $y \neq 0$.

(4) The negation of " $a > 0$ and $b > 0$ " is $a \leq 0$ or $b \leq 0$.

3.3.4 Universal Quantifier and Existential Quantifier

Some propositions do not talk about only one object, but about "all" objects in a class or "at least one" object in a class. At this time we use quantifiers.

Quantifiers solve the problem of "how many objects must be checked". "For all" means no exceptions are allowed; "there exists" means finding one example is enough. Therefore, the way we judge truth values is also different: for a universal proposition, guard against counterexamples; for an existential proposition, look for an example.

Process for Judging Quantified Propositions

- To judge "for all ..." as true, you need to explain that all objects satisfy the condition; to judge it as false, one counterexample is enough.
- To judge "there exists ..." as true, one qualifying example is enough; to judge it as false, you need to explain that all objects fail the condition.

Two Common Quantifiers

- $\forall$ is read as "for all" or "for every", and is called the **universal quantifier** (全称量词);
- $\exists$ is read as "there exists", and is called the **existential quantifier** (存在量词).

For example:

$$\forall x \in \mathbb{R}, \quad x^2 \geq 0$$

is read as "for every real number x , $x^2 \geq 0$ ". This is a true proposition.

$$\exists x \in \mathbb{R}, \quad x^2 = 2$$

is read as "there exists a real number x such that $x^2 = 2$ ". This is also a true proposition, for example $x = \sqrt{2}$.

Example 3.13 Understanding Propositions with Quantifiers

Explain each proposition in ordinary language and determine whether it is true or false.

- (1) $\forall x \in \mathbb{R}, x^2 + 1 > 0$;
- (2) $\exists x \in \mathbb{Z}, x^2 = 2$;
- (3) $\forall n \in \mathbb{N}, n + 1 > n$;
- (4) $\exists x \in \mathbb{R}, x < 0$ and $x^2 = 4$.

Solution:

- (1) For every real number x , $x^2 + 1 > 0$. Since $x^2 \geq 0$, we have $x^2 + 1 > 0$, so it is true.
- (2) There exists an integer x such that $x^2 = 2$. There is no such integer, so it is false.
- (3) For every natural number n , $n + 1 > n$. This is true.
- (4) There exists a real number x satisfying $x < 0$ and $x^2 = 4$. Taking $x = -2$ works, so it is true.

3.3.5 Negating Propositions with Quantifiers

When negating quantifiers, the core rule is: **for all becomes there exists, there exists becomes for all, and then negate the condition that follows.**

This rule comes from asking "how can the original statement fail?" For "everyone passed" to fail, we only need to find one person who did not pass. For "at least one person passed" to fail, everyone must have failed. Therefore, when taking a negation, not only the condition after the quantifier must be negated, but the quantifier itself must also change.

Rules for Negating Quantifiers

$$\neg(\forall x \in D, P(x)) \iff \exists x \in D, \neg P(x),$$

$$\neg(\exists x \in D, P(x)) \iff \forall x \in D, \neg P(x).$$

Intuitively:

- the negation of "everyone passed" is not "everyone failed", but "at least one person failed";
- the negation of "there exists one person who passed" is not "there exists one person who failed", but "everyone failed".

Example 3.14 Negating Quantified Propositions

Write the negation of each proposition.

- (1) For every real number x , $x^2 \geq 0$.
- (2) There exists an integer n such that $n^2 = 5$.
- (3) For every $x \in \mathbb{R}$, there exists $y \in \mathbb{R}$ such that $y > x$.

- (4) There exists $x \in \mathbb{R}$ such that $x > 0$ and $x^2 < 1$.

Solution:

- (1) The original proposition is $\forall x \in \mathbb{R}, x^2 \geq 0$. Its negation is

$$\exists x \in \mathbb{R}, x^2 < 0.$$

In words: ``There exists a real number x such that $x^2 < 0$."

- (2) The original proposition is $\exists n \in \mathbb{Z}, n^2 = 5$. Its negation is

$$\forall n \in \mathbb{Z}, n^2 \neq 5.$$

- (3) The original proposition is $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, y > x$. When negating, both quantifiers must change, and finally $y > x$ is negated to $y \leq x$:

$$\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, y \leq x.$$

In words: ``There exists a real number x such that every real number y satisfies $y \leq x$."

- (4) The original proposition is $\exists x \in \mathbb{R}, (x > 0 \wedge x^2 < 1)$. Its negation is

$$\forall x \in \mathbb{R}, x \leq 0 \text{ or } x^2 \geq 1.$$

3.3.6 Sufficient Conditions and Necessary Conditions (充分条件与必要条件)

This subsection deals with the question ``which condition can imply which condition". When solving equations and inequalities or writing proofs, we often write down many conditions. Some conditions, once true, are guaranteed to make another condition true; others are not. Sufficient and necessary conditions are the language used to describe such implication relationships accurately.

If p implies q , we write $p \Rightarrow q$. At this time p is a sufficient condition for q , and q is a necessary condition for p . If $p \Rightarrow q$ and $q \Rightarrow p$, then p and q are sufficient and necessary conditions for each other, written as $p \Leftrightarrow q$.

This section is a higher step in the logic part of this chapter. During the first pass, you only need to master one sentence: $p \Rightarrow q$ means ``once  $p$  holds,  $q$  must hold". After you are fluent with this, then distinguish the names ``sufficient" and ``necessary".

To understand these two words, do not first memorize the definitions. First look at the direction of the arrow:

$$p \Rightarrow q.$$

This means that as long as p holds, it is enough to guarantee q , so p is **sufficient** for q . At the same time, since q must hold whenever p holds, if we want p to hold, q at least cannot fail, so q is **necessary** for p .

Steps for Judging Sufficient and Necessary Conditions

- (1) First check $p \Rightarrow q$. If it holds, then p is a sufficient condition for q . If it does not hold, find a counterexample that satisfies p but not q .
- (2) Then check $q \Rightarrow p$. If it holds, then p is a necessary condition for q . If it does not hold, find a counterexample that satisfies q but not p .
- (3) If both directions hold, it is a necessary and sufficient condition. If neither direction holds, it is neither sufficient nor necessary.

Example 3.15 Relations Between Conditions

Judge the relations between the following conditions:

- (1) $p : x = 2, q : x^2 = 4$;
- (2) $p : x > 3, q : x > 0$;
- (3) $p : x$ is even, $q : x^2$ is even, where $x \in \mathbb{Z}$.

Solution:

- (1) If $x = 2$, then $x^2 = 4$, so p is a sufficient condition for q . But when $x^2 = 4$, x may also be -2 , so $x = 2$ does not necessarily hold. Therefore p is not a necessary condition for q .
- (2) If $x > 3$, then certainly $x > 0$, so p is a sufficient condition for q . But $x > 0$ does not necessarily imply $x > 3$, so it is not necessary.
- (3) If the integer x is even, then x^2 is even. Conversely, if x^2 is even, then x must also be even. Therefore p and q are necessary and sufficient conditions for each other.

In-Class Practice 3.3.1 Foundations of Logic

- (1) Decide whether " $x + 1 = 0$ " is a proposition, and explain why.
- (2) Write the negation of " $x \leq 1$ ".
- (3) Write the negation of " $x > 0$ and $y > 0$ ".
- (4) Write the negation of " $x = 0$ or $y = 0$ ".
- (5) Write the negation of "for every real number x , $x^2 + 1 > 0$ ".
- (6) Write the negation of "there exists an integer n such that $n^2 = 3$ ".
- (7) Decide whether " $x = 1$ " is a sufficient condition or a necessary condition for " $x^2 = 1$ ".
- (8) Decide whether " $ab = 0$ " and " $a = 0$ or $b = 0$ " are necessary and sufficient conditions for each other.

3.4 Cartesian Coordinate System (平面直角坐标系)

Coordinates convert geometric positions into ordered pairs, and are the common language of function graphs and analytic geometry.

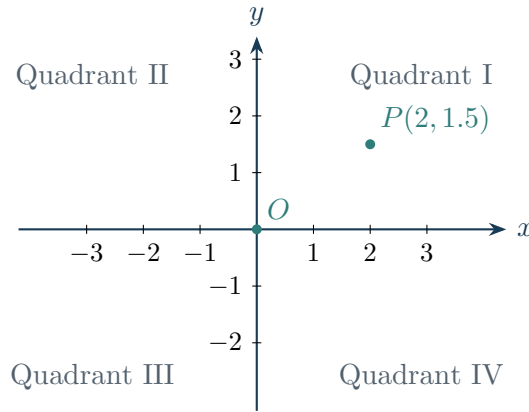
3.4.1 Coordinate Axes, Quadrants and Plotting Points (坐标轴、象限与描点)

When we studied the number line earlier, one real number was enough to represent a point on a line. For example, 3 represents the position 3 units to the right of the origin, and -2 represents the position 2 units to the left of the origin. This is suitable for describing positions on a one-dimensional line.

But for a point in the plane, saying only "left or right" is not enough; we must also say "up or down". Therefore, the Cartesian coordinate system uses two numbers to locate a point: the first number records the horizontal position, and the second number records the vertical position. In other words, a number line uses one number to represent a one-dimensional position; a coordinate system uses an ordered pair (x, y) to represent a position in the two-dimensional plane.

The horizontal axis is the x -axis, the vertical axis is the y -axis, and their intersection is the origin $O(0,0)$. For a point $P(x, y)$, x is the x -coordinate and y is the y -coordinate.

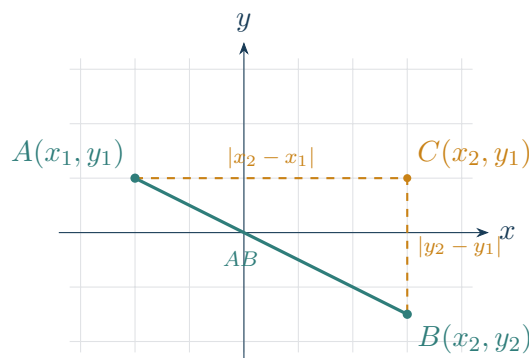
When plotting a point, first move along the x -axis direction to the position corresponding to the x -coordinate, and then move vertically to the position corresponding to the y -coordinate. The signs determine direction: a positive x -coordinate means right, a negative one means left; a positive y -coordinate means up, and a negative one means down.



3.4.2 Distance Between Two Points and Midpoint (两点间距离与中点)

This subsection deals with two basic measurement problems in the coordinate plane: how far apart are two points? Where is the exact middle of the segment connecting two points? The first is used to measure length, and the second is used to locate the center of a segment. They will be used repeatedly later when studying lines, circles and analytic geometry.

The distance formula between two points comes from the Pythagorean theorem. Let $A(x_1, y_1)$, $B(x_2, y_2)$. Draw a horizontal line from A , and a vertical line from B , meeting at $C(x_2, y_1)$. Then the length of AC is the absolute value of the change in the x -coordinate, $|x_2 - x_1|$, and the length of BC is the absolute value of the change in the y -coordinate, $|y_2 - y_1|$. The segment AB is the hypotenuse of the right triangle.



By the Pythagorean theorem,

$$AB^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2,$$

so

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

The midpoint formula comes from ‘‘averaging’’. The x -coordinate of the midpoint of a segment should be halfway between x_1 and x_2 , and the y -coordinate should be halfway between y_1 and y_2 . Therefore the coordinates of the midpoint M are

$$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right).$$

Example 3.16 Coordinate Calculations

Given $A(-2, 1)$, $B(4, -3)$:

- (1) determine the quadrants in which the two points lie;
- (2) find the midpoint of segment AB ;
- (3) find the length of AB .

Solution:

- (1) $A(-2, 1)$ has negative x -coordinate and positive y -coordinate, so it is in Quadrant II.
 $B(4, -3)$ has positive x -coordinate and negative y -coordinate, so it is in Quadrant IV.

- (2) The midpoint is

$$\left(\frac{-2 + 4}{2}, \frac{1 + (-3)}{2}\right) = (1, -1).$$

- (3) The distance is

$$AB = \sqrt{(4 - (-2))^2 + (-3 - 1)^2} = \sqrt{6^2 + (-4)^2} = \sqrt{52} = 2\sqrt{13}.$$

3.5 The Concept of Function (函数的概念)

A function is not a formula, but a correspondence in which every input has only one output. A formula is only one way to describe such a correspondence. Tables, graphs and verbal rules can also describe functions.

3.5.1 Function, Domain and Function Value (函数、定义域与函数值)

First look at several correspondences:

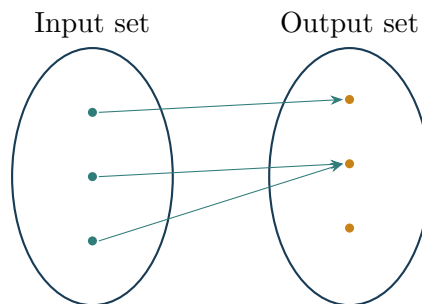
Correspondence	Input	Is It a Function?
Each student corresponds to their own student ID number	student	Yes. Each student has only one student ID number
Each real number x corresponds to x^2	real number x	Yes. Once x is given, x^2 is uniquely determined
Finding y from x in the relation $y^2 = x$	nonnegative real number x	Not necessarily. For example, when $x = 4$, $y = 2$ or $y = -2$

This shows that when judging a function, the key is not whether there is a formula, but whether each input can produce a unique output.

Suppose there are two variables x, y in a changing process. For every x in a given set D , if a certain definite rule gives a unique y corresponding to it, then y is called a **function (函数)** of x , written as $y = f(x)$. D is the **domain (定义域)**.

Core Criterion for Judging a Function

Every allowed input x must have an output, and can have only one output. Different inputs may have the same output.



Different Inputs May Have the Same Output

A function does not require different inputs to give different outputs. For example, for $f(x) = x^2$, $f(2) = 4$ and $f(-2) = 4$, but this is still a function. A function only forbids "one same input gives multiple outputs".

3.5.2 Basic Restrictions on the Domain

The domain comes before every calculation. If a number is not in the domain, it cannot be substituted into the function. Even if some algebraic simplification seems to produce a result after substitution, the original function is still meaningless at that point.

Common Restrictions

- ▶ A denominator cannot be zero.
- ▶ The expression under an even root must be greater than or equal to zero.
- ▶ In real-world problems, the quantity must also fit its practical meaning.

The three restrictions can be remembered using the smallest examples:

$$\frac{1}{x-2} \text{ requires } x \neq 2, \quad \sqrt{x+1} \text{ requires } x \geq -1, \quad \text{side length } x \text{ requires } x > 0.$$

The first two come from the expression itself, while the third comes from practical meaning.

Example 3.17 Finding the Domain

Find the domains of the following functions:

$$(1) f(x) = \frac{x+1}{x^2-4};$$

$$(2) \quad g(x) = \sqrt{3 - x};$$

$$(3) \quad h(x) = \frac{\sqrt{x + 2}}{x - 1}.$$

Solution: When finding the domain, do not rush to simplify. First find, one by one, the places where substitution is not allowed.

(1) Require $x^2 - 4 \neq 0$, so $x \neq \pm 2$. The domain is $\{x \in \mathbb{R} \mid x \neq \pm 2\}$.

(2) Require $3 - x \geq 0$, so $x \leq 3$. The domain is $(-\infty, 3]$.

(3) Require $x + 2 \geq 0$ and $x - 1 \neq 0$, so $x \geq -2$ and $x \neq 1$. The domain is $[-2, 1) \cup (1, +\infty)$.

Preparation for Function Graphs: Analytic Expressions, Tables and Plotting Points

If two variables x, y are related by an analytic expression, for example

$$y = x^2 - 2x + 3,$$

then each allowed value of x determines a corresponding value of y . Write each pair of corresponding values as an ordered pair (x, y) , and then plot these points in the Cartesian coordinate plane. The whole collection of these points forms the graph of this correspondence.

This step connects the three languages before: the domain tells us which x -values may be chosen; the function rule tells us the corresponding y -values; the coordinate plane turns each pair (x, y) into a point on the graph.

Basic Steps for Drawing a Graph from an Analytic Expression

- (1) First determine which x -values can be substituted.
- (2) Choose several representative x -values.
- (3) Use the analytic expression to calculate the corresponding y -values.
- (4) Write each pair of numbers as a point (x, y) .
- (5) Plot these points in the coordinate plane.
- (6) If the points show a continuous trend, connect them with a smooth curve or line segment.

Example 3.18 Plotting and Connecting Points for a Linear Function

Given the linear function $y = 2x + 1$, draw its graph by making a table, plotting points and connecting them.

Solution: First choose several x -values that are easy to calculate, and substitute them into

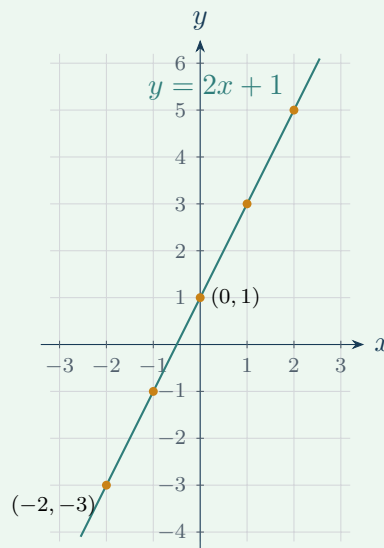
the expression one by one:

x	-2	-1	0	1	2
$y = 2x + 1$	-3	-1	1	3	5

Therefore we obtain five points:

$$(-2, -3), \quad (-1, -1), \quad (0, 1), \quad (1, 3), \quad (2, 5).$$

Plot these points in the coordinate plane and connect them to obtain the graph of $y = 2x + 1$. We can see that the graph of a linear function is a straight line.



Example 3.19 Plotting and Connecting Points for an Inverse Proportional Function

Given the inverse proportional function $y = \frac{4}{x}$, draw its graph by making a table, plotting points and connecting them.

Solution: In the domain, $x \neq 0$, so we cannot take $x = 0$ when making the table. Choose points on the left and right sides of 0:

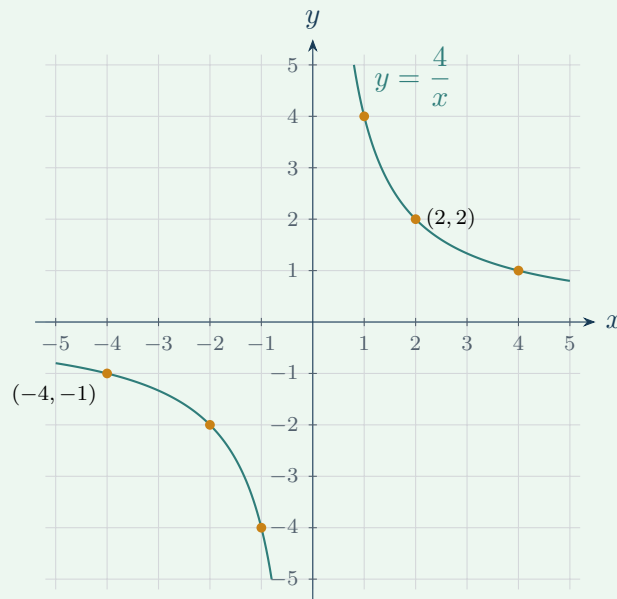
x	-4	-2	-1	1	2	4
$y = \frac{4}{x}$	-1	-2	-4	4	2	1

Therefore we obtain six points:

$$(-4, -1), \quad (-2, -2), \quad (-1, -4), \quad (1, 4), \quad (2, 2), \quad (4, 1).$$

Plot these points in the coordinate plane. Since $x = 0$ cannot be substituted, the graph will not pass through the y -axis. The points on the left form one branch, and the points on the right form another branch. When connecting, use smooth curves, and do not connect the two branches across the y -axis. We can see that the graph of an inverse proportional function

consists of two curves, also called a hyperbola. You may also try plotting more points to draw the function graph more accurately.



Common Mistake

A graph is not merely the few points in the table, but the whole collection of all points satisfying the analytic expression. The more points we plot, the easier it is to see the overall trend; but the final graph represents ``all points satisfying the expression''.

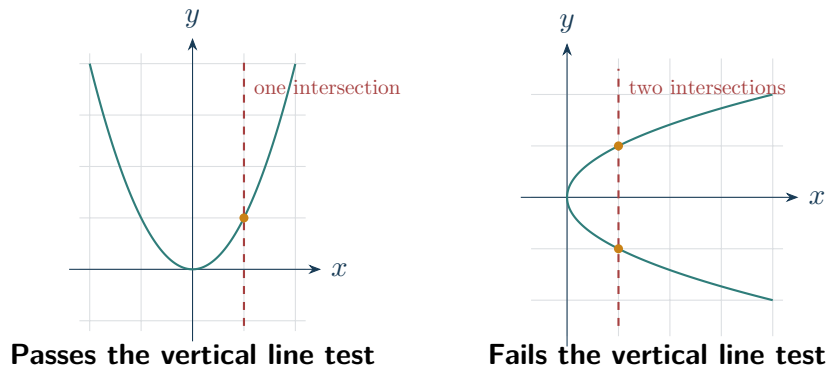
3.5.3 Range and Graphical Test (值域与图像判定)

The set of all possible outputs of a function is called the **range (值域)**. The domain asks ``which  $x$ -values are allowed as inputs'', while the range asks ``which y -values these inputs can finally produce''. Therefore, when finding the range, we cannot only look at the formula; we must also look at the domain.

From a table, the range is the set of all output values in the right column. From an analytic expression, the range is all values that $f(x)$ can take. From a graph, the domain is the projection of the graph onto the x -axis, and the range is the projection of the graph onto the y -axis.

If a vertical line intersects a graph at two or more points, then the graph is not the graph of y as a function of x . This is called the **vertical line test (竖线检验)**.

The reason for the vertical line test is exactly the definition of a function: when an x -value is fixed, all points on a vertical line have the same x -coordinate. If this vertical line intersects the graph at two points, it means the same input x corresponds to two different outputs y , which does not meet the requirement of a function.

**Example 3.20 Function Values and Range**

Let $f(x) = 2x + 1$.

- (1) Find $f(0)$, $f(-1)$, and $f(a+1)$.
- (2) Find its range on $\mathbb{R}$.
- (3) Find its range when $x \in [0, 3]$.
- (4) A segment of the graph of a quadratic function on a certain domain is given below. Without calculating, read its domain and range only from the graph.

Solution:

(1)

$$f(0) = 1, \quad f(-1) = -1, \quad f(a+1) = 2(a+1) + 1 = 2a + 3.$$

(2) When x varies freely in $\mathbb{R}$, $2x + 1$ can also take any real value. Therefore the range of $f(x)$ on $\mathbb{R}$ is $\mathbb{R}$.

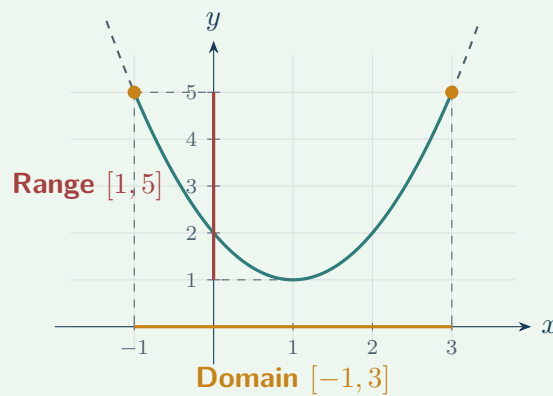
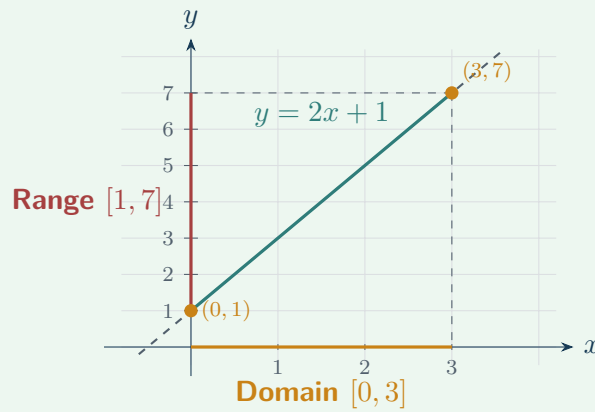
(3) When $x \in [0, 3]$, $f(x) = 2x + 1$ increases as x increases. The endpoint values are

$$f(0) = 1, \quad f(3) = 7.$$

Therefore the range is $[1, 7]$.

(4) Look at the projection of the graph onto the x -axis: the domain is $[-1, 3]$. Look at the projection of the graph onto the y -axis: the lowest point is the vertex, whose y -coordinate is 1; the highest points are at the two endpoints of this segment of the graph, whose y -coordinate is 5. Therefore the range is $[1, 5]$.

From a graph, if we study only a segment of a function graph, then the domain is the projection of this segment onto the x -axis, and the range is the projection of this segment onto the y -axis.

**In-Class Practice 3.5.1 Concept of Function**

- (1) Does the relation $y^2 = x$ define y as a function of x ?
- (2) Let $f(x) = 2x - 3$. Find $f(4)$ and $f(a + 1)$.
- (3) Find the domain of $y = \frac{1}{x^2 - x}$.
- (4) Find the domain of $y = \sqrt{2x - 1}$.
- (5) Find the range of $y = |x - 2|$ on $[-1, 4]$.

3.6 Chapter Summary

Main Knowledge Thread

- (1) Sets are used to describe the scope of study; the roster method, set-builder notation and interval notation are only different ways to write the same range.
- (2) Intersection, union and complement of sets correspond to "and", "or" and "not"; interval operations can be read from a number line first.
- (3) Logical language is used to judge truth values of propositions, write negations, and analyze whether a condition can imply a conclusion.
- (4) In $p \Rightarrow q$, p is enough to guarantee q , so p is a sufficient condition for q ; q is a condition that must hold when p holds, so q is a necessary condition for p .
- (5) Coordinates write positions of points as ordered pairs; the distance formula comes from the Pythagorean theorem, and the midpoint formula comes from averaging the x - and y -coordinates separately.
- (6) A function requires every input to have a unique output; the domain comes before every calculation, and the range is the set of all possible outputs.
- (7) A function graph consists of all points satisfying the analytic expression. Plotting points and connecting them helps us see the overall trend, and the vertical line test comes from "one input can have only one output".

3.7 Chapter Exercises

How to Use These Exercises

Group A checks the minimum mastery for this chapter. If you make mistakes, return to the corresponding section and reread the examples. Group B is for integrated practice before a quiz. Group C contains extension problems; during a first self-study pass, you may skip it without affecting your entry into the next chapter.

Group A: Basic Check

Difficulty: Basic

- (1) Represent $\{x \in \mathbb{Z} \mid -3 < x \leq 2\}$ by the roster method.
- (2) Let $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$. Find $A \cap B$ and $A \cup B$.
- (3) Let $A = (-2, 3)$, $B = [1, 5]$. Find $A \cap B$.
- (4) Write the negation of the proposition " $x > 1$ and $x \leq 4$ ".
- (5) Determine what kind of condition $x = 0$ is for $x^2 = 0$.
- (6) Find the midpoint and distance between $A(-1, 3)$ and $B(5, -1)$.

(7) Let $f(x) = x^2 - 3x$. Find $f(-2)$ and $f(a+1)$.

(8) Find the domain of $y = \frac{x+1}{x^2-9}$.

(9) Find the domain of $y = \sqrt{4-2x}$.

Group B: Quiz Level

Difficulty: Core

(1) $A = \{x \mid -2 < x < 3\}$, $B = \{x \mid x \geq a\}$. Find all a such that $A \cap B = \emptyset$.

(2) The universal set is $\mathbb{R}$. Find $\complement_{\mathbb{R}}((-\infty, -1] \cup (2, 4))$.

(3) Let $p : x > 2$, $q : x^2 > 4$. Judge the sufficient and necessary relationship between p and q .

(4) Find the domain of $y = \frac{\sqrt{x+1}}{x^2-4}$.

(5) Find the range of $y = |x-1| + |x+2|$ on $[-3, 2]$.

Group C: Extension and Thinking

Difficulty: Extension

- (1) Prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. You may use a Venn diagram.
- (2) Write the negation of ``for every real number x , there exists a real number y such that $y > x$ ''.

Self-Check

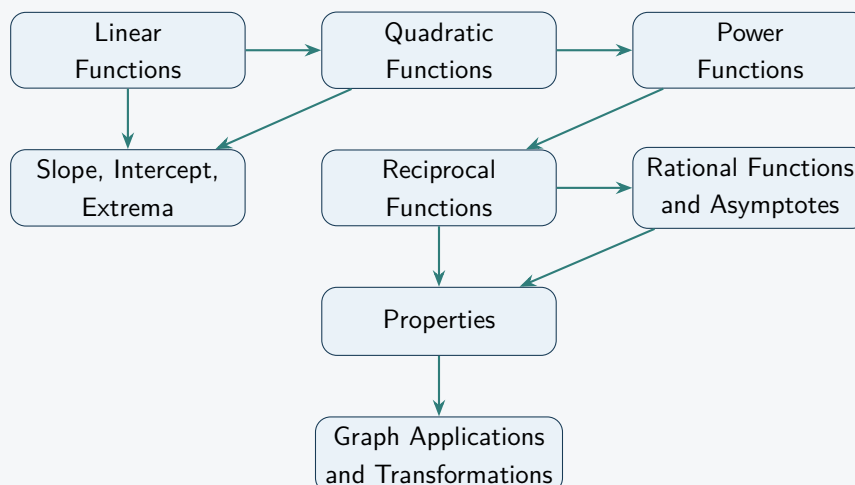
Skill	Can do	Mostly can	Not yet
Represent sets in three ways	☐	☐	☐
Compute intersections, unions and complements of sets	☐	☐	☐
Correctly negate a simple proposition	☐	☐	☐
Judge sufficient and necessary conditions	☐	☐	☐
Plot points in the coordinate plane and compute distances	☐	☐	☐
Determine whether a correspondence is a function	☐	☐	☐
Find domains and ranges of common functions	☐	☐	☐

Elementary Functions (基本函数)

Learning Goals

After studying this chapter, you should be able to:

- ▶ understand slope (斜率) and intercept (截距), draw graphs of linear functions and determine monotonicity;
- ▶ determine the analytic expression of a linear function from points, slope or intercepts;
- ▶ convert a quadratic function into vertex form, and find its axis of symmetry, vertex, zeros and extrema;
- ▶ determine the monotonicity and range of a quadratic function on a specified interval;
- ▶ recognize the graphs, domains, parity and monotonicity of common power functions;
- ▶ study the domain, range and monotonicity of reciprocal and rational functions;
- ▶ find the horizontal and vertical asymptotes (渐近线) and the center of symmetry (对称中心) of a rational function;
- ▶ summarize monotonicity, parity and periodicity from concrete function graphs in general language;
- ▶ use function graphs to discuss roots of equations and solution sets of inequalities;
- ▶ understand composite functions and graph transformations.

Knowledge Map

This chapter first builds the ability to read graphs from lines and parabolas, then expands our understanding of domains, symmetry and asymptotes through power functions and rational functions. After these concrete functions have been studied, monotonicity, parity and periodicity are organized into general language, and finally graphs are used to handle equations, inequalities and transformations.

Learning Route of This Chapter

When studying functions, look at three things at the same time:

- (1) the analytic expression tells us how to calculate y from x ;
- (2) the graph tells us how the function changes as a whole;
- (3) properties organize rising, falling, symmetry and repetition in the graph into general language.

This chapter first studies several most basic types of functions, and then organizes their properties uniformly. When studying by yourself, whenever you meet a new function, read it according to these questions: domain and range, graph shape, key points or asymptotes, monotonicity and symmetry.

Prerequisites and Minimum Mastery

To study this chapter, you need equations from Chapter 2 and the foundations of function graphs from Chapter 3. The minimum goal for the first pass is: draw sketches of linear functions, quadratic functions and reciprocal functions; read domains, ranges, monotonic intervals and intersections from graphs; and use completing the square to find the vertex of a quadratic function. Rational functions, composite functions and scaling transformations can be treated as second-pass priorities.

Common Self-Study Sticking Points in This Chapter

In this chapter, do not only memorize analytic expressions. Every function should be viewed together with a graph. When you meet "asymptote", "parity", "periodicity", "composite function" or "scaling", if it is unfamiliar on the first pass, first learn how to read it from the graph, and then come back to handle the symbolic definition.

4.1 Linear Functions (一次函数)

The graph of a linear function is a straight line. The slope determines the direction of tilt, and the intercept determines the intersection with an axis.

4.1.1 Slope and Intercept

The general form of a linear function is

$$y = kx + b \quad (k \neq 0).$$

Here k is the **slope**, and b is the y -intercept. The graph intersects the y -axis at $(0, b)$. When $k \neq 0$, it intersects the x -axis at $(-b/k, 0)$.

Linear functions are often used to describe relationships in which "whenever the input increases by 1 unit, the result increases or decreases by the same amount". For example, when walking at a constant speed, the distance increases by a fixed length every minute; in taxi fare, when the distance traveled increases by 1 kilometer, the fare increases by a fixed amount. This "fixed change" is the slope.

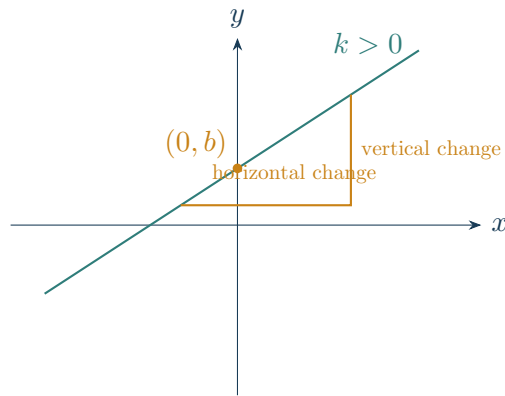
If two points on a line are (x_1, y_1) and (x_2, y_2) , with $x_1 \neq x_2$, then

$$k = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{change in } y}{\text{change in } x}.$$

The slope measures "how much the y -coordinate changes on average when the x -coordinate increases by 1". When moving from (x_1, y_1) to (x_2, y_2) , the horizontal change is $x_2 - x_1$, and the vertical change is $y_2 - y_1$. Therefore their ratio describes the tilt of the line. The tilt of a line is the same everywhere, so the slope computed from any two points is the same.

Reading Slope, Rising and Falling from a Graph

- ▶ $k > 0$: the graph rises from left to right; later we will say it is **increasing** on $\mathbb{R}$;
- ▶ $k < 0$: the graph falls from left to right; later we will say it is **decreasing** on $\mathbb{R}$;
- ▶ $k = 0$: this gives the constant function $y = b$.



4.1.2 Determining the Analytic Expression of a Linear Function

This subsection deals with the problem of "recovering a line". That is, the problem does not directly give $y = kx + b$, but gives a point, slope, intercept or a real-world rate of change. We need to translate this information into k and b .

If a line passes through (x_0, y_0) and has slope k , then any point (x, y) on the line satisfies: when moving from (x_0, y_0) to (x, y) ,

$$\frac{y - y_0}{x - x_0} = k \quad (x \neq x_0).$$

After rearranging, we obtain the point-slope form:

$$y - y_0 = k(x - x_0).$$

The meaning of the point-slope form is direct: one point fixes the position of the line, and the slope fixes the direction of the line.

Two points determine a line. The reason is: once two distinct points are known, the direction of the line is known; after the direction is known, requiring the line to pass through one of the points fixes its position. Therefore, the essence of "finding a line from two points" is: **first use the two points to compute the slope, and then substitute one point and the slope into point-slope form.**

Suppose a line passes through two distinct points

$$P_1(x_1, y_1), \quad P_2(x_2, y_2).$$

If $x_1 \neq x_2$, this line is not vertical, and its slope is

$$k = \frac{y_2 - y_1}{x_2 - x_1}.$$

Substituting $P_1(x_1, y_1)$ and the slope k into point-slope form gives

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1).$$

This is the basic form of the line obtained from two points. If we rearrange both sides, it can also be written in the common two-point form:

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1} \quad (y_1 \neq y_2).$$

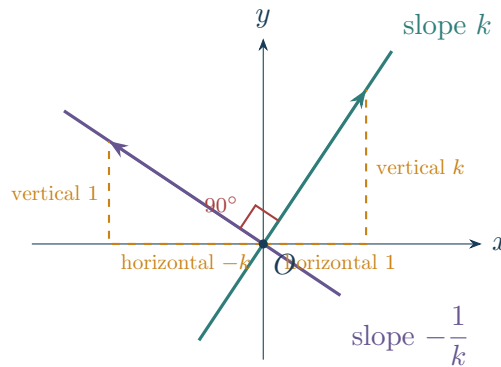
When $y_1 = y_2$, the line is horizontal, so we directly write $y = y_1$. When $x_1 = x_2$, the line is the vertical line $x = x_1$, and it is not the graph of a linear function.

Parallel and Perpendicular Lines

For two non-vertical lines with slopes k_1, k_2 :

$$k_1 = k_2 \Rightarrow \text{parallel or coincident}, \quad k_1 k_2 = -1 \Rightarrow \text{perpendicular}.$$

Parallel lines have the same direction, so they have the same slope. Why does $k_1 k_2 = -1$ for perpendicular lines? First look at the figure below. A line with slope k can be represented by the direction "right 1, up k ". A direction perpendicular to it can be viewed as rotating this direction by 90° , giving "left k , up 1".



Reading slope from the figure, the slope of the first line is

$$\frac{\text{vertical change}}{\text{horizontal change}} = \frac{k}{1} = k.$$

For the perpendicular direction, the horizontal change is $-k$, and the vertical change is 1, so the slope of the perpendicular line is

$$\frac{1}{-k} = -\frac{1}{k}.$$

Therefore the slopes of two non-horizontal and non-vertical perpendicular lines satisfy

$$k \cdot \left(-\frac{1}{k}\right) = -1.$$

A horizontal line and a vertical line are also perpendicular, but a vertical line has no slope, so it is not included in this product formula.

Example 4.1 Finding a Line from Two Points

Find the analytic expression of the linear function passing through $(-1, -1)$ and $(1, 3)$, and write the intercepts on the two coordinate axes.

Solution: The slope is

$$k = \frac{3 - (-1)}{1 - (-1)} = \frac{4}{2} = 2.$$

Let the analytic expression be $y = 2x + b$. Substituting $(1, 3)$, we get $3 = 2 + b$, so $b = 1$.

Therefore

$$y = 2x + 1.$$

The intersection with the y -axis is $(0, 1)$, so the y -intercept is 1. Letting $y = 0$, we get $2x + 1 = 0$, so $x = -\frac{1}{2}$. Therefore the intersection with the x -axis is $\left(-\frac{1}{2}, 0\right)$, and the x -intercept is $-\frac{1}{2}$.

Example 4.2 Parallel and Perpendicular Lines

Find the line passing through $(1, 2)$ and parallel to $y = 2x + 1$. Then find the line passing through the same point and perpendicular to that line.

The perpendicular line is a second-level requirement in this section. During the first pass, first make sure you can find parallel lines; perpendicular relationships will be used systematically again in Chapter 7 on analytic geometry.

Solution: A line parallel to $y = 2x + 1$ still has slope 2. Let $y = 2x + b$. Substituting $(1, 2)$:

$$2 = 2 \cdot 1 + b,$$

so $b = 0$. Therefore the parallel line is $y = 2x$.

The perpendicular line has slope $-\frac{1}{2}$. Let $y = -\frac{1}{2}x + b$. Substituting $(1, 2)$:

$$2 = -\frac{1}{2} + b,$$

so $b = \frac{5}{2}$. Therefore the perpendicular line is

$$y = -\frac{1}{2}x + \frac{5}{2}.$$

In-Class Practice 4.1.1 Linear Functions

- (1) Write the slope, x -intercept and y -intercept of $y = -3x + 6$.
- (2) Find the line passing through $(2, -1)$ with slope 4.
- (3) Find the line passing through $(-2, 3)$ and $(4, 0)$.
- (4) Find the line passing through $(0, -1)$ and parallel to $y = 3x - 2$.
- (5) Find the line passing through $(0, -1)$ and perpendicular to $y = 3x - 2$.
- (6) A taxi fare consists of a starting fee and a mileage fee. A trip of 3 km costs 18 yuan, and a trip of 8 km costs 33 yuan. Build a linear function model for fare and mileage.

4.2 Quadratic Functions (二次函数)

The graph of a quadratic function is a parabola (抛物线). Standard form is suitable for reading coefficients, vertex form is suitable for reading symmetry and extrema, and factored form is suitable for reading zeros.

4.2.1 Three Common Forms

A quadratic function is a function given by a quadratic polynomial in x . It usually has the following three forms.

Representations of a Quadratic Function

$$\text{Standard form: } y = ax^2 + bx + c,$$

$$\text{Vertex form: } y = a(x - h)^2 + k,$$

$$\text{Zero form: } y = a(x - x_1)(x - x_2),$$

where $a \neq 0$. The zero form requires the function to have real zeros.

What Each Form Reads Most Easily

Form	Information Read Most Easily
$y = ax^2 + bx + c$	y -intercept $(0, c)$, and a, b, c needed for completing the square
$y = a(x - h)^2 + k$	vertex (h, k) , axis of symmetry $x = h$, opening direction and range
$y = a(x - x_1)(x - x_2)$	zeros x_1, x_2 , namely the intersections with the x -axis

4.2.2 Opening, Axis of Symmetry and Vertex (开口、对称轴与顶点)

When seeing a quadratic function, first determine whether it opens upward or downward, where its center line is, and where the lowest or highest point is. This information determines the rough position of the graph and the range of its values.

For $y = ax^2 + bx + c$, we have

$$\text{axis of symmetry } x = -\frac{b}{2a}, \quad \text{vertex } \left(-\frac{b}{2a}, -\frac{\Delta}{4a}\right).$$

If $a > 0$, it opens upward and the vertex gives the minimum value. If $a < 0$, it opens downward and the vertex gives the maximum value.

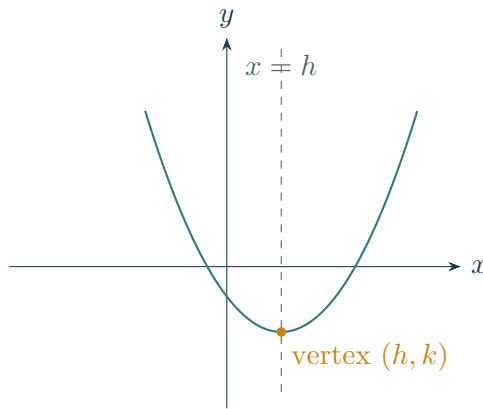
This formula comes from completing the square:

$$y = a\left(x^2 + \frac{b}{a}x\right) + c = a\left[\left(x + \frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2\right] + c.$$

After simplifying,

$$y = a\left(x + \frac{b}{2a}\right)^2 + \frac{4ac - b^2}{4a}.$$

The square term has minimum 0, and it is symmetric about $x = -\frac{b}{2a}$, so the axis of symmetry is $x = -\frac{b}{2a}$, and the y -coordinate of the vertex is $\frac{4ac - b^2}{4a} = -\frac{\Delta}{4a}$. We can see that the information of the vertex $(h, k) = \left(-\frac{b}{2a}, -\frac{\Delta}{4a}\right)$ is contained in this form, so it is also called vertex form.



Example 4.3 Completing the Square and Graphing Information

Convert $y = 2x^2 - 8x + 5$ into vertex form, and write its opening direction, axis of symmetry, vertex, extremum, and intersection with the y -axis.

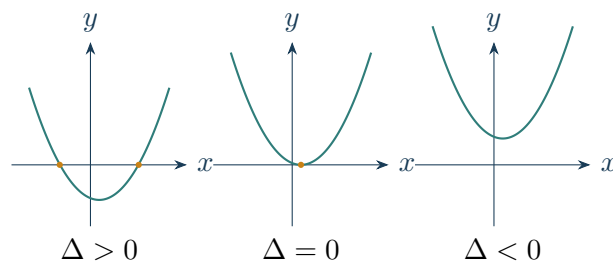
Solution:

$$y = 2(x^2 - 4x) + 5 = 2[(x - 2)^2 - 4] + 5 = 2(x - 2)^2 - 3.$$

Since the quadratic coefficient $2 > 0$, the parabola opens upward. The axis of symmetry is $x = 2$, the vertex is $(2, -3)$, the minimum value is -3 , and there is no maximum value. Letting $x = 0$, we get $y = 5$, so the intersection with the y -axis is $(0, 5)$.

4.2.3 Zeros, Intersections and Discriminant (零点、交点与判别式)

The x -coordinates of the intersections of $y = ax^2 + bx + c$ with the x -axis are exactly the real roots of the equation $ax^2 + bx + c = 0$. The three cases $\Delta > 0$, $\Delta = 0$, and $\Delta < 0$ correspond respectively to two, one and zero intersections.



The discriminant is not a new graph rule; it is the graphical expression of the number of roots of an equation.

Example 4.4 Parabola and Coordinate Axes

Study $y = x^2 - 5x + 4$:

- (1) find its intersections with the two coordinate axes;
- (2) write its axis of symmetry and vertex;
- (3) draw a sketch, and use the graph to write the range of x when $y < 0$.

Solution:

- (1) To find intersections with the x -axis, let $y = 0$:

$$x^2 - 5x + 4 = (x - 1)(x - 4) = 0,$$

giving $(1, 0)$ and $(4, 0)$. To find the intersection with the y -axis, let $x = 0$, giving $(0, 4)$.

- (2) The axis of symmetry is

$$x = -\frac{b}{2a} = \frac{5}{2}.$$

The y -coordinate of the vertex is

$$f\left(\frac{5}{2}\right) = \frac{25}{4} - \frac{25}{2} + 4 = -\frac{9}{4},$$

so the vertex is $\left(\frac{5}{2}, -\frac{9}{4}\right)$.

- (3) The parabola opens upward. The part below the x -axis lies between the two zeros, so when $y < 0$,

$$1 < x < 4.$$

4.2.4 Reading Rising, Falling and Interval Extrema from a Graph

The graph of a quadratic function is a parabola. Its axis of symmetry divides the graph into left and right halves. If $a > 0$, the parabola opens upward; from left to right, it first falls to the vertex, and then rises from the vertex. If $a < 0$, the situation is reversed: it first rises to the vertex, and then falls from the vertex.

Using the x -coordinate of the vertex $x = -\frac{b}{2a}$, this means: when $a > 0$, the function is decreasing on $(-\infty, -b/2a]$ and increasing on $[-b/2a, +\infty)$; when $a < 0$, the opposite holds.

To find the extrema on a closed interval $[m, n]$, compare the function values at the endpoints, and also compare the value at the vertex if the x -coordinate of the vertex belongs to the interval.

Example 4.5 Extrema on Closed Intervals

Find the maximum and minimum values of $f(x) = -x^2 + 2x + 3$ on $[0, 2]$ and on $[2, 4]$.

Solution: First complete the square:

$$f(x) = -(x^2 - 2x) + 3 = -(x - 1)^2 + 4.$$

The vertex is $(1, 4)$, and the parabola opens downward.

On $[0, 2]$, the vertex $x = 1$ lies inside the interval, so the maximum value is 4. The endpoint values are $f(0) = 3$ and $f(2) = 3$, so the minimum value is 3.

On $[2, 4]$, the farther the function is from the vertex, the smaller it becomes. The endpoint values are $f(2) = 3$ and $f(4) = -16 + 8 + 3 = -5$. Therefore the maximum value is 3, and the minimum value is -5 .

4.2.5 Reading Symmetry of Quadratic Functions from Graphs

Every parabola has an axis of symmetry. When the axis of symmetry is the y -axis, for example $f(x) = ax^2 + c$, the left and right sides appear in pairs with respect to the y -axis. Later we will call it an **even function**.

For a general $ax^2 + bx + c$, the axis of symmetry is $x = -\frac{b}{2a}$. Only when this axis is exactly the y -axis, that is, when $b = 0$, is it an even function. A nonzero quadratic function cannot be an odd function.

In-Class Practice 4.2.1 Quadratic Functions

- (1) Find the opening direction, axis of symmetry and vertex of $y = 3x^2 - 9x + 1$.
- (2) Find the range of $y = -x^2 + 4x - 1$ on $\mathbb{R}$.
- (3) Find the range of $y = -x^2 + 4x - 1$ on $[0, 3]$.
- (4) Find the intersections of the parabola $y = x^2 + 2x - 3$ with the coordinate axes.
- (5) A quadratic function has zeros 1, 3, and passes through $(0, 3)$. Find its analytic expression.
- (6) Given that $y = ax^2 + bx - 3$ passes through $(-4, 5)$ and has axis of symmetry $x = -1$, find a, b .

4.3 Power Functions (幂函数)

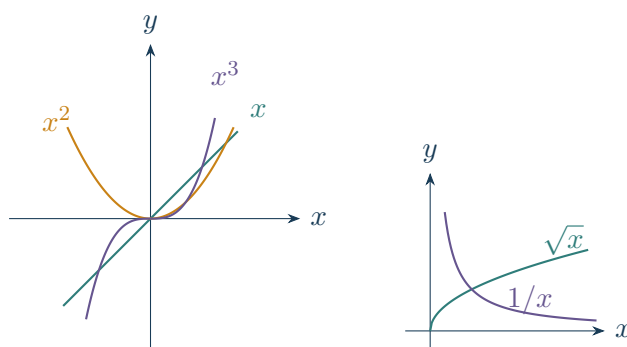
Functions of the form $y = x^\alpha$ are called power functions. When the exponent changes, the domain and graph shape may change completely. First remember from the graph shape how they rise or fall, and whether they are symmetric about the origin or the y -axis.

4.3.1 Common Power Functions with Integer Exponents

This subsection deals with the most common graphs of power functions. Power functions all look like $y = x^\alpha$, but the sign and parity of the exponent change the domain, symmetry and direction of the graph. When studying, first build intuition from graph shapes, and then return to memorize the properties.

Function	Domain	Range	Parity	Basic Monotonicity
$y = x$	$\mathbb{R}$	$\mathbb{R}$	odd	increasing
$y = x^2$	$\mathbb{R}$	$[0, +\infty)$	even	decreasing on the left, increasing on the right
$y = x^3$	$\mathbb{R}$	$\mathbb{R}$	odd	increasing
$y = x^{-1}$	$\mathbb{R} \setminus \{0\}$	$\mathbb{R} \setminus \{0\}$	odd	decreasing on each branch
$y = x^{-2}$	$\mathbb{R} \setminus \{0\}$	$(0, +\infty)$	even	increasing on the left, decreasing on the right

First understand this table through graphs: graphs such as x , x^3 , and x^5 rise all the way from left to right; graphs such as x^2 , x^4 , and x^{-2} appear in left-right pairs, so rising and falling often have to be considered on the two sides separately; graphs such as x^{-1} split into two branches, and monotonicity is discussed only inside each branch.



4.3.2 Power Functions with Fractional Exponents

Power functions with fractional exponents deal with the fact that "radicals can also be written as powers". For example, $\sqrt{x}$ can be written as $x^{1/2}$, and $\sqrt[3]{x}$ can be written as $x^{1/3}$. They belong to the same family of power functions as integer-exponent power functions, but their domains are especially restricted by the meaning of radicals.

The domain of $y = \sqrt{x} = x^{1/2}$ is $[0, +\infty)$, while the domain of $y = \sqrt[3]{x} = x^{1/3}$ is $\mathbb{R}$. When handling $x^{m/n}$, pay attention to whether n is odd or even, and reduce the exponent when appropriate.

Common Mistake

In the real-number system, $x^{1/2}$ requires $x \geq 0$, while $x^{-1/2}$ also requires the denominator to be nonzero, so we must have $x > 0$.

Example 4.6 Properties of Power Functions

Write the domain, range, parity and monotonic intervals of $y = x^4$, $y = x^5$, $y = x^{-2}$, and $y = x^{1/2}$.

Solution:

Function	Domain	Range	Parity and Monotonicity
$y = x^4$	$\mathbb{R}$	$[0, +\infty)$	even; decreasing on $(-\infty, 0]$, increasing on $[0, +\infty)$
$y = x^5$	$\mathbb{R}$	$\mathbb{R}$	odd; increasing on $\mathbb{R}$
$y = x^{-2}$	$(-\infty, 0) \cup (0, +\infty)$	$(0, +\infty)$	even; increasing on $(-\infty, 0)$, decreasing on $(0, +\infty)$
$y = x^{1/2}$	$[0, +\infty)$	$[0, +\infty)$	neither odd nor even; increasing on $[0, +\infty)$

In-Class Practice 4.3.1 Power Functions

- (1) Compare $2^{1/2}$ and $3^{1/2}$, and compare $(-2)^3$ and $(-1)^3$.
- (2) Find the domain of $y = x^{-3}$ and determine its parity.
- (3) Find the domain and range of $y = x^{2/3}$.
- (4) Use the basic graph to explain how the graph of $y = (x - 2)^3 + 1$ is obtained from $y = x^3$.

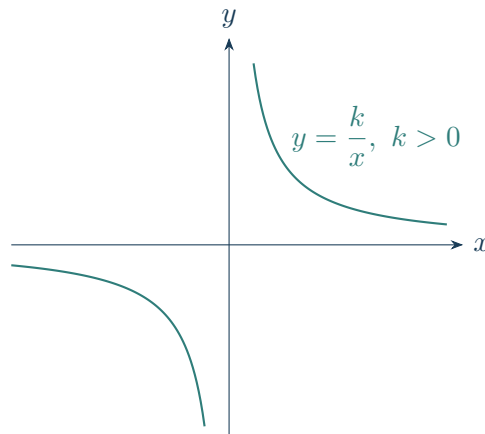
4.4 Reciprocal Functions (反比例函数)

The two branches of the reciprocal function $y = k/x$ approach the coordinate axes indefinitely, but never intersect them.

4.4.1 Graph and Basic Properties

For $y = k/x$ ($k \neq 0$): the domain and range are both $\mathbb{R} \setminus \{0\}$; the graph is symmetric about the origin, meaning that if a point (x, y) is on the graph, then the point $(-x, -y)$ obtained by a 180° rotation is still on the graph; $x = 0$ and $y = 0$ are asymptotes.

Reciprocal relationships also come from real-world problems. If the distance is fixed at 120 km, speed v and time t satisfy $vt = 120$, so $t = \frac{120}{v}$. The greater the speed, the less time is needed; but speed cannot be 0, and time will not be 0 either.



When $k > 0$, the graph lies in Quadrants I and III, and each branch falls from left to right. When $k < 0$, the graph lies in Quadrants II and IV, and each branch rises from left to right.

Common Mistake

We cannot say that $(-\infty, 0) \cup (0, +\infty)$ as a whole is one monotonic interval. For example, $y = 1/x$ is decreasing on each branch separately, but we cannot cross 0 and discuss monotonicity on the whole set as one interval.

Example 4.7 Determining a Reciprocal Function from a Point

The graph of a reciprocal function passes through $(-2, 3)$.

- (1) Find its analytic expression.
- (2) Determine the quadrants where the graph lies and its monotonic intervals.
- (3) Decide whether the point $(3, -2)$ lies on the graph.

Solution:

- (1) Let the reciprocal function be $y = \frac{k}{x}$. Substituting $(-2, 3)$:

$$3 = \frac{k}{-2},$$

giving $k = -6$, so the analytic expression is

$$y = -\frac{6}{x}.$$

(2) Since $k < 0$, the graph lies in Quadrants II and IV; it is increasing on both $(-\infty, 0)$ and $(0, +\infty)$.

(3) Substituting the point $(3, -2)$:

$$-\frac{6}{3} = -2,$$

which holds, so $(3, -2)$ lies on the graph.

4.5 Rational Functions (分式函数)

The most basic rational functions can be rewritten by algebraic transformation as "a constant plus a reciprocal function", and their graphs are translated hyperbolas.

4.5.1 Functions of the Form $\frac{ax+b}{cx+d}$

This subsection deals with the simplest type of rational function: both numerator and denominator are linear expressions. Their graphs are usually not lines, but hyperbolas obtained from reciprocal functions by translation and scaling. The learning goal is to see from the analytic expression which x -value is excluded, which lines the graph approaches, and roughly where the graph is.

When $c \neq 0$ and $ad - bc \neq 0$,

$$y = \frac{ax+b}{cx+d} = \frac{a}{c} + \frac{bc-ad}{c(cx+d)}.$$

Therefore the vertical asymptote (竖直渐近线) and horizontal asymptote (水平渐近线) are respectively

$$x = -\frac{d}{c}, \quad y = \frac{a}{c}.$$

The intersection $(-d/c, a/c)$ of the two asymptotes is the center of symmetry of the graph.

The idea of this transformation is "make the numerator contain the denominator". Since $cx+d$ is the denominator, if we first make $\frac{a}{c}(cx+d)$ appear in the numerator, we can separate out the constant $\frac{a}{c}$, and the remaining part is a constant divided by $cx+d$. In this way, the rational function is connected with the reciprocal function, and its graph can also be viewed as a translated hyperbola.

Example 4.8 Converting to a Translated Reciprocal Function

Study the function

$$y = \frac{2x+1}{x-3}.$$

Find its domain, range, two asymptotes, center of symmetry, intersections with the coordinate axes, and draw a sketch.

Solution: First convert it to a translated reciprocal function:

$$\frac{2x+1}{x-3} = \frac{2(x-3)+7}{x-3} = 2 + \frac{7}{x-3}.$$

The domain is $x \neq 3$. Since $\frac{7}{x-3}$ can never be 0, we have $y \neq 2$, so the range is $(-\infty, 2) \cup (2, +\infty)$.

The asymptotes are

$$x = 3, \quad y = 2,$$

and the center of symmetry is their intersection $(3, 2)$.

For the intersection with the y -axis, let $x = 0$, giving $y = -\frac{1}{3}$, so the point is $(0, -\frac{1}{3})$. For the intersection with the x -axis, let $y = 0$, giving $2x + 1 = 0$, so the point is $(-\frac{1}{2}, 0)$.

4.5.2 Reading Monotonicity and Range from a Translated Hyperbola

If a function can be written as

$$y = A + \frac{B}{x-h}, \quad B \neq 0,$$

then it is a hyperbola obtained by translating a reciprocal function. The vertical asymptote $x = h$ divides the graph into two branches, and the horizontal asymptote $y = A$ is a height that the graph approaches forever but never reaches. Therefore the domain excludes h , and the range excludes A . When $B > 0$, each branch falls from left to right. When $B < 0$, each branch rises from left to right.

Example 4.9 Properties of a Rational Function

Convert $y = \frac{x+2}{x-1}$ into the form $A + B/(x-h)$, and write its domain, range, asymptotes, center of symmetry and monotonic intervals.

Solution:

$$\frac{x+2}{x-1} = \frac{(x-1)+3}{x-1} = 1 + \frac{3}{x-1}.$$

Therefore the domain is $x \neq 1$, and the range is $y \neq 1$. The asymptotes are

$$x = 1, \quad y = 1,$$

and the center of symmetry is $(1, 1)$.

Since the coefficient $3 > 0$ in $y = 1 + \frac{3}{x-1}$, the reciprocal part is decreasing on $(-\infty, 1)$ and $(1, +\infty)$. Therefore the original function is also decreasing on both intervals.

4.5.3 Domains of More Complicated Rational Functions

For $y = P(x)/Q(x)$, the domain first excludes all points satisfying $Q(x) = 0$. Even if a factor disappears after cancellation, these points still cannot be added back to the domain of the original function; the graph leaves "holes".

Example 4.10 Cancellation and Holes

Study

$$f(x) = \frac{x^2 - 1}{x - 1}.$$

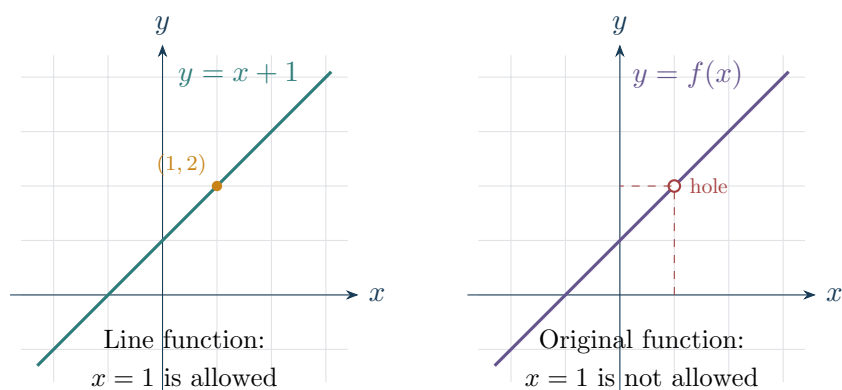
Simplify the analytic expression, write the domain of the original function, and explain how its graph differs from the graph of the line $y = x + 1$.

Solution: The original rational expression requires $x - 1 \neq 0$, so the domain is $x \neq 1$. When $x \neq 1$,

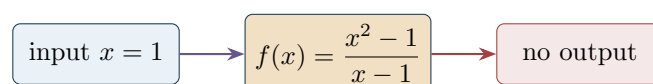
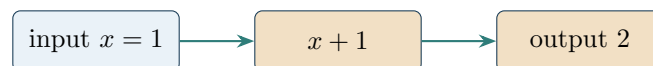
$$f(x) = \frac{(x-1)(x+1)}{x-1} = x+1.$$

Therefore its graph is basically the same as the line $y = x + 1$, but the original function is undefined at $x = 1$. The corresponding point on the line is $(1, 2)$, so the graph of the original function is the line $y = x + 1$ with the point $(1, 2)$ removed.

The following two figures illustrate the same idea: although the simplified expression is $x + 1$, whether the function allows the input $x = 1$ is different.



The line function allows 1 to enter



The original denominator is 0,
so 1 is excluded

In-Class Practice 4.5.1 Reciprocal and Rational Functions

- (1) $y = k/x$ passes through $(4, -2)$. Find k and determine the quadrants where the graph lies.
- (2) Write the domain, range, parity and monotonic intervals of $y = -3/x$.
- (3) Convert $y = \frac{3x-2}{x+1}$ into $A + B/(x-h)$.
- (4) Find the asymptotes, center of symmetry, domain and range of the function in the previous problem.
- (5) Study the monotonic intervals of $y = \frac{x+1}{2x-4}$.
- (6) Explain the difference between the graphs of $y = \frac{x^2-4}{x-2}$ and $y = x+2$.

4.6 Basic Properties of Functions (函数的基本性质)

Earlier we have seen lines, parabolas, power functions, reciprocal functions and rational functions. Now we organize "rising or falling from left to right", "appearing in pairs about an axis or a point", and "repeating after a fixed distance" into general language.

4.6.1 Looking Back at Properties from Concrete Functions

When observing a function graph, the first thing we see is not the symbols, but the shape: does the curve rise or fall from left to right? Do the left and right sides appear in pairs? If the graph is rotated half a turn about some point, does it coincide with itself? These intuitive phenomena correspond respectively to monotonicity, axial symmetry and central symmetry. A function symmetric about the y -axis is called an even function, and a function symmetric about the origin is called an odd function.

Review of Properties of Linear, Quadratic and Rational Functions

- The graph of a linear function $y = kx + b$ is a straight line. When $k > 0$, the whole line rises from left to right, so it is increasing on $\mathbb{R}$. When $k < 0$, the whole line falls from left to right, so it is decreasing on $\mathbb{R}$. If $b = 0$ and $k \neq 0$, the graph is symmetric about the origin and is an odd function. If $k = 0$, it is a constant function, symmetric about the y -axis, and is an even function; among these, $y = 0$ is also an odd function.
- The graph of a quadratic function $y = ax^2 + bx + c$ ($a \neq 0$) is a parabola, with axis of symmetry $x = -\frac{b}{2a}$. When $a > 0$, the graph falls on the left side of the axis of symmetry and rises on the right side. When $a < 0$, the graph rises on the left side and falls on the right side. Only when the axis of symmetry is the y -axis, that is, when $b = 0$, is a quadratic function even. A nonzero quadratic function is not odd.
- If a rational function can be written as $y = A + \frac{B}{x-h}$, its graph is a translated hyperbola, with center of symmetry (h, A) . When $B > 0$, each branch falls; when $B < 0$, each branch rises. A translated hyperbola is generally neither odd nor even. Only in a case like $y = \frac{B}{x}$, where the center of symmetry is the origin and the domain is also symmetric about the origin, does it appear as an odd function.

4.6.2 Monotonicity (单调性)

For a function $f(x)$ on an interval (a, b) , if whenever $x_1 < x_2$ we always have $f(x_1) < f(x_2)$, then the function is **increasing** on (a, b) . If whenever $x_1 < x_2$ we always have $f(x_1) > f(x_2)$, then the function is **decreasing** on (a, b) .

Using logical symbols, this can be written as:

$$\forall x_1, x_2 \in (a, b), \quad x_1 < x_2 \Rightarrow f(x_1) < f(x_2),$$

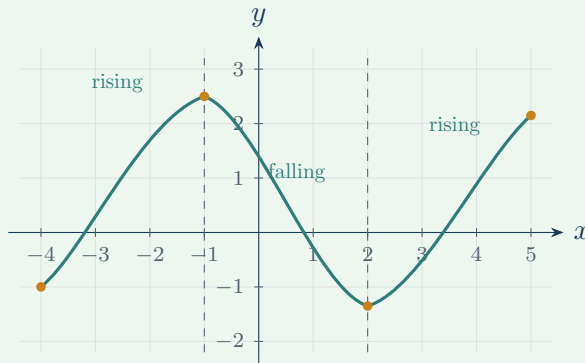
then f is **increasing** on (a, b) ;

$$\forall x_1, x_2 \in (a, b), \quad x_1 < x_2 \Rightarrow f(x_1) > f(x_2),$$

then f is **decreasing** on (a, b) .

Example 4.11 Reading Monotonic Intervals from a Graph

Observe the graph below, and write the **increasing** intervals and **decreasing** intervals of the function.



Solution: Read the graph from left to right: the curve rises on $[-4, -1]$, falls on $[-1, 2]$, and rises again on $[2, 5]$.

Rising corresponds to being **increasing**, and falling corresponds to being **decreasing**. Therefore the **increasing** intervals are

$$[-4, -1] \quad \text{and} \quad [2, 5],$$

and the **decreasing** interval is

$$[-1, 2].$$

4.6.3 Parity (奇偶性)

When the domain is symmetric about the origin:

$f(-x) = f(x) \Rightarrow f$ is even, and its graph is symmetric about the y -axis;

$f(-x) = -f(x) \Rightarrow f$ is odd, and its graph is symmetric about the origin.

Steps for Determining Parity

First check whether the domain is symmetric about the origin, then compute $f(-x)$, and finally compare it with $f(x)$ and $-f(x)$.

Look at the Domain First

If the domain is not symmetric about the origin, the function cannot be judged as odd or even. For example, the domain of $f(x) = \sqrt{x}$ is $[0, +\infty)$. The number 1 is in the domain, but -1 is not in the domain, so it is neither odd nor even. Do not only look at whether the graph "looks like half of something".

Example 4.12 Determining Parity

Determine the parity of the following functions:

(1) $f(x) = x^3 - x$;

(2) $g(x) = \frac{1}{x^2 + 1}$;

(3) $h(x) = (x - 1)^2$.

Solution: The domains of all three functions are $\mathbb{R}$, so we may directly compare $f(-x)$.

(1)

$$f(-x) = (-x)^3 - (-x) = -x^3 + x = -(x^3 - x) = -f(x),$$

so f is an odd function.

(2)

$$g(-x) = \frac{1}{(-x)^2 + 1} = \frac{1}{x^2 + 1} = g(x),$$

so g is an even function.

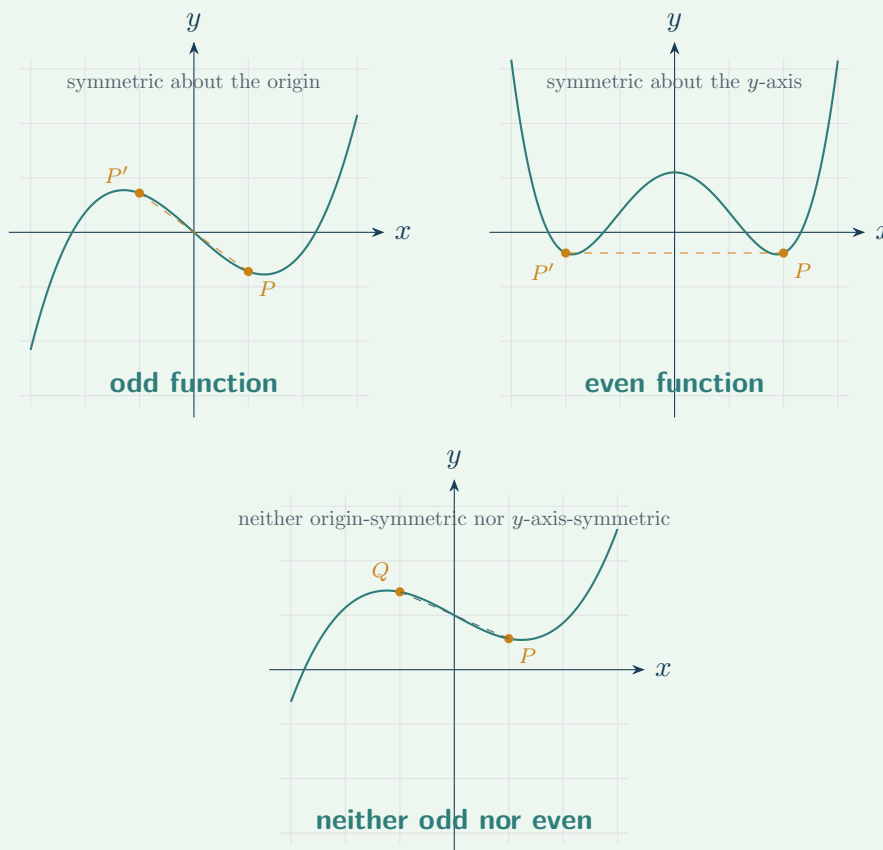
(3)

$$h(-x) = (-x - 1)^2 = (x + 1)^2,$$

which is neither equal to $(x - 1)^2$ nor equal to $-(x - 1)^2$. Therefore h is neither odd nor even.

Example 4.13 Reading Parity from Graphs

For some functions, the analytic expression is more complicated, but the symmetry of the graph is clear. When judging, first check whether the domain is symmetric about the origin, and then check whether the graph has the corresponding symmetry.



In the left graph, every point $P(x, y)$ has a corresponding point $P'(-x, -y)$ on the graph, so it is an **odd function**. In the middle graph, every point $P(x, y)$ has a corresponding point $P'(-x, y)$ on the graph, so it is an **even function**. The lower graph has neither of these two

whole-graph symmetries, so it is **neither odd nor even**.

Example 4.14 Both Odd and Even

Explain: if a function f is both odd and even, then $f(x) = 0$ must hold throughout its domain.

Solution: Since f is even,

$$f(-x) = f(x).$$

Since f is also odd,

$$f(-x) = -f(x).$$

Combining the two equations gives

$$f(x) = -f(x),$$

so

$$2f(x) = 0, \quad f(x) = 0.$$

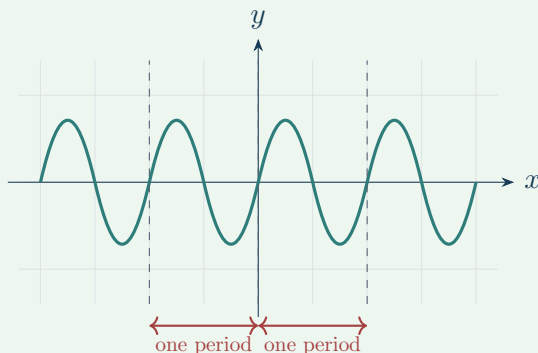
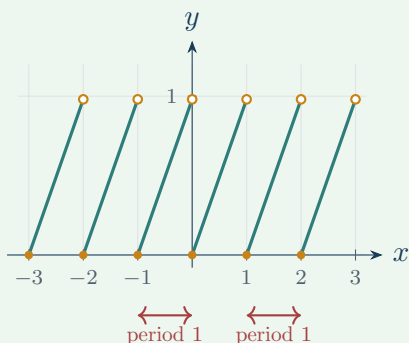
Therefore, if a function is both odd and even, it can only be the zero function. Conversely, the zero function $f(x) = 0$ is both odd and even on any domain symmetric about the origin.

4.6.4 Periodicity (周期性)

If there exists a nonzero constant T such that every x in the domain satisfies $f(x+T) = f(x)$, then f is a periodic function. The smallest positive such T is called the smallest positive period.

Example 4.15 Reading Periodicity from Graphs

Periodicity means the graph repeats after a fixed horizontal distance. Although the following two graphs have different shapes, both show that "after shifting one period to the right, the graph coincides with the original one".



In-Class Practice 4.6.1 Function Properties

- (1) Determine the parity of $f(x) = |x|$.
- (2) Determine the parity of $g(x) = x^2 + x$.
- (3) If $f(x + 4) = f(x)$, and $f(1) = 3$, find $f(9)$.

4.7 Equations and Inequalities from Graphs (图像与方程、不等式)

The solutions of the equation $f(x) = g(x)$ are the x -coordinates of the intersections of the two graphs; the solutions of the inequality $f(x) > g(x)$ are the intervals where the graph of f is above the graph of g .

Example 4.16 Intersections of a Line and a Parabola

Discuss the number of intersections between the line $y = kx + 1$ and the parabola $y = x^2$, and convert the problem into a discriminant problem for a quadratic equation in x .

Solution: At an intersection, the two function values are equal:

$$x^2 = kx + 1,$$

that is,

$$x^2 - kx - 1 = 0.$$

This is a quadratic equation in x , and its discriminant is

$$\Delta = (-k)^2 - 4 \cdot 1 \cdot (-1) = k^2 + 4.$$

Since $k^2 + 4 > 0$ for every real number k , the equation always has two distinct real roots. Therefore the line $y = kx + 1$ and the parabola $y = x^2$ always have two intersections.

Example 4.17 Solving Inequalities with Function Graphs

Given $f(x) = x^2 - 5x + 4$, draw a sketch, and use the graph to write:

- (1) the solution set of $f(x) > 0$;
- (2) the solution set of $f(x) \leq 0$;
- (3) the solution set of $f(x) > x - 2$.

Solution: First factor:

$$f(x) = x^2 - 5x + 4 = (x - 1)(x - 4).$$

The parabola opens upward, and its zeros are 1, 4. Therefore:

(1)

$$f(x) > 0 \Rightarrow x < 1 \text{ or } x > 4,$$

namely $(-\infty, 1) \cup (4, +\infty)$.

(2)

$$f(x) \leq 0 \Rightarrow 1 \leq x \leq 4,$$

namely $[1, 4]$.

(3) To solve $f(x) > x - 2$:

$$x^2 - 5x + 4 > x - 2 \Rightarrow x^2 - 6x + 6 > 0.$$

The roots of $x^2 - 6x + 6 = 0$ are

$$x = 3 \pm \sqrt{3}.$$

The parabola opens upward, so it is greater than 0 outside the two roots. Hence the solution set is

$$(-\infty, 3 - \sqrt{3}) \cup (3 + \sqrt{3}, +\infty).$$

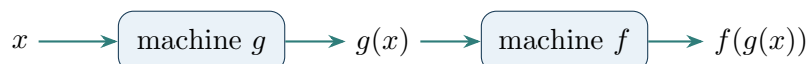
4.8 Composite Functions and Graph Transformations (复合函数与函数图像变换)

A composite function uses the output of one function as the input of another function; graph transformations help us obtain new graphs from familiar ones.

4.8.1 Composite Functions

If $u = g(x)$ and $y = f(u)$, then $y = f(g(x))$, written as $f \circ g$. When composing functions, we must not only substitute, but also ensure that the output of the inner function lies in the domain of the outer function.

When first learning composite functions, you can regard them as "two machines working in sequence": x first enters machine g , producing output $g(x)$; this output then enters machine f , giving $f(g(x))$. Therefore, the easiest mistake in composite functions is not substitution, but forgetting to check whether the inner output can enter the outer domain.



Example 4.18 Composite Functions

Let $f(x) = \sqrt{x}$, $g(x) = x^2 - 1$.

(1) Write $f(g(x))$ and find its domain.

(2) Write $g(f(x))$ and find its domain.

(3) Compare $f \circ g$ and $g \circ f$.

Solution:

(1)

$$f(g(x)) = \sqrt{x^2 - 1}.$$

We require $x^2 - 1 \geq 0$, so $x \leq -1$ or $x \geq 1$. The domain is $(-\infty, -1] \cup [1, +\infty)$.

(2)

$$g(f(x)) = (\sqrt{x})^2 - 1 = x - 1.$$

Here we must first ensure that $f(x) = \sqrt{x}$ is meaningful, so $x \geq 0$. Therefore the domain of $g(f(x))$ is $[0, +\infty)$.

(3) The two composite functions have different analytic expressions and different domains, so in general $f \circ g \neq g \circ f$.

4.8.2 Translation, Reflection and Scaling (平移、对称与伸缩)

Learning Order for Graph Transformations

On the first pass, first master translations: all points move up, down, left or right together. On the second pass, master reflections: points flip to the other side of a coordinate axis. Finally study scaling: coordinates in a certain direction change by a ratio. Do not combine translation, reflection and scaling from the very beginning.

Obtaining New Graphs from $y = f(x)$

Analytic Expression	Graph Transformation
$y = f(x - a)$	shift right by a units
$y = f(x + a)$	shift left by a units
$y = f(x) + b$	shift up by b units
$y = f(x) - b$	shift down by b units
$y = -f(x)$	reflect about the x -axis
$y = f(-x)$	reflect about the y -axis
$y = Af(x)$	multiply the y -coordinate by A
$y = f(Bx)$	horizontal scale becomes $1/ B $ of the original

Common Mistake

The inside of the parentheses controls left-right translation, and the direction looks ``opposite'': $f(x - 2)$ shifts right, while $f(x + 2)$ shifts left.

How Point Coordinates Change

If a point (x, y) lies on the original graph, common transformations can be read directly from the position change of the point:

New Graph	Where the Original Point (x, y) Goes
$y = f(x - a)$	$(x + a, y)$, the whole graph shifts right by a
$y = f(x) + b$	$(x, y + b)$, the whole graph shifts up by b
$y = -f(x)$	$(x, -y)$, reflection about the x -axis
$y = f(-x)$	$(-x, y)$, reflection about the y -axis
$y = Af(x)$	(x, Ay) , the y -coordinate is multiplied by A
$y = f(Bx)$	$(\frac{x}{B}, y)$, the x -coordinate is divided by B

Example 4.19 Translation: Every Point Moves Together

In a translation, every point on the graph moves in the same direction by the same distance.

- (1) Shift the linear function $f(x) = 2x + 1$ to the right by 2 units:

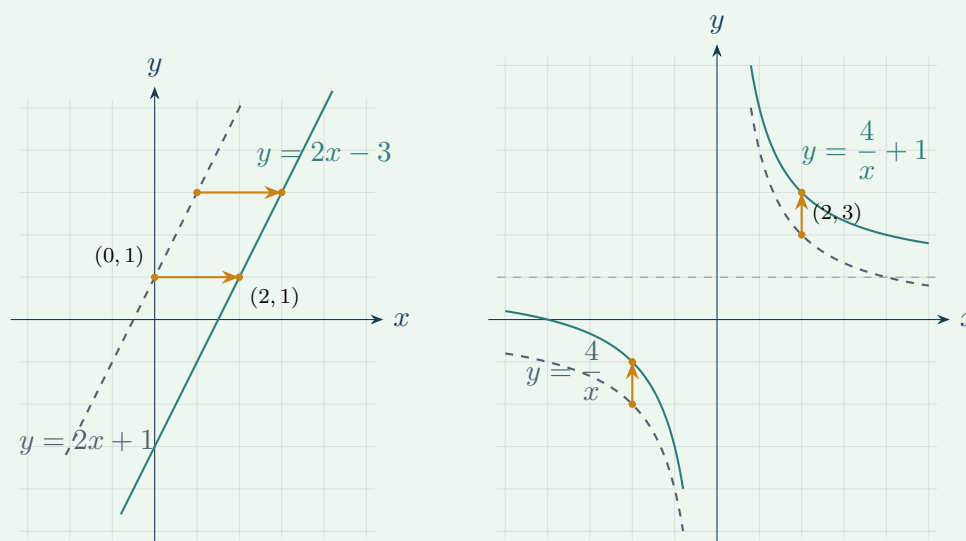
$$y = f(x - 2) = 2(x - 2) + 1 = 2x - 3.$$

For example, the point $(0, 1)$ on the original graph becomes $(2, 1)$, and the point $(1, 3)$ becomes $(3, 3)$.

- (2) Shift the reciprocal function $g(x) = \frac{4}{x}$ upward by 1 unit:

$$y = g(x) + 1 = \frac{4}{x} + 1.$$

For example, the point $(2, 2)$ on the original graph becomes $(2, 3)$, and the point $(-2, -2)$ becomes $(-2, -1)$.



Example 4.20 Reflection: Distance to the Axis Is Unchanged

In a reflection, the distance from a point to the axis of reflection stays the same, but the point moves to the other side of the axis.

- (1) Reflect the linear function $f(x) = 2x + 1$ about the x -axis:

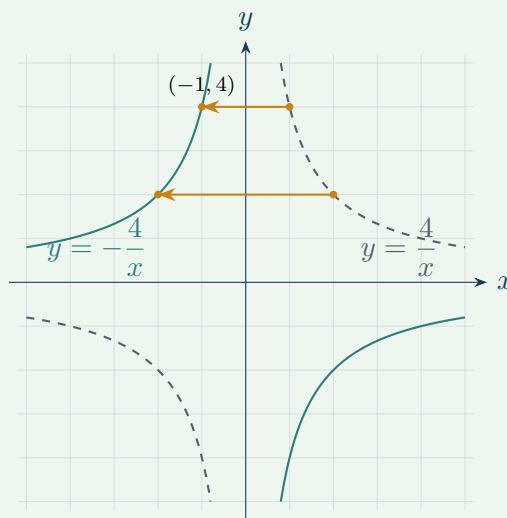
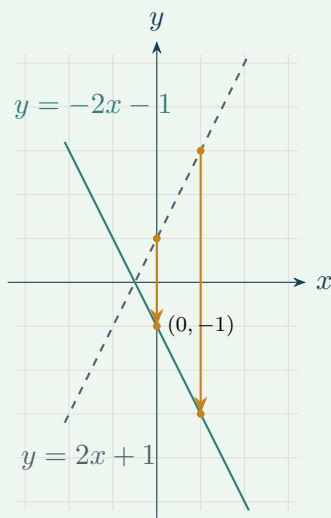
$$y = -f(x) = -(2x + 1) = -2x - 1.$$

The point $(0, 1)$ becomes $(0, -1)$, and the point $(1, 3)$ becomes $(1, -3)$.

- (2) Reflect the reciprocal function $g(x) = \frac{4}{x}$ about the y -axis:

$$y = g(-x) = \frac{4}{-x} = -\frac{4}{x}.$$

The point $(1, 4)$ becomes $(-1, 4)$, and the point $(2, 2)$ becomes $(-2, 2)$.

**Example 4.21 Scaling: Coordinates Change by a Ratio**

Scaling is not moving the whole graph. Instead, coordinates in a certain direction are changed by a ratio.

- (1) Vertically stretch the linear function $f(x) = 2x + 1$ to 2 times its original height:

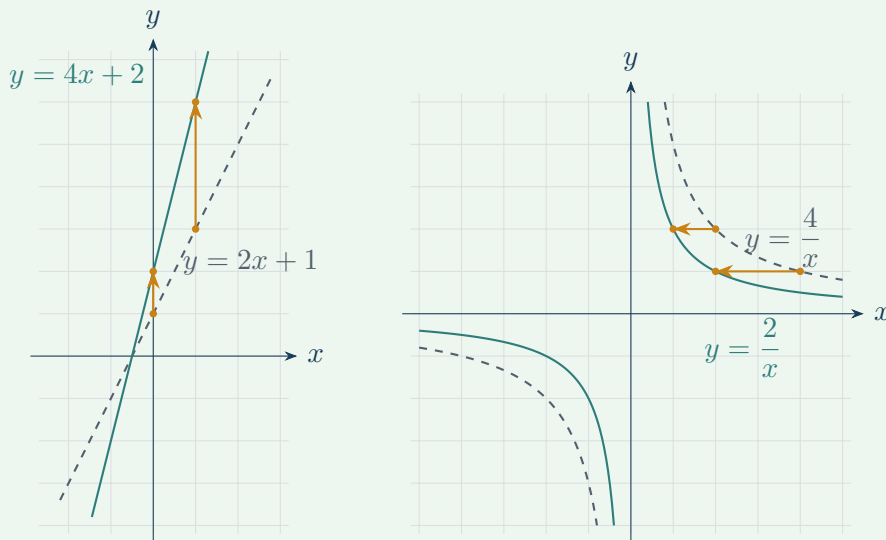
$$y = 2f(x) = 2(2x + 1) = 4x + 2.$$

The point $(0, 1)$ becomes $(0, 2)$, and the point $(1, 3)$ becomes $(1, 6)$. The x -coordinate is unchanged, and the y -coordinate becomes 2 times the original.

- (2) Horizontally compress the reciprocal function $g(x) = \frac{4}{x}$ to $\frac{1}{2}$ of its original width:

$$y = g(2x) = \frac{4}{2x} = \frac{2}{x}.$$

The point $(2, 2)$ becomes $(1, 2)$, and the point $(4, 1)$ becomes $(2, 1)$. The y -coordinate is unchanged, and the x -coordinate becomes $\frac{1}{2}$ of the original.


Example 4.22 Reading Successive Transformations from an Analytic Expression

Starting from $y = f(x)$, the function $y = -2f(x-1)$ can be understood from the inside outward:

$f(x-1)$ means shifting right by 1 unit;

$2f(x-1)$ means vertically stretching to 2 times the original;

$-2f(x-1)$ then means reflecting about the x -axis.

The same rule applies both to lines and to reciprocal functions, but after translating a reciprocal function, we must also pay attention to the excluded x -value and the positions of asymptotes.

In-Class Practice 4.8.1 Composite Functions and Transformations

- (1) $f(x) = \sqrt{x}$, $g(x) = x^2 - 1$. Write the analytic expression and domain of $f(g(x))$.
- (2) Shift $y = x^2$ right by 3 and down by 2. Write the new function.
- (3) Explain the graph relationship between $y = |x|$ and $y = |x+1| + 2$.
- (4) How is $y = -2f(x-1)$ obtained from $y = f(x)$?

4.9 Chapter Summary

Main Knowledge Thread

- (1) A linear function is determined by its slope and one point; the sign of the slope determines monotonicity.
- (2) The three forms of a quadratic function respectively emphasize coefficients, vertex and zeros.
- (3) To find extrema on a closed interval, check the endpoints and the vertex inside the interval.
- (4) The exponent of a power function (函数叫幂函数) determines its domain, parity and graph shape.
- (5) For reciprocal and rational functions, focus on domain, range, asymptotes and center of symmetry.
- (6) Monotonicity describes rising and falling from left to right; parity is special symmetry about the origin or the y -axis; periodicity describes repetition of the graph at fixed intervals.
- (7) The properties of linear, quadratic and rational functions can be read from graphs, and can also be restated using general definitions.
- (8) Intersections of graphs unify functions, equations and inequalities.
- (9) For composite functions, pay attention to both substitution order and domain restrictions; graph transformations help us obtain new functions from familiar ones.

Order for Reading Function Graphs

First read the domain and range, then read intersections and key points. Next look from left to right for rising and falling. Finally look for symmetry, periodicity, translation and scaling. In comprehensive problems, draw a sketch first; the graph can help you judge whether the answer is reasonable.

4.10 Chapter Exercises

How to Use These Exercises

Group A checks the minimum mastery for this chapter. If you make mistakes, return to the corresponding section and reread the examples. Group B is for integrated practice before a quiz. Group C contains extension problems; during a first self-study pass, you may skip it without affecting your entry into the next chapter.

Difficulty: Basic

- (1) Find the line passing through $(1, 1)$ with slope 3.
- (2) Find the line passing through $(-1, -2)$ and $(3, 2)$.
- (3) Find the axis of symmetry, vertex and extremum of $y = x^2 - 4x + 2$.
- (4) Find the monotonic intervals and range of $y = -2x^2 + 8x - 5$.
- (5) Find the intersections of $y = x^2 - 2x - 3$ with the two coordinate axes.
- (6) Find the maximum and minimum values of $y = -x^2 + 2x + 3$ on $[0, 2]$.
- (7) Write the domain, range, parity and monotonicity of $y = x^5$.
- (8) Write the domain, range and parity of $y = x^{-2}$.
- (9) $y = k/x$ passes through $(-3, 4)$. Find its analytic expression.
- (10) Study the asymptotes and center of symmetry of $y = \frac{2x - 1}{x + 2}$.
- (11) Find the domain of $y = \frac{x^2 - 9}{x - 3}$ and explain the hole in the graph.
- (12) Determine the parity of $x^4 + 2$, $x^3 + x$, and $x^2 + x$.
- (13) If f has period 3, and $f(2) = -1$, find $f(11)$.
- (14) Shift $y = |x|$ left by 2 and up by 1.

Group B: Quiz Level

Difficulty: Core

- (1) Find the line passing through $(0, -1)$ and perpendicular to $y = 3x - 2$.
- (2) A parabola has zeros $-1, 3$, and intersects the y -axis at $(0, -3)$. Find its analytic expression and vertex.
- (3) The line $y = kx + 2k + 7$ and the parabola $y = x^2 - 2x$ have exactly one intersection. Find k .
- (4) Given that the quadratic function satisfies $f(1) = f(3) = 0$, and its minimum value is -2 , find $f(x)$.
- (5) Find the minimum value of $y = x^2 - 4x + a$ on $[0, 3]$, and discuss how it changes with a .
- (6) Compare $(-2)^5$, $(-1)^5$, 0^5 , and 2^5 , and explain using the graph.
- (7) Find the domain, range, asymptotes and monotonic intervals of $y = \frac{3x+1}{2x-4}$.
- (8) If an odd function f satisfies $f(2) = 5$, find $f(-2)$. If an even function g satisfies $g(-3) = 4$, find $g(3)$.
- (9) Discuss the number of intersections between the line $y = x + m$ and the parabola $y = x^2$.
- (10) $f(x) = \sqrt{x}$, $g(x) = x^2 - 4$. Find the domain of $f(g(x))$.
- (11) Given the graph of $y = f(x)$, reflect it about the y -axis first, then shift it down by 3. Write the new function.

Group C: Extension and Thinking

Difficulty: Extension

- (1) A quadratic function intersects the x -axis at $(-1, 0)$ and $(3, 0)$, and intersects the y -axis at C . If the area of the triangle is 6, find the equation of the parabola.

- (2) Find all m such that the line $y = mx + 1$ and $y = x + 1/x$ have exactly one common point.
- (3) Let $f(x) = \frac{x+1}{x-1}$. Find $f(f(x))$, and explain the corresponding relationship on the graph.
- (4) Determine whether $y = x + 1/x$ is increasing everywhere on $(0, +\infty)$, and explain using concrete values or another method.
- (5) Given $f(x+1) = 2x+3$, find $f(x)$.
- (6) Let $f(x) = \frac{x-1}{x+1}$. Find $f(f(x))$, and write the domain restrictions completely.
- (7) If a function f is both odd and even, explain what condition $f(x)$ must satisfy throughout its domain.

Self-Check

Skill	Can do	Mostly can	Not yet
Determine a linear function from a point and slope	☐	☐	☐
Find the vertex, zeros and extrema of a quadratic function	☐	☐	☐
Find properties of a quadratic function on an interval	☐	☐	☐
Recognize graphs and properties of common power functions	☐	☐	☐
Study the graph of a reciprocal function	☐	☐	☐
Find asymptotes and centers of symmetry of rational functions	☐	☐	☐
Determine monotonicity, parity and periodicity	☐	☐	☐
Use graphs to understand equations and inequalities	☐	☐	☐
Perform basic graph transformations	☐	☐	☐
Handle simple composite functions	☐	☐	☐

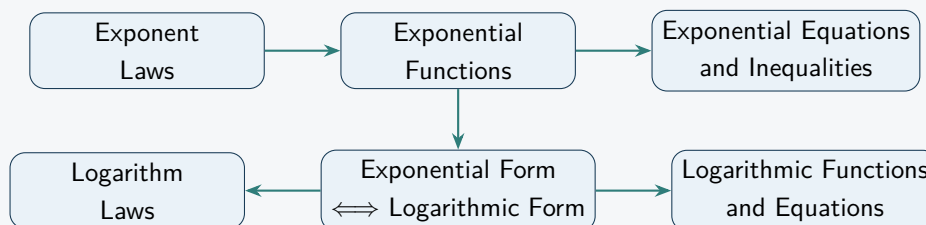
Exponents and Logarithms (指数与对数)

Learning Goals

After studying this chapter, you should be able to:

- ▶ fluently use the laws of integer exponents, fractional exponents and negative exponents;
- ▶ understand the domain, range, monotonicity and graph of an exponential function (指数函数) ;
- ▶ understand the definition of logarithm and convert between exponential form and logarithmic form;
- ▶ use the product, quotient and power laws of logarithms and the change-of-base formula;
- ▶ study the domain, range, monotonicity and graph of a logarithmic function;
- ▶ use monotonicity to compare exponential and logarithmic expressions;
- ▶ solve basic exponential and logarithmic equations and inequalities;
- ▶ use substitution to handle exponential or logarithmic problems that can be converted into quadratic equations.

Knowledge Map



Exponents describe repeated multiplication. Logarithms answer "to what power must the base be raised?" They are inverse operations, and their graphs are symmetric about the line $y = x$.

Learning Route of This Chapter

This chapter first systematizes the exponent operations from Chapter 1, and then studies the function $y = a^x$, where the exponent changes. When the unknown appears in the exponent position, ordinary algebraic transformations are not enough, so logarithms are introduced: a logarithm asks the reverse question, "to what power must the base be raised to obtain this number?" After that, all logarithm laws are derived from exponent laws.

When studying by yourself, grasp two dividing points: when the base $a > 1$, the function is increasing; when $0 < a < 1$, the function is decreasing. Also, the argument of a logarithm must be greater than 0. These two conditions determine every step in comparisons, equations and inequalities.

Prerequisites and Minimum Mastery

To study this chapter, you need to know exponent laws, domains and ranges of functions, graph translations, and quadratic equations in one variable. The minimum goal for the first pass is: convert between exponential and logarithmic forms; determine the monotonicity of exponential and logarithmic functions; and when solving logarithm problems, write the restriction that each argument is greater than 0 as the first step.

Common Self-Study Sticking Points in This Chapter

The most common mistakes in this chapter come from restrictions: the base of an exponential function must satisfy $a > 0, a \neq 1$; a logarithm $\log_a N$ requires $a > 0, a \neq 1, N > 0$. Whenever a logarithm appears in a problem, first write the restrictions on the arguments, and then simplify algebraically.

5.1 Extension and Operations of Exponents (指数的扩充与运算)

After exponents are extended from positive integers to zero, negative integers and rational numbers, the original operation laws still hold, but the meaning of the base and the expression must be checked.

5.1.1 Zero Exponent and Negative Exponent (零指数与负指数)

A positive integer exponent represents repeated multiplication, but subtraction of exponents naturally appears in division, and may produce 0 or a negative number. To keep the laws of powers with the same base unified, we need to give zero exponents and negative exponents a reasonable meaning.

For $a \neq 0$,

$$a^0 = 1, \quad a^{-n} = \frac{1}{a^n}.$$

In particular, 0^0 is undefined in this course, and 0^{-n} is also meaningless.

The definitions of zero and negative exponents are made so that the rule for dividing powers with

the same base continues to hold. For example,

$$\frac{a^3}{a^3} = 1,$$

while subtracting exponents would give $a^{3-3} = a^0$, so we define $a^0 = 1$. Also,

$$\frac{a^2}{a^5} = \frac{1}{a^3},$$

while subtracting exponents would give a^{-3} , so we define $a^{-3} = \frac{1}{a^3}$.

5.1.2 Fractional Exponent (分数指数)

Square roots and cube roots originally belong to the language of radicals. Fractional exponents include them in the language of exponents, so that radicals and powers can be handled using the same laws.

For $a > 0$, $m \in \mathbb{Z}$, $n \in \mathbb{N}^*$,

$$a^{1/n} = \sqrt[n]{a}, \quad a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m.$$

Exponent Laws

Whenever each expression is meaningful:

$$\begin{aligned} a^m a^n &= a^{m+n}, & \frac{a^m}{a^n} &= a^{m-n}, \\ (a^m)^n &= a^{mn}, & (ab)^n &= a^n b^n, \\ \left(\frac{a}{b}\right)^n &= \frac{a^n}{b^n}. \end{aligned}$$

Common Mistake

$(-2)^4$ and -2^4 are different. $(a+b)^n$ is generally not equal to $a^n + b^n$. If the base of a fractional exponent is allowed to be negative, it must be judged separately using the meaning of radicals.

Example 5.1 Exponent Calculations

Compute:

- (1) $3^4 \cdot 3^{-3}$;
- (2) $(4^3)^2 \div 4^4$;
- (3) $\left(\frac{1}{2}\right)^{-3} 2^{-3}$;
- (4) $\left(\frac{27}{8}\right)^{-2/3}$.

Solution:

- (1) $3^4 \cdot 3^{-3} = 3^{4-3} = 3$.

$$(2) \quad (4^3)^2 \div 4^4 = 4^6 \div 4^4 = 4^2 = 16.$$

$$(3) \quad \left(\frac{1}{2}\right)^{-3} 2^{-3} = 2^3 \cdot 2^{-3} = 1.$$

$$(4) \quad \left(\frac{27}{8}\right)^{-2/3} = \left[\left(\frac{3}{2}\right)^3\right]^{-2/3} = \left(\frac{3}{2}\right)^{-2} = \left(\frac{2}{3}\right)^2 = \frac{4}{9}.$$

Example 5.2 Exponential Expressions with Letters

Simplify under the condition that all letters are positive:

$$(1) \quad \frac{a^2(\sqrt{a})^5}{a^{3/2}};$$

$$(2) \quad \frac{(2x^2y^{-1})^3}{4xy^{-2}};$$

$$(3) \quad \sqrt[3]{a^6b^{-3}}(a^2b^4)^{1/2}.$$

Solution: Since all letters are positive, fractional exponents can be used directly.

$$(1) \quad \frac{a^2(\sqrt{a})^5}{a^{3/2}} = \frac{a^2(a^{1/2})^5}{a^{3/2}} = a^{2+5/2-3/2} = a^3.$$

$$(2) \quad \frac{(2x^2y^{-1})^3}{4xy^{-2}} = \frac{8x^6y^{-3}}{4xy^{-2}} = 2x^5y^{-1} = \frac{2x^5}{y}.$$

$$(3) \quad \sqrt[3]{a^6b^{-3}}(a^2b^4)^{1/2} = a^2b^{-1} \cdot ab^2 = a^3b.$$

In-Class Practice 5.1.1 Exponent Operations

$$(1) \text{ Fill in the blanks: } a^0 = \underline{\hspace{2cm}}, a^{-n} = \underline{\hspace{2cm}}, a^{m/n} = \underline{\hspace{2cm}}.$$

$$(2) \text{ Compute } 9^0 + 9^{1/2} + 9^1.$$

$$(3) \text{ Compute } 8^{2/3} \cdot 2^{-2}.$$

$$(4) \text{ Simplify } (x^{-2}y^3)^{-2}/(x^{-1}y).$$

$$(5) \text{ Compare } (-3)^4 \text{ and } -3^4.$$

5.2 Exponential Functions (指数函数)

When the base is fixed and the exponent changes, we obtain an exponential function. It is always positive, and grows or decays very fast.

5.2.1 Definition and Graph

Exponential functions are often used to describe changes where "after every equal time interval, the quantity is multiplied by the same factor". For example, bacteria may double every hour, remaining medicine may become 0.8 times the original amount every hour, and savings may grow by a fixed proportion. The common feature of these problems is not adding the same number each time, but multiplying by the same number each time.

For example, suppose a certain bacterial population is initially 100, and doubles every hour:

Time x (hours)	0	1	2	3	4
Population $100 \cdot 2^x$	100	200	400	800	1600

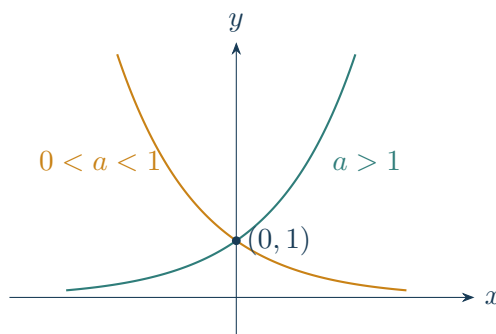
The table lists only integer times, while the function $100 \cdot 2^x$ extends this "change by a factor" rule to all real times.

A function of the form

$$y = a^x \quad (a > 0, a \neq 1)$$

is called an exponential function. Its domain is $\mathbb{R}$, its range is $(0, +\infty)$, its graph passes through $(0, 1)$, and $y = 0$ is its horizontal asymptote (水平渐近线).

The condition $a > 0$ is required so that arbitrary real exponents have a stable meaning. The condition $a \neq 1$ is required because 1^x is always 1, and no longer shows growth or decay.



Monotonicity of Exponential Functions

- When $a > 1$, $y = a^x$ is increasing on $\mathbb{R}$.
- When $0 < a < 1$, $y = a^x$ is decreasing on $\mathbb{R}$.

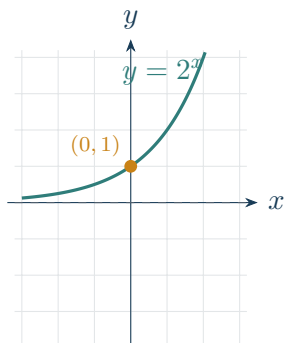
The intuitive reason for monotonicity is: when $a > 1$, every time the exponent increases by 1, the function value is multiplied by a number greater than 1, so it becomes larger and larger. When $0 < a < 1$, every time the exponent increases by 1, the function value is multiplied by a positive number less than 1, so it becomes smaller and smaller.

5.2.2 Graph Transformations and Range

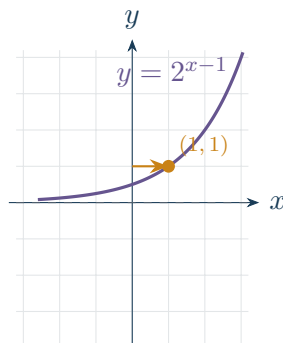
Expressions such as $2^{x-1} - 3$ and $-3^x + 4$ often appear in problems. Understanding them means understanding translations, reflections, scalings and new asymptotes.

The graph of $y = Aa^{x-h} + k$ is obtained from $y = a^x$ by translation, scaling or reflection. The horizontal asymptote becomes $y = k$. If $A > 0$, the range is $(k, +\infty)$. If $A < 0$, the range is $(-\infty, k)$.

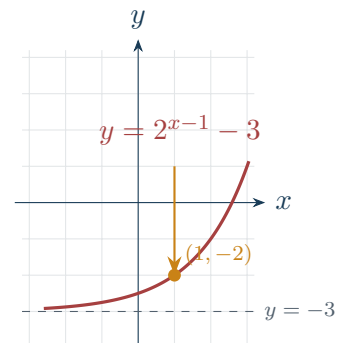
Take $y = 2^{x-1} - 3$ as an example. First look at $x - 1$ inside the exponent: it means shifting $y = 2^x$ to the right by 1 unit. Then look at the outside -3 : it means shifting the whole graph down by 3 units. During a translation, every point moves together, and the horizontal asymptote moves together as well.



Step 1: basic graph
asymptote $y = 0$



Step 2: shift right by 1
asymptote still $y = 0$



Step 3: shift down by 3
asymptote becomes $y = -3$

Example 5.3 Properties of Exponential Functions

Study the domain, range, monotonicity, horizontal asymptote and graph transformations of $f(x) = 2^{x-1} - 3$ and $g(x) = -2^{x+1} + 4$.

Solution: The domain of 2^x is $\mathbb{R}$, its range is $(0, +\infty)$, and it is increasing.

For $f(x) = 2^{x-1} - 3$, the domain is $\mathbb{R}$. Since $2^{x-1} > 0$, we have $f(x) > -3$, so the range is $(-3, +\infty)$. It is obtained from $y = 2^x$ by shifting right by 1 unit and then down by 3 units. Its horizontal asymptote is $y = -3$, and it remains increasing.

For $g(x) = -2^{x+1} + 4$, the domain is $\mathbb{R}$. Since $2^{x+1} > 0$, we have $-2^{x+1} < 0$, so $g(x) < 4$. The range is $(-\infty, 4)$. It is obtained from $y = 2^x$ by shifting left by 1 unit, reflecting about the x -axis, and then shifting up by 4 units. Its horizontal asymptote is $y = 4$, and it is decreasing.

5.2.3 Comparing Sizes

Using the monotonicity of exponential functions, we can compare expressions containing exponents.

Comparing Exponential Expressions

When comparing a^u and a^v , first state the range of the base. If $a > 1$, the larger exponent gives the larger power. If $0 < a < 1$, the larger exponent gives the smaller power.

Example 5.4 Comparing by Monotonicity

Compare without using a calculator:

- (1) $3^{1.2}$ and $3^{0.8}$;

(2) 0.4^{-2} and 0.4^{-1} ;

(3) $2^{\sqrt{2}}$ and $2^{1.4}$.

Solution:(1) Since $3 > 1$, the larger exponent gives the larger power. Since $1.2 > 0.8$, $3^{1.2} > 3^{0.8}$.(2) Since $0 < 0.4 < 1$, the larger exponent gives the smaller power. Since $-2 < -1$, $0.4^{-2} > 0.4^{-1}$.(3) Since $2 > 1$, we only need to compare exponents. $\sqrt{2} \approx 1.414 > 1.4$, so $2^{\sqrt{2}} > 2^{1.4}$.**In-Class Practice 5.2.1 Exponential Functions**(1) Complete the table of values for $y = 2^x$ at $x = -2, -1, 0, 1, 2$, and draw a sketch.(2) Write the domain, range, monotonicity and asymptote of $y = 0.5^x$.(3) Find the range and horizontal asymptote of $y = 3^{x+2} - 1$.(4) Compare $2^{-0.3}$, 1 , $2^{0.2}$.(5) Compare $(1/3)^{-2}$ and $(1/3)^{-3}$.

5.3 Exponential Equations and Inequalities (指数方程与不等式)

Exponential equations and inequalities are equations and inequalities in which the unknown appears in the exponent.

5.3.1 Converting to the Same Base

The most basic exponential equation has the form $a^{u(x)} = a^{v(x)}$. Since an exponential function is monotonic,

$$a^{u(x)} = a^{v(x)} \iff u(x) = v(x).$$

The most basic exponential inequality has the form $a^{u(x)} > a^{v(x)}$. By monotonicity, when $a > 1$, this inequality is equivalent to $u(x) > v(x)$. When $0 < a < 1$, it is equivalent to $u(x) < v(x)$.

Reverse Direction When the Base Is Less Than 1

For example,

$$\left(\frac{1}{2}\right)^x > \left(\frac{1}{2}\right)^3.$$

Since $y = \left(\frac{1}{2}\right)^x$ is decreasing, the solution is $x < 3$, not $x > 3$.

Example 5.5 Exponential Equations

Solve:

(1) $3^{x+1} = 9$;

(2) $2^{3x-1} = 4^{x+2}$;

(3) $0.25^{-x} = 2^{x+1}$.

Solution:

(1) $3^{x+1} = 9 = 3^2 \Rightarrow x + 1 = 2 \Rightarrow x = 1$.

(2) $2^{3x-1} = 4^{x+2} = 2^{2x+4} \Rightarrow 3x - 1 = 2x + 4 \Rightarrow x = 5$.

(3) $0.25^{-x} = \left(\frac{1}{4}\right)^{-x} = (2^{-2})^{-x} = 2^{2x} = 2^{x+1} \Rightarrow 2x = x + 1 \Rightarrow x = 1$.

5.3.2 Substitution (换元法)

This subsection deals with problems where a^x and a^{2x} appear at the same time. Viewed directly, the unknowns all seem to be in exponent positions. But if a^x is regarded as one whole, then $a^{2x} = (a^x)^2$, and the problem becomes a familiar quadratic equation or quadratic inequality.

Let $t = a^x$. Then $t > 0$. The equation $a^{2x} + ba^x + c = 0$ can be converted into a quadratic equation in t . At the end, roots satisfying $t \leq 0$ must be rejected, and then we substitute back.

Example 5.6 Exponential Substitution

Solve the equation or inequality:

(1) $2^{2x} - 5 \cdot 2^x + 4 = 0$;

(2) $3^{2x} - 10 \cdot 3^x + 9 \leq 0$.

Solution:

(1) Let $t = 2^x$. Then $t > 0$, and the original equation becomes

$$t^2 - 5t + 4 = 0 = (t - 1)(t - 4).$$

Therefore $t = 1$ or $t = 4$, namely $2^x = 1$ or $2^x = 4$, so $x = 0$ or $x = 2$.

(2) Let $u = 3^x$. Then $u > 0$, and we get

$$u^2 - 10u + 9 \leq 0 = (u - 1)(u - 9) \leq 0.$$

Therefore $1 \leq u \leq 9$, namely

$$1 \leq 3^x \leq 9 = 3^2.$$

Since 3^x is increasing, we obtain $0 \leq x \leq 2$.

In-Class Practice 5.3.1 Exponential Equations and Inequalities

(1) Solve $5^{x+4} = 25^{2x-1}$.

(2) Solve $4^{x+1} = 8^{x-1}$.

(3) Solve $2^x > 4$.

(4) Solve $(1/3)^{2x-1} \geq (1/3)^{x+2}$.

(5) Solve $4^x - 5 \cdot 2^x + 4 = 0$.

(6) Solve $9^x - 3^{x+1} - 18 \leq 0$.

5.4 The Concept and Operations of Logarithms (对数的概念与运算)

Logarithms bring the exponent position "down", turning multiplication into addition and powers into multiplication.

5.4.1 Definition of Logarithm (对数的定义)

If $a > 0, a \neq 1, N > 0$, then $\log_a N$ is defined as "the power to which a must be raised to get N ", that is,

$$a^b = N \iff \log_a N = b.$$

a is the base, and N is the argument. The common logarithm $\log_{10} N$ is written as $\lg N$. The natural logarithm $\log_e N$ is written as $\ln N$.

When reading a logarithm, always read it as a question:

$$\log_2 8$$

is read as ``what power of 2 equals 8?'' The answer is 3, so $\log_2 8 = 3$. Reading it this way is easier than treating logarithms as unfamiliar symbols.

The restrictions on logarithms come from the exponential function itself: when $a > 0, a \neq 1$, the value of a^x is always positive. Therefore, when solving for the exponent in reverse, the argument N must be greater than 0. Hence $\log_a 0$ and $\log_a(-2)$ are meaningless in this course.

This is not a new operation rule, but two ways to say the same thing:

$$2^5 = 32$$

means ``2 to the fifth power equals 32''. Written as

$$\log_2 32 = 5$$

it means ``with base 2, the exponent must be 5 to obtain 32''. Therefore logarithms are specifically used to find exponents.

Immediate Consequences of the Definition

$$\log_a 1 = 0, \quad \log_a a = 1, \quad a^{\log_a N} = N, \quad \log_a(a^x) = x.$$

Example 5.7 Converting Between Exponential Form and Logarithmic Form

Complete the conversions and computations:

- (1) $2^5 = 32$;
- (2) $10^{-3} = 0.001$;
- (3) $\log_3 9$;
- (4) $\log_{1/4} 2$;
- (5) $\log_2 \frac{1}{4}$.

Solution: The exponential form $a^b = N$ is equivalent to the logarithmic form $\log_a N = b$.

- (1) $2^5 = 32 \iff \log_2 32 = 5$.
- (2) $10^{-3} = 0.001 \iff \log_{10} 0.001 = -3$.
- (3) $\log_3 9 = 2$.
- (4) $\left(\frac{1}{4}\right)^{-1/2} = 2 \Rightarrow \log_{1/4} 2 = -\frac{1}{2}$.
- (5) $\frac{1}{4} = 2^{-2} \Rightarrow \log_2 \frac{1}{4} = -2$.

5.4.2 Logarithm Laws (对数运算法则)

This subsection deals with simplifying and combining logarithmic expressions. Logarithms convert multiplication, division and powers into addition, subtraction and multiplication, so they are especially suitable for breaking apart complicated product-quotient structures, or for combining several logarithms into one logarithm.

For $M > 0, N > 0$,

Product, Quotient and Power Laws

$$\log_a(MN) = \log_a M + \log_a N,$$

$$\log_a \frac{M}{N} = \log_a M - \log_a N,$$

$$\log_a(M^r) = r \log_a M.$$

These laws come from exponent laws. Let

$$\log_a M = u, \quad \log_a N = v.$$

Then $M = a^u$, $N = a^v$. Therefore

$$MN = a^u a^v = a^{u+v},$$

so $\log_a(MN) = u + v = \log_a M + \log_a N$. The quotient law and the power law are similar; they come respectively from division of powers with the same base and raising a power to a power.

Common Mistake

Logarithm laws do not apply to addition or subtraction: $\log_a(M + N)$ is generally not equal to $\log_a M + \log_a N$. Every logarithmic argument must be positive.

Fixed First Step for Logarithm Problems

- (1) First list the argument of every logarithm.
- (2) Write all restrictions requiring the arguments to be > 0 .
- (3) Use logarithm laws or same-base conversion inside this domain.
- (4) After obtaining candidate solutions, substitute them back into the restrictions to check.

5.4.3 Change-of-Base Formula (换底公式)

This subsection handles what to do when the base is inconvenient. Some problems have different bases, and some calculators do not have a logarithm key for an arbitrary base. The change-of-base formula allows us to convert logarithms to a familiar base, and then compare or compute.

$$\log_a b = \frac{\log_c b}{\log_c a} \quad (c > 0, c \neq 1).$$

Therefore $\log_a b \cdot \log_b a = 1$, and $\log_a b \cdot \log_b c = \log_a c$.

The change-of-base formula is a second-level tool. During the first pass, it is enough to know that it can convert an unfamiliar base into a familiar base. After becoming fluent, use it to handle products of logarithms or calculator problems.

One practical use of the change-of-base formula is calculator computation. Many calculators provide only $\lg$ or $\ln$. To calculate $\log_2 7$, we can write

$$\log_2 7 = \frac{\ln 7}{\ln 2}.$$

The derivation of the change-of-base formula also uses only the definition. Let $x = \log_a b$. Then $a^x = b$. Taking logarithms with base c on both sides,

$$\log_c(a^x) = \log_c b.$$

Using the power law gives

$$x \log_c a = \log_c b,$$

so

$$x = \frac{\log_c b}{\log_c a}.$$

Example 5.8 Logarithm Calculations

Compute:

- (1) $\log_5 100 - \log_5 4$;
- (2) $\lg 25 + \lg 4$;
- (3) $\frac{\ln 8}{\ln 2}$;
- (4) $\log_3 5 \cdot \log_5 7 \cdot \log_7 9$.

Solution:

- (1) $\log_5 100 - \log_5 4 = \log_5 \frac{100}{4} = \log_5 25 = 2.$
- (2) $\lg 25 + \lg 4 = \lg(25 \cdot 4) = \lg 100 = 2.$
- (3) $\frac{\ln 8}{\ln 2} = \log_2 8 = 3.$
- (4) $\log_3 5 \cdot \log_5 7 \cdot \log_7 9 = \frac{\ln 5}{\ln 3} \cdot \frac{\ln 7}{\ln 5} \cdot \frac{\ln 9}{\ln 7} = \frac{\ln 9}{\ln 3} = 2.$

In-Class Practice 5.4.1 Logarithm Operations

- (1) Compute $\log_3 9$, $\log_2(1/8)$, and $\log_{1/4} 2$.
- (2) Compute $\log_6 2 + \log_6 18$.
- (3) Compute $\log_2 3 \cdot \log_3 8$.
- (4) Expand $\log_a(M^2\sqrt{N}/P)$.
- (5) Combine $2\ln x - \ln(x - 1)$ into one logarithm, and write the domain of the original expression.

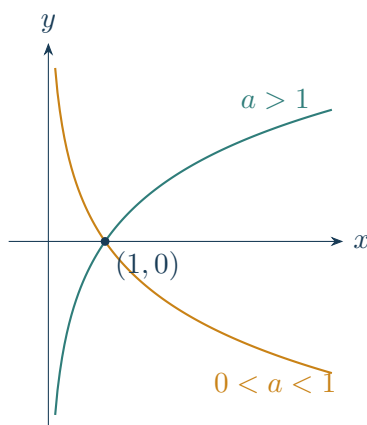
5.5 Logarithmic Functions (对数函数)

The logarithmic function $y = \log_a x$ and the exponential function $y = a^x$ are reverse to each other: the exponential function finds the result from the exponent, while the logarithmic function finds the exponent from the result.

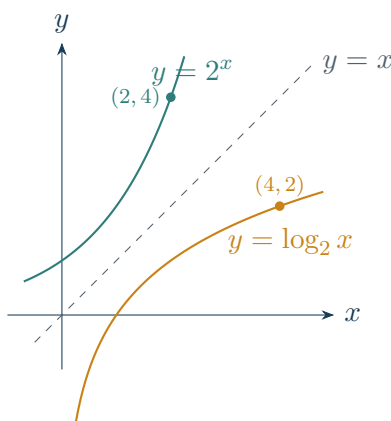
5.5.1 Graph and Properties

This subsection deals with the overall graph of logarithmic functions. Since a logarithmic function is the reverse relation of an exponential function, its graph can be obtained from the exponential graph by "swapping the horizontal and vertical coordinates". This understanding is more stable than separately memorizing the domain, range and asymptote.

The domain of $y = \log_a x$ is $(0, +\infty)$, the range is $\mathbb{R}$, the graph passes through $(1, 0)$, and the vertical asymptote (竖直渐近线) is $x = 0$.



When $a > 1$, the function is increasing. When $0 < a < 1$, the function is decreasing. The graphs of $y = a^x$ and $y = \log_a x$ are symmetric about $y = x$. This symmetry comes from swapping inputs and outputs. If $a^u = v$, then the exponential graph contains the point (u, v) . The same relationship can be written as $\log_a v = u$, so the logarithmic graph contains the point (v, u) . The two points have swapped coordinates, so they are symmetric about the line $y = x$. Mathematically, two functions whose graphs are symmetric about $y = x$ are called inverse functions of each other. Thus the logarithmic function $y = \log_a x$ is the inverse function of the same-base exponential function $y = a^x$, and vice versa.



5.5.2 Domains and Ranges of Composite Logarithmic Functions

For $y = \log_a g(x)$, the domain requires $g(x) > 0$. To find the range, first study the range of $g(x)$, and then use the monotonicity of the logarithmic function.

Reading Order for Composite Logarithms

First look at the argument $g(x)$ and write $g(x) > 0$. Then study which positive values $g(x)$ can take. Finally, according to whether the base $a > 1$ or $0 < a < 1$, determine the monotonicity and range of the logarithmic function.

Example 5.9 Domain and Range

Find:

- (1) the domain and range of $y = \log_2(2x - 1)$;

(2) the domain and range of $y = \log_3(-x^2 + 6x - 5)$;

(3) the monotonic interval and asymptote of $y = \log_{1/2}(x + 1) - 2$.

Solution:

(1) Require $2x - 1 > 0$, so $x > \frac{1}{2}$. The domain is $\left(\frac{1}{2}, +\infty\right)$. When $2x - 1$ takes all values in $(0, +\infty)$, the logarithm takes all values in $\mathbb{R}$. Therefore the range is $\mathbb{R}$.

(2) Require $-x^2 + 6x - 5 > 0$, that is,

$$(x - 1)(x - 5) < 0,$$

so the domain is $(1, 5)$. Also,

$$-x^2 + 6x - 5 = -(x - 3)^2 + 4.$$

On $(1, 5)$, the argument has range $(0, 4]$. Therefore the range of $y = \log_3(\text{argument})$ is $(-\infty, \log_3 4]$.

(3) Require $x + 1 > 0$, so the domain is $(-1, +\infty)$. Since the base $\frac{1}{2}$ lies in $(0, 1)$, $\log_{1/2}(x + 1)$ is decreasing. Subtracting 2 does not change monotonicity, so the function is decreasing on $(-1, +\infty)$. Its asymptote is $x = -1$.

5.5.3 Comparing Sizes

Logarithmic expressions with the same base can be compared using monotonicity by comparing their arguments. When bases differ, use change of base or transformation. Before comparing, the arguments must be confirmed to be positive.

Example 5.10 Comparing Logarithmic Expressions

Compare without using a calculator:

(1) $\log_2 1.2$, $\log_2 0.3$, 0;

(2) $\log_{1/3} 2$ and $\log_{1/3} 5$.

Solution:

(1) Since the base $2 > 1$, the logarithmic function is increasing, and $\log_2 1 = 0$. Therefore

$$0.3 < 1 < 1.2 \Rightarrow \log_2 0.3 < 0 < \log_2 1.2.$$

(2) Since the base $\frac{1}{3} \in (0, 1)$, the logarithmic function is decreasing. Since $2 < 5$,

$$\log_{1/3} 2 > \log_{1/3} 5.$$

In-Class Practice 5.5.1 Logarithmic Functions

- (1) Write the domain, range, monotonicity and asymptote of $y = \log_2 x$.
 - (2) Find the domain of $y = \log_2(x - 3)$.
 - (3) Find the domain of $y = \log_{1/2}(x^2 - 3x + 2)$.
 - (4) Compare $\log_3 5$ and $\log_3 7$.
 - (5) Find the domain and range of $y = \log_2(4 - x^2)$.
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-
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5.6 Logarithmic Equations and Inequalities (对数方程与不等式)

First write the domain, then simplify using logarithm properties; finally check whether the obtained solutions make every argument positive.

5.6.1 Basic Logarithmic Equations

This subsection deals with equations in which the unknown appears inside a logarithmic argument. The first thing to do when solving such equations is not simplification, but asking "does every logarithm have meaning?" That is, first write all restrictions that the arguments are greater than zero.

$$\log_a f(x) = b \iff f(x) = a^b, \quad f(x) > 0.$$

With the same base, $\log_a f(x) = \log_a g(x)$ can be converted to $f(x) = g(x)$, but $f(x) > 0, g(x) > 0$ must be kept.

Solution Format for Logarithmic Equations

For every logarithmic equation, the first line of the solution should be:

Domain: all arguments > 0 .

Then combine logarithms or convert to exponential form. Finally substitute candidate solutions back into the domain restrictions and delete illegal values.

Example 5.11 Logarithmic Equations

Solve:

(1) $\log_8(3x - 4) = 1;$

(2) $\log_2(x - 3) + \log_2(x - 1) = 3;$

(3) $\lg x + \lg(x - 21) = 2.$

Solution:(1) Require $3x - 4 > 0$. From $\log_8(3x - 4) = 1$, we get

$$3x - 4 = 8,$$

so $x = 4$, which satisfies the restriction.(2) Require $x - 3 > 0$ and $x - 1 > 0$, so $x > 3$. Combine the logarithms:

$$\log_2((x - 3)(x - 1)) = 3,$$

giving

$$(x - 3)(x - 1) = 8.$$

This simplifies to $x^2 - 4x - 5 = 0$, giving $x = 5$ or $x = -1$. Combining with $x > 3$, we take $x = 5$.(3) Require $x > 21$. Combine the logarithms:

$$\lg(x(x - 21)) = 2,$$

so $x(x - 21) = 100$, namely $x^2 - 21x - 100 = 0$. Solving gives $x = 25$ or $x = -4$. Combining with $x > 21$, we take $x = 25$.**5.6.2 Logarithmic Inequalities**

When $a > 1$, the inequality direction is preserved. When $0 < a < 1$, the inequality direction is reversed. At the same time, restrictions that the arguments are greater than zero must be included.

This is because the monotonicity of a logarithmic function is determined by its base. When the base is greater than 1, the larger the argument, the larger the logarithm. When the base is between 0 and 1, the larger the argument, the smaller the logarithm, so the inequality direction must be reversed.

Example 5.12 Logarithmic Inequalities

Solve:

(1) $\log_2(2x - 4) \leq 3;$

(2) $\log_{1/2}(x + 1) > -2;$

$$(3) \log_3(x+4) + \log_3(x-4) \geq 2.$$

Solution:

(1) Require $2x - 4 > 0$, that is, $x > 2$. Since the base $2 > 1$,

$$\log_2(2x - 4) \leq 3 \Rightarrow 2x - 4 \leq 8.$$

Hence $x \leq 6$. Combining with the domain, the solution set is $(2, 6]$.

(2) Require $x + 1 > 0$, that is, $x > -1$. Since the base $\frac{1}{2} \in (0, 1)$, the inequality direction reverses:

$$\log_{1/2}(x+1) > -2 \Rightarrow x+1 < \left(\frac{1}{2}\right)^{-2} = 4.$$

Hence $x < 3$. Combining with the domain, the solution set is $(-1, 3)$.

(3) Require $x + 4 > 0$ and $x - 4 > 0$, so $x > 4$. Combine the logarithms:

$$\log_3((x+4)(x-4)) \geq 2.$$

Since the base $3 > 1$,

$$x^2 - 16 \geq 9 \Rightarrow x^2 \geq 25.$$

Combining with $x > 4$, we get $x \geq 5$. The solution set is $[5, +\infty)$.

In-Class Practice 5.6.1 Logarithmic Equations and Inequalities

- (1) Solve $\log_2(2x) = -2$.
 - (2) Solve $\log_3(x - 1) = \log_3(5 - x)$.
 - (3) Solve $\log_2 x > 4$.
 - (4) Solve $\log_{1/3}(2x - 1) \leq -1$.
 - (5) Solve $\log_2(x - 1) + \log_2(x + 1) = 3$.
 - (6) Solve $\log_2(3x - 2) - \log_2(x - 2) \leq 2$.
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5.7 Chapter Summary**Main Knowledge Thread**

- (1) A negative exponent represents a reciprocal, and a fractional exponent connects powers and radicals.
- (2) The domain of an exponential function is $\mathbb{R}$, and its range is positive numbers; a logarithmic function restricts the input in the opposite way.
- (3) When the base is greater than 1, the function is increasing; when the base lies in $(0, 1)$, the function is decreasing.
- (4) For exponential equations, first convert to the same base or let $t = a^x > 0$. For logarithmic equations, first write argument restrictions.
- (5) Logarithms turn multiplication and division into addition and subtraction, and turn powers into coefficients; they cannot split addition or subtraction.

Formula List for This Chapter

$$a^{-n} = \frac{1}{a^n}, \quad a^{m/n} = \sqrt[n]{a^m}, \quad a^u = a^v \iff u = v.$$

$$a^b = N \iff \log_a N = b, \quad \log_a(MN) = \log_a M + \log_a N, \quad \log_a b = \frac{\log_c b}{\log_c a}.$$

5.8 Chapter Exercises**How to Use These Exercises**

Group A checks the minimum mastery for this chapter. If you make mistakes, return to the corresponding section and reread the examples. Group B is for integrated practice before a quiz. Group C contains extension problems; during a first self-study pass, you may skip it without affecting your entry into the next chapter.

Group A: Basic Check

Difficulty: Basic

- (1) Compute $3^4 \cdot 3^{-3}$ and $(4^3)^2/4^4$.
- (2) Compute $27^{2/3}$ and $16^{-3/4}$.
- (3) Write the common properties and different properties of $y = 2^x$ and $y = (1/2)^x$.
- (4) Solve $3^{x+1} = 27$.
- (5) Solve $2^{2x} - 5 \cdot 2^x + 4 = 0$.
- (6) Compute $\log_3 9$, $\log_2(1/4)$, and $\log_{1/3} 9$.
- (7) Compute $\lg 8 + \lg 125$.
- (8) Compute $\log_2 3 \cdot \log_3 8$.
- (9) Find the domain and range of $y = \log_2(3x - 1)$.
- (10) Solve $\log_2(x - 1) = 3$.
- (11) Solve $\log_2 x > 4$.

Group B: Quiz Level

Difficulty: Core

- (1) Simplify $\frac{(a^{1/2}b^{-1})^4}{a^{-1}b^2}$ over the positive numbers.
- (2) Find the domain, range, monotonicity and asymptote of $y = -3^{x-2} + 4$.
- (3) Solve $3^{2x+1} = 27^{x-1}$.
- (4) Solve $9^x - 10 \cdot 3^x + 9 \leq 0$.
- (5) Expand $\log_a \frac{x^2\sqrt{y}}{z^3}$.
- (6) Find the domain of $y = \log_{1/2}(x^2 - 3x + 2)$.
- (7) Solve $\log_2(x - 3) + \log_2(x - 1) = 3$.
- (8) Solve $\log_3(x + 4) + \log_3(x - 4) \geq 2$.
- (9) Find the domain and range of $y = \log_3(-x^2 + 6x - 5)$.

Group C: Extension and Thinking

Difficulty: Extension

- (1) Solve $2^x + \sqrt{2^x + 1} = 10$.
- (2) Solve $4^x + 2^x = 18$.
- (3) Solve $2x - 2\log_{1/2} x = 6$, and note that x appears in both an algebraic expression and a

logarithmic expression. (Hint: does the left side of the equation have some monotonicity? If the equation has solutions, how many are there?)

(4) Study how the parameter m affects the number of real roots of $2^x + 2^{-x} = m$.

(5) Explain why the graphs of $\log_a x$ and a^x are symmetric about the line $y = x$.

Self-Check

Skill	Can do	Mostly can	Not yet
Correctly use exponent laws	☐	☐	☐
Study graphs and properties of exponential functions	☐	☐	☐
Solve exponential equations and inequalities	☐	☐	☐
Convert between exponential and logarithmic forms	☐	☐	☐
Correctly use logarithm laws	☐	☐	☐
Use the change-of-base formula	☐	☐	☐
Study graphs and properties of logarithmic functions	☐	☐	☐
Solve logarithmic equations and inequalities and check domains	☐	☐	☐

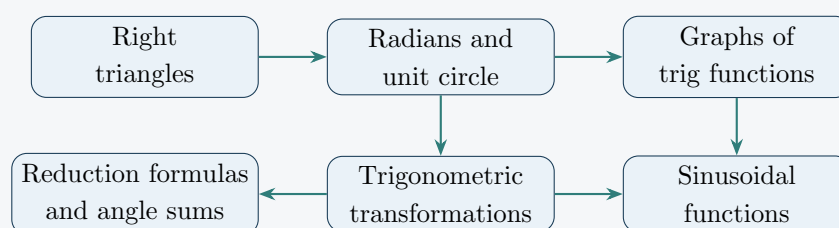
Trigonometric Functions and Trigonometric Identities (三角函数与三角变换)

Learning Goals

After studying this chapter, you should be able to:

- ▶ use sine, cosine, and tangent in right triangles to find sides and angles;
- ▶ remember and use special-angle trigonometric values and the Law of Cosines;
- ▶ convert between degrees and radians;
- ▶ define trigonometric functions of arbitrary angles with the unit circle and determine their signs;
- ▶ draw the graphs of sine, cosine, and tangent and read their properties;
- ▶ use reduction formulas, sum-and-difference formulas, and double-angle formulas;
- ▶ combine $a \sin x + b \cos x$ with the auxiliary-angle formula;
- ▶ study the amplitude, period, extrema, symmetry, and monotonicity of $y = A \sin(\omega x + \varphi) + k$;
- ▶ understand the basic meanings of cotangent, secant, cosecant, and common inverse trigonometric functions.

Knowledge Map



Trigonometric functions grow from "side ratios in triangles" into "coordinates on the unit circle," and then into periodic functions and transformation tools.

Learning Route for This Chapter

Trigonometry starts from side ratios in right triangles, but triangles alone only let us discuss acute angles. To discuss arbitrary angles, we place angles on the unit circle: cosine becomes the x -coordinate, sine becomes the y -coordinate, and tangent becomes the ratio of the y -coordinate to the x -coordinate. The unit circle further explains signs, periods, symmetry, and graphs of trigonometric functions.

The trigonometric identities in the second half of this chapter are not mainly for memorizing formulas. Their core purpose is to turn unfamiliar angles into familiar ones, and to combine or split trigonometric expressions so that computation and graphing become easier.

Prerequisites and Minimum Goals

This chapter uses the Pythagorean theorem, similar triangles, coordinate systems, and basic function graphs. For a first pass, the minimum goals are: use sine, cosine, and tangent in right triangles; convert common angles between degrees and radians; use the unit circle to determine signs of trigonometric functions; and read the basic graphs of sine, cosine, and tangent.

Levels of Formulas in This Chapter

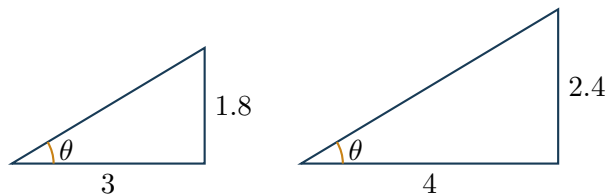
You must master: the definitions of acute-angle trigonometric functions, the special-angle table, the unit-circle definitions, sign determination in reduction formulas, and the basic graphs of sine and cosine. The second level is to master sum-and-difference formulas and double-angle formulas. The extension level is the auxiliary-angle formula and comprehensive work with sinusoidal functions.

6.1 Trigonometric Ratios of Acute Angles (锐角三角函数)

In a right triangle, an acute angle determines fixed ratios among the three sides.

6.1.1 Sine, Cosine, and Tangent (正弦、余弦、正切)

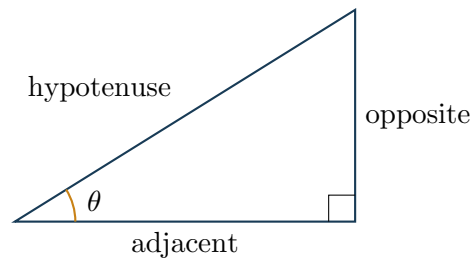
The key idea in trigonometric functions is not the size of the triangle, but the ratios of side lengths. As long as two right triangles have one same acute angle, they are similar, so the ratios of corresponding sides are the same. Therefore we can use these ratios to describe the angle.



After enlarging the triangle, opposite/adjacent is still $1.8/3 = 2.4/4$.

Suppose an acute angle in a right triangle is θ . Then

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}, \quad \cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \quad \tan \theta = \frac{\text{opposite}}{\text{adjacent}}.$$



Basic Relations

$$\sin^2 \theta + \cos^2 \theta = 1, \quad \tan \theta = \frac{\sin \theta}{\cos \theta} \quad (\cos \theta \neq 0).$$

The first relation comes from the Pythagorean theorem. If the hypotenuse is 1, then the opposite side has length $\sin \theta$, and the adjacent side has length $\cos \theta$. Hence

$$\sin^2 \theta + \cos^2 \theta = 1.$$

The second relation comes from dividing two ratios:

$$\frac{\sin \theta}{\cos \theta} = \frac{\text{opposite/hypotenuse}}{\text{adjacent/hypotenuse}} = \frac{\text{opposite}}{\text{adjacent}} = \tan \theta.$$

Example 6.1 Solving a Right Triangle

In a right triangle, the hypotenuse is 10, and the side opposite an acute angle θ is 6. Find the adjacent side and $\sin \theta$, $\cos \theta$, $\tan \theta$.

Solution: By the Pythagorean theorem, the adjacent side is

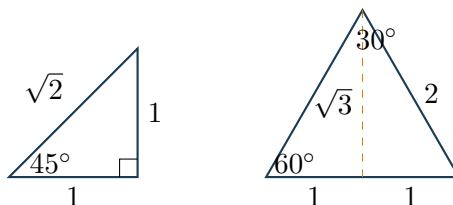
$$\sqrt{10^2 - 6^2} = \sqrt{64} = 8.$$

Therefore

$$\begin{aligned} \sin \theta &= \frac{\text{opposite}}{\text{hypotenuse}} = \frac{6}{10} = \frac{3}{5}, & \cos \theta &= \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{8}{10} = \frac{4}{5}, \\ \tan \theta &= \frac{\text{opposite}}{\text{adjacent}} = \frac{6}{8} = \frac{3}{4}. \end{aligned}$$

6.1.2 Special-Angle Trigonometric Values (特殊角的三角函数值)

Special-angle values come from two basic triangles: an isosceles right triangle gives 45° , and cutting an equilateral triangle in half gives 30° and 60° . Understand their origin first; then the table is much easier to remember.



The two basic triangles explain the square roots in the special-angle table.

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined

Example 6.2 Special-Angle Computations

Compute:

(1) $2 \sin 30^\circ - \cos 60^\circ$;

(2) $\frac{\tan 45^\circ + \tan 30^\circ}{\cos 30^\circ}$;

(3) $\sin^2 45^\circ + \cos^2 45^\circ$.

Solution:

(1) $2 \sin 30^\circ - \cos 60^\circ = 2 \cdot \frac{1}{2} - \frac{1}{2} = \frac{1}{2}$.

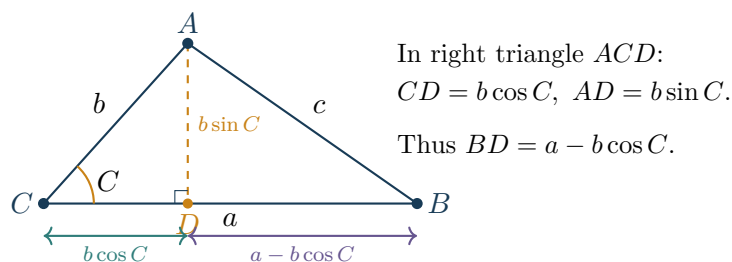
(2) $\frac{\tan 45^\circ + \tan 30^\circ}{\cos 30^\circ} = \frac{1 + \frac{\sqrt{3}}{3}}{\frac{\sqrt{3}}{2}} = \frac{2(\sqrt{3} + 1)}{3}$.

(3) $\sin^2 45^\circ + \cos^2 45^\circ = 1$.

6.1.3 The Law of Cosines in Triangles (三角形中的余弦定理)

This subsection answers the question: how do we find sides and angles when the triangle is not a right triangle? Acute-angle trigonometric functions depend directly on right triangles. If a triangle has no right angle, we need a formula that connects the three sides with an included angle. The Law of Cosines can be viewed as an extension of the Pythagorean theorem: when the included angle is 90° , it becomes the Pythagorean theorem.

We first derive it from the Pythagorean theorem. Suppose angle C is included between the two sides $CA = b$ and $CB = a$, and the opposite side is $AB = c$. Drop a perpendicular from A to line CB , and let the foot be D . This cuts out a right triangle ABD , so we can use the Pythagorean theorem.



Now apply the Pythagorean theorem to right triangle ABD :

$$c^2 = AB^2 = BD^2 + AD^2.$$

Substitute $BD = a - b \cos C$ and $AD = b \sin C$:

$$\begin{aligned} c^2 &= (a - b \cos C)^2 + (b \sin C)^2 \\ &= a^2 - 2ab \cos C + b^2 \cos^2 C + b^2 \sin^2 C \\ &= a^2 + b^2 - 2ab \cos C. \end{aligned}$$

In a triangle, if side c is opposite angle C , and the two sides adjacent to angle C are a, b , then

$$c^2 = a^2 + b^2 - 2ab \cos C.$$

Equivalently,

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}.$$

Example 6.3 Law of Cosines

In a triangle:

- (1) Two sides of a triangle are 5, 7, and the included angle is 60° . Find the third side.
- (2) The three sides of a triangle are 3, 4, 6. Find the cosine of the largest angle.

Solution:

- (1) Let the third side be c . By the Law of Cosines,

$$c^2 = 5^2 + 7^2 - 2 \cdot 5 \cdot 7 \cos 60^\circ = 25 + 49 - 70 \cdot \frac{1}{2} = 39.$$

Therefore $c = \sqrt{39}$.

- (2) When the side lengths are 3, 4, 6, the largest angle is opposite the longest side 6. Let the largest angle be C . Then

$$6^2 = 3^2 + 4^2 - 2 \cdot 3 \cdot 4 \cos C.$$

Hence

$$36 = 25 - 24 \cos C \Rightarrow \cos C = -\frac{11}{24}.$$

In-Class Practice 6.1.1 Trigonometric Ratios of Acute Angles (锐角三角函数)

- (1) In a right triangle, the hypotenuse is 15, and $\sin \theta = 4/5$. Find the two legs.
- (2) Given $\cos \theta = 3/5$ and θ is acute, find $\sin \theta, \tan \theta$.
- (3) Compute $\sin 30^\circ \cos 60^\circ + \cos 30^\circ \sin 60^\circ$.
- (4) Two sides are 8, 10, and the included angle is 120° . Find the third side.
- (5) The three side lengths are 5, 5, 6. Find the cosine of the vertex angle.

6.2 Radian Measure and Arbitrary Angles (弧度制与任意角)

Radians connect the size of an angle directly with arc length; this is the most natural unit for studying trigonometric functions.

6.2.1 Radian Measure

In a circle of radius r , if the arc length cut off by a central angle is l , then the radian measure of the central angle is

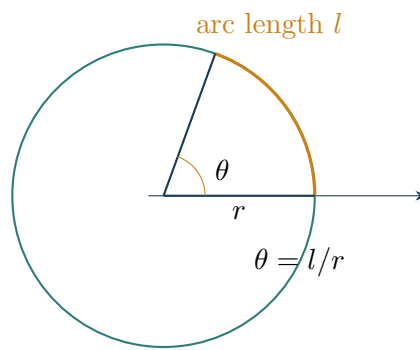
$$\theta = \frac{l}{r}.$$

In other words, radian measure tells us "how many radii long the arc is." When the radius is 1, the radian measure of the angle is exactly the corresponding arc length, which makes trigonometric graphs and later calculations more natural. For example, a semicircle with radius 2 has arc length 2π , and a semicircle with radius 5 has arc length 5π , but in both cases the arc length divided by the radius is π . Thus a semicircle corresponds to π radians, independent of the size of the circle. Therefore

$$180^\circ = \pi \text{ rad}, \quad 1^\circ = \frac{\pi}{180} \text{ rad}.$$

The arc length and sector area formulas are

$$l = r|\theta|, \quad S = \frac{1}{2}r^2|\theta|.$$



Example 6.4 Converting Between Degrees and Radians (角度与弧度互化)

(1) Convert $30^\circ, 135^\circ, 300^\circ, -45^\circ$ to radians.

(2) Convert $\pi/6, 5\pi/4, -2\pi/3$ to degrees.

Solution:

(1) To convert degrees to radians, multiply by $\frac{\pi}{180^\circ}$:

$$30^\circ = \frac{\pi}{6}, \quad 135^\circ = \frac{3\pi}{4}, \quad 300^\circ = \frac{5\pi}{3}, \quad -45^\circ = -\frac{\pi}{4}.$$

(2) To convert radians to degrees, multiply by $\frac{180^\circ}{\pi}$:

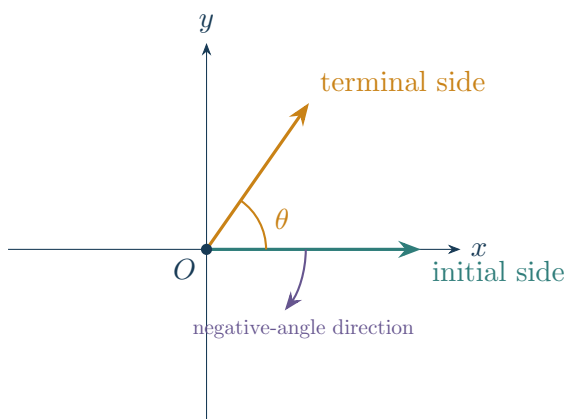
$$\frac{\pi}{6} = 30^\circ, \quad \frac{5\pi}{4} = 225^\circ, \quad -\frac{2\pi}{3} = -120^\circ.$$

6.3 Unit-Circle Definition (单位圆定义)

The unit circle assigns each real angle to a point on the circle, thereby extending trigonometric functions to arbitrary angles.

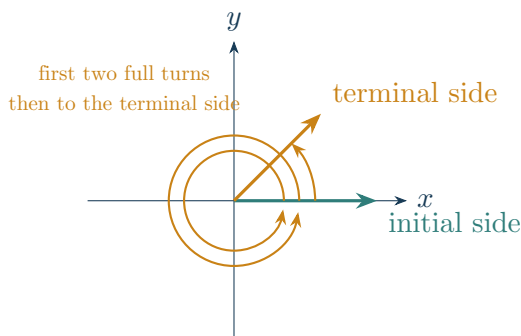
6.3.1 Initial Side, Terminal Side, and Coterminal Angles

When placing an angle in the coordinate plane, we usually put the vertex at the origin and place the starting ray along the positive x -axis. This starting ray is called the **initial side** (始边). The ray where the rotation ends is called the **terminal side** (终边). Counterclockwise rotation gives a positive angle, while clockwise rotation gives a negative angle.

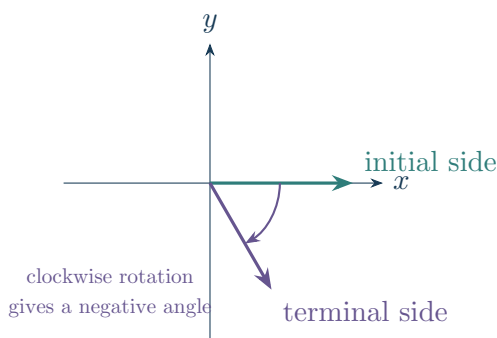


An angle in standard position: vertex at the origin,
initial side on the positive x -axis.

Trigonometric functions only depend on the direction of the terminal side. For example, after an angle rotates more or fewer full turns, the path of rotation is different, but if the terminal side is the same, then the trigonometric values are also the same. The following two graphs show "rotating several full turns" and "a negative angle." Notice: the size of the angle records the process of rotation, while the terminal side records the final direction.



For example $2 \cdot 2\pi + \frac{\pi}{4}$:
after several turns, the terminal side is still in Quadrant I.



For example $-\frac{\pi}{3}$:
rotate clockwise from the initial side to the terminal side.

All angles coterminal with α are

$$\alpha + 2k\pi \quad (k \in \mathbb{Z}).$$

To determine the quadrant, we may add or subtract 2π to move the angle into $[0, 2\pi)$.

Here $2k\pi$ means whole turns: $k = 1$ means one extra full turn, and $k = -1$ means one fewer full turn in the opposite direction. In degrees, the corresponding expression is $\alpha + 360^\circ k$.

Example 6.5 Quadrants and Coterminal Angles (象限与同终边角)

For each angle, determine where its terminal side lies, and write a coterminal angle in $[0, 2\pi)$:

(1) $-7\pi/6$;

(2) $13\pi/4$;

(3) $2027\pi/2$.

Solution: Add or subtract 2π to move the angle into $[0, 2\pi)$.

(1)

$$-\frac{7\pi}{6} + 2\pi = \frac{5\pi}{6},$$

so the terminal side lies in Quadrant II.

(2)

$$\frac{13\pi}{4} - 2\pi = \frac{5\pi}{4},$$

so the terminal side lies in Quadrant III.

(3)

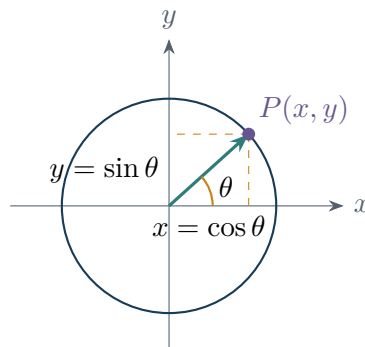
$$\frac{2027\pi}{2} = 1013\pi + \frac{\pi}{2}.$$

Since $1013\pi = 506 \cdot 2\pi + \pi$, a coterminal angle is $\pi + \frac{\pi}{2} = \frac{3\pi}{2}$, and the terminal side lies on the negative y -axis.

6.3.2 Trigonometric Functions of Arbitrary Angles

Let an angle θ be in standard position. Its terminal side intersects the unit circle at point $P(x, y)$. Define

$$\cos \theta = x, \quad \sin \theta = y, \quad \tan \theta = \frac{y}{x} \quad (x \neq 0).$$



This definition agrees with the right-triangle definition. When θ is acute, the point $P(x, y)$ lies in

Quadrant I, and the right triangle formed has hypotenuse 1. Therefore

$$\cos \theta = \frac{\text{adjacent}}{1}, \quad \sin \theta = \frac{\text{opposite}}{1}.$$

The unit-circle definition is more powerful because it also handles obtuse angles, negative angles, and angles larger than 2π .

6.3.3 Signs in Quadrants (象限符号)

The signs of trigonometric functions are exactly the signs of coordinates:

$$\cos \theta = x, \quad \sin \theta = y, \quad \tan \theta = \frac{y}{x}.$$

Quadrant	$\sin \theta$	$\cos \theta$	$\tan \theta$
I	+	+	+
II	+	−	−
III	−	−	+
IV	−	+	−

Example 6.6 Finding the Other Trigonometric Values from One Value

Given $\sin \theta = 3/5$, and θ is in Quadrant II, find $\cos \theta$ and $\tan \theta$; then discuss how the answer changes if θ is in Quadrant I.

Solution: Using $\sin^2 \theta + \cos^2 \theta = 1$,

$$\cos^2 \theta = 1 - \left(\frac{3}{5}\right)^2 = \frac{16}{25}.$$

In Quadrant II, cosine is negative, so

$$\cos \theta = -\frac{4}{5}, \quad \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{3/5}{-4/5} = -\frac{3}{4}.$$

If θ is in Quadrant I, then both cosine and tangent are positive:

$$\cos \theta = \frac{4}{5}, \quad \tan \theta = \frac{3}{4}.$$

In-Class Practice 6.3.1 Radians and Unit Circle

- (1) Reduce the angle $17\pi/6$ to $[0, 2\pi)$ and determine its quadrant.
- (2) Write the coordinates of the point on the unit circle corresponding to angle $2\pi/3$.
- (3) If $\cos \theta = -\frac{12}{13}$, and θ is in Quadrant III, find $\sin \theta$, $\tan \theta$.
- (4) In a circle of radius 6, the central angle is $5\pi/6$. Find the arc length and the sector area.

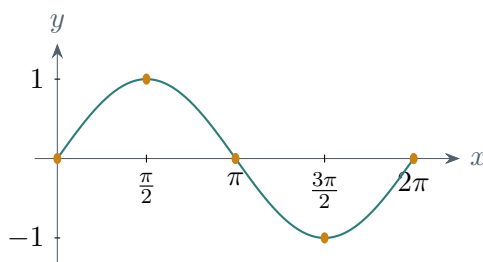
6.4 Graphs of Sine, Cosine, and Tangent (正弦、余弦和正切函数的图像)

A trigonometric function turns continuous rotation into a function graph. The graph lets us see periodicity, symmetry, monotonicity, and maximum and minimum values.

6.4.1 The Sine Function

On the unit circle, $\sin x$ is the y -coordinate of the point reached by rotating through angle x . As x increases from 0 to 2π , the point makes one full turn, and the y -coordinate goes from 0 up to 1, back to 0, down to -1 , and finally back to 0. This gives the basic shape of the sine graph.

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$\sin x$	0	1	0	-1	0



Plot the five key points, then connect them with a smooth curve.

Why can the same shape be repeated to the left and right? After the angle increases by 2π , the

terminal side returns to the same position, so the y -coordinate is unchanged:

$$\sin(x + 2\pi) = \sin x.$$

More generally, if two angles differ by a whole number of full turns,

$$x + 2k\pi \quad (k \in \mathbb{Z}),$$

then they have the same terminal side. Hence

$$\sin(x + 2k\pi) = \sin x, \quad \cos(x + 2k\pi) = \cos x.$$

For tangent, as long as both sides are defined,

$$\tan(x + k\pi) = \tan x.$$

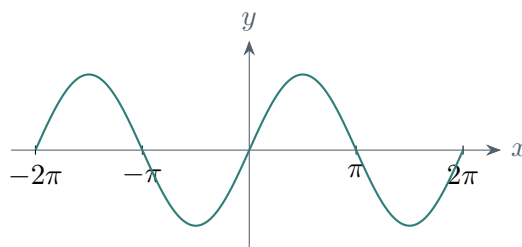
Periodic Function

If there exists a nonzero constant T such that

$$f(x + T) = f(x)$$

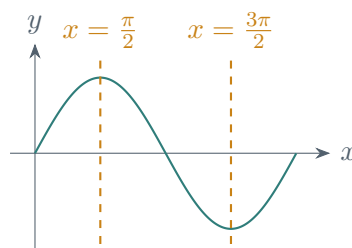
for every allowed x in the domain, then $f(x)$ is called a periodic function, and T is called a period.

For the sine function, 2π is a period. Therefore, after drawing the part on $[0, 2\pi]$, copy this part to the left and right every 2π to obtain the whole sine graph.

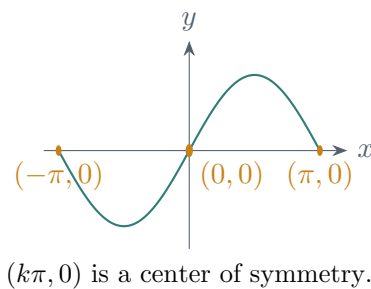


The sine graph repeats every 2π .

The symmetry of sine can be read from the graph. The peaks and troughs give axes of symmetry; the points where the curve crosses the midline in the same shape give centers of symmetry.



$x = \frac{\pi}{2} + k\pi$ is an axis of symmetry of the sine graph.



For the sine function:

domain: $\mathbb{R}$, range: $[-1, 1]$, period: 2π ;

$\sin(-x) = -\sin x$, so sine is odd;

axes of symmetry: $x = \frac{\pi}{2} + k\pi$, centers of symmetry: $(k\pi, 0)$;

maximum 1 occurs at $x = \frac{\pi}{2} + 2k\pi$, minimum -1 occurs at $x = \frac{3\pi}{2} + 2k\pi$;

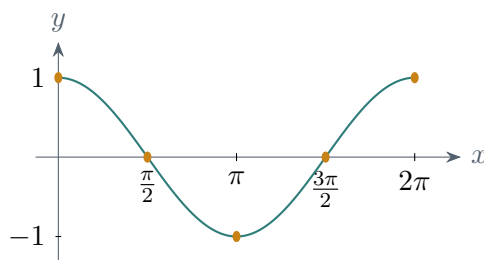
increasing on $\left[-\frac{\pi}{2} + 2k\pi, \frac{\pi}{2} + 2k\pi\right]$,

decreasing on $\left[\frac{\pi}{2} + 2k\pi, \frac{3\pi}{2} + 2k\pi\right]$, $k \in \mathbb{Z}$.

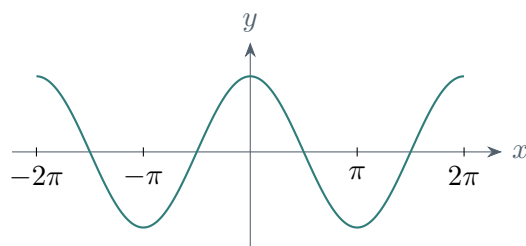
6.4.2 The Cosine Function

On the unit circle, $\cos x$ is the x -coordinate of the point reached by rotating through angle x . When x goes from 0 to 2π , the x -coordinate starts at 1, decreases to 0, reaches -1 , returns to 0, and finally returns to 1.

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$\cos x$	1	0	-1	0	1



Five-point graphing: start from the highest point, go to the lowest point, then return.



The cosine graph also repeats every 2π .

For the cosine function:

domain: $\mathbb{R}$, range: $[-1, 1]$, period: 2π ;

$\cos(-x) = \cos x$, so cosine is even;

axes of symmetry: $x = k\pi$, centers of symmetry: $\left(\frac{\pi}{2} + k\pi, 0\right)$;

maximum 1 occurs at $x = 2k\pi$, minimum -1 occurs at $x = (2k + 1)\pi$;

increasing on $[(2k - 1)\pi, 2k\pi]$, decreasing on $[2k\pi, (2k + 1)\pi]$, $k \in \mathbb{Z}$.

6.4.3 The Tangent Function

The tangent function is defined by

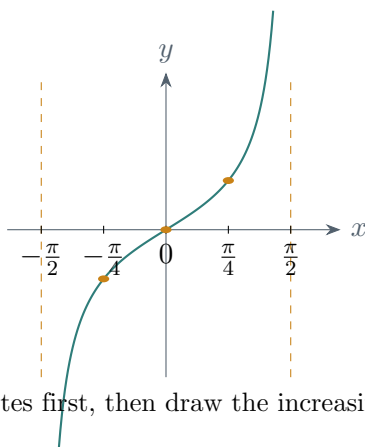
$$\tan x = \frac{\sin x}{\cos x}.$$

Thus tangent is undefined when $\cos x = 0$, that is, when

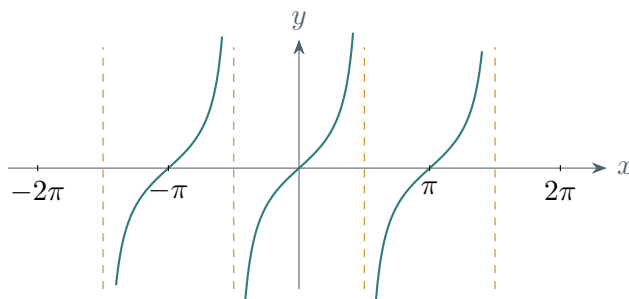
$$x = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}.$$

These x -values become vertical asymptotes of the tangent graph. When drawing the tangent graph, first draw the vertical asymptotes, then plot several key points between two neighboring asymptotes.

x	$-\frac{\pi}{3}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{3}$
$\tan x$	$-\sqrt{3}$	-1	0	1	$\sqrt{3}$



Draw the vertical asymptotes first, then draw the increasing curve between them.



The tangent graph repeats every π .

For the tangent function:

$$\text{domain: } \left\{x \mid x \neq \frac{\pi}{2} + k\pi, k \in \mathbb{Z}\right\}, \quad \text{range: } \mathbb{R}, \quad \text{period: } \pi;$$

$$\tan(-x) = -\tan x, \text{ so tangent is odd;}$$

$$\text{tangent is increasing on each interval } \left(-\frac{\pi}{2} + k\pi, \frac{\pi}{2} + k\pi\right);$$

$$\text{centers of symmetry: } (k\pi, 0), \quad \text{no axes of symmetry, no maximum, and no minimum.}$$

In-Class Practice 6.4.1 Graphs of Trigonometric Functions

(1) Write the maximum point and minimum point of $y = \sin x$ on $[0, 2\pi]$.

(2) Write the monotonic intervals of $y = \cos x$ on $[-\pi, \pi]$.

(3) Find the domain and zeros of $y = \tan x$ on $[-\pi, \pi]$.

(4) Compare $\sin 40^\circ$ and $\sin 160^\circ$.

(5) If $f(x) = \sin x$, find $f(2026\pi + \pi/6)$.

6.5 Reduction Formulas (诱导公式)

Reduction formulas turn angles such as $\pi + x$, $\pi - x$, and $-x$ into expressions involving x . Their essence is the symmetry and periodicity of the unit circle.

6.5.1 Four Basic Relations

Reduction formulas are easiest to understand from the unit circle. We mainly use four types of relations.

Type 1: periodicity After an angle rotates by a whole number of full turns, its terminal side is unchanged, so the trigonometric values repeat:

$$\sin(x + 2k\pi) = \sin x, \quad \cos(x + 2k\pi) = \cos x, \quad \tan(x + k\pi) = \tan x.$$

Here $k \in \mathbb{Z}$. Notice that sine and cosine have period 2π , while tangent has period π .

Type 2: parity The terminal sides of x and $-x$ are symmetric about the x -axis. The y -coordinates are opposite, while the x -coordinates are the same. Therefore

$$\sin(-x) = -\sin x, \quad \cos(-x) = \cos x, \quad \tan(-x) = -\tan x.$$

This also shows that sine and tangent are odd functions, while cosine is an even function.

Type 3: $\pi - x$ The terminal sides of x and $\pi - x$ are symmetric about the y -axis. The x -coordinates are opposite, while the y -coordinates are the same. Hence

$$\sin(\pi - x) = \sin x, \quad \cos(\pi - x) = -\cos x, \quad \tan(\pi - x) = -\tan x.$$

This type is often used for angles in Quadrant II.

Type 4: $\frac{\pi}{2} - x$ The angles $\frac{\pi}{2} - x$ and x are complementary. On the unit circle, their x - and y -coordinates are exchanged, so

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x, \quad \cos\left(\frac{\pi}{2} - x\right) = \sin x.$$

If both sides are defined, then

$$\tan\left(\frac{\pi}{2} - x\right) = \frac{\cos x}{\sin x} = \frac{1}{\tan x}.$$

The core of this type is that sine and cosine are exchanged.

Four Basic Reduction Relations

Most reduction formulas can be derived from these four ideas:

$$\text{periodicity}, \quad \text{parity}, \quad \pi - x, \quad \frac{\pi}{2} - x.$$

When using a formula, first identify the related acute or basic angle, then determine the sign by the quadrant.

6.5.2 Deriving Other Forms from the Four Relations

Other common forms do not need to be memorized separately. They can be derived from the four basic relations.

Example 6.7 Deriving Other Reduction Formulas

Derive the following:

- (1) $\sin(\pi + x) = -\sin x$;
- (2) $\cos(2\pi - x) = \cos x$;
- (3) $\sin\left(\frac{\pi}{2} + x\right) = \cos x$;
- (4) $\cos\left(\frac{3\pi}{2} - x\right) = -\sin x$.

Solution:

- (1) View $\pi + x$ as $\pi - (-x)$. Use the third type first, then parity:

$$\sin(\pi + x) = \sin(\pi - (-x)) = \sin(-x) = -\sin x.$$

- (2) First use periodicity to subtract 2π , then use parity:

$$\cos(2\pi - x) = \cos(-x) = \cos x.$$

- (3) View $\frac{\pi}{2} + x$ as $\frac{\pi}{2} - (-x)$. Use the fourth type first, then parity:

$$\sin\left(\frac{\pi}{2} + x\right) = \sin\left(\frac{\pi}{2} - (-x)\right) = \cos(-x) = \cos x.$$

- (4) First rewrite the angle as $\pi + \left(\frac{\pi}{2} - x\right)$, and then turn $\pi + u$ into $\pi - (-u)$:

$$\cos\left(\frac{3\pi}{2} - x\right) = \cos\left(\pi + \left(\frac{\pi}{2} - x\right)\right) = \cos\left(\pi - \left(x - \frac{\pi}{2}\right)\right).$$

By the third type,

$$\cos\left(\pi - \left(x - \frac{\pi}{2}\right)\right) = -\cos\left(x - \frac{\pi}{2}\right).$$

Also,

$$\cos\left(x - \frac{\pi}{2}\right) = \cos\left(-\left(\frac{\pi}{2} - x\right)\right) = \cos\left(\frac{\pi}{2} - x\right) = \sin x,$$

so

$$\cos\left(\frac{3\pi}{2} - x\right) = -\sin x.$$

Example 6.8 Computations with Reduction Formulas

Compute:

(1) $\sin \frac{5\pi}{6};$

(2) $\cos \frac{5\pi}{4};$

(3) $\tan\left(-\frac{2\pi}{3}\right);$

(4) $\sin \frac{11\pi}{6}.$

Solution:

- (1)

$$\sin \frac{5\pi}{6} = \sin\left(\pi - \frac{\pi}{6}\right) = \sin \frac{\pi}{6} = \frac{1}{2}.$$

- (2) First rewrite $\pi + \frac{\pi}{4}$ as $\pi - \left(-\frac{\pi}{4}\right)$:

$$\cos \frac{5\pi}{4} = \cos\left(\pi + \frac{\pi}{4}\right) = \cos\left(\pi - \left(-\frac{\pi}{4}\right)\right) = -\cos\left(-\frac{\pi}{4}\right) = -\cos \frac{\pi}{4} = -\frac{\sqrt{2}}{2}.$$

(3) First use parity for the negative angle, then use $\pi - x$:

$$\tan\left(-\frac{2\pi}{3}\right) = -\tan\frac{2\pi}{3} = -\tan\left(\pi - \frac{\pi}{3}\right) = -(-\tan\frac{\pi}{3}) = \sqrt{3}.$$

(4) First subtract 2π by periodicity:

$$\sin\frac{11\pi}{6} = \sin\left(-\frac{\pi}{6}\right) = -\sin\frac{\pi}{6} = -\frac{1}{2}.$$

6.6 Angle-Sum, Angle-Difference, Double-Angle, and Auxiliary-Angle Formulas (和差角、二倍角与辅助角)

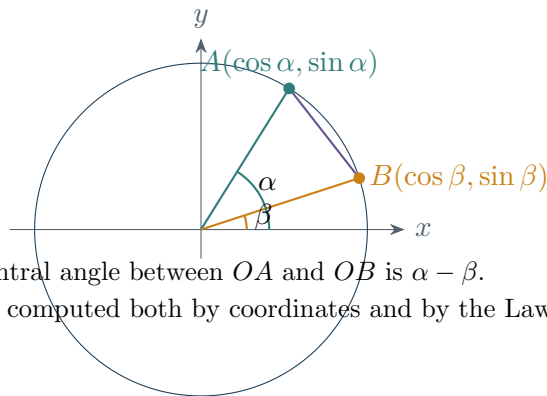
Angle formulas let us calculate trigonometric values of combined angles and transform trigonometric expressions.

6.6.1 Sum and Difference Formulas (和差角公式)

The most fundamental angle formula is the cosine difference formula:

$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta.$$

We prove it on the unit circle using the Law of Cosines.



The central angle between OA and OB is $\alpha - \beta$.

The distance AB can be computed both by coordinates and by the Law of Cosines.

Since $A(\cos\alpha, \sin\alpha)$ and $B(\cos\beta, \sin\beta)$,

$$AB^2 = (\cos\alpha - \cos\beta)^2 + (\sin\alpha - \sin\beta)^2.$$

Expanding and using $\sin^2 x + \cos^2 x = 1$, we get

$$AB^2 = 2 - 2(\cos\alpha\cos\beta + \sin\alpha\sin\beta).$$

On the other hand, $OA = OB = 1$, and the included angle is $\alpha - \beta$. By the Law of Cosines,

$$AB^2 = 1^2 + 1^2 - 2\cos(\alpha - \beta) = 2 - 2\cos(\alpha - \beta).$$

Comparing the two expressions for AB^2 gives

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta.$$

The first formula has now been proved. We derive the other five formulas from it.

Replace β by $-\beta$, and use parity:

$$\cos(\alpha + \beta) = \cos(\alpha - (-\beta)) = \cos \alpha \cos(-\beta) + \sin \alpha \sin(-\beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta.$$

Next use the reduction formula $\sin u = \cos\left(\frac{\pi}{2} - u\right)$ to change sine into cosine:

$$\begin{aligned} \sin(\alpha + \beta) &= \cos\left(\frac{\pi}{2} - (\alpha + \beta)\right) \\ &= \cos\left(\left(\frac{\pi}{2} - \alpha\right) - \beta\right) \\ &= \cos\left(\frac{\pi}{2} - \alpha\right) \cos \beta + \sin\left(\frac{\pi}{2} - \alpha\right) \sin \beta \\ &= \sin \alpha \cos \beta + \cos \alpha \sin \beta. \end{aligned}$$

Replacing β by $-\beta$ gives

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta.$$

Finally use the idea of "changing tangent into sine and cosine," namely

$$\tan u = \frac{\sin u}{\cos u}.$$

From the sine and cosine sum formulas,

$$\tan(\alpha + \beta) = \frac{\sin \alpha \cos \beta + \cos \alpha \sin \beta}{\cos \alpha \cos \beta - \sin \alpha \sin \beta}.$$

When all quantities involved are defined and $\cos \alpha \cos \beta \neq 0$, divide the numerator and denominator by $\cos \alpha \cos \beta$ to obtain

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}.$$

Replacing β by $-\beta$ gives

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}.$$

Sum and Difference Formulas

$$\begin{aligned} \sin(\alpha \pm \beta) &= \sin \alpha \cos \beta \pm \cos \alpha \sin \beta, \\ \cos(\alpha \pm \beta) &= \cos \alpha \cos \beta \mp \sin \alpha \sin \beta, \\ \tan(\alpha \pm \beta) &= \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}. \end{aligned}$$

The tangent formulas require the denominators to be nonzero, and the tangent values involved must be defined.

Example 6.9 Sum and Difference for Special Angles

Compute:

- (1) $\sin 75^\circ$;
 (2) $\cos 15^\circ$;
 (3) $\tan 105^\circ$.

Solution:

(1)

$$\sin 75^\circ = \sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

(2)

$$\cos 15^\circ = \cos(45^\circ - 30^\circ) = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

(3)

$$\tan 105^\circ = \tan(60^\circ + 45^\circ) = \frac{\sqrt{3} + 1}{1 - \sqrt{3}} = -(2 + \sqrt{3}).$$

Example 6.10 Recognizing Formula Structure

Simplify:

- (1) $\sin 15^\circ \cos 45^\circ + \cos 15^\circ \sin 45^\circ$;
 (2) $\cos 75^\circ \cos 15^\circ + \sin 75^\circ \sin 15^\circ$;
 (3) $\frac{\tan 25^\circ + \tan 20^\circ}{1 - \tan 25^\circ \tan 20^\circ}$.

Solution:

- (1) This matches the sine sum formula:

$$\sin 15^\circ \cos 45^\circ + \cos 15^\circ \sin 45^\circ = \sin 60^\circ = \frac{\sqrt{3}}{2}.$$

- (2) This matches the cosine difference formula:

$$\cos 75^\circ \cos 15^\circ + \sin 75^\circ \sin 15^\circ = \cos(75^\circ - 15^\circ) = \cos 60^\circ = \frac{1}{2}.$$

- (3) This matches the tangent sum formula:

$$\frac{\tan 25^\circ + \tan 20^\circ}{1 - \tan 25^\circ \tan 20^\circ} = \tan 45^\circ = 1.$$

6.6.2 Double-Angle Formulas (二倍角公式)Putting $\alpha = \beta = x$ in the sum formulas gives the double-angle formulas:

Double-Angle Formulas

$$\sin 2x = 2 \sin x \cos x,$$

$$\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x,$$

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}.$$

Example 6.11 Finding Double-Angle Values from One Value

Suppose $\sin x = \frac{3}{5}$, and x is in Quadrant II. Find $\sin 2x$, $\cos 2x$, $\tan 2x$.

Solution: Since x is in Quadrant II, $\cos x < 0$. From $\sin^2 x + \cos^2 x = 1$,

$$\cos x = -\frac{4}{5}.$$

Thus

$$\sin 2x = 2 \sin x \cos x = 2 \cdot \frac{3}{5} \cdot \left(-\frac{4}{5}\right) = -\frac{24}{25},$$

$$\cos 2x = \cos^2 x - \sin^2 x = \frac{16}{25} - \frac{9}{25} = \frac{7}{25},$$

$$\tan 2x = \frac{\sin 2x}{\cos 2x} = -\frac{24}{7}.$$

6.6.3 Auxiliary-Angle Formula (辅助角公式)

Expressions such as $a \sin x + b \cos x$ often appear when studying trigonometric functions. The auxiliary-angle formula combines them into one sine function, making maximum and minimum values and graphs much easier to read.

Let

$$R = \sqrt{a^2 + b^2}.$$

Choose an angle φ such that

$$\cos \varphi = \frac{a}{R}, \quad \sin \varphi = \frac{b}{R}.$$

Then

$$a \sin x + b \cos x = R(\cos \varphi \sin x + \sin \varphi \cos x) = R \sin(x + \varphi).$$

Pay attention to the quadrant of φ , because $\cos \varphi$ and $\sin \varphi$ must have the same signs as a and b .

Example 6.12 Combining into One Sine

Rewrite the following expressions in the form $R \sin(x + \varphi)$:

(1) $\sin x + \cos x$;

(2) $\sqrt{3} \sin x + \cos x$;

(3) $3 \sin x - 4 \cos x$.

Also write the maximum and minimum of each expression.

Solution: Suppose $a \sin x + b \cos x = R \sin(x + \varphi)$, where $R = \sqrt{a^2 + b^2}$.

(1) For $\sin x + \cos x$, $a = 1, b = 1$, so

$$R = \sqrt{1^2 + 1^2} = \sqrt{2}.$$

Therefore we need to choose φ such that

$$\cos \varphi = \frac{a}{R} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}, \quad \sin \varphi = \frac{b}{R} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}.$$

Both cosine and sine are positive, so φ lies in Quadrant I. From the special-angle table, $\cos \frac{\pi}{4} = \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$, so we may take $\varphi = \frac{\pi}{4}$. Thus

$$\sin x + \cos x = \sqrt{2} \sin \left(x + \frac{\pi}{4} \right).$$

The maximum is $\sqrt{2}$, and the minimum is $-\sqrt{2}$.

(2) For $\sqrt{3} \sin x + \cos x$, $a = \sqrt{3}, b = 1$, so

$$R = \sqrt{(\sqrt{3})^2 + 1^2} = 2.$$

Therefore

$$\cos \varphi = \frac{a}{R} = \frac{\sqrt{3}}{2}, \quad \sin \varphi = \frac{b}{R} = \frac{1}{2}.$$

Both cosine and sine are positive, so φ lies in Quadrant I. From the special-angle table, $\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$, $\sin \frac{\pi}{6} = \frac{1}{2}$, so we may take $\varphi = \frac{\pi}{6}$. Thus

$$\sqrt{3} \sin x + \cos x = 2 \sin \left(x + \frac{\pi}{6} \right).$$

The maximum is 2, and the minimum is -2.

(3) For $3 \sin x - 4 \cos x$, $a = 3, b = -4$, so

$$R = \sqrt{3^2 + (-4)^2} = 5.$$

Therefore

$$\cos \varphi = \frac{a}{R} = \frac{3}{5}, \quad \sin \varphi = \frac{b}{R} = -\frac{4}{5}.$$

Here cosine is positive and sine is negative, so φ lies in Quadrant IV. First choose an acute angle θ such that

$$\cos \theta = \frac{3}{5}, \quad \sin \theta = \frac{4}{5}.$$

This means $\tan \theta = \frac{4}{3}$. This angle is not a special angle, and is usually written as $\theta = \arctan \frac{4}{3}$, called the inverse tangent function. Then the angle in Quadrant IV satisfying

the required signs may be taken as $\varphi = -\theta$. In other words, it can be written in the form

$$\varphi = -\arctan \frac{4}{3}.$$

Therefore

$$3 \sin x - 4 \cos x = 5 \sin\left(x - \arctan \frac{4}{3}\right).$$

The maximum is 5, and the minimum is -5 .

In-Class Practice 6.6.1 Trigonometric Transformations (三角变换)

- (1) Compute $\cos(45^\circ + 30^\circ)$.
- (2) Simplify $\sin 80^\circ \cos 20^\circ - \cos 80^\circ \sin 20^\circ$.
- (3) Given $\tan \alpha = 1/3, \tan \beta = 1/2$, find $\tan(\alpha + \beta)$.
- (4) Simplify $2 \sin x \cos x$ and $1 - 2 \sin^2 x$.
- (5) Rewrite $\sin x - \sqrt{3} \cos x$ as one sine function.

6.7 Graphs and Properties of Sinusoidal Functions (正弦型函数的图像与性质)

A sinusoidal function can be viewed as the graph of $y = \sin x$ after scaling and translation. First understand the parameters through graph transformations, then use substitution to read the properties uniformly.

6.7.1 Meanings of the Parameters and Graph Transformations

We already know that the graph of $y = \sin x$ rises and falls repeatedly, with highest value 1, lowest value -1 , and period 2π . Real periodic phenomena do not necessarily oscillate around the x -axis, nor do they necessarily have height 1 or period 2π . For example, temperature, water level, and alternating voltage may all have different heights, periods, and midlines. Therefore we use

$$y = A \sin(\omega x + \varphi) + k \quad (A\omega \neq 0)$$

to describe functions that "rise and fall periodically like the sine function."

When reading this type of function, first rewrite the expression inside the parentheses as

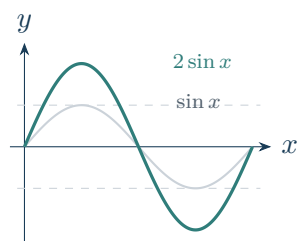
$$\omega x + \varphi = \omega(x - h), \quad h = -\frac{\varphi}{\omega}.$$

Then we can see that h controls horizontal translation, k controls vertical translation, A controls vertical scaling, and ω controls horizontal scaling.

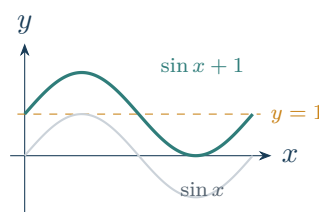
Graphical Meaning of the Four Parameters (四个参数的图像意义)

Parameter	Graphical meaning
A	The distance from the graph to the midline becomes $ A $ times the original distance, so the amplitude is $ A $. If $A < 0$, the graph is also reflected across the midline.
ω	The graph is compressed or stretched horizontally, and the period becomes $T = \frac{2\pi}{ \omega }$.
φ	Together with ω , it determines horizontal translation. First rewrite the inside as $\omega(x - h)$; then the graph shifts right by h .
k	The whole graph shifts vertically, and the midline becomes $y = k$.

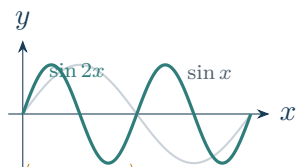
The following four small graphs each change only one parameter. Comparing them makes the effect of each parameter much clearer than memorizing formulas alone.



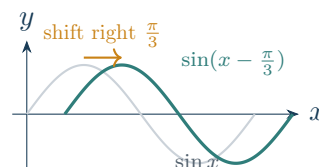
A : vertical scaling



k : vertical translation



ω : horizontal scaling



φ : horizontal translation

General Order for Graphing Sinusoidal Functions (正弦型函数作图的一般顺序)

To draw $y = A \sin(\omega x + \varphi) + k$, use the following order:

- (1) Rewrite the phase as $\omega(x - h)$, and first read the horizontal translation h .
- (2) Use ω to get the period $T = \frac{2\pi}{|\omega|}$, and determine the length of one period.
- (3) Use A to get the amplitude $|A|$, and determine how far the highest and lowest points are from the midline.
- (4) Use k to get the midline $y = k$, and finally shift the whole graph vertically.

Example 6.13 Drawing a Sinusoidal Function Step by Step (分步骤画出一个正弦型函数)

Draw

$$y = 2 \sin \left(2x - \frac{\pi}{3} \right) - 1.$$

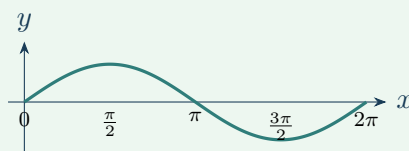
Solution: First rewrite the expression inside the parentheses as ``a constant times $(x - h)$ ':

$$2x - \frac{\pi}{3} = 2 \left(x - \frac{\pi}{6} \right),$$

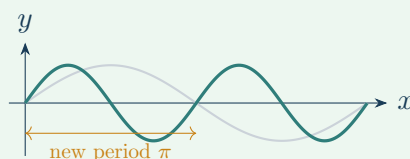
so this function can be viewed as

$$y = 2 \sin \left(2 \left(x - \frac{\pi}{6} \right) \right) - 1.$$

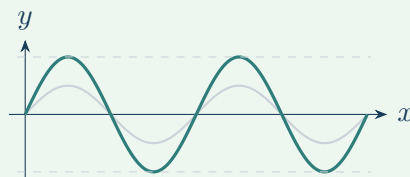
Starting from $y = \sin x$, do the following four steps.



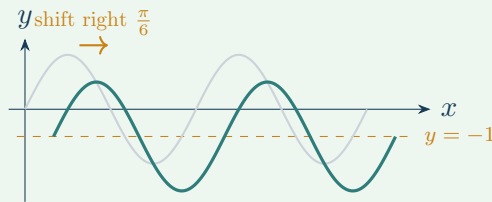
Step 1: $y = \sin x$



Step 2: compress horizontally to one half, giving $y = \sin 2x$



Step 3: stretch vertically by a factor of 2, giving $y = 2 \sin 2x$



Step 4: first shift right by $\frac{\pi}{6}$, then shift down by 1, giving the target function

If we use the five-point method to draw one period, let

$$u = 2x - \frac{\pi}{3} = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi.$$

Solving gives

$$x = \frac{\pi}{6}, \frac{5\pi}{12}, \frac{2\pi}{3}, \frac{11\pi}{12}, \frac{7\pi}{6},$$

and the corresponding function values are

$$-1, 1, -1, -3, -1.$$

Plot these five points, then extend periodically to both sides with period π , and we obtain the whole graph.

6.7.2 Reading Properties by Substitution

The properties of a sinusoidal function come from $y = \sin u$. Let

$$u = \omega x + \varphi.$$

Then the problem becomes: for which u -values does $\sin u$ increase, decrease, reach a maximum or minimum, have axes of symmetry, or have centers of symmetry; after that, convert the conditions on u back into conditions on x .

Reading from u Back to x (从 u 换回 x 的读法)

For example, for

$$y = A \sin(\omega x + \varphi) + k, \quad u = \omega x + \varphi$$

read as follows:

- (1) Period: when u increases by 2π , the function value repeats once, so $T = \frac{2\pi}{|\omega|}$.
- (2) Maximum and minimum values: $\sin u = 1$ gives the highest or lowest value, and $\sin u = -1$ gives the other end.
- (3) Axes of symmetry: the axes of symmetry of the sine graph come from $u = \frac{\pi}{2} + n\pi$.
- (4) Centers of symmetry: the centers of the sine graph come from $u = n\pi$, and the corresponding points all lie on the midline $y = k$.
- (5) Monotonicity: first find increasing and decreasing intervals on the u -axis, then solve inequalities in x .

More specifically, if $\omega > 0$ and $A > 0$, then

$$\text{increasing intervals: } -\frac{\pi}{2} + 2n\pi \leq \omega x + \varphi \leq \frac{\pi}{2} + 2n\pi,$$

$$\text{decreasing intervals: } \frac{\pi}{2} + 2n\pi \leq \omega x + \varphi \leq \frac{3\pi}{2} + 2n\pi.$$

Here $n \in \mathbb{Z}$. If $A < 0$, vertical reflection swaps increasing and decreasing intervals; if $\omega < 0$, the direction of the inequality changes when solving. For beginners, it is often best to first rewrite the expression so that $\omega > 0$, then read the properties.

Example 6.14 Reading Complete Properties by Substitution (用换元读完整性质)

Study

$$y = 2 \sin\left(2x - \frac{\pi}{3}\right) - 1$$

and find its period, range, maximum and minimum values, axes of symmetry, centers of symmetry, and monotonic intervals.

Solution: Let

$$u = 2x - \frac{\pi}{3}.$$

Then this function is $y = 2 \sin u - 1$.

The period is

$$T = \frac{2\pi}{2} = \pi.$$

Since $-1 \leq \sin u \leq 1$,

$$-3 \leq 2 \sin u - 1 \leq 1,$$

so the range is $[-3, 1]$.

When $\sin u = 1$, that is,

$$u = \frac{\pi}{2} + 2n\pi,$$

the function attains its maximum value 1. Converting back to x ,

$$2x - \frac{\pi}{3} = \frac{\pi}{2} + 2n\pi \Rightarrow x = \frac{5\pi}{12} + n\pi.$$

When $\sin u = -1$, that is,

$$u = \frac{3\pi}{2} + 2n\pi,$$

the function attains its minimum value -3 . Converting back to x ,

$$2x - \frac{\pi}{3} = \frac{3\pi}{2} + 2n\pi \Rightarrow x = \frac{11\pi}{12} + n\pi.$$

Axes of symmetry come from

$$u = \frac{\pi}{2} + n\pi,$$

so

$$2x - \frac{\pi}{3} = \frac{\pi}{2} + n\pi \Rightarrow x = \frac{5\pi}{12} + \frac{n\pi}{2}.$$

Centers of symmetry come from

$$u = n\pi,$$

so

$$2x - \frac{\pi}{3} = n\pi \Rightarrow x = \frac{\pi}{6} + \frac{n\pi}{2}.$$

The corresponding centers are

$$\left(\frac{\pi}{6} + \frac{n\pi}{2}, -1\right).$$

Because here $A = 2 > 0, \omega = 2 > 0$, increasing intervals are obtained from

$$-\frac{\pi}{2} + 2n\pi \leq 2x - \frac{\pi}{3} \leq \frac{\pi}{2} + 2n\pi$$

as

$$-\frac{\pi}{12} + n\pi \leq x \leq \frac{5\pi}{12} + n\pi.$$

Decreasing intervals are obtained from

$$\frac{\pi}{2} + 2n\pi \leq 2x - \frac{\pi}{3} \leq \frac{3\pi}{2} + 2n\pi$$

as

$$\frac{5\pi}{12} + n\pi \leq x \leq \frac{11\pi}{12} + n\pi.$$

In-Class Practice 6.7.1 Sinusoidal Functions (正弦型函数)

- (1) Find the range, period, and midline of $y = 3 \sin x + 2$.
- (2) Explain how $y = \sin(2x)$ and $y = \sin(x - \pi/4)$ are transformed from $y = \sin x$, respectively.
- (3) Use the five-point method to draw one period of $y = 2 \sin(2x) - 1$.
- (4) Find the period, range, and five key points in one period of $y = 2 \sin(3x - \pi/2) + 1$.
- (5) Find the increasing intervals, decreasing intervals, and centers of symmetry of $y = \sin(2x)$ on $[0, \pi]$.

6.8 Other Trigonometric Functions and Inverse Trigonometric Functions

Besides sine, cosine, and tangent, we will sometimes meet cotangent, secant, cosecant, and inverse trigonometric functions. This section is only an introduction; the goal is to know roughly what they express.

6.8.1 Cotangent, Secant, and Cosecant

Tangent can be regarded as the ratio of sine to cosine:

$$\tan x = \frac{\sin x}{\cos x}.$$

Similarly, we can define three more functions:

$$\cot x = \frac{\cos x}{\sin x}, \quad \sec x = \frac{1}{\cos x}, \quad \csc x = \frac{1}{\sin x}.$$

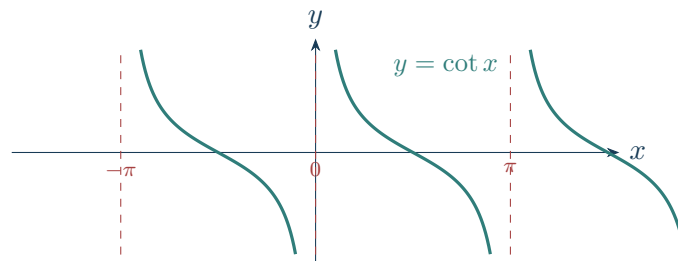
Their names are cotangent (余切), secant (正割), and cosecant (余割). Because denominators cannot be 0, their domains must exclude the corresponding points.

Function	Meaning	Excluded x -values
$\cot x$	$\frac{\cos x}{\sin x}$, namely the reciprocal of $\tan x$	$x = n\pi$
$\sec x$	$\frac{1}{\cos x}$, namely the reciprocal of $\cos x$	$x = \frac{\pi}{2} + n\pi$
$\csc x$	$\frac{1}{\sin x}$, namely the reciprocal of $\sin x$	$x = n\pi$

where $n \in \mathbb{Z}$. For example,

$$\cot \frac{\pi}{4} = 1, \quad \sec \frac{\pi}{3} = 2, \quad \csc \frac{\pi}{6} = 2.$$

These functions also repeat periodically, and vertical asymptotes appear where the denominator is 0.



The cotangent function is undefined at $x = n\pi$.

6.8.2 Inverse Trigonometric Functions

The equation

$$\sin x = \frac{1}{2}$$

has many solutions, for example $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \dots$. If we want to turn "given a trigonometric value, find the angle" into a function, we must first decide to look for the answer only within a certain interval. The functions obtained in this way are called inverse trigonometric functions (反三角函数).

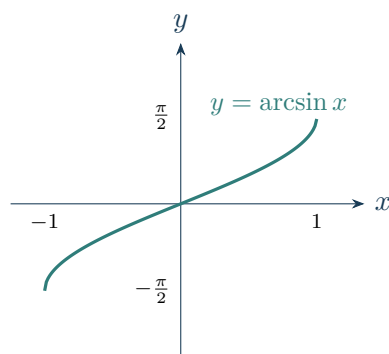
Inverse trigonometric function	Meaning	Domain	Range
$y = \arcsin x$	$\sin y = x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$y = \arccos x$	$\cos y = x$	$[-1, 1]$	$[0, \pi]$
$y = \arctan x$	$\tan y = x$	$\mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Read this table as a sentence. For example, $\arcsin \frac{1}{2}$ does not mean "all angles satisfying $\sin x = \frac{1}{2}$ ", but rather "the angle in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ satisfying $\sin y = \frac{1}{2}$." Therefore

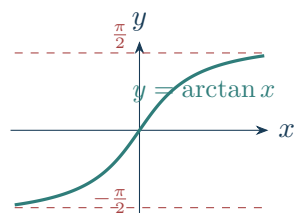
$$\arcsin \frac{1}{2} = \frac{\pi}{6}.$$

Similarly,

$$\arccos \frac{1}{2} = \frac{\pi}{3}, \quad \arctan 1 = \frac{\pi}{4}.$$



The range is fixed as $[-\frac{\pi}{2}, \frac{\pi}{2}]$.



The range is $(-\frac{\pi}{2}, \frac{\pi}{2})$.

Example 6.15 Simple Inverse Trigonometric Equations (简单反三角方程)

Solve the following equations:

(1) $\arcsin x = \frac{\pi}{6}$;

(2) $\arccos x = \frac{2\pi}{3}$;

(3) $\arctan x = -\frac{\pi}{4}$.

Solution:

(1) $\arcsin x = \frac{\pi}{6}$ means $\sin \frac{\pi}{6} = x$, so

$$x = \frac{1}{2}.$$

(2) $\arccos x = \frac{2\pi}{3}$ means $\cos \frac{2\pi}{3} = x$, so

$$x = -\frac{1}{2}.$$

(3) $\arctan x = -\frac{\pi}{4}$ means $\tan\left(-\frac{\pi}{4}\right) = x$, so

$$x = -1.$$

In-Class Practice 6.8.1 Other Trigonometric Functions and Inverse Trigonometric Functions (其他三角函数与反三角函数)

- (1) Compute $\cot \frac{\pi}{3}, \sec \frac{\pi}{4}, \csc \frac{\pi}{2}$.
- (2) State where $y = \cot x$ and $y = \sec x$ are undefined.
- (3) Compute $\arcsin\left(-\frac{1}{2}\right), \arccos 0, \arctan \sqrt{3}$.
- (4) Solve $\arccos x = \frac{\pi}{3}$.

6.9 Chapter Summary

Main Ideas

- (1) In a right triangle, sine, cosine, and tangent are side ratios; the Law of Cosines handles general triangles.
- (2) Radian measure satisfies $180^\circ = \pi$, and the unit circle gives the definition for arbitrary angles.
- (3) Sine and cosine have period 2π ; tangent has period π and vertical asymptotes.
- (4) Reduction formulas handle quadrants and periods; sum-and-difference formulas and double-angle formulas change angles.
- (5) The auxiliary-angle formula combines a sum of sine and cosine into one sinusoidal function.
- (6) The four parameters of a sinusoidal function control amplitude, period, phase, and midline.
- (7) Cotangent, secant, and cosecant come from ratios or reciprocals of sine, cosine, and tangent; inverse trigonometric functions express "finding the angle in the prescribed range from a trigonometric value."

Difficulty: Core

Difficulty: Extension

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changes as m changes.

Self-Check

Skill	Yes	Mostly	Not Yet
Use acute-angle trigonometric functions to solve right triangles	☐	☐	☐
Use special-angle values and the Law of Cosines	☐	☐	☐
Convert between degrees and radians	☐	☐	☐
Use the unit circle to determine function values and signs	☐	☐	☐
Read the graph properties of the three trigonometric functions	☐	☐	☐
Use reduction formulas, sum-and-difference formulas, and double-angle formulas	☐	☐	☐
Use the auxiliary-angle formula	☐	☐	☐
Study graphs and properties of sinusoidal functions	☐	☐	☐
Understand cotangent, secant, cosecant, and inverse trigonometric functions	☐	☐	☐

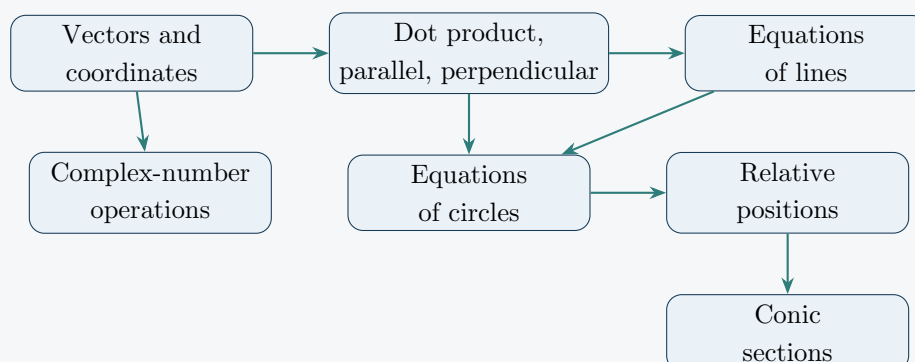
Vectors, Complex Numbers, and Plane Analytic Geometry

Learning Goals

After studying this chapter, you should be able to:

- ▶ understand the magnitude and direction of a vector, and perform vector addition, subtraction, and scalar multiplication;
- ▶ represent vectors with coordinates, and compute vector lengths and dot products;
- ▶ use vectors to describe parallelism, perpendicularity, and angles;
- ▶ understand the real part (实部), imaginary part (虚部), conjugate, and modulus of a complex number (复数);
- ▶ perform addition, subtraction, multiplication, and division of complex numbers;
- ▶ master point-slope, slope-intercept, two-point, and general forms of a line;
- ▶ compute the distance from a point to a line, and determine whether two lines are parallel or perpendicular;
- ▶ master the standard and general equations of a circle, and study the relative position of a line and a circle;
- ▶ get an initial understanding of the standard equations and basic properties of ellipses, hyperbolas, and parabolas.

Knowledge Map



Vectors turn direction and length into coordinate operations; analytic geometry turns lines and circles into equations; complex numbers extend the real line into the complex plane.

Learning Route for This Chapter

The central theme of this chapter is ``using algebra to study geometry.''

- (1) Vectors write displacement, direction, and length as coordinates, so geometric relations become coordinate calculations.
- (2) Complex numbers extend the real line into a plane; in computations, treat $i^2 = -1$ as the basic rule.
- (3) Lines, circles, and conic sections (它们统称圆锥曲线) can all be described by equations satisfied by point coordinates, so intersections and relative positions become equation problems.

While reading this chapter, keep translating back and forth between graphical language and algebraic language.

Prerequisites and Minimum Goals

This chapter uses coordinate systems, the distance formula, linear functions, quadratic equations, and completing the square. For a first pass, study it in three rounds: first vector coordinate operations, then complex-number arithmetic, and finally introductory lines, circles, and conic sections. The minimum goals are to compute vector coordinates and lengths, carry out the four operations on complex numbers, write basic equations of lines and circles, identify standard equations of ellipses, hyperbolas, and parabolas, and use the discriminant to determine the number of intersections between a line and a conic section.

Common Self-Study Sticking Points

Distance from a point to a line, relative positions of a line and a circle, vector proofs, and comprehensive complex-number problems are second-level content. Do not let these comprehensive problems block your first pass. First make sure you can answer: how are coordinates represented, how are equations written, and how are operations performed?

7.1 Vectors and Their Linear Operations (向量及其线性运算)

For some quantities, knowing only ``how much'' is not enough; we must also know ``in which direction.'' Vectors are the mathematical language for such quantities.

7.1.1 Vectors, Magnitude, and Equal Vectors (向量、模与相等向量)

Start with these two examples:

temperature is 25°C , wind speed is 5 meters per second, blowing northeast.

Temperature only needs one number, but wind speed requires both ``5 meters per second'' and a direction. Similarly, when reading a map, if someone says ``walk 300 meters'' from school to a bookstore, you still do not know where to go. Only when they say ``walk 300 meters east, then 100 meters north'' is the change of position clear.

Quantities with both magnitude and direction are common in life:

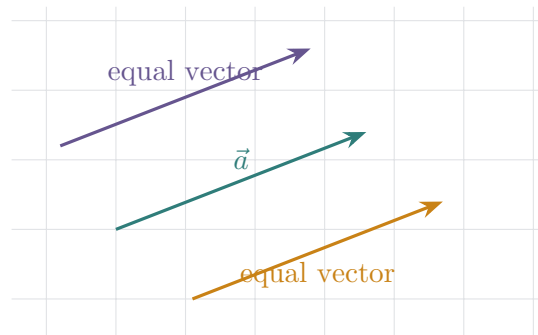
- **Displacement**: moving from one intersection to another involves both distance and direction.
- **Velocity**: driving 60 km/h east and driving 60 km/h west are completely different.
- **Force**: when pushing a desk, the size of the force matters, and so does its direction.
- **Wind direction**: wind of the same speed has different effects when it is tailwind or headwind.

Mathematics abstracts the common features: keep only length and direction. An object with both magnitude and direction is called a **vector** (向量). A vector may be written as $\vec{a}$, or represented by a directed segment $\overrightarrow{AB}$. The length of a vector is called its **magnitude** (模), written $|\vec{a}|$. A vector of length 0 is called the **zero vector** (零向量), written $\vec{0}$.

Abstract Definition of a Vector

A vector is a quantity with both magnitude and direction. Two vectors are considered equal if they have the same magnitude and the same direction.

At first, you may understand a vector as "the displacement from one position to another." For example, walking 3 units east and 2 units north gives the same displacement whether you start from the classroom or from the dormitory, so it represents the same vector. Abstractly, a vector does not care about its starting point; it only cares about magnitude and direction.



Different starting points, but the same length and direction, so they represent the same vector.

7.1.2 Addition, Subtraction, and Scalar Multiplication (加法、减法与数乘)

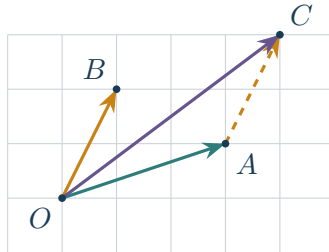
We are already familiar with addition, subtraction, multiplication, and division of numbers. How should vector operations be defined?

Addition: consecutive displacements can be combined. Suppose Ming walks in city blocks: first 3 blocks east, then 2 blocks north. If we only care about the total change from the starting point to the final position, we can temporarily ignore the intermediate path. The displacement pointing directly from the starting point to the final point is the sum of the two displacements. In general, first walking vector $\vec{a}$ and then vector $\vec{b}$ gives the total displacement $\vec{a} + \vec{b}$. If Ming starts from A , goes to B , and then goes from B to C , then the overall displacement is from A to C . In vector language,

$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}.$$

This is the triangle rule (向量加法可用三角形法则) of vector addition.

If the two vectors to be added do not already connect head-to-tail, we use the parallelogram rule (或平行四边形法则). For example, in the figure below, consider $\vec{OA} + \vec{OB}$. Since a vector is determined only by magnitude and direction, not by its starting point, we may translate $\vec{OB}$ to $\vec{AC}$, so that $\vec{OA}$ and $\vec{AC}$ connect head-to-tail. Therefore $\vec{OA} + \vec{OB} = \vec{OC}$. At this moment, translating $\vec{OB}$ to $\vec{AC}$ makes O, A, C, B a parallelogram, and the sum vector $\vec{OC}$ is one diagonal of the parallelogram. This is why the method is called the **parallelogram rule**.



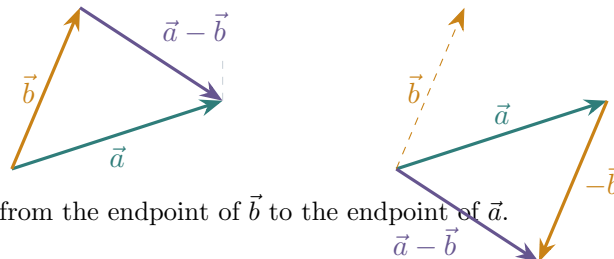
First walk $\vec{a}$, then $\vec{b}$; the total displacement is the diagonal of the parallelogram.

Subtraction: comparing two endpoints. Suppose two people start from the same intersection O . One walks by $\vec{a}$ and reaches A ; the other walks by $\vec{b}$ and reaches B . Then $\vec{b}$ plus ``the displacement from  $B$  to  $A$ '' becomes  $\vec{a}$ . Therefore,  $\vec{a} - \vec{b}$  should be defined as ``the displacement from B to A .'' When $\vec{a}$ and $\vec{b}$ are drawn from the same starting point, $\vec{a} - \vec{b}$ is the vector from the endpoint of $\vec{b}$ to the endpoint of $\vec{a}$.

Subtraction can also be understood as adding the opposite vector:

$$\vec{a} - \vec{b} = \vec{a} + (-\vec{b}).$$

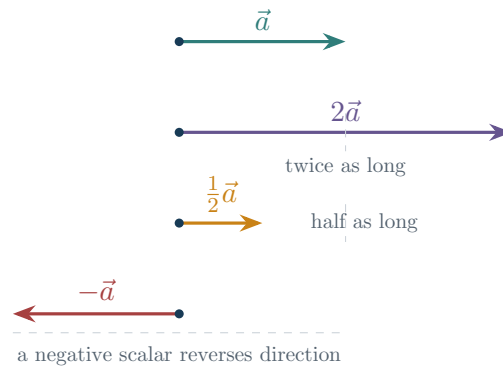
Here $-\vec{b}$ has the same length as $\vec{b}$, but the opposite direction.



Same starting point: from the endpoint of $\vec{b}$ to the endpoint of $\vec{a}$.

Head-to-tail form: $\vec{a} - \vec{b} = \vec{a} + (-\vec{b})$.

Scalar multiplication: stretch, shrink, or reverse along the same direction. In real life, we often say that speed becomes 2 times as large, that someone walks only half the original distance, or that someone returns in the opposite direction. Mathematically, scalar multiplication describes these changes. For example, $2\vec{a}$ means moving twice as far in the direction of $\vec{a}$, $\frac{1}{2}\vec{a}$ means moving half as far in the same direction, and $-\vec{a}$ means moving the same distance in the opposite direction.



Scalar multiplication changes length and direction, but keeps the vector on the same line.

Geometric Meaning of Linear Operations

Vector addition can be drawn with the triangle rule or the parallelogram rule. Subtraction is defined by

$$\vec{a} - \vec{b} = \vec{a} + (-\vec{b}).$$

The magnitude of $\lambda\vec{a}$ is $|\lambda||\vec{a}|$. If $\lambda > 0$, the direction is the same; if $\lambda < 0$, the direction is opposite.

Rules for Linear Operations

$$\vec{a} + \vec{b} = \vec{b} + \vec{a}, \quad (\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c}),$$

$$\lambda(\vec{a} + \vec{b}) = \lambda\vec{a} + \lambda\vec{b}, \quad (\lambda + \mu)\vec{a} = \lambda\vec{a} + \mu\vec{a}.$$

Example 7.1 Vector Linear Combinations

Given $\vec{a}, \vec{b}$, simplify:

(1) $2(\vec{a} - \vec{b}) - 3(\vec{a} + 2\vec{b})$;

(2) $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD}$.

Solution:

(1)

$$2(\vec{a} - \vec{b}) - 3(\vec{a} + 2\vec{b}) = 2\vec{a} - 2\vec{b} - 3\vec{a} - 6\vec{b} = -\vec{a} - 8\vec{b}.$$

(2) Head-to-tail vectors can be combined:

$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC},$$

so

$$\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} = \overrightarrow{AC} + \overrightarrow{CD} = \overrightarrow{AD}.$$

7.2 Vector Coordinates and Dot Product (向量的坐标与内积)

Coordinates record ``how far right'' and ``how far up'' as an ordered pair; the dot product records whether two directions are close, opposite, or perpendicular.

7.2.1 Vector Coordinates

After we have a rectangular coordinate system, a two-dimensional displacement can be decomposed into two basic directions:

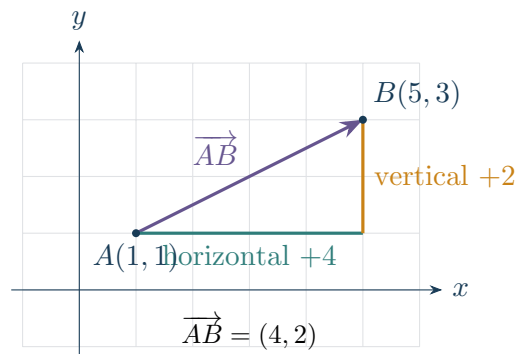
horizontal change, vertical change.

For example, "3 units east and 2 units north" can be written as $(3, 2)$; "1 unit west and 4 units south" can be written as $(-1, -4)$. Vector coordinates must be distinguished from point coordinates. The coordinates of a two-dimensional vector do not say "where a point is." They say "from the starting point to the endpoint, how much did the x -coordinate change, and how much did the y -coordinate change?"

If $A(x_1, y_1)$ and $B(x_2, y_2)$, then

$$\overrightarrow{AB} = (x_2 - x_1, y_2 - y_1).$$

This is because moving from A to B changes the horizontal coordinate by $x_2 - x_1$, and the vertical coordinate by $y_2 - y_1$. Vector coordinates record the change from start to finish, not the start and finish themselves.



With coordinates, vector addition, subtraction, and scalar multiplication become very simple. If $\vec{a} = (a_1, a_2)$ and $\vec{b} = (b_1, b_2)$, then

$$\vec{a} \pm \vec{b} = (a_1 \pm b_1, a_2 \pm b_2), \quad \lambda \vec{a} = (\lambda a_1, \lambda a_2).$$

The meaning is direct: horizontal changes combine with horizontal changes, and vertical changes combine with vertical changes; when a vector is scaled by λ , both directional components are scaled by λ .

If $\vec{a} = (a_1, a_2)$, its length comes from the Pythagorean theorem:

$$|\vec{a}| = \sqrt{a_1^2 + a_2^2}.$$

Example 7.2 Coordinate Operations

Given $A(-1, 2)$, $B(3, -2)$, $\vec{a} = (2, 1)$, and $\vec{b} = (-1, 4)$, find:

(1) $\overrightarrow{AB}$;

$$(2) |\vec{AB}|;$$

$$(3) 2\vec{a} - 3\vec{b}.$$

Solution:

$$(1) \vec{AB} = B - A = (3 - (-1), -2 - 2) = (4, -4).$$

$$(2) |\vec{AB}| = \sqrt{4^2 + (-4)^2} = 4\sqrt{2}.$$

$$(3) 2\vec{a} - 3\vec{b} = 2(2, 1) - 3(-1, 4) = (4, 2) - (-3, 12) = (7, -10).$$

7.2.2 Dot Product (向量内积)

Addition, subtraction, and scalar multiplication mainly describe how displacements are combined. Next we introduce an operation that describes the relation between two directions. For example:

- ▶ when pushing a box, the more the force points in the direction of motion, the more effective it is;
- ▶ when cycling into the wind, the wind direction is opposite the direction of motion, so it is tiring; with a tailwind, it is easier;
- ▶ if a force is vertical while the box moves horizontally, that force has no effect on the horizontal motion.

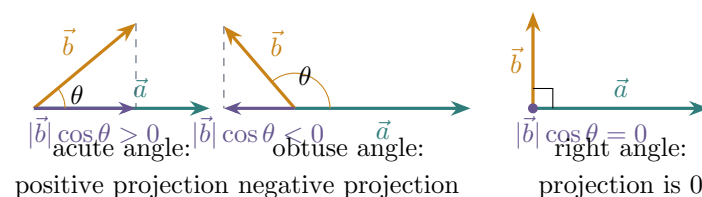
These examples show that the relation between two vectors depends not only on their lengths, but also on the angle (夹角) between them. To describe this relation, we define the **dot product** or **inner product** (内积).

For nonzero vectors $\vec{a}, \vec{b}$, let their angle be θ . Define

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta.$$

The dot product of the zero vector with any vector is defined to be 0.

This definition can be understood as "the projection length of one vector in the direction of the other, multiplied by the length of the other vector." The smaller the angle, the longer the projection and the larger the dot product. If the angle is obtuse, the projection is in the opposite direction and the dot product is negative. If the angle is right, one direction has no projection on the other, and the dot product is 0.



Real-World Meaning of Dot Product: Effective Action

If a force $\vec{F}$ produces displacement $\vec{s}$, then the effective component of the force in the displacement direction is $|\vec{F}| \cos \theta$. The work can be written as

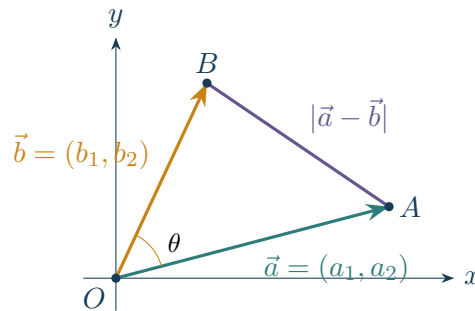
$$W = \vec{F} \cdot \vec{s} = |\vec{F}| |\vec{s}| \cos \theta.$$

This shows that the dot product is not an arbitrary definition: it calculates how much one direction effectively acts along another direction.

7.2.3 Coordinate Formula for the Dot Product

The geometric definition contains $\cos \theta$, but in coordinate problems, we want to compute directly from coordinates. We now prove, using the Law of Cosines, that

$$\vec{a} = (a_1, a_2), \vec{b} = (b_1, b_2) \implies \vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2.$$

**Deriving the Coordinate Formula by the Law of Cosines**

Let $\vec{OA} = \vec{a} = (a_1, a_2)$, $\vec{OB} = \vec{b} = (b_1, b_2)$, and let the angle between them be θ . In triangle OAB ,

$$OA = |\vec{a}|, \quad OB = |\vec{b}|, \quad AB = |\vec{a} - \vec{b}|.$$

By the Law of Cosines,

$$|\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2|\vec{a}||\vec{b}| \cos \theta.$$

On the other hand, by the coordinate distance formula,

$$\begin{aligned} |\vec{a} - \vec{b}|^2 &= (a_1 - b_1)^2 + (a_2 - b_2)^2 \\ &= a_1^2 + a_2^2 + b_1^2 + b_2^2 - 2(a_1 b_1 + a_2 b_2) \\ &= |\vec{a}|^2 + |\vec{b}|^2 - 2(a_1 b_1 + a_2 b_2). \end{aligned}$$

Comparing the two expressions gives

$$|\vec{a}||\vec{b}| \cos \theta = a_1 b_1 + a_2 b_2.$$

By the geometric definition $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta$, so

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2.$$

Therefore, for nonzero vectors,

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{a_1b_1 + a_2b_2}{\sqrt{a_1^2 + a_2^2}\sqrt{b_1^2 + b_2^2}}$$

This formula can be used to find the angle between two vectors.

7.2.4 Parallel and Perpendicular (平行与垂直)

The two most important relations between vector directions are parallelism and perpendicularity. If two vectors are parallel, they have the same or opposite direction, possibly with different lengths; algebraically, one is a scalar multiple of the other. If two vectors are perpendicular, their angle is 90° , so their dot product is 0. Therefore, for nonzero vectors:

Coordinate Tests for Parallel and Perpendicular Vectors

$$\vec{a} \parallel \vec{b} \iff \text{there exists a real number } \lambda \text{ such that } \vec{a} = \lambda \vec{b} \iff a_1b_2 - a_2b_1 = 0,$$

$$\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0 \iff a_1b_1 + a_2b_2 = 0.$$

In particular,

$$\vec{a} \parallel \vec{b} \implies \vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \text{ or } -|\vec{a}||\vec{b}|,$$

with the positive value for the same direction and the negative value for opposite directions.

Example 7.3 Angle, Parallelism, and Perpendicularity

Let $\vec{a} = (1, 2)$, $\vec{b} = (3, -1)$, and $\vec{c} = (4, 8)$.

- (1) Find $\vec{a} \cdot \vec{b}$ and the cosine of the angle between them.
- (2) Determine whether $\vec{a}$ and $\vec{b}$ are perpendicular.
- (3) Determine whether $\vec{a}$ and $\vec{c}$ are parallel.

Solution:

(1)

$$\vec{a} \cdot \vec{b} = 1 \cdot 3 + 2 \cdot (-1) = 1.$$

Also $|\vec{a}| = \sqrt{5}$, $|\vec{b}| = \sqrt{10}$, so

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{1}{\sqrt{5}\sqrt{10}} = \frac{1}{5\sqrt{2}} = \frac{\sqrt{2}}{10}.$$

- (2) Since $\vec{a} \cdot \vec{b} = 1 \neq 0$, $\vec{a}$ and $\vec{b}$ are not perpendicular.
- (3) Since $\vec{c} = (4, 8) = 4(1, 2) = 4\vec{a}$, $\vec{a}$ and $\vec{c}$ are parallel.

In-Class Practice 7.2.1 Vectors

- (1) Given $A(2, -1)$ and $B(-3, 5)$, find $\overrightarrow{AB}$ and its length.
- (2) Given $\vec{a} = (2, -3)$ and $\vec{b} = (-1, 4)$, find $3\vec{a} + 2\vec{b}$.
- (3) Find the angle between $(1, \sqrt{3})$ and $(2, 0)$.
- (4) Find m such that $(m, 2)$ is perpendicular to $(1, -3)$.
- (5) Find m such that $(m, 6)$ is parallel to $(2, 3)$.
- (6) Prove that the triangle with vertices $A(0, 0)$, $B(4, 0)$, $C(2, 2)$ is isosceles.

7.3 Complex Numbers (复数)

After introducing $i^2 = -1$, every quadratic equation has roots in the complex-number system.

7.3.1 Basic Concepts of Complex Numbers

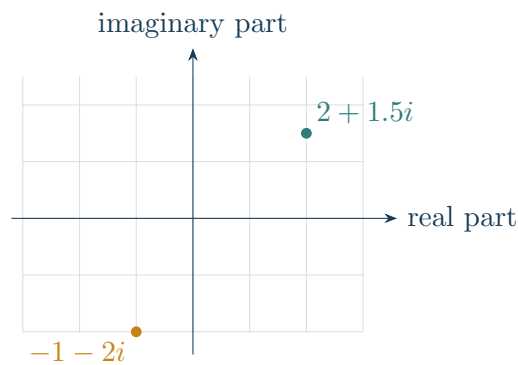
In the real-number system, the equation $x^2 + 1 = 0$ has no solution, because the square of a real number cannot be -1 . To make such equations solvable, we introduce a new number i , defined by $i^2 = -1$. This number is called the imaginary unit. This is a new rule in an enlarged number system; it is not derived from the rules of real numbers.

A number of the form $z = a + bi$ is called a complex number. Here a is the real part, written $\operatorname{Re} z$, and b is the imaginary part, written $\operatorname{Im} z$. If $b = 0$, the number is real. If $a = 0$ and $b \neq 0$, the number is purely imaginary.

Two complex numbers are equal if and only if their real parts and imaginary parts are respectively equal:

$$a + bi = c + di \iff a = c, b = d.$$

A complex number can also be viewed as a point in a plane: $a + bi$ corresponds to the point (a, b) . The horizontal coordinate is the real part, and the vertical coordinate is the imaginary part. In this way, the real number line is just the horizontal axis in the complex plane where the imaginary part is 0.



7.3.2 Addition, Subtraction, and Multiplication

We now study the four arithmetic operations for complex numbers. Addition and subtraction combine real parts with real parts and imaginary parts with imaginary parts. Multiplication is expanded just like multiplying polynomials, and then $i^2 = -1$ is used to rewrite the result in the form $a + bi$.

$$(a + bi) \pm (c + di) = (a \pm c) + (b \pm d)i.$$

For multiplication,

$$(a + bi)(c + di) = ac + adi + bci + bdi^2 = (ac - bd) + (ad + bc)i.$$

The powers of i repeat with period 4:

$$i^2 = -1, \quad i^3 = -i, \quad i^4 = 1, \quad i^5 = i, \quad \dots$$

In general,

$$i^{4k} = 1, \quad i^{4k+1} = i, \quad i^{4k+2} = -1, \quad i^{4k+3} = -i.$$

Example 7.4 Addition, Subtraction, and Multiplication of Complex Numbers

Compute:

(1) $(3 - 2i) + (1 + 5i)$;

(2) $(2 + i)(3 - 4i)$;

(3) $(1 - i)^3$;

(4) i^{2026} .

Solution:

(1) $(3 - 2i) + (1 + 5i) = 4 + 3i$.

(2) $(2 + i)(3 - 4i) = 6 - 8i + 3i - 4i^2 = 10 - 5i$.

(3) First,

$$(1 - i)^2 = 1 - 2i + i^2 = -2i,$$

so

$$(1 - i)^3 = -2i(1 - i) = -2 - 2i.$$

(4) Since $i^4 = 1$, and $2026 = 4 \cdot 506 + 2$,

$$i^{2026} = i^2 = -1.$$

7.3.3 Conjugate, Modulus, and Division (共轭、模与除法)

For $z = a + bi$, its complex conjugate (共轭复数) is

$$\bar{z} = a - bi,$$

and its modulus is

$$|z| = \sqrt{a^2 + b^2}.$$

Moreover,

$$z\bar{z} = |z|^2 = a^2 + b^2.$$

Complex-number division is done by multiplying both numerator and denominator by the conjugate of the denominator. The purpose is to turn the denominator into a real number:

$$\frac{a + bi}{c + di} = \frac{(a + bi)(c - di)}{(c + di)(c - di)} = \frac{(ac + bd) + (bc - ad)i}{c^2 + d^2} = \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2}i.$$

Example 7.5 Complex-Number Division

Write each result in the form $a + bi$:

(1) $\frac{3 + 2i}{1 - i};$

(2) $\frac{1}{2 + 3i};$

(3) the conjugate and modulus of $z = -3 + 4i$.

Solution:

(1)

$$\frac{3 + 2i}{1 - i} \cdot \frac{1 + i}{1 + i} = \frac{(3 + 2i)(1 + i)}{1 - i^2} = \frac{1 + 5i}{2} = \frac{1}{2} + \frac{5}{2}i.$$

(2)

$$\frac{1}{2 + 3i} \cdot \frac{2 - 3i}{2 - 3i} = \frac{2 - 3i}{4 + 9} = \frac{2}{13} - \frac{3}{13}i.$$

(3) If $z = -3 + 4i$, then its conjugate is

$$\bar{z} = -3 - 4i,$$

and its modulus is

$$|z| = \sqrt{(-3)^2 + 4^2} = 5.$$

In-Class Practice 7.3.1 Complex-Number Operations

- (1) Write the real part, imaginary part, conjugate, and modulus of $2 - 5i$.
- (2) Compute $(4 + i) - (2 - 3i)$.
- (3) Compute $(1 + 2i)(3 - i)$.
- (4) Simplify i^{35} .
- (5) Compute $(2 + i)/(1 + 2i)$.
- (6) Solve $x^2 + 4 = 0$ and $x^2 - 2x + 5 = 0$ in the complex-number system.

7.4 Equations of Lines (直线方程)

A line can be determined by one point and one direction. Different forms are suitable for different given information.

7.4.1 Common Forms of the Equation of a Line

From this subsection onward, we enter plane analytic geometry. The purpose of this branch of mathematics is to use coordinates and algebraic computation to study geometric objects. We begin with the simplest geometric object: a line. A line can be represented by a linear equation in two variables, commonly in the following forms.

Four Forms

$$\begin{aligned}
 \text{point-slope form: } & y - y_0 = k(x - x_0), \\
 \text{slope-intercept form: } & y = kx + b, \\
 \text{two-point form: } & \frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}, \\
 \text{general form: } & Ax + By + C = 0 \quad (A, B \text{ not both } 0).
 \end{aligned}$$

A vertical line (竖直直线) is $x = x_0$, and it has no finite slope. A horizontal line (水平直线) is $y = y_0$, and its slope is 0. Remember that except for vertical lines, nonvertical lines are similar to

the linear functions studied earlier.

Choosing a Form from the Given Information

Given information	Recommended form
One point and slope	point-slope form $y - y_0 = k(x - x_0)$
Slope and y -intercept	slope-intercept form $y = kx + b$
Two points	first find the slope, then use point-slope form; two-point form is also possible
Parallel or perpendicular to a line in general form	keep or use the proportional relation among A, B
Vertical line	directly write $x = x_0$; do not force a slope

Example 7.6 Different Forms of a Line

Find:

- (1) the line passing through $(2, -1)$ with direction vector $(3, 2)$;
- (2) the line passing through $(-1, 3)$ and $(4, -2)$;
- (3) the line parallel to $2x - 3y + 5 = 0$ and passing through the origin.

Solution:

- (1) The direction vector is $(3, 2)$, so the slope is $\frac{2}{3}$. Thus

$$y + 1 = \frac{2}{3}(x - 2).$$

- (2) The slope determined by the two points is

$$k = \frac{-2 - 3}{4 - (-1)} = -1,$$

so

$$y - 3 = -(x + 1),$$

that is, $x + y - 2 = 0$.

- (3) A parallel line has proportional x, y coefficients in general form, so let it be $2x - 3y + c = 0$. Passing through the origin gives $c = 0$. Hence

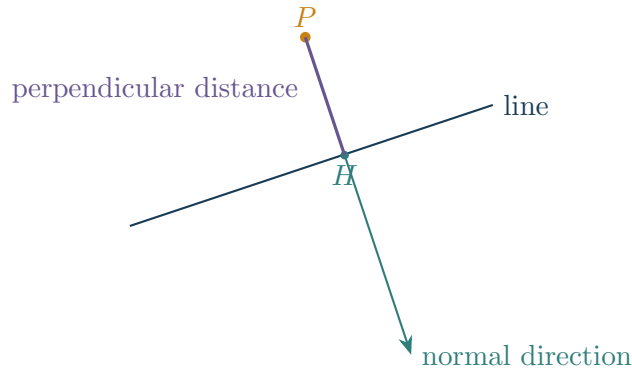
$$2x - 3y = 0.$$

7.4.2 Distance from a Point to a Line (点到直线距离)

This subsection is second-level content in analytic geometry. On a first pass, make sure you can determine whether a point lies on a line and write equations of lines; after the distance formula

becomes familiar, use it for the relative position of a line and a circle.

The distance from a point to a line is not the distance from the point to any random point on the line. It is the **shortest distance** from the point to the line. Geometrically, the shortest segment from a point P to a line l must be perpendicular to l . If the foot of the perpendicular is H , then the length PH is the distance from P to l .



To compute this length, we first examine the directional information hidden inside the general equation

$$Ax + By + C = 0 \quad (A, B \text{ not both } 0).$$

Direction Vector and Normal Vector

A vector along the direction of a line is called a **direction vector** of the line. A vector perpendicular to the line is called a **normal vector** of the line.

From the general equation $Ax + By + C = 0$, we can read off one normal vector:

$$\vec{n} = (A, B).$$

Here is why. Take any two points $M(x_1, y_1)$ and $N(x_2, y_2)$ on the line. Since both lie on the line,

$$Ax_1 + By_1 + C = 0, \quad Ax_2 + By_2 + C = 0.$$

Subtracting the two equations gives

$$A(x_2 - x_1) + B(y_2 - y_1) = 0.$$

But $\overrightarrow{MN} = (x_2 - x_1, y_2 - y_1)$ is a direction vector of the line, so

$$(A, B) \cdot (x_2 - x_1, y_2 - y_1) = 0.$$

This shows that (A, B) is perpendicular to the direction of the line, hence (A, B) is a normal vector. Therefore we may also choose

$$\vec{v} = (B, -A) \quad \text{or} \quad (-B, A)$$

as a direction vector of the line, because

$$(A, B) \cdot (B, -A) = AB - AB = 0.$$

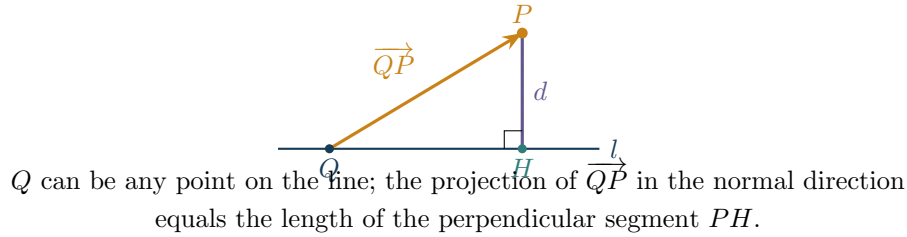
Now derive the distance formula. Let the point be $P(x_0, y_0)$, and let the line be

$$l : Ax + By + C = 0.$$

Choose any point $Q(x_1, y_1)$ on line l . The vector from Q to P is

$$\overrightarrow{QP} = (x_0 - x_1, y_0 - y_1).$$

The distance from the point to the line is the projection length of $\overrightarrow{QP}$ in the normal-vector direction.



The normal vector is $\vec{n} = (A, B)$, so the projection length is

$$d = \frac{|\overrightarrow{QP} \cdot \vec{n}|}{|\vec{n}|}.$$

Compute numerator and denominator separately:

$$|\vec{n}| = \sqrt{A^2 + B^2},$$

$$\overrightarrow{QP} \cdot \vec{n} = A(x_0 - x_1) + B(y_0 - y_1) = Ax_0 + By_0 - (Ax_1 + By_1).$$

Since $Q(x_1, y_1)$ lies on the line,

$$Ax_1 + By_1 + C = 0, \quad \text{that is,} \quad Ax_1 + By_1 = -C.$$

Substitute this into the expression above:

$$\overrightarrow{QP} \cdot \vec{n} = Ax_0 + By_0 + C.$$

Therefore, the distance from $P(x_0, y_0)$ to the line $Ax + By + C = 0$ is

$$d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}.$$

Example 7.7 Distance

Find the distance from point $(2, -1)$ to the line $3x - 4y + 5 = 0$, and find the distance between the two parallel lines $3x - 4y + 5 = 0$ and $3x - 4y - 10 = 0$.

Solution: The distance formula from a point to a line is

$$d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}.$$

Thus the distance from $(2, -1)$ to the line is

$$d = \frac{|3 \cdot 2 - 4(-1) + 5|}{\sqrt{3^2 + (-4)^2}} = \frac{15}{5} = 3.$$

The distance between the two parallel lines is

$$d = \frac{|5 - (-10)|}{\sqrt{3^2 + (-4)^2}} = \frac{15}{5} = 3.$$

In-Class Practice 7.4.1 Equations of Lines

- (1) Find the general form of the line passing through $(1, 2)$ and $(3, -2)$.
- (2) Find the line passing through $(-1, 4)$ and parallel to the x -axis.
- (3) Find the line passing through $(2, 3)$ and perpendicular to $2x - y + 1 = 0$.
- (4) Find the intersection of $x + y - 1 = 0$ and $2x - y - 4 = 0$.
- (5) Find the distance from the origin to $5x + 12y - 13 = 0$.

7.5 Equations of Circles (圆的方程)

All points on a circle are at the same distance from the center. Squaring the distance formula gives the standard equation of a circle.

7.5.1 Standard Form (标准方程)

Another common geometric object is the circle. Consider a circle with center (a, b) and radius $r > 0$. Every point (x, y) on the circle has distance r from the center (a, b) . By the distance formula,

$$\sqrt{(x - a)^2 + (y - b)^2} = r.$$

Squaring both sides gives the standard equation of a circle:

$$(x - a)^2 + (y - b)^2 = r^2.$$

The main difference from the equation of a line is that this is a quadratic equation in two variables;

both x and y appear squared.

Example 7.8 Finding a Circle from Its Center and a Point

Find the equation of the circle with center $(2, -1)$ passing through $(5, 3)$. Determine whether point $(2, 4)$ is inside the circle, on the circle, or outside the circle.

Solution: The radius is the distance from the center to $(5, 3)$:

$$r = \sqrt{(5-2)^2 + (3-(-1))^2} = \sqrt{3^2 + 4^2} = 5.$$

Thus the equation is

$$(x-2)^2 + (y+1)^2 = 25.$$

The distance from $(2, 4)$ to the center is

$$\sqrt{(2-2)^2 + (4+1)^2} = 5.$$

This equals the radius, so $(2, 4)$ lies on the circle.

7.5.2 General Form (一般方程)

The general equation of a circle is

$$x^2 + y^2 + Dx + Ey + F = 0.$$

Completing the square converts it into standard form:

$$\left(x + \frac{D}{2}\right)^2 + \left(y + \frac{E}{2}\right)^2 = \frac{D^2 + E^2}{4} - F.$$

Hence the center is $(-D/2, -E/2)$, and the square of the radius is

$$r^2 = \frac{D^2 + E^2}{4} - F.$$

Notice that only when $r^2 > 0$ does the equation represent a circle.

Example 7.9 Converting General Form to Standard Form

Convert

$$x^2 + y^2 - 4x + 6y - 12 = 0$$

to standard form, and find the center and radius.

Solution: Complete the square for x, y separately:

$$x^2 - 4x = (x-2)^2 - 4, \quad y^2 + 6y = (y+3)^2 - 9.$$

Substitute into the original equation:

$$(x-2)^2 - 4 + (y+3)^2 - 9 - 12 = 0,$$

so

$$(x - 2)^2 + (y + 3)^2 = 25.$$

The center is $(2, -3)$, and the radius is 5.

7.5.3 Relative Position of a Line and a Circle (直线与圆的位置关系)

After learning equations of lines and circles, we now consider the relative position of a line and a circle. Geometrically, a line may intersect a circle in two points, be tangent to it in one point, or have no intersection. Since every point on a circle is at distance r from the center, the geometric method is to compare the distance d from the center to the line with the radius r . Algebraically, we may also substitute the line equation into the circle equation and check how many real roots the resulting quadratic equation has.

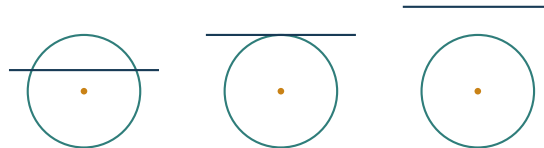
Let d be the distance from the center to the line, and let r be the radius.

Relative Position

$$d < r \Rightarrow \text{two intersections}, \quad d = r \Rightarrow \text{tangent}, \quad d > r \Rightarrow \text{no intersection}.$$

Alternatively, substitute the line equation into the circle equation, obtain a quadratic equation in one variable, and use its discriminant:

$$\Delta > 0 \Rightarrow \text{two intersections}, \quad \Delta = 0 \Rightarrow \text{tangent}, \quad \Delta < 0 \Rightarrow \text{no intersection}.$$



$d < r$: two intersections $d = r$: tangent $d > r$: no intersection

Example 7.10 Line and Circle

Determine the relative position of the line $3x + 4y - 20 = 0$ and the circle $x^2 + y^2 = 16$. If they are tangent, find the tangent point.

Solution: Method 1: distance method.

The center is $(0, 0)$, and the radius is 4. The distance from the center to the line is

$$d = \frac{|3 \cdot 0 + 4 \cdot 0 - 20|}{\sqrt{3^2 + 4^2}} = \frac{20}{5} = 4.$$

The distance equals the radius, so the line is tangent to the circle and has only one intersection.

Method 2: discriminant method.

From the line equation,

$$y = 5 - \frac{3}{4}x.$$

Substitute into the circle:

$$x^2 + \left(5 - \frac{3}{4}x\right)^2 = 16.$$

Multiply both sides by 16:

$$16x^2 + (20 - 3x)^2 = 256.$$

Simplifying gives

$$25x^2 - 120x + 144 = 0.$$

The discriminant is

$$\Delta = (-120)^2 - 4 \cdot 25 \cdot 144 = 0,$$

so there is only one real root, and the line is tangent.

Thus $x = \frac{12}{5}$. Substitute into the line:

$$y = 5 - \frac{3}{4} \cdot \frac{12}{5} = \frac{16}{5}.$$

The tangent point is

$$\left(\frac{12}{5}, \frac{16}{5} \right).$$

Example 7.11 An Intersecting Example

Determine the relative position of the line $y = 1$ and the circle $x^2 + y^2 = 4$, and find the intersections.

Solution: Distance method.

The center is $(0, 0)$, and the radius is 2. The line $y = 1$ can be written as $y - 1 = 0$, so the distance from the center to the line is

$$d = \frac{|0 - 1|}{1} = 1.$$

Since $d < r$, the line intersects the circle in two points.

Discriminant method.

Substitute $y = 1$ into the circle:

$$x^2 + 1 = 4,$$

that is,

$$x^2 - 3 = 0.$$

The discriminant is

$$\Delta = 0^2 - 4 \cdot 1 \cdot (-3) = 12 > 0,$$

so there are two real roots. From $x^2 = 3$,

$$x = \pm\sqrt{3}.$$

The intersections are

$$(\sqrt{3}, 1), \quad (-\sqrt{3}, 1).$$

Example 7.12 A Non-Intersecting Example

Determine the relative position of the line $y = 3$ and the circle $x^2 + y^2 = 4$.

Solution: Distance method.

The center is $(0, 0)$, and the radius is 2. The line $y = 3$ can be written as $y - 3 = 0$, so the distance from the center to the line is

$$d = \frac{|0 - 3|}{1} = 3.$$

Since $d > r$, the line has no intersection with the circle.

Discriminant method.

Substitute $y = 3$ into the circle:

$$x^2 + 9 = 4,$$

that is,

$$x^2 + 5 = 0.$$

The discriminant is

$$\Delta = 0^2 - 4 \cdot 1 \cdot 5 = -20 < 0,$$

so there is no real root and the line has no intersection with the circle.

Why We Still Need the Discriminant Method

For circles, comparing the distance d from the center to the line with the radius r is often the fastest method. But the discriminant method must also be mastered: its essence is "solve the system of equations and count the intersections." In the next topic, conic sections, the relative position of a line with a parabola (抛物线), ellipse (椭圆), or hyperbola (双曲线) is usually determined by this algebraic method.

In-Class Practice 7.5.1 Circles

- (1) Write the equation of the circle with center $(-2, 3)$ and radius 4.
- (2) Find the equation of the circle whose diameter has endpoints $(1, 2)$ and $(5, 4)$.
- (3) Convert $x^2 + y^2 + 6x - 2y - 6 = 0$ to standard form.
- (4) Determine the relative position of $y = x + 1$ and $x^2 + y^2 = 4$.
- (5) Find the tangent line to $x^2 + y^2 = 25$ at $(3, 4)$.

7.6 Introduction to Conic Sections (圆锥曲线简介)

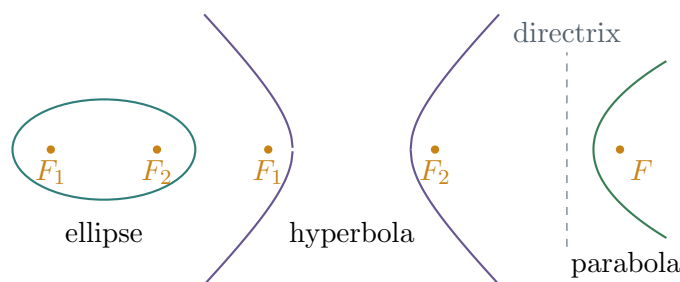
Conic sections are curves obtained by cutting a cone with a plane. This section only introduces the basic ideas, standard equations, and line-position problems for ellipses, hyperbolas, and parabolas as preparation for later study.

A circle is the simplest quadratic curve. In this section, we briefly introduce three other important quadratic curves: the ellipse, the hyperbola, and the parabola. Together they are called conic sections.

7.6.1 Basic Ideas of Three Conic Sections

A circle is the set of points whose distance from one fixed point is constant. Conic sections can also be understood by distance relations, but their distance relations are richer.

Curve	Basic idea	Common shape
Ellipse	The locus of points whose sum of distances to two fixed points is constant. The two fixed points are called foci F_1, F_2 .	A closed flattened oval
Hyperbola	The locus of points whose absolute difference of distances to two fixed points is constant. The two fixed points are called foci F_1, F_2 .	Two separate branches
Parabola	The locus of points equidistant from one fixed point and one fixed line. The fixed point is called the focus F , and the fixed line is called the directrix.	One open curve



7.6.2 Standard Equations

Here are the standard equations of the three conic sections.

Three Standard Equations

ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a > b > 0).$

The foci lie on the x -axis: $F_1(-c, 0), F_2(c, 0)$, where $c^2 = a^2 - b^2$. The number c is called the focal half-distance.

hyperbola: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad (a > 0, b > 0).$

The foci lie on the x -axis: $F_1(-c, 0), F_2(c, 0)$, where $c^2 = a^2 + b^2$. The number c is called the focal half-distance; the asymptotes are $y = \pm \frac{b}{a}x$.

parabola: $y^2 = 2px \quad (p > 0).$

The focus is $\left(\frac{p}{2}, 0\right)$, the directrix is $x = -\frac{p}{2}$, the parabola opens to the right, and its axis of symmetry is $y = 0$, that is, the x -axis.

Example 7.13 Identifying Conic Sections

Determine which conic section each equation represents, and write its key quantities.

(1) $\frac{x^2}{25} + \frac{y^2}{9} = 1;$

(2) $\frac{x^2}{4} - \frac{y^2}{9} = 1;$

(3) $y^2 = 8x.$

Solution:

(1) This is an ellipse. Here $a^2 = 25, b^2 = 9$, so $a = 5, b = 3$. The focal parameter satisfies

$$c^2 = a^2 - b^2 = 25 - 9 = 16,$$

so the foci are $(-4, 0), (4, 0)$.

(2) This is a hyperbola. Here $a^2 = 4, b^2 = 9$, so $a = 2, b = 3$. The focal parameter satisfies

$$c^2 = a^2 + b^2 = 4 + 9 = 13.$$

The foci are $(-\sqrt{13}, 0), (\sqrt{13}, 0)$, and the asymptotes are

$$y = \pm \frac{3}{2}x.$$

(3) This is a parabola. Comparing with $y^2 = 2px$, we get $2p = 8$, so $p = 4$. The focus is $(2, 0)$, and the directrix is $x = -2$.

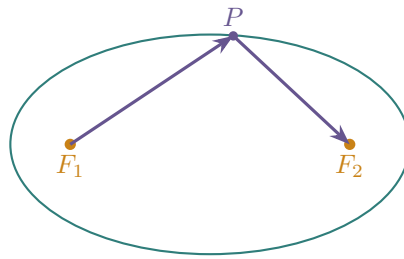
7.6.3 Optical Properties

Conic sections have important optical properties. This section states the results without proof.

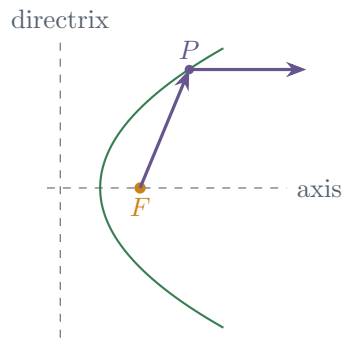
Optical Properties

- **Ellipse:** a ray emitted from one focus reflects off the ellipse and passes through the other focus.
- **Parabola:** a ray emitted from the focus reflects off the parabola and travels parallel to the axis of symmetry; conversely, a ray entering parallel to the axis reflects to the focus.
- **Hyperbola:** a ray coming from the direction of one focus reflects off the hyperbola and appears to come from the other focus.

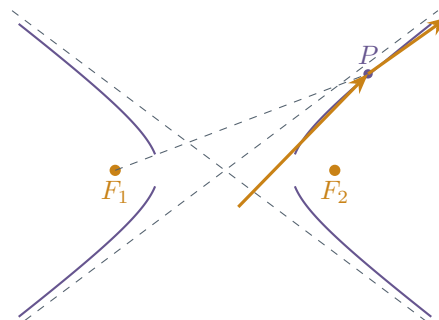
These properties explain the mathematics behind elliptical whispering galleries, parabolic antennas, and car-headlight reflectors.



Ellipse: a ray from one focus reflects through the other focus.



Parabola: a ray from the focus reflects parallel to the axis of symmetry.



Hyperbola: the extension of the reflected ray passes through the other focus, so it appears to come from that focus.

7.6.4 Relative Position of a Line and a Conic Section

To determine the relative position of a line and a conic section, the basic idea is still to solve the system of equations. After substituting the line equation into the conic equation, we usually obtain a quadratic equation in one variable:

$$Ax^2 + Bx + C = 0.$$

Then check the discriminant

$$\Delta = B^2 - 4AC.$$

Discriminant Method

$\Delta > 0 \Rightarrow$ two intersections,

$\Delta = 0 \Rightarrow$ one intersection, tangent,

$\Delta < 0 \Rightarrow$ no real intersection.

For a circle, the distance method is often quick; for conic sections, the discriminant method is more common. In later deeper study of conic sections, the discriminant method will appear repeatedly.

Example 7.14 Relative Position of a Line and a Conic Section

Determine the relative position in each case. If there are intersections or a tangent point, find the coordinates.

(1) The line $y = x + 1$ and the ellipse $x^2 + \frac{y^2}{4} = 1$.

(2) The line $x - y + 1 = 0$ and the parabola $y^2 = 4x$.

(3) The line $y = x$ and the hyperbola $\frac{x^2}{4} - y^2 = 1$.

Solution:

(1) Substitute $y = x + 1$ into the ellipse:

$$x^2 + \frac{(x+1)^2}{4} = 1.$$

Simplifying gives

$$5x^2 + 2x - 3 = 0.$$

Since

$$\Delta = 2^2 + 4 \cdot 5 \cdot 3 = 64 > 0,$$

there are two real solutions $x = -1, \frac{3}{5}$. Thus the line intersects the ellipse at two points:

$$(-1, 0), \quad \left(\frac{3}{5}, \frac{8}{5}\right).$$

(2) From $x - y + 1 = 0$, we get $x = y - 1$. Substitute into the parabola:

$$y^2 = 4(y - 1),$$

that is,

$$y^2 - 4y + 4 = 0.$$

The discriminant is

$$\Delta = (-4)^2 - 4 \cdot 1 \cdot 4 = 0.$$

Therefore the line is tangent to the parabola. Since

$$y^2 - 4y + 4 = (y - 2)^2 = 0,$$

we have $y = 2$, and then from $x = y - 1$ we get $x = 1$. The tangent point is

$$(1, 2).$$

(3) Substitute $y = x$ into the hyperbola:

$$\frac{x^2}{4} - x^2 = 1,$$

that is,

$$-\frac{3}{4}x^2 - 1 = 0.$$

Viewed as a quadratic equation in x , its discriminant is

$$\Delta = 0^2 - 4\left(-\frac{3}{4}\right)(-1) = -3 < 0.$$

Thus there is no real intersection between the line and this hyperbola, so there are no coordinates to find.

In-Class Practice 7.6.1 Conic Sections

- (1) Determine which conic section $\frac{x^2}{16} + \frac{y^2}{7} = 1$ is, and find its foci.
- (2) Determine which conic section $\frac{x^2}{9} - \frac{y^2}{4} = 1$ is, and write its asymptotes.
- (3) Write the focus and directrix of the parabola $y^2 = 6x$.
- (4) Determine the number of intersections between $y = 2$ and $y^2 = 4x$.

7.7 Chapter Summary

Main Thread

- (1) Vector addition, subtraction, and scalar multiplication are performed component by component; vector magnitude comes from the distance formula.
- (2) The dot product connects coordinate computation with angles; a zero dot product represents perpendicularity.
- (3) Complex-number multiplication uses $i^2 = -1$, and division uses conjugate rationalization.
- (4) A line is determined by a point and a direction; a circle is determined by its center and radius.
- (5) Distance from a point to a line can be used to determine whether a line intersects, is tangent to, or misses a circle.
- (6) Ellipses, hyperbolas, and parabolas can all be defined by distance relations; the relative position of a line and a conic section is often determined by the discriminant.

7.8 Chapter Exercises

How to Use These Exercises

Group A checks the minimum mastery for this chapter. If you make mistakes, return to the corresponding subsection and reread the examples. Group B is for integrated quiz preparation. Group C is for extension; you may skip it during a first self-study pass without affecting your ability to enter the next chapter.

Group A: Basic Practice

Difficulty: Basic

- (1) Given $A(-2, 1)$ and $B(3, 5)$, find $\overrightarrow{AB}$ and its magnitude.
- (2) Given $\vec{a} = (1, -2)$ and $\vec{b} = (3, 4)$, find $2\vec{a} - \vec{b}$ and $\vec{a} \cdot \vec{b}$.
- (3) Determine whether $(2, 3)$ and $(-3, 2)$ are perpendicular.
- (4) Compute $(3 - 2i) + (1 + 4i)$ and $(2 + i)(1 - 3i)$.
- (5) Compute $(3 + i)/(2 - i)$.
- (6) Find the line passing through $(1, -1)$ and $(4, 5)$.
- (7) Find the distance from point $(-2, 3)$ to $4x - 3y + 1 = 0$.
- (8) Write the circle with center $(1, -2)$ and radius 3.
- (9) Find the center and radius of $x^2 + y^2 - 2x + 4y - 4 = 0$.
- (10) Determine the relative position of $x + y = 1$ and $x^2 + y^2 = 1$.

- (11) Determine which conic section $y^2 = 10x$ is, and write its focus and directrix.

Group B: Quiz Level

Difficulty: Core

- (1) Find m such that vectors $(m, 1)$ and $(2, m)$ are perpendicular.
- (2) Given $|\vec{a}| = 2$, $|\vec{b}| = 3$, and the angle between them is 60° , find $|\vec{a} + \vec{b}|$.
- (3) Solve $z^2 - 2z + 5 = 0$ in the complex-number system.
- (4) Find the line passing through $(2, 1)$ and perpendicular to $3x - 4y + 7 = 0$.
- (5) Find the circle whose center lies on $x - y = 0$ and which passes through $(1, 0)$, $(0, 1)$.
- (6) Find k such that $y = kx + 2$ is tangent to $x^2 + y^2 = 4$.
- (7) Find the common points of $x^2 + y^2 - 4x + 2y - 4 = 0$ and $x + y = 2$.
- (8) Determine the relative position of $y = x$ and $y^2 = 4x$, and find the common points.

Group C: Extension and Thinking

Difficulty: Extension

- (1) Use vectors to prove that the three medians of a triangle meet at one point, and find the coordinate formula for the centroid.
- (2) Suppose $z + 1/z$ is real. Discuss possible conditions on the complex number z .
- (3) Find the two tangent lines from the exterior point $(3, 1)$ to the circle $x^2 + y^2 = 5$.
- (4) Find the equation of the circle obtained by reflecting $x^2 + y^2 - 2x - 4y - 4 = 0$ across the line $x = y$.
- (5) Determine the relative position of $y = x + 1$ and the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$.

Self-Check

Skill	Yes	Almost	Not yet
Perform vector addition, subtraction, scalar multiplication, and coordinate operations	☐	☐	☐
Use dot product to find angles and test perpendicularity	☐	☐	☐
Use coordinates to test vector parallelism	☐	☐	☐
Perform the four operations on complex numbers	☐	☐	☐
Write equations of lines from given conditions	☐	☐	☐
Compute the distance from a point to a line	☐	☐	☐
Write the standard equation of a circle and complete the square	☐	☐	☐
Determine the relative position of a line and a circle	☐	☐	☐
Identify three conic sections and use the discriminant to count intersections	☐	☐	☐

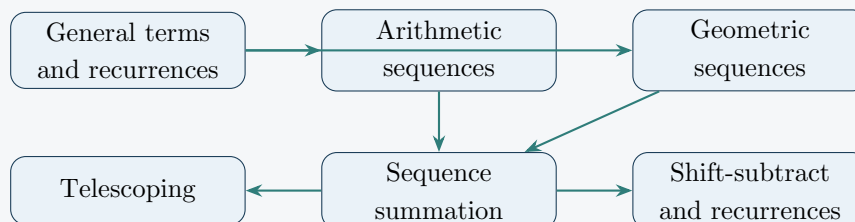
Sequences (数列)

Learning Goals

After studying this chapter, you should be able to:

- ▶ understand sequences, terms, indices, general terms, and recurrence relations;
- ▶ find simple general-term patterns from the first several terms;
- ▶ use the general-term formula and the first- n -terms sum formula for arithmetic sequences;
- ▶ use the general-term formula and the first- n -terms sum formula for geometric sequences;
- ▶ determine monotonicity of a sequence and solve simple application problems;
- ▶ use Gaussian summation, telescoping, and shift-and-subtract summation;
- ▶ find a_n from S_n , and handle grouped or combined sequences;
- ▶ find general terms of simple first-order linear recurrence sequences.

Knowledge Map



A sequence (数列) is an ordered list of numbers. A general term (通项公式) directly gives a term from its index; a recurrence relation (递推公式) generates later terms from earlier terms.

Learning Route for This Chapter

A sequence can be viewed as a function defined on positive integer indices, but it emphasizes order. First distinguish two representations: a general-term formula directly computes the n th term, while a recurrence relation starts from initial terms and generates terms step by step. Then we focus on two most basic patterns: arithmetic sequences, where neighboring differences are constant, and geometric sequences, where neighboring ratios are constant.

All summation methods are essentially about "creating cancellation" or "using structure": arithmetic summation uses first-last pairing, geometric summation uses shift-and-subtract, and telescoping uses cancellation between neighboring terms.

Prerequisites and Minimum Goals

This chapter uses function notation, exponent rules, equations, and simple summation. The minimum goals for a first pass are: write the first several terms from a general formula; write the first several terms from a recurrence; use the general-term and first- n -terms sum formulas for arithmetic and geometric sequences; and check $n = 1$ separately when finding a_n from S_n .

Common Self-Study Sticking Points

Telescoping, shift-and-subtract, and first-order linear recurrences are second-level techniques. If you get stuck on a first pass, first make arithmetic sequences, geometric sequences, and basic partial sums completely fluent; technique-heavy problems can be trained together on a second pass.

8.1 Concept and Representation of Sequences (数列的概念与表示)

The order of terms in a sequence cannot be exchanged; the same number may appear in different positions.

8.1.1 Terms and General Terms (项与通项公式)

A sequence is an ordered list of numbers, such as $1, 2, 3, 4, 5, \dots$, or $1, 4, 9, 16, \dots$. A sequence is usually written as $\{a_n\}$, and its n th term is a_n . If a_n can be expressed directly in terms of n , that expression is called a general term. For the two examples above, the general terms are $a_n = n$ and $a_n = n^2$.

You may think of a general term as a machine: input the index n , and it outputs the n th term a_n . For example, when $a_n = 2n + 1$, input $n = 1, 2, 3$, and the outputs are $3, 5, 7$. From the function viewpoint, a sequence is a function whose domain is the positive integers: an ordinary function may take continuous real-number inputs, while a sequence only takes $1, 2, 3, \dots$. Therefore, the graph of a sequence is usually a set of discrete points, not a continuous curve.

Example 8.1 Writing the First Several Terms from a General Term

Write the first five terms of each sequence:

(1) $a_n = 2n - 1$;

(2) $b_n = (-1)^{n-1}n$;

(3) $c_n = \frac{n}{n+1}$.

Solution: Substitute $n = 1, 2, 3, 4, 5$.

(1) $a_n : 1, 3, 5, 7, 9$.

(2) $b_n : 1, -2, 3, -4, 5$.

(3) $c_n : \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}$.

8.1.2 Recurrence Relations (递推公式)

Some sequences are not easy to describe directly by the n th term, but it is easy to describe how to get the next term from previous terms. In such cases, a recurrence relation is more natural than a general-term formula. A recurrence relation defines later terms using earlier terms. For example,

$$a_1 = 2, \quad a_n = a_{n-1} + 3 \quad (n \geq 2)$$

generates 2, 5, 8, ... A recurrence relation must also provide enough initial terms. A rule without an initial term is like knowing how to take the next step without knowing where to start; it cannot determine the whole sequence.

$$\boxed{a_1} \longrightarrow \boxed{a_2} \longrightarrow \boxed{a_3} \longrightarrow \cdots$$

The recurrence relation tells us how the arrows move; the initial term tells us what is in the first box. Both are necessary.

Example 8.2 Writing a Sequence from a Recurrence

Write the first six terms:

(1) $a_1 = 1, a_n = 2a_{n-1} + 1;$

(2) $b_1 = b_2 = 1, b_n = b_{n-1} + b_{n-2}.$

Solution:

(1)

$$a_1 = 1, a_2 = 2 \cdot 1 + 1 = 3, a_3 = 7, a_4 = 15, a_5 = 31, a_6 = 63.$$

(2)

$$b_1 = 1, b_2 = 1, b_3 = 2, b_4 = 3, b_5 = 5, b_6 = 8.$$

8.2 Arithmetic Sequences (等差数列)

First look at some very simple lists:

$$2, 5, 8, 11, 14, \dots$$

Each step from left to right adds 3. Next,

$$20, 17, 14, 11, 8, \dots$$

Each step adds -3 , that is, decreases by 3. Also,

$$6, 6, 6, 6, \dots$$

Each step adds 0. These sequences look different in direction, but they share one feature: **the difference between neighboring terms is always the same.**

Such patterns are common in life. For example, if each later row in a theater has 2 more seats than

the previous row, the number of seats may be

$$20, 22, 24, 26, \dots$$

If someone deposits 10 yuan every day, the accumulated amount may be

$$10, 20, 30, 40, \dots$$

A sequence whose neighboring differences are constant is called an arithmetic sequence (等差数列).

Recognizing an Arithmetic Sequence

For a sequence $a_1, a_2, a_3, \dots$, compute

$$a_2 - a_1, \quad a_3 - a_2, \quad a_4 - a_3, \dots$$

If these differences are all equal, the sequence is arithmetic.

This definition gives the recurrence

$$a_{n+1} = a_n + d,$$

where d is called the common difference.

8.2.1 General Term

The general term follows directly from the recurrence definition. If $a_n - a_{n-1} = d$, then

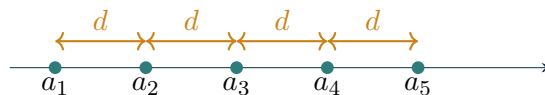
$$a_n = a_1 + (n - 1)d.$$

More generally,

$$a_n = a_m + (n - m)d.$$

The reason is that from the first term to the n th term, we add the common difference $n - 1$ times:

$$a_n = a_1 + \underbrace{d + d + \dots + d}_{n-1 \text{ times}} = a_1 + (n - 1)d.$$



Example 8.3 Finding the First Term, Common Difference, and General Term

In an arithmetic sequence, $a_3 = 7$ and $a_5 + a_7 = 26$. Find a_1, d, a_n .

Solution: The general term is $a_n = a_1 + (n - 1)d$. From $a_3 = 7$,

$$a_1 + 2d = 7.$$

Also,

$$a_5 + a_7 = (a_1 + 4d) + (a_1 + 6d) = 2a_1 + 10d = 26.$$

Multiplying the first equation by 2 gives $2a_1 + 4d = 14$. Subtracting gives $6d = 12$, so $d = 2$. Hence $a_1 = 7 - 4 = 3$, and

$$a_n = 3 + 2(n - 1) = 2n + 1.$$

Also, from the definition, a sequence with general term $a_n = dn + b$ is always arithmetic with common difference d . Thus an arithmetic sequence may be viewed as a "linear function defined on positive integers."

8.2.2 Arithmetic Mean and Properties (等差中项与性质)

This subsection studies the sum relation between symmetric positions in an arithmetic sequence. Since each step adds the same number, the middle term equals the average of the two endpoint terms; this is often used to quickly find a middle term or prove sum relations.

If a, b, c form an arithmetic sequence, then

$$b - a = c - b.$$

Rearranging gives

$$2b = a + c.$$

In this case, b is called the arithmetic mean of a and c , and

$$b = \frac{a + c}{2}.$$

In an arithmetic sequence $\{a_n\}$, if $m + n = k + l$, then

$$a_m + a_n = a_k + a_l = 2a_1 + (m + n - 2)d.$$

Indeed,

$$a_m + a_n = [a_1 + (m - 1)d] + [a_1 + (n - 1)d] = 2a_1 + (m + n - 2)d.$$

Therefore, as long as two pairs of indices have the same sum, the corresponding sums of terms are equal. For example,

$$a_2 + a_8 = a_3 + a_7 = a_4 + a_6 = 2a_5.$$

This shows that terms symmetric about the middle position in an arithmetic sequence always have the same sum.

8.2.3 First n Terms Sum (前 n 项和)

Sometimes we care not only about each term of a sequence, but also about the sum of its first n terms. We usually use S_n to denote the first n terms sum.

Start with the classic arithmetic sum:

$$1 + 2 + 3 + \cdots + 98 + 99 + 100.$$

Adding directly is long. Instead, write the sum once forward and once backward:

$$\begin{aligned} S &= 1 + 2 + 3 + \cdots + 98 + 99 + 100, \\ S &= 100 + 99 + 98 + \cdots + 3 + 2 + 1. \end{aligned}$$

Adding corresponding terms vertically, every column equals 101:

$$2S = (1 + 100) + (2 + 99) + (3 + 98) + \cdots + (100 + 1).$$

There are 100 columns, so

$$2S = 100 \cdot 101, \quad S = \frac{100 \cdot 101}{2} = 5050.$$

This is called Gaussian summation: **write forward once, write backward once, add vertically, then divide by 2**. This method works for all arithmetic sequences. It does not work for arbitrary sequences.

Four Steps of Gaussian Summation

- (1) Let the first n terms sum be $S_n = a_1 + a_2 + \cdots + a_n$.
- (2) Write the same sum again in reverse order.
- (3) Add the two lines vertically, using the equal first-last pair sums in an arithmetic sequence.
- (4) After obtaining $2S_n$, divide both sides by 2.

Now apply this method to a general arithmetic sequence. Write the sum twice:

$$S_n = a_1 + a_2 + \cdots + a_{n-1} + a_n,$$

$$S_n = a_n + a_{n-1} + \cdots + a_2 + a_1.$$

Since the sequence is arithmetic, all first-last pair sums are equal:

$$a_1 + a_n = a_2 + a_{n-1} = a_3 + a_{n-2} = \cdots = 2a_1 + (n-1)d.$$

Thus each column is $a_1 + a_n$, and there are n columns:

$$2S_n = n(a_1 + a_n).$$

Dividing by 2, we get

$$S_n = \frac{n(a_1 + a_n)}{2} = \frac{n}{2}[2a_1 + (n-1)d].$$

Example 8.4 Summing an Arithmetic Sequence

For the arithmetic sequence $-1, 1, 3, 5, \dots$:

- (1) find the general term;

- (2) find the sum of the first 50 terms;
- (3) determine whether 101 is a term of the sequence.

Solution:

- (1) The first term is $a_1 = -1$, and the common difference is $d = 2$, so

$$a_n = -1 + 2(n - 1) = 2n - 3.$$

- (2)

$$S_{50} = \frac{50}{2}(a_1 + a_{50}).$$

Since $a_{50} = 2 \cdot 50 - 3 = 97$,

$$S_{50} = 25(-1 + 97) = 2400.$$

- (3) To determine whether 101 is a term, solve $2n - 3 = 101$. This gives $n = 52$, a positive integer, so 101 is the 52nd term.

8.2.4 Monotonicity

As with functions, we can discuss monotonicity of a sequence. If $a_{n+1} > a_n$ for all positive integers n , the sequence is increasing; if $a_{n+1} < a_n$ for all positive integers n , the sequence is decreasing.

For an arithmetic sequence, the change from one term to the next is always the common difference d , so we only need to check the sign of d . If $d > 0$, the sequence is increasing; if $d < 0$, it is decreasing; if $d = 0$, it is constant.

In-Class Practice 8.2.1 Arithmetic Sequences

- (1) Given $a_1 = 4, d = -3$, find a_{20} and S_{20} .
- (2) Given $a_4 = 5, a_{10} = 23$, find a_1, d .
- (3) Given $a_5 + a_9 = 30$, find a_7 .
- (4) Insert 5 numbers between 7 and 31 so that all seven numbers form an arithmetic sequence.
- (5) A theater has 20 seats in the first row, and each later row has 2 more seats than the previous row. There are 18 rows. Find the total number of seats.

8.3 Geometric Sequences (等比数列)

An arithmetic sequence adds the same number each time; a geometric sequence (等比数列) multiplies by the same number each time.

Observe the following sequences:

$$3, 6, 12, 24, 48, \dots$$

Each step from left to right multiplies by 2. Next,

$$64, 32, 16, 8, 4, \dots$$

Each step multiplies by $\frac{1}{2}$. Also,

$$1, -2, 4, -8, 16, \dots$$

Each step multiplies by -2 , so the signs alternate. The common feature is not "how much is added each time," but rather: **the ratio of a later term to the previous term is always the same.**

This kind of "multiply by the same number each time" pattern is also common. For example, if a sheet of paper is folded repeatedly, the number of layers is

$$1, 2, 4, 8, 16, \dots$$

with each step multiplying by 2. If money grows by 5% compound interest each year, the amount at the end of each year is 1.05 times the amount at the end of the previous year. To determine whether a sequence is geometric, check whether "later term divided by previous term" is always the same nonzero number.

Recognizing a Geometric Sequence

For a nonzero sequence $a_1, a_2, a_3, \dots$, compute

$$\frac{a_2}{a_1}, \quad \frac{a_3}{a_2}, \quad \frac{a_4}{a_3}, \dots$$

If these ratios are all equal, the sequence is geometric; this fixed ratio is called the common ratio (公比).

A sequence in which, from the second term onward, the ratio of each term to the previous term is the same nonzero constant is called a geometric sequence.

8.3.1 General Term

As with arithmetic sequences, the definition of a geometric sequence gives a recurrence:

$$a_n = qa_{n-1}, \quad q \neq 0.$$

Here q is called the common ratio. Geometric sequences usually require all terms to be nonzero. When $q > 1$ and $a_1 > 0$, the sequence increases; but if negative signs appear or $q < 0$, monotonicity must be checked more carefully. Arithmetic sequences add the same number each time, while

geometric sequences multiply by the same number each time:

$$a_1 \xrightarrow{\times q} a_2 \xrightarrow{\times q} a_3 \xrightarrow{\times q} \cdots \xrightarrow{\times q} a_n.$$

From the first term to the n th term, we multiply by q a total of $n - 1$ times. Hence

$$a_n = a_1 q^{n-1}.$$

Example 8.5 Finding a Geometric Sequence

In a decreasing positive geometric sequence, $a_2 = 6$ and $a_1 + a_2 + a_3 = 19$. Find a_1, q, a_n .

Solution: Let the first term be a_1 , and the common ratio be q . From $a_2 = a_1 q = 6$, we get $a_1 = \frac{6}{q}$, and $a_3 = 6q$. Substitute into the sum:

$$\frac{6}{q} + 6 + 6q = 19.$$

Multiplying both sides by q ,

$$6q^2 - 13q + 6 = 0 = (3q - 2)(2q - 3).$$

Thus $q = \frac{2}{3}$ or $\frac{3}{2}$. Since the sequence is positive and decreasing, $0 < q < 1$, so $q = \frac{2}{3}$. Therefore

$$a_1 = \frac{6}{2/3} = 9, \quad a_n = 9 \left(\frac{2}{3} \right)^{n-1}.$$

8.3.2 Geometric Mean and Properties (等比中项与性质)

This subsection studies product relations between symmetric positions in a geometric sequence. Since each step multiplies by the same number, the square of the middle term equals the product of the two endpoint terms; this is often used to quickly find a middle term or prove product relations.

If nonzero numbers a, b, c form a geometric sequence, then

$$b^2 = ac.$$

If $m + n = k + l$, then

$$a_m a_n = a_k a_l = a_1^2 q^{m+n-2}.$$

8.3.3 First n Terms Sum

Geometric sums cannot be found by simple first-last pairing as arithmetic sums can. However, if the whole sum is multiplied by the common ratio, similar terms shift by one position; subtracting then cancels many terms.

When $q = 1$, we have

$$S_n = na_1.$$

When $q \neq 1$, let

$$S_n = a_1 + a_1 q + \cdots + a_1 q^{n-1}.$$

Multiplying both sides by q increases every exponent by 1:

$$qS_n = a_1q + a_1q^2 + \cdots + a_1q^n.$$

The key is to align like terms vertically. The first line starts with a_1 , while the second line starts with a_1q , so the second line is shifted one place to the right:

$$\begin{array}{rcccccc} S_n & = & a_1 & +a_1q & +a_1q^2 & +\cdots & +a_1q^{n-1} \\ qS_n & = & & a_1q & +a_1q^2 & +\cdots & +a_1q^{n-1} & +a_1q^n \end{array}$$

Now subtract the second line from the first. To make the subtraction visible, write $-qS_n$ underneath:

$$\begin{array}{rcccccc} S_n & = & a_1 & +a_1q & +a_1q^2 & +\cdots & +a_1q^{n-1} \\ -qS_n & = & & -a_1q & -a_1q^2 & -\cdots & -a_1q^{n-1} & -a_1q^n \\ \hline (1-q)S_n & = & a_1 & 0 & 0 & \cdots & 0 & -a_1q^n \end{array}$$

The line below the horizontal rule shows that the middle terms $a_1q, a_1q^2, \dots, a_1q^{n-1}$ cancel in positive-negative pairs, leaving only the first and last terms:

$$(1-q)S_n = a_1 - a_1q^n.$$

Therefore

$$S_n = a_1 \frac{1-q^n}{1-q} = a_1 \frac{q^n-1}{q-1}.$$

This is the shift-and-subtract method.

Example 8.6 Summing a Geometric Sequence

For the sequence 3, 6, 12, 24, ...:

- (1) find the general term and S_n ;
- (2) find the sum of the first 10 terms;
- (3) find the smallest n such that $a_n > 1000$.

Solution:

- (1) This is a geometric sequence with $a_1 = 3$ and $q = 2$. Thus

$$a_n = 3 \cdot 2^{n-1}.$$

The first n terms sum is

$$S_n = \frac{3(2^n - 1)}{2 - 1} = 3(2^n - 1).$$

- (2)

$$S_{10} = 3(2^{10} - 1) = 3(1024 - 1) = 3069.$$

- (3) We need $3 \cdot 2^{n-1} > 1000$. Check $n = 9$: $a_9 = 3 \cdot 2^8 = 768$. Check $n = 10$: $a_{10} = 1536$. Thus the smallest n is 10.

In-Class Practice 8.3.1 Geometric Sequences

- (1) Given $a_1 = 2, q = 3$, find a_8, S_8 .
- (2) Given $a_2 = 6, a_3 = 12$, find a_1, q, S_{10} .
- (3) Given $a_2 a_6 = 64$, find a_4^2 .
- (4) Insert three positive numbers between 2 and 162 so that the five numbers form a geometric sequence.
- (5) A principal grows by 5% compound interest each year. Write the amount at the end of the n th year.

8.4 Sequences and First n Terms Sums (数列与前 n 项和)

The partial sum $S_n = a_1 + \cdots + a_n$ changes with n , so it is itself a sequence.

8.4.1 Finding a_n from S_n

This subsection answers: if the first n terms sum is known, how do we recover each term? Since S_n has exactly one more term than S_{n-1} , subtracting them isolates a_n .

Relation Between Terms and Partial Sums

$$a_1 = S_1, \quad a_n = S_n - S_{n-1} \quad (n \geq 2).$$

When using a unified formula, always check $n = 1$ separately.

Always Check $n = 1$ Separately

The formula $a_n = S_n - S_{n-1}$ only makes sense for $n \geq 2$, because S_0 is not defined by default in this chapter. When finding a general term from S_n , first write $a_1 = S_1$, then handle $n \geq 2$.

Example 8.7 Finding the General Term from Partial Sums

Given $S_n = n^2 + 2n$, find a_n , and check whether the formula obtained is valid for $n = 1$.

Solution: For $n \geq 2$,

$$a_n = S_n - S_{n-1}.$$

Since

$$S_{n-1} = (n-1)^2 + 2(n-1) = n^2 - 1,$$

we have

$$a_n = (n^2 + 2n) - (n^2 - 1) = 2n + 1.$$

For $n = 1$, $a_1 = S_1 = 1^2 + 2 = 3$, and the formula gives $2 \cdot 1 + 1 = 3$, so it works. Therefore

$$a_n = 2n + 1 \quad (n \geq 1).$$

8.4.2 Summation Notation

This subsection introduces a new writing tool: the summation symbol $\sum$. It is not a new operation rule. Its purpose is simply to write "many terms added together" more compactly.

For example,

$$\sum_{k=1}^5 k = 1 + 2 + 3 + 4 + 5.$$

Here k is a counting letter. The lower limit $k = 1$ means start from 1, and the upper limit 5 means continue through 5. Replace k successively by 1, 2, 3, 4, 5, then add the results.

Here are a few more examples:

$$\sum_{k=1}^4 2^k = 2^1 + 2^2 + 2^3 + 2^4,$$

$$\sum_{k=3}^6 a_k = a_3 + a_4 + a_5 + a_6,$$

$$\sum_{k=1}^n (2k+1) = 3 + 5 + 7 + \cdots + (2n+1).$$

How to Read Summation Notation

When you see

$$\sum_{k=m}^n f(k),$$

read it in three steps:

- (1) let k start from the lower limit m ;
- (2) each time, substitute k into $f(k)$ to write one term;
- (3) continue until the upper limit n , then add all the terms.

Thus $\sum$ mainly saves space, especially when writing the first n terms sum.

The letter inside summation notation is only a temporary counting letter. Changing the letter does not change the meaning. For example,

$$\sum_{k=1}^n k = \sum_{i=1}^n i = 1 + 2 + \cdots + n.$$

Later, in grouped summation, telescoping, and summation methods for recurrences, we will often use this notation to write long sums in one line.

8.4.3 Grouped Summation

This subsection handles sums whose terms mix several simple structures. When a term contains "arithmetic part + geometric part + constant part," do not search for a new formula. Split the sum first, then use the known formulas separately.

If a_n is the sum of several simple sequences, we may sum them separately. For example,

$$a_n = 2^{-n} + 4n + 2$$

has a partial sum that can be split into a geometric part and an arithmetic part.

When grouping, first name each group:

$$S_n = \underbrace{\sum_{k=1}^n 2^{-k}}_{\text{geometric part}} + \underbrace{4 \sum_{k=1}^n k}_{\text{arithmetic part}} + \underbrace{2n}_{\text{constant part}}.$$

After naming the groups, apply the appropriate formula to each part. This prevents the computation from becoming mixed together.

Example 8.8 Grouped Summation

Find the first n terms sum of $a_n = 2^{-n} + 4n + 2$.

Solution:

$$S_n = \sum_{k=1}^n (2^{-k} + 4k + 2) = \sum_{k=1}^n 2^{-k} + 4 \sum_{k=1}^n k + 2n.$$

The first part is geometric:

$$\sum_{k=1}^n 2^{-k} = \frac{\frac{1}{2}(1 - (1/2)^n)}{1 - \frac{1}{2}} = 1 - 2^{-n}.$$

The second part is

$$4 \sum_{k=1}^n k = 4 \cdot \frac{n(n+1)}{2} = 2n(n+1).$$

Therefore

$$S_n = 1 - 2^{-n} + 2n(n+1) + 2n = 2n^2 + 4n + 1 - 2^{-n}.$$

8.5 Telescoping Method (裂项相消法)

Split each term into the difference of neighboring terms; after summation, most middle terms cancel.

Second-Level Technique

Telescoping is not something you must master on the first pass. It is suitable for fractional sums whose denominators contain neighboring factors. When studying it, first expand the first four or five terms and see the middle terms cancel with your own eyes; then memorize the formulas.

8.5.1 Common Telescoping Forms

The goal of splitting is not to make the expression more complicated, but to make neighboring terms cancel. For example,

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$$

leads to

$$\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots,$$

where $\frac{1}{2}, \frac{1}{3}, \dots$ cancel in positive-negative pairs.

Writing the first few terms in full makes it clearer:

$$\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) = 1 - \frac{1}{5}.$$

The middle terms cancel pairwise, leaving only the first positive term and the last negative term.

Common Forms

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1},$$

$$\frac{1}{k(k+a)} = \frac{1}{a} \left(\frac{1}{k} - \frac{1}{k+a} \right),$$

$$\frac{1}{\sqrt{k} + \sqrt{k+1}} = \sqrt{k+1} - \sqrt{k},$$

$$\frac{1}{\sqrt{k} + \sqrt{k+a}} = \frac{1}{a} (\sqrt{k+a} - \sqrt{k}).$$

Example 8.9 Telescoping

Find the sums:

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{(n-1)n},$$

$$\frac{1}{10 \cdot 12} + \frac{1}{12 \cdot 14} + \cdots + \frac{1}{58 \cdot 60},$$

$$\frac{1}{1 + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \cdots + \frac{1}{\sqrt{15} + \sqrt{16}}.$$

Solution: For the first sum,

$$\sum_{k=1}^{n-1} \frac{1}{k(k+1)} = \sum_{k=1}^{n-1} \left(\frac{1}{k} - \frac{1}{k+1} \right) = 1 - \frac{1}{n}.$$

For the second sum, use

$$\frac{1}{k(k+2)} = \frac{1}{2} \left(\frac{1}{k} - \frac{1}{k+2} \right).$$

Thus

$$\frac{1}{2} \left[\left(\frac{1}{10} - \frac{1}{12} \right) + \left(\frac{1}{12} - \frac{1}{14} \right) + \cdots + \left(\frac{1}{58} - \frac{1}{60} \right) \right] = \frac{1}{2} \left(\frac{1}{10} - \frac{1}{60} \right) = \frac{1}{24}.$$

For the third sum, use

$$\frac{1}{\sqrt{k} + \sqrt{k+1}} = \sqrt{k+1} - \sqrt{k}.$$

Then

$$\sum_{k=1}^{15} \frac{1}{\sqrt{k} + \sqrt{k+1}} = \sum_{k=1}^{15} (\sqrt{k+1} - \sqrt{k}) = \sqrt{16} - \sqrt{1} = 4 - 1 = 3.$$

In-Class Practice 8.5.1 Partial Sums and Telescoping

- (1) Given $S_n = 2n^2 - n$, find a_n .
- (2) Find the first n terms sum of $a_n = 3n + 2^{-n}$.
- (3) Find $\sum_{k=1}^n \frac{1}{k(k+1)}$.
- (4) Find $\sum_{k=1}^n \frac{1}{(2k-1)(2k+1)}$.
- (5) Find $\sum_{k=1}^n \frac{3}{(3k+1)(3k+4)}$.

8.6 Shift-and-Subtract Method (错位相减法)

A sum with an arithmetic factor multiplied by a geometric factor is usually handled by shift-and-subtract.

Second-Level Technique

Shift-and-subtract is used for sums of the form "arithmetic term times geometric term," and the steps are longer. On a first pass, it is enough to understand this idea: after multiplying by the common ratio, the powers shift by one position; subtracting the two equations makes most terms become like terms.

8.6.1 Arithmetico-Geometric Sequences and Summation Problems

This section studies a new type of sequence: each term is the product of an arithmetic part and a geometric part. Such a sequence is often called an arithmetico-geometric sequence.

For example,

$$1 \cdot 2, \quad 2 \cdot 2^2, \quad 3 \cdot 2^3, \quad 4 \cdot 2^4, \dots$$

has coefficients $1, 2, 3, 4, \dots$, an arithmetic sequence, and powers $2, 2^2, 2^3, 2^4, \dots$, a geometric sequence. Its n th term is

$$a_n = n \cdot 2^n.$$

Another example is

$$1, \quad 2x, \quad 3x^2, \quad 4x^3, \dots$$

whose n th term is

$$a_n = nx^{n-1}.$$

The general term of this kind of sequence is usually not hard to write. The common problem is to find the first n terms sum:

$$S_n = 1 \cdot 2 + 2 \cdot 2^2 + 3 \cdot 2^3 + \dots + n \cdot 2^n,$$

or more generally,

$$S_n = 1q + 2q^2 + 3q^3 + \dots + nq^n.$$

Such a sum cannot directly use the arithmetic sum formula or the geometric sum formula, because each term contains both an increasing coefficient and a power changing by a fixed ratio. The idea of shift-and-subtract is: multiply the whole sum by the common ratio q , shifting the powers by one position; then subtract, turning the increasing coefficients into constants and reducing the problem to a geometric sum.

8.6.2 Standard Steps

Let

$$S_n = 1q + 2q^2 + 3q^3 + \dots + nq^n.$$

Multiply both sides by q , so that like powers shift by one position. Then subtract the new equation from the original one. A geometric sum will appear.

When writing the work, do not just place two lines randomly. Align powers of the same degree vertically. The first line begins with q , while the second line is multiplied by q , so it shifts one place

to the right:

$$S_n = q + 2q^2 + 3q^3 + \cdots + (n-1)q^{n-1} + nq^n,$$

$$qS_n = \quad q^2 + 2q^3 + \cdots + (n-2)q^{n-1} + (n-1)q^n + nq^{n+1}.$$

This "shift" can be viewed as shifting the coefficients one place to the right:

power	q	q^2	q^3	$\cdots$	q^n	q^{n+1}
coefficient in S_n	1	2	3	$\cdots$	n	0
coefficient in qS_n	0	1	2	$\cdots$	$n-1$	n
coefficient in $S_n - qS_n$	1	1	1	$\cdots$	1	$-n$

Thus subtraction is not cancellation by intuition; it is subtraction of coefficients for each same power:

$$S_n - qS_n = q + (2-1)q^2 + (3-2)q^3 + \cdots + [n - (n-1)]q^n - nq^{n+1}$$

$$= q + q^2 + q^3 + \cdots + q^n - nq^{n+1}.$$

The difficulty of increasing coefficients has now been converted into an ordinary geometric sum.

Checklist for Shift-and-Subtract

- (1) Write the first and last terms of the original sum clearly.
- (2) Multiply by the common ratio and align powers of the same degree vertically.
- (3) When subtracting, pay attention to the term left at the end.
- (4) Apply the geometric sum formula to the remaining sum.
- (5) Solve for S_n at the end.

Example 8.10 Arithmetic Times Geometric

Find

$$S_n = 1 \cdot 2 + 2 \cdot 2^2 + 3 \cdot 2^3 + \cdots + n \cdot 2^n.$$

Solution: Write

$$S_n = 1 \cdot 2 + 2 \cdot 2^2 + 3 \cdot 2^3 + \cdots + n \cdot 2^n.$$

Multiplying by 2 increases every exponent by 1. Align like powers vertically:

$$S_n = 1 \cdot 2 + 2 \cdot 2^2 + 3 \cdot 2^3 + \cdots + (n-1)2^{n-1} + n2^n,$$

$$2S_n = \quad 1 \cdot 2^2 + 2 \cdot 2^3 + \cdots + (n-2)2^{n-1} + (n-1)2^n + n2^{n+1}.$$

Now subtract the first line from the second. To see the subtraction clearly, write $-S_n$ below

and add a horizontal rule:

$$\begin{array}{rcccccc}
 2S_n = & & 1 \cdot 2^2 & + 2 \cdot 2^3 & + \cdots & + (n-1)2^n & + n2^{n+1} \\
 -S_n = & -1 \cdot 2 & -2 \cdot 2^2 & -3 \cdot 2^3 & - \cdots & & -n2^n \\
 \hline
 S_n = & -2 & -2^2 & -2^3 & - \cdots & -2^n & + n2^{n+1}
 \end{array}$$

Each column below the horizontal rule comes from subtracting coefficients of the same power. For example, the coefficient of 2^2 is $1 - 2 = -1$, the coefficient of 2^3 is $2 - 3 = -1$, and this continues through 2^n , whose coefficient is $(n-1) - n = -1$.

Since

$$2 + 2^2 + \cdots + 2^n = 2(2^n - 1),$$

we get

$$S_n = n \cdot 2^{n+1} - 2(2^n - 1) = (n-1)2^{n+1} + 2.$$

Example 8.11 Alternating Shift-and-Subtract Sum

Find

$$1 - 2x + 3x^2 - 4x^3 + \cdots + (-1)^{n-1}nx^{n-1} \quad (x \neq -1).$$

Solution: Let

$$S = 1 - 2x + 3x^2 - 4x^3 + \cdots + (-1)^{n-1}nx^{n-1}.$$

Multiply both sides by $-x$:

$$-xS = -x + 2x^2 - 3x^3 + \cdots + (-1)^n nx^n.$$

Use $S - (-xS) = S + xS$:

$$(1+x)S = 1 - x + x^2 - \cdots + (-1)^{n-1}x^{n-1} + (-1)^{n-1}nx^n.$$

The first n terms form a geometric sum:

$$1 - x + x^2 - \cdots + (-1)^{n-1}x^{n-1} = \frac{1 - (-x)^n}{1 + x}.$$

Therefore

$$S = \frac{1 - (-x)^n}{(1+x)^2} + \frac{(-1)^{n-1}nx^n}{1+x} = \frac{1 - (-x)^n + (-1)^{n-1}nx^n(1+x)}{(1+x)^2}.$$

8.7 First-Order Linear Recurrence Sequences (一阶线性递推数列)

A recurrence of the form $a_n = pa_{n-1} + q$ can be turned into a geometric sequence by finding an equilibrium value.

8.7.1 Constant-Coefficient Recurrences

This subsection handles recurrence sequences of the form $a_n = pa_{n-1} + q$. Compared with a geometric sequence, it has an extra constant q , so the geometric formula cannot be applied directly. The key is to find an equilibrium value and remove the constant term.

Consider

$$a_n = pa_{n-1} + q.$$

When $p \neq 1$, find an equilibrium value L satisfying $L = pL + q$, so $L = q/(1 - p)$. Then

$$a_n - L = p(a_{n-1} - L),$$

so $\{a_n - L\}$ is a geometric sequence.

The number L can be understood as an equilibrium value: if one term is exactly L , then the next term is still L . After subtracting this equilibrium value from every term, the extra constant in the recurrence disappears, and the problem becomes geometric.

The fixed move for finding the equilibrium value is: assume the previous term and the next term are both the same number L , substitute into the recurrence

$$L = pL + q,$$

solve for L , and then study $a_n - L$.

Example 8.12 Constructing a Geometric Sequence

Given $a_1 = 1$ and $a_n = 2a_{n-1} + 3$, find a_n .

Solution: We want to construct

$$a_n + c = 2(a_{n-1} + c).$$

Expanding the right side gives

$$a_n + c = 2a_{n-1} + 2c.$$

Comparing with $a_n = 2a_{n-1} + 3$, we get

$$2a_{n-1} + 3 + c = 2a_{n-1} + 2c \Rightarrow c = 3.$$

Thus

$$a_n + 3 = 2(a_{n-1} + 3).$$

Let $b_n = a_n + 3$. Then $b_n = 2b_{n-1}$, and $b_1 = 4$. Therefore

$$b_n = 4 \cdot 2^{n-1} = 2^{n+1},$$

so

$$a_n = 2^{n+1} - 3.$$

8.7.2 Summation Method (累加法)

This subsection handles recurrences where the difference between neighboring terms is known. If it can be written as $a_n - a_{n-1} = f(n)$, then add the changes from the second term to the n th term; the middle terms cancel step by step, leaving only $a_n - a_1$.

For

$$a_n - a_{n-1} = f(n),$$

sum from 2 to n :

$$a_n - a_1 = \sum_{k=2}^n f(k).$$

This is adding up the neighboring differences:

$$(a_2 - a_1) + (a_3 - a_2) + \cdots + (a_n - a_{n-1}) = a_n - a_1.$$

All middle terms on the left cancel, leaving only the difference between the last and first terms.

Example 8.13 Finding a General Term by Summation

Given $a_1 = -1$ and $a_n = a_{n-1} - 2n + 1$ ($n \geq 2$), find a_n .

Solution: Rewrite the recurrence as

$$a_n - a_{n-1} = -2n + 1.$$

Sum from 2 to n :

$$a_n - a_1 = \sum_{k=2}^n (-2k + 1).$$

Thus

$$a_n = -1 + \left(-2 \sum_{k=2}^n k + \sum_{k=2}^n 1 \right).$$

Here

$$\sum_{k=2}^n k = \frac{n(n+1)}{2} - 1, \quad \sum_{k=2}^n 1 = n - 1.$$

Therefore

$$a_n = -1 - 2 \left(\frac{n(n+1)}{2} - 1 \right) + (n - 1) = -n^2.$$

When $n = 1$, $a_1 = -1 = -1^2$, so $a_n = -n^2$.

In-Class Practice 8.7.1 General Terms of Recurrences

- (1) Given $a_1 = 2, a_n = 3a_{n-1} - 4$, find the general term.
- (2) Given $a_1 = 5, a_n = \frac{1}{2}a_{n-1} + 3$, find the general term.
- (3) Given $a_1 = 1, a_n - a_{n-1} = 2n - 1$, find the general term.
- (4) Given $a_1 = 2, a_n - a_{n-1} = 3 \cdot 2^{n-1}$, find the general term.

8.8 Chapter Summary**Main Thread**

- (1) A general-term formula gives terms directly; a recurrence generates terms step by step.
- (2) Arithmetic sequences are recognized by differences; geometric sequences are recognized by ratios. Both have general-term and sum formulas.
- (3) $a_n = S_n - S_{n-1}$ connects a single term with partial sums.
- (4) Telescoping is suitable for sums whose denominators are products of neighboring factors.
- (5) Shift-and-subtract is suitable for "arithmetic term times geometric term."
- (6) A first-order linear recurrence can be turned into a geometric sequence by shifting, or solved by summing neighboring differences.

Formula List for This Chapter

$$a_n = a_1 + (n-1)d, \quad S_n = \frac{n(a_1 + a_n)}{2},$$

$$a_n = a_1 q^{n-1}, \quad S_n = a_1 \frac{1-q^n}{1-q} \quad (q \neq 1),$$

$$a_1 = S_1, \quad a_n = S_n - S_{n-1} \quad (n \geq 2).$$

How to Use These Exercises

Group A: Basic Practice

- (1) Write the first six terms of $a_n = 2n + (-1)^n$.
- (2) For an arithmetic sequence with $a_1 = -1, a_4 = 5$, find d, a_n, S_6 .
- (3) In an arithmetic sequence, $a_3 = 7, a_5 + a_7 = 26$. Find the general term.
- (4) In a geometric sequence, $a_2 = 6, a_3 = 12$. Find a_1, q, S_{10} .
- (5) Find $1 + 3 + 5 + \cdots + (2n - 1)$.
- (6) Find $3 + 6 + 12 + \cdots + 3 \cdot 2^{n-1}$.
- (7) Given $S_n = n^2 + 3n$, find a_n .
- (8) Find $\sum_{k=1}^n 1/[k(k+1)]$.
- (9) Given $a_1 = 1, a_n = 2a_{n-1} + 1$, find a_n .

Difficulty: Core

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- (2) In a positive geometric sequence, $a_2 + a_4 = 20$, $a_3 = 8$. Find a_1, q .
- (3) Find the first n terms sum of $a_n = 2^{-n} + 4n + 2$.
- (4) Find $\sum_{k=1}^n \frac{1}{(2k-1)(2k+1)}$.
- (5) Use shift-and-subtract to find $1 \cdot 3 + 2 \cdot 3^2 + \cdots + n \cdot 3^n$.
- (6) Given $a_1 = -1$, $a_n = a_{n-1} - 2n + 1$, find the general term.
- (7) Given $a_1 = 4$, $a_n = \frac{1}{2}a_{n-1} + 2$, find the general term and explain which number a_n approaches when n is large.
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Group C: Extension and Thinking

Difficulty: Extension

- (1) Prove that in an arithmetic sequence, $S_{2n} = n(a_n + a_{n+1})$.
- (2) A positive sequence satisfies $a_{n+1} = \sqrt{2a_n}$, $a_1 = 4$. Take logarithms on both sides and find the general term.
- (3) Find $1^2 + 2^2 + \cdots + n^2$. You may try comparing $(k+1)^3 - k^3$.
- (4) A sequence satisfies $a_{n+1} = \frac{a_n}{1+a_n}$, $a_1 = 1$. Take reciprocals of the recurrence and find the general term.
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Self-Check

Skill	Yes	Almost	Not yet
Understand general-term formulas and recurrence relations	☐	☐	☐
Use arithmetic-sequence general-term and sum formulas	☐	☐	☐
Use geometric-sequence general-term and sum formulas	☐	☐	☐
Find a_n from S_n	☐	☐	☐
Use grouped summation and Gaussian summation	☐	☐	☐
Use telescoping	☐	☐	☐
Use shift-and-subtract	☐	☐	☐
Find general terms of first-order linear recurrences	☐	☐	☐

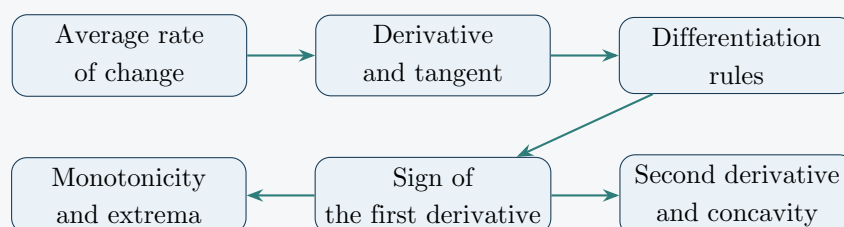
Derivatives and Applications

Learning Goals

After studying this chapter, you should be able to:

- ▶ understand secant slopes through average rates of change, and tangent slopes through derivatives;
- ▶ compute derivatives of constant, linear, and quadratic functions from the definition;
- ▶ fluently use the basic derivative table and the sum, difference, product, and quotient rules;
- ▶ use the chain rule to differentiate common composite functions;
- ▶ find the tangent line to a curve at a given point;
- ▶ use first derivatives to find intervals of monotonicity, local extrema, and absolute extrema on closed intervals;
- ▶ use second derivatives to determine convexity/concavity and find inflection points;
- ▶ solve comprehensive derivative application problems.

Knowledge Map



A derivative turns "how a function changes near one point" into a number. First learn how to differentiate, then use the sign of the derivative to study the whole graph.

Learning Route for This Chapter

This chapter starts from a concrete question: given the graph of a function, how can we describe whether it is rising or falling at a point, and how fast it is changing? We first use the slope of a secant line through two points to describe average change on an interval. Then we let the two points get infinitely close, obtaining the tangent slope, which is the derivative. After understanding the definition, we cannot compute every derivative from the definition, so we need a basic derivative table and differentiation rules. Finally, we bring derivatives back to graphs: use f' to determine monotonicity and local highs/lows, and use f'' to determine bending direction.

Prerequisites and Minimum Goals

This chapter uses function graphs, slope, linear/quadratic/exponential/logarithmic/trigonometric functions, and algebraic simplification. The minimum goals for a first pass are: understand a derivative as an instantaneous rate of change (处的瞬时变化率); use the definition to differentiate constant functions, linear functions, and x^2 ; use basic derivative formulas and the chain rule; and use the sign of f' to determine monotonicity.

Common Self-Study Sticking Points

Limits, the chain rule, absolute extrema on closed intervals, and second derivatives may all be difficult at first. On a first pass, hold onto the main thread: a derivative is the slope of the tangent line. Tangents through an external point can be treated as second-pass or extension content.

9.1 Rate of Change, Secant Lines, and Tangent Lines (变化率、割线与切线)

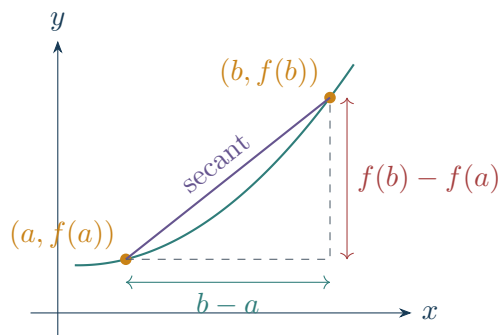
Average rate of change describes change over an interval; instantaneous rate of change describes change at one point.

9.1.1 Average Rate of Change (平均变化率)

For a given function $f(x)$, how can we describe how fast the function value changes? We can divide the total change in function value by the total change in the input. This gives the average amount by which the function value changes per unit change in the input. When $y = f(x)$ changes from $x = a$ to $x = b$, the average rate of change is defined as

$$\frac{f(b) - f(a)}{b - a}.$$

It is the slope of the secant line passing through the two graph points $(a, f(a))$ and $(b, f(b))$.



In the figure, the secant slope is "vertical change divided by horizontal change":

$$\frac{\text{vertical change}}{\text{horizontal change}} = \frac{f(b) - f(a)}{b - a}.$$

Average rate of change is not only a graphical concept. If $s(t)$ denotes the distance traveled by a car after t hours, then

$$\frac{s(3) - s(1)}{3 - 1}$$

is the average speed from hour 1 to hour 3. If $T(t)$ denotes the temperature of a place, then

$$\frac{T(10) - T(6)}{10 - 6}$$

is the average temperature increase per hour from 6 o'clock to 10 o'clock. The numerator is the change in function value, and the denominator is the change in the input, so average rate of change means "average function-value change per unit input change."

Example 9.1 Average Rate of Change

Find the average rate of change of $f(x) = x^2$ on $[1, 3]$, $[1, 1.5]$, and $[1, 1.1]$. Observe how the result changes as the right endpoint approaches 1.

Solution: The average rate of change is

$$\frac{f(b) - f(a)}{b - a}.$$

On $[1, 3]$:

$$\frac{f(3) - f(1)}{3 - 1} = \frac{9 - 1}{2} = 4.$$

On $[1, 1.5]$:

$$\frac{f(1.5) - f(1)}{1.5 - 1} = \frac{2.25 - 1}{0.5} = 2.5.$$

On $[1, 1.1]$:

$$\frac{f(1.1) - f(1)}{1.1 - 1} = \frac{1.21 - 1}{0.1} = 2.1.$$

The closer the right endpoint is to 1, the closer the average rate of change is to 2. This is the intuitive origin of the tangent slope at $x = 1$.

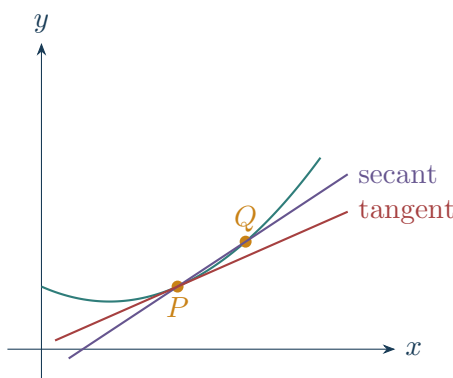
9.1.2 Slope of a Tangent Line (切线斜率)

Fix the point $P(a, f(a))$, and let another point $Q(a + h, f(a + h))$ move along the curve toward P . The secant slope is

$$\frac{f(a + h) - f(a)}{h}.$$

If this quotient approaches a definite number as $h \rightarrow 0$, that number is the slope of the tangent line to the curve at P , also called the instantaneous rate of change.

We cannot directly set $h = 0$, because then P and Q coincide and the denominator of the difference quotient is 0. The key idea of the derivative is: first compute secant slopes for $h \neq 0$, then observe whether these slopes stabilize toward one number as h gets closer and closer to 0.



Graphically, the closer Q is to P , the more the secant line looks like the tangent line through P . The notation $h \rightarrow 0$ in the derivative definition is the algebraic way to say "let Q approach P along the curve."

9.2 Definition and Geometric Meaning of the Derivative (导数的定义与几何意义)

A derivative is the limit of a difference quotient. The symbol $f'(x)$ denotes the derivative function; $f'(a)$ denotes the derivative value at one point.

9.2.1 Definition of Derivative

In this chapter, we use an intuitive meaning of "limit": when h can be positive or negative and gets closer and closer to 0, if a difference quotient gets closer and closer to the same definite number, then that number is called the limit of the difference quotient. In more advanced mathematics, limits will be defined more rigorously. Here, the focus is on computing and understanding them as instantaneous rates of change.

For example, for $f(x) = x^2$ near $x = 1$, the difference quotient is

$$\frac{f(1+h) - f(1)}{h} = 2 + h.$$

When $h = 0.1, 0.01, -0.1, -0.01$, the quotient is 2.1, 2.01, 1.9, 1.99, all approaching 2. So we say the limit of this difference quotient as $h \rightarrow 0$ is 2.

The derivative of f at $x = a$ is defined as

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

If this can be done for all x in an interval of the domain, we obtain the derivative function

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

9.2.2 Using the Definition to Differentiate Simple Functions

For polynomial functions, we can use the definition to compute derivatives.

Example 9.2 Derivative of a Linear Function

Use the definition to find the derivative function of $f(x) = mx + b$.

Solution: By definition,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

For $f(x) = mx + b$,

$$\frac{f(x+h) - f(x)}{h} = \frac{m(x+h) + b - (mx+b)}{h} = \frac{mh}{h} = m.$$

Thus

$$f'(x) = m.$$

This agrees with the graph of a linear function: the tangent slope at every point is exactly the line slope m .

Example 9.3 Derivative of a Quadratic Function

Use the definition to find the derivative function of $f(x) = x^2$, and find the tangent slope at $x = 2$.

Solution: By definition,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

For $f(x) = x^2$,

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h.$$

Letting $h \rightarrow 0$, we get

$$f'(x) = 2x.$$

Therefore the tangent slope at $x = 2$ is

$$f'(2) = 4.$$

Example 9.4 General Quadratic Function

Use the definition or known results to find the derivative of $f(x) = ax^2 + bx + c$, and explain why the vertex x -coordinate satisfies $f'(x) = 0$.

Solution: By basic differentiation rules,

$$f'(x) = (ax^2 + bx + c)' = 2ax + b.$$

The vertex x -coordinate of a quadratic function is

$$x = -\frac{b}{2a} \quad (a \neq 0).$$

Substitute into the derivative:

$$f' \left(-\frac{b}{2a} \right) = 2a \left(-\frac{b}{2a} \right) + b = -b + b = 0.$$

Geometrically, the tangent line at the vertex of a parabola is horizontal, so its slope is 0.

Common Mistake

$f'(a)$ is a number, while $f'(x)$ is a function. $f'(x)$ is not the reciprocal of $f(x)$, and it is not an exponent.

9.2.3 Geometric and Practical Meaning of the Derivative

$f'(a)$ is the tangent slope, and it is also the instantaneous rate of change of the function at a . If $s(t)$ denotes position, then $s'(t)$ is instantaneous velocity (瞬时速度). If we later study how velocity itself changes, we obtain acceleration, which is related to the second derivative (二阶导数) discussed later.

In-Class Practice 9.2.1 Definition and Meaning

- (1) Write and simplify the difference quotient of $f(x) = 3x + 1$ at $x = a$.
- (2) Use the definition to find the derivative of the constant function $f(x) = 5$.
- (3) Given $s(t) = t^2 + 2t$, find the instantaneous velocity at $t = 3$.
- (4) The derivative of a graph at $(2, 5)$ is -3 . Write the tangent line at that point.

9.3 Basic Derivatives and Differentiation Rules (基本导数与四则运算法则)

When differentiating, first identify the structure, then choose the appropriate formula.

9.3.1 Basic Derivative Table

This subsection gives derivatives of common basic functions. The derivative definition can compute derivatives, but starting from the definition every time is too slow. The derivative table organizes basic results as tools; later complicated functions will be broken down into these basic blocks.

Derivatives to Master

$$\begin{array}{ll}
(C)' = 0, & (x^n)' = nx^{n-1}, \\
(e^x)' = e^x, & (a^x)' = a^x \ln a, \\
(\ln x)' = \frac{1}{x}, & (\log_a x)' = \frac{1}{x \ln a}, \\
(\sin x)' = \cos x, & (\cos x)' = -\sin x, \\
(\tan x)' = \frac{1}{\cos^2 x}.
\end{array}$$

These derivatives should be memorized, but their full derivations require deeper mathematics and are not required in this course.

Memorize the Derivative Table in Three Layers

Layer 1: $(C)' = 0$, $(x^n)' = nx^{n-1}$; use these first for polynomials. Layer 2: $e^x, a^x, \ln x, \log_a x$; use these for exponential and logarithmic functions. Layer 3: $\sin x, \cos x, \tan x$; use these for trigonometric functions. Do not memorize the table as an unstructured list all at once.

9.3.2 Linearity, Product Rule, and Quotient Rule (线性、乘积与商法则)

This subsection handles functions made from adding, subtracting, multiplying, or dividing basic functions. Differentiation rules let us split a complicated function into simpler functions, differentiate each part, and then combine the results according to the structure.

Differentiation Rules

$$\begin{aligned}
(cf)' &= cf', \\
(f+g)' &= f' + g', \\
(f-g)' &= f' - g', \\
(fg)' &= f'g + fg', \\
\left(\frac{f}{g}\right)' &= \frac{f'g - fg'}{g^2} \quad (g \neq 0).
\end{aligned}$$

Proofs of the Arithmetic Rules

We now use the derivative definition to explain why these formulas are true. To reduce notation, fix some x , and assume f, g are differentiable at x . Differentiable functions have nearby function values that stably approach the original function value; that is, as $h \rightarrow 0$,

$$f(x+h) \rightarrow f(x), \quad g(x+h) \rightarrow g(x).$$

First consider the constant multiple rule:

$$\begin{aligned}(cf)'(x) &= \lim_{h \rightarrow 0} \frac{cf(x+h) - cf(x)}{h} \\ &= \lim_{h \rightarrow 0} c \frac{f(x+h) - f(x)}{h} \\ &= cf'(x).\end{aligned}$$

Thus a constant factor may be pulled outside the derivative.

For the sum rule:

$$\begin{aligned}(f+g)'(x) &= \lim_{h \rightarrow 0} \frac{[f(x+h) + g(x+h)] - [f(x) + g(x)]}{h} \\ &= \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} + \frac{g(x+h) - g(x)}{h} \right] \\ &= f'(x) + g'(x).\end{aligned}$$

This says the derivative of a sum is the sum of derivatives.

The difference rule is similar:

$$\begin{aligned}(f-g)'(x) &= \lim_{h \rightarrow 0} \frac{[f(x+h) - g(x+h)] - [f(x) - g(x)]}{h} \\ &= \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} - \frac{g(x+h) - g(x)}{h} \right] \\ &= f'(x) - g'(x).\end{aligned}$$

Therefore

$$(f-g)' = f' - g'.$$

Next consider the product rule. When taking the difference quotient of $f(x)g(x)$, we add and subtract an intermediate term so that the expression can be split into two difference quotients:

$$\begin{aligned}(fg)'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \left[g(x+h) \frac{f(x+h) - f(x)}{h} + f(x) \frac{g(x+h) - g(x)}{h} \right] \\ &= g(x)f'(x) + f(x)g'(x).\end{aligned}$$

Thus

$$(fg)' = f'g + fg'.$$

This proof also explains why the derivative of a product is not $f'g'$: when a product changes, there is change from f and change from g , and both parts must be counted.

Finally, consider the quotient rule. Suppose $g(x) \neq 0$, and $g(x+h) \neq 0$ when h is sufficiently close

to 0. By definition,

$$\begin{aligned}
 \left(\frac{f}{g}\right)'(x) &= \lim_{h \rightarrow 0} \frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x) - f(x)g(x+h)}{h g(x+h)g(x)} \\
 &= \lim_{h \rightarrow 0} \frac{g(x)[f(x+h) - f(x)] - f(x)[g(x+h) - g(x)]}{h g(x+h)g(x)} \\
 &= \lim_{h \rightarrow 0} \frac{g(x) \frac{f(x+h)-f(x)}{h} - f(x) \frac{g(x+h)-g(x)}{h}}{g(x+h)g(x)} \\
 &= \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}.
 \end{aligned}$$

Therefore

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

Common Mistake

The derivative of a product is not the product of derivatives. For the quotient rule, remember the numerator order as "top derivative times bottom, minus top times bottom derivative." For example, $(x \cdot x)' = (x^2)' = 2x$. If one incorrectly writes $x' \cdot x' = 1 \cdot 1 = 1$, the result is obviously wrong.

Identify Structure Before Differentiating

When you meet a function, do not rush to apply formulas term by term. First ask: what is the outermost structure? Is it a sum, product, quotient, or a function inside another function? Once the structure is clear, the calculation usually becomes substitution into formulas and simplification.

Example 9.5 Polynomials and Trigonometric Functions

Differentiate:

(1) $x^4 - 3x^2 + 5x - 7$;

(2) $\sin x + 2 \cos x$;

(3) $5x + 8 \ln x$.

Solution:

(1) $(x^4 - 3x^2 + 5x - 7)' = 4x^3 - 6x + 5$.

(2) $(\sin x + 2 \cos x)' = \cos x - 2 \sin x$.

(3) $(5x + 8 \ln x)' = 5 + \frac{8}{x} \quad (x > 0)$.

Example 9.6 Product and Quotient

Differentiate:

(1) $x^3 e^x$;

(2) $e^x \sin x$;

(3) $\frac{\ln x}{x}$;

(4) $\frac{\sin x}{e^x}$.

Solution:

(1) $(x^3 e^x)' = 3x^2 e^x + x^3 e^x = e^x(x^3 + 3x^2)$.

(2) $(e^x \sin x)' = e^x \sin x + e^x \cos x = e^x(\sin x + \cos x)$.

(3)
$$\left(\frac{\ln x}{x}\right)' = \frac{x \cdot \frac{1}{x} - \ln x}{x^2} = \frac{1 - \ln x}{x^2}.$$

(4)
$$\left(\frac{\sin x}{e^x}\right)' = \frac{e^x \cos x - \sin x \cdot e^x}{e^{2x}} = e^{-x}(\cos x - \sin x).$$

In-Class Practice 9.3.1 Basic Differentiation

(1) Differentiate $f(x) = 3x^5 - 2x^3 + x - 8$.

(2) Differentiate $f(x) = \cos x + \sin x$.

(3) Differentiate $f(x) = x^2 \ln x$.

(4) Differentiate $f(x) = e^x(x^2 + 1)$.

(5) Differentiate $f(x) = \frac{x+1}{x-1}$.

(6) Differentiate $f(x) = \frac{\ln x}{x^2}$.

9.4 Composite Functions and the Chain Rule (复合函数求导)

Differentiate the outer function with respect to the inner function, then multiply by the derivative of the inner function.

9.4.1 Chain Rule (链式法则)

The change of a composite function happens in two steps: x first changes the inner quantity $u = g(x)$, and then u changes the outer quantity $y = f(u)$. Therefore, the total rate of change should multiply "the rate of change of the outer function with respect to the inner quantity" by "the rate of change of the inner quantity with respect to x ."

If $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = f'(g(x))g'(x).$$

In words, "differentiate the outside, then multiply by the derivative of the inside," but in writing, each layer should be made clear.

Explanation of the Chain Rule

The chain rule can be understood through multiplication of changes. Let

$$u = g(x), \quad y = f(u).$$

When x changes from x to $x + \Delta x$, u also changes:

$$\Delta u = g(x + \Delta x) - g(x).$$

At the same time, the change in y is

$$\Delta y = f(u + \Delta u) - f(u).$$

Thus the total rate of change can be split into two parts:

$$\frac{\Delta y}{\Delta x} = \frac{\Delta y}{\Delta u} \cdot \frac{\Delta u}{\Delta x}.$$

This means: x first affects u , and u then affects y , so the rate of change of y with respect to x should equal the rate of change of y with respect to u times the rate of change of u with respect to x .

As $\Delta x \rightarrow 0$, we also have $\Delta u \rightarrow 0$. If both rates of change approach definite values, then

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \left(\lim_{\Delta u \rightarrow 0} \frac{\Delta y}{\Delta u} \right) \left(\lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} \right).$$

Here

$$\lim_{\Delta u \rightarrow 0} \frac{\Delta y}{\Delta u} = f'(u), \quad \lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} = g'(x).$$

Therefore

$$\frac{dy}{dx} = f'(u)g'(x) = f'(g(x))g'(x).$$

This is the chain rule. Its core is not merely the phrase "outside derivative times inside derivative," but the fact that a composite function changes in two steps: first from x to u , then from u to y .

Marking Inner and Outer Layers

First temporarily write the function as $y = f(u)$, $u = g(x)$, then differentiate. For example:

$$(2x + 1)^3 : \quad u = 2x + 1, \quad y = u^3;$$

$$\ln(1 + x^2) : \quad u = 1 + x^2, \quad y = \ln u;$$

$$\sin(x^2 + 2x + 1) : \quad u = x^2 + 2x + 1, \quad y = \sin u.$$

After marking inner and outer layers, compute by "differentiate the outer layer while keeping the inner expression, then multiply by the derivative of the inner layer."

Example 9.7 Composites with Powers and Radicals

Differentiate:

$$(1) (2x + 1)^3;$$

$$(2) \sqrt{2x + 1};$$

$$(3) (x^2 + 2x + 1)^{-2}.$$

Solution:

$$(1) ((2x + 1)^3)' = 3(2x + 1)^2 \cdot 2 = 6(2x + 1)^2.$$

$$(2)$$

$$(\sqrt{2x + 1})' = ((2x + 1)^{1/2})' = \frac{1}{2}(2x + 1)^{-1/2} \cdot 2 = \frac{1}{\sqrt{2x + 1}}.$$

$$(3) ((x^2 + 2x + 1)^{-2})' = -2(x^2 + 2x + 1)^{-3}(2x + 2).$$

Example 9.8 Exponential, Logarithmic, and Trigonometric Composites

Differentiate:

$$(1) e^{2x};$$

$$(2) \ln(1 + x^2);$$

$$(3) \sin(x^2 + 2x + 1);$$

$$(4) (2 \ln x - 1)^5.$$

Solution:

$$(1) (e^{2x})' = 2e^{2x}.$$

$$(2) (\ln(1 + x^2))' = \frac{2x}{1 + x^2}.$$

$$(3) (\sin(x^2 + 2x + 1))' = \cos(x^2 + 2x + 1)(2x + 2).$$

$$(4) \ ((2 \ln x - 1)^5)' = 5(2 \ln x - 1)^4 \cdot \frac{2}{x} = \frac{10(2 \ln x - 1)^4}{x}.$$

9.4.2 Multiple Layers

This subsection handles functions with more than one layer of composition. For expressions such as $e^{\sin(x^2)}$, do not rush to use one large formula. Start from the outermost layer and write "outer derivative times inner derivative" one layer at a time.

For example,

$$y = e^{\sin(x^2)}$$

contains, from outside to inside, the exponential function, the sine function, and the square function.

Writing Multi-Layer Composite Differentiation

First split the layers, then differentiate from outside to inside. For

$$y = e^{\sin(x^2)},$$

write

$$v = x^2, \quad u = \sin v, \quad y = e^u.$$

Then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}.$$

Compute each part:

$$\frac{dy}{du} = e^u, \quad \frac{du}{dv} = \cos v, \quad \frac{dv}{dx} = 2x.$$

Substitute $u = \sin(x^2)$, $v = x^2$ back:

$$y' = e^{\sin(x^2)} \cos(x^2) \cdot 2x.$$

After practice, you may not need to write u, v , but at the beginning, writing every layer makes it much harder to forget an inner derivative.

Example 9.9 Differentiating Multi-Layer Composites

Differentiate:

$$(1) \ y = e^{\sin(x^2)};$$

$$(2) \ y = \sqrt{\ln(1+x^2)}.$$

Solution:

(1) Split layers:

$$v = x^2, \quad u = \sin v, \quad y = e^u.$$

Differentiate from outside to inside:

$$y' = e^u \cdot \cos v \cdot 2x.$$

Substitute $u = \sin(x^2)$, $v = x^2$:

$$y' = 2xe^{\sin(x^2)} \cos(x^2).$$

(2) Split layers:

$$v = 1 + x^2, \quad u = \ln v, \quad y = \sqrt{u} = u^{1/2}.$$

Differentiate from outside to inside:

$$y' = \frac{1}{2}u^{-1/2} \cdot \frac{1}{v} \cdot 2x.$$

Substitute $u = \ln(1 + x^2)$, $v = 1 + x^2$:

$$y' = \frac{x}{(1 + x^2)\sqrt{\ln(1 + x^2)}}.$$

In-Class Practice 9.4.1 Chain Rule

- (1) Differentiate $y = (3x - 2)^6$.
- (2) Differentiate $y = \sqrt{x^2 + 1}$.
- (3) Differentiate $y = \ln(3x + 1)$.
- (4) Differentiate $y = e^{-x^2}$.
- (5) Differentiate $y = \cos(2x - \pi/3)$.
- (6) Differentiate $y = e^{2x}(\cos x + \sin x)$.

9.5 Tangent Lines (切线)

The tangent slope of a curve at $(x_0, f(x_0))$ is $f'(x_0)$.

9.5.1 Equation of a Tangent Line

This subsection answers: given a curve and a tangent point, how do we write the tangent line? The derivative gives the tangent slope, and the tangent point gives the point the line passes through. Together they give the point-slope form of the tangent line:

$$y - f(x_0) = f'(x_0)(x - x_0).$$

Three Steps for Tangent-Line Problems

Step 1: find the tangent point. If the problem gives x_0 , first compute $y_0 = f(x_0)$. If the problem gives the point directly, still check that it lies on the curve. Step 2: find the slope by computing $f'(x_0)$. Step 3: substitute into point-slope form:

$$y - y_0 = f'(x_0)(x - x_0).$$

Example 9.10 Tangent to a Quadratic Function

Find the tangent line to $y = x^2 - 5x + 4$ at $(2, -2)$.

Solution: First differentiate:

$$y' = 2x - 5.$$

At $x = 2$, the tangent slope is

$$k = y'(2) = 4 - 5 = -1.$$

The tangent line passes through $(2, -2)$, so

$$y + 2 = -(x - 2),$$

that is, $y = -x$.

Example 9.11 Tangents to General Functions

(1) Find the tangent line to $y = x \ln x$ at (e, e) .

(2) Find the tangent line to $y = e^x \sin x$ at $x = 0$.

Solution:

(1) For $y = x \ln x$,

$$y' = \ln x + 1.$$

At $x = e$, the slope is 2, and the tangent point is (e, e) . Hence

$$y - e = 2(x - e).$$

(2) For $y = e^x \sin x$,

$$y' = e^x \sin x + e^x \cos x = e^x(\sin x + \cos x).$$

When $x = 0$, $y = 0$, and the slope is 1. Hence the tangent line is

$$y = x.$$

9.5.2 Tangents Through an External Point

This type of problem is extension content. Ordinary tangent-line problems give the x -coordinate of the tangent point; an external-point tangent problem does not tell us where the tangent point is, so we first set the tangent point, then require the tangent line to pass through the given external point.

If the tangent point has x -coordinate t , then the tangent point is $(t, f(t))$, and the tangent line is

$$y - f(t) = f'(t)(x - t).$$

Substitute the known external point into this line and solve for t .

Example 9.12 Tangents Through an External Point

Find all lines passing through $(0, -1)$ that are tangent to the parabola $y = x^2 - 5x + 4$.

Solution: Let the tangent point have x -coordinate t . Then the tangent point is

$$(t, t^2 - 5t + 4).$$

Since $y' = 2x - 5$, the tangent slope is $2t - 5$. The tangent line is

$$y - (t^2 - 5t + 4) = (2t - 5)(x - t).$$

This line passes through $(0, -1)$, so

$$-1 - (t^2 - 5t + 4) = (2t - 5)(-t).$$

Simplifying,

$$-t^2 + 5t - 5 = -2t^2 + 5t,$$

so $t^2 = 5$, and $t = \pm\sqrt{5}$.

The corresponding tangent lines are

$$y + 1 = (2\sqrt{5} - 5)x$$

and

$$y + 1 = (-2\sqrt{5} - 5)x.$$

In-Class Practice 9.5.1 Tangent Lines

- (1) Find the tangent line to $y = x^3 - 3x$ at $x = 2$.
- (2) Find the tangent line to $y = \ln x$ at $(1, 0)$.
- (3) Find the tangent line to $y = \sqrt{x}$ at $x = 4$.
- (4) Find the line tangent to $y = x^2$ with slope 4.
- (5) Find all tangent lines to $y = x^2$ passing through $(0, -4)$.

9.6 Derivatives and Monotonicity (导数与单调性)

When a function rises, its derivative is usually positive; when it falls, its derivative is usually negative. Critical points divide the domain into intervals.

9.6.1 Intervals of Monotonicity (单调区间)

On an interval, if $f'(x) > 0$, then f is increasing; if $f'(x) < 0$, then f is decreasing.

This applies the idea of tangent slope to a whole part of the graph. If every nearby tangent slope on an interval is positive, the graph as a whole rises to the right; if every slope is negative, the graph falls to the right. Points where the derivative is 0 or undefined may change the rising/falling behavior, so we first use these points to cut the domain into intervals, then determine the derivative sign on each interval.

Steps for Finding Monotonic Intervals

- (1) Write the domain of the function.
- (2) Find $f'(x)$.
- (3) Find critical points where $f'(x) = 0$ or $f'(x)$ does not exist.
- (4) Use the critical points to split the domain into intervals, and determine the sign of f' on each interval.
- (5) Write the increasing and decreasing intervals.

Sign of derivative	Graph trend	Conclusion
$f'(x) > 0$	rises to the right	function is increasing
$f'(x) < 0$	falls to the right	function is decreasing
$f'(x) = 0$	may be horizontal	check the signs on both sides

Example 9.13 Monotonicity of a Polynomial

Find the intervals of monotonicity of $f(x) = x^3 - 3x^2 - 9x + 1$.

Solution:

$$f'(x) = 3x^2 - 6x - 9 = 3(x - 3)(x + 1).$$

The critical points are $x = -1, 3$. The sign table is

$$(-\infty, -1) : + \quad (-1, 3) : - \quad (3, +\infty) : +.$$

Therefore $f(x)$ is increasing on $(-\infty, -1)$ and $(3, +\infty)$, and decreasing on $(-1, 3)$.

Example 9.14 Functions with Exponentials or Logarithms

Find the monotonic intervals:

(1) $f(x) = xe^{-x}$;

(2) $g(x) = \ln x/x$.

Solution:

(1)

$$f'(x) = e^{-x} + x(-e^{-x}) = e^{-x}(1 - x).$$

Since $e^{-x} > 0$, the sign of $f'(x)$ is determined by $1 - x$. Hence f is increasing on $(-\infty, 1)$ and decreasing on $(1, +\infty)$.

(2) The domain is $x > 0$.

$$g'(x) = \frac{x \cdot \frac{1}{x} - \ln x}{x^2} = \frac{1 - \ln x}{x^2}.$$

Since $x^2 > 0$, the sign of $g'(x)$ is determined by $1 - \ln x$. The critical point is $x = e$. Therefore g is increasing on $(0, e)$ and decreasing on $(e, +\infty)$.

In-Class Practice 9.6.1 Monotonicity

- (1) Find the monotonic intervals of $f(x) = x^3 - 3x$.
- (2) Find the monotonic intervals of $f(x) = x^4 - 2x^2$.
- (3) Find the monotonic intervals of $f(x) = x \ln x$.
- (4) Find the derivative of $f(x) = x^2 - 2a \ln x$ ($x > 0$), and discuss the monotonic intervals when $a > 0$.

9.7 Local Extrema and Absolute Extrema (极值与最值)

A local extremum is a local high or low; an absolute extremum is the highest or lowest value on a specified range.

9.7.1 First Derivative Test (一阶导数判别)

This subsection studies whether a function forms a local high point or local low point near certain points. If the derivative sign changes from positive to negative, the graph first rises and then falls. If it changes from negative to positive, the graph first falls and then rises.

A local extremum is compared only with nearby points; it is not necessarily the maximum or minimum over the entire domain. For example, a small hilltop on a slope is locally highest, but there may be a higher hilltop farther away. The first derivative test looks exactly at the change in rising/falling behavior on the two sides of a critical point.

If f' is positive on the left of x_0 and negative on the right, then x_0 is a local maximum point. If f' is negative on the left and positive on the right, then x_0 is a local minimum point. Having $f'(x_0) = 0$

alone is not enough to guarantee a local extremum.

f' left of x_0	f' right of x_0	Conclusion
+	−	increasing then decreasing; local high at x_0
−	+	decreasing then increasing; local low at x_0
+	+	increasing throughout; no local high/low
−	−	decreasing throughout; no local high/low

Example 9.15 Local Extrema

Find all local extrema points and local extrema values of $f(x) = x^3 - 6x^2 + 9x + 1$.

Solution:

$$f'(x) = 3x^2 - 12x + 9 = 3(x - 1)(x - 3).$$

The critical points are $x = 1, 3$. The sign table is

$$(-\infty, 1) : + \quad (1, 3) : - \quad (3, +\infty) : +.$$

Thus at $x = 1$, the function changes from increasing to decreasing, so it has a local maximum:

$$f(1) = 1 - 6 + 9 + 1 = 5.$$

At $x = 3$, the function changes from decreasing to increasing, so it has a local minimum:

$$f(3) = 27 - 54 + 27 + 1 = 1.$$

The local maximum point is $(1, 5)$, and the local minimum point is $(3, 1)$.

9.7.2 Absolute Extrema on a Closed Interval (闭区间最值)

This subsection studies the highest and lowest values on a specified closed interval. Unlike local highs and lows, absolute extrema on a closed interval must be compared over the whole interval. Candidate positions can only be endpoints or critical points inside the interval.

For a continuous function on a closed interval $[a, b]$, compare the function values at all interior critical points and at the two endpoints.

Example 9.16 Absolute Extrema on a Closed Interval

Find the maximum and minimum values of $f(x) = x^3 - 3x$ on $[-2, 3]$.

Solution:

$$f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1).$$

The critical points are $x = -1, 1$, both in $[-2, 3]$. For absolute extrema on a closed interval, compare endpoints and critical points:

$$f(-2) = -8 + 6 = -2, \quad f(-1) = -1 + 3 = 2,$$

$$f(1) = 1 - 3 = -2, \quad f(3) = 27 - 9 = 18.$$

Therefore the maximum value is 18, attained at $x = 3$; the minimum value is -2 , attained at $x = -2$ and $x = 1$.

Common Mistake

When writing an extremum point, give both x - and y -coordinates. A "local minimum point" and a "local minimum value" are not the same concept.

In-Class Practice 9.7.1 Local and Absolute Extrema

- (1) Find the local extrema of $x^3 - 3x^2 + 2$.
- (2) Find the local extrema of $x^4 - 4x^2$.
- (3) Find the absolute extrema of xe^{-x} on $[0, 3]$.
- (4) Find the local extrema of $\ln x/x$.
- (5) A rectangle has perimeter 20. Use derivatives to find its maximum area.

9.8 Second Derivative, Concavity/Convexity, and Inflection Points (二阶导数、凹凸性与拐点)

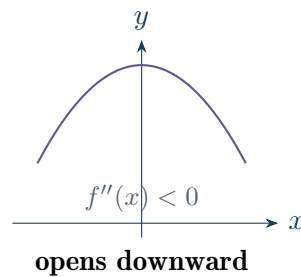
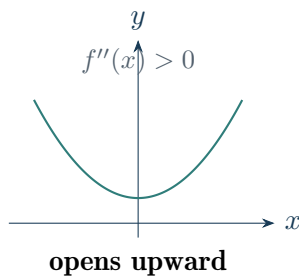
The first derivative describes slope; the second derivative describes how the slope itself changes.

9.8.1 Second Derivative

The derivative of $f'(x)$ is called the second derivative, written $f''(x)$. Since $f'(x)$ represents slope, $f''(x)$ describes how the slope itself changes. When slopes are increasing, the graph bends upward; when slopes are decreasing, the graph bends downward. In this text we use the following terminology:

- $f''(x) > 0$: the graph opens upward, called convex downward (下凸) in the Chinese convention used here;
- $f''(x) < 0$: the graph opens downward, called convex upward (上凸) in the Chinese convention used here.

Use the two simplest parabolas to remember the directions:



At first, understand the second derivative as "the rate of change of slope." For example, moving from left to right along a graph, if the tangent slope changes from -3 to -1 , then to 1 , the slope is increasing and the graph usually bends upward. If the tangent slope changes from 3 to 1 , then to -1 , the slope is decreasing and the graph usually bends downward.

9.8.2 Inflection Points

This subsection studies where the bending direction of a graph changes. The condition $f''(x) = 0$ is only a candidate signal; a true inflection point (拐点) requires the concavity/convexity to actually change on the two sides.

A point where the graph changes its bending direction is called an inflection point. Candidate points usually come from $f''(x) = 0$ or where $f''(x)$ does not exist, but we must check that the sign changes on the two sides.

Example 9.17 Convexity/Concavity Intervals and Inflection Point

For $f(x) = x^3 - 6x^2 + 9x + 1$:

- (1) find $f''(x)$;
- (2) find intervals where the graph opens upward and downward;

(3) find the inflection point.

Solution:

(1) First find the first and second derivatives:

$$f'(x) = 3x^2 - 12x + 9, \quad f''(x) = 6x - 12 = 6(x - 2).$$

(2) When $x < 2$, $f''(x) < 0$, so the graph opens downward. When $x > 2$, $f''(x) > 0$, so the graph opens upward.

(3) Since $f''(x)$ changes sign across $x = 2$, $x = 2$ is the x -coordinate of an inflection point. The y -coordinate is

$$f(2) = 8 - 24 + 18 + 1 = 3.$$

Thus the inflection point is $(2, 3)$.

Example 9.18 Concavity of a Logarithmic Function

Study the concavity/convexity intervals and inflection point of $f(x) = \ln x/x$.

Solution: The domain is $x > 0$.

$$f'(x) = \frac{1 - \ln x}{x^2}.$$

Continue differentiating, writing $f'(x) = (1 - \ln x)x^{-2}$:

$$f''(x) = -\frac{1}{x}x^{-2} + (1 - \ln x)(-2x^{-3}) = \frac{2 \ln x - 3}{x^3}.$$

Since $x^3 > 0$, the sign of $f''(x)$ is determined by $2 \ln x - 3$.

$$2 \ln x - 3 = 0 \Rightarrow x = e^{3/2}.$$

When $0 < x < e^{3/2}$, $f''(x) < 0$, so the graph opens downward. When $x > e^{3/2}$, $f''(x) > 0$, so the graph opens upward. The inflection point is

$$\left(e^{3/2}, \frac{\ln(e^{3/2})}{e^{3/2}} \right) = \left(e^{3/2}, \frac{3}{2e^{3/2}} \right).$$

In-Class Practice 9.8.1 Second Derivative

- (1) Find the second derivative of $f(x) = x^4 - 3x^2$.
- (2) Find the concavity/convexity intervals and inflection point of $f(x) = x^3 - 3x$.
- (3) Determine the concavity/convexity of $f(x) = e^x$ and $f(x) = \ln x$.
- (4) Does $f''(x_0) = 0$ necessarily imply that $(x_0, f(x_0))$ is an inflection point? Give an example or explanation.

9.9 Sketching Graphs with Derivative Information

The first derivative tells us where the graph rises or falls; the second derivative tells us where the graph bends upward or downward.

This section does not pursue exact graphing. Instead, it practices an important way of thinking: first use algebraic information to determine the large-scale shape of the graph, then draw a reasonable sketch. In general, proceed as follows.

Basic Steps for Derivative-Based Sketching

- (1) Check the domain, symmetry, and possible asymptotes.
- (2) Find $f'(x)$, and determine monotonic intervals and key points.
- (3) Find $f''(x)$, and determine upward/downward bending intervals and inflection points.
- (4) Combine key points, asymptotes, monotonicity, and concavity/convexity to draw a sketch.

Example 9.19 Sketch of a Bell-Shaped Curve

Sketch

$$f(x) = e^{-x^2/2}.$$

Solution: The domain is $\mathbb{R}$, and $f(-x) = f(x)$, so the graph is symmetric about the y -axis.

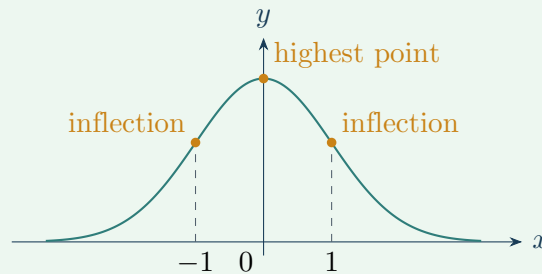
$$f'(x) = -xe^{-x^2/2}.$$

Since $e^{-x^2/2} > 0$, the sign of $f'(x)$ is determined by $-x$. Thus the function is increasing when $x < 0$, decreasing when $x > 0$, and has its highest point at $x = 0$, where $f(0) = 1$.

Continue to the second derivative:

$$f''(x) = (x^2 - 1)e^{-x^2/2}.$$

Thus $x = \pm 1$ are possible points where the bending direction changes. When $|x| < 1$, $f''(x) < 0$, so the graph opens downward. When $|x| > 1$, $f''(x) > 0$, so the graph opens upward. Therefore the sketch is a bell-shaped curve: high in the middle, low on both sides, and symmetric left-right.



Example 9.20 Sketch of the Reciprocal-Plus-Linear Function

Sketch

$$f(x) = x + \frac{1}{x}.$$

Solution: The domain is $x \neq 0$, so $x = 0$ is a special position. The function satisfies $f(-x) = -f(x)$, so the graph is symmetric about the origin.

$$f'(x) = 1 - \frac{1}{x^2} = \frac{x^2 - 1}{x^2}.$$

Since $x^2 > 0$, the sign of $f'(x)$ is determined by $x^2 - 1$:

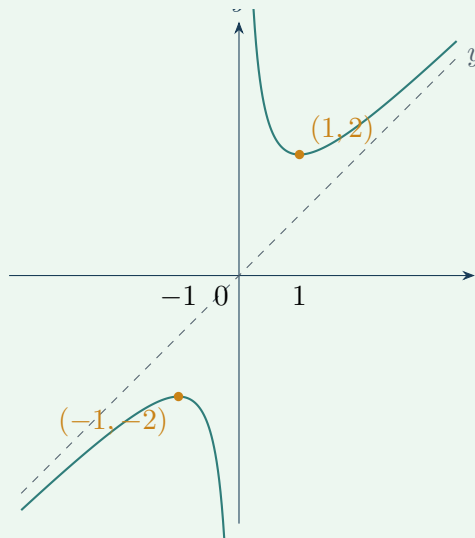
$(-\infty, -1)$ increasing, $(-1, 0)$ decreasing, $(0, 1)$ decreasing, $(1, \infty)$ increasing.

There is a local high point $(-1, -2)$ at $x = -1$, and a local low point $(1, 2)$ at $x = 1$.

The second derivative is

$$f''(x) = \frac{2}{x^3}.$$

When $x < 0$, $f''(x) < 0$, so the graph opens downward. When $x > 0$, $f''(x) > 0$, so the graph opens upward. Also, near $x = 0$, the function values tend to infinity in magnitude; when $|x|$ is large, $\frac{1}{x}$ is small and the graph approaches the line $y = x$. Combining this information gives the sketch.



Example 9.21 Sketch of $\frac{1}{1+x^2}$

Sketch

$$f(x) = \frac{1}{1+x^2}.$$

Solution: The domain is $\mathbb{R}$, and $f(-x) = f(x)$, so the graph is symmetric about the y -axis.

$$f'(x) = \frac{-2x}{(1+x^2)^2}.$$

The denominator is always positive, so the sign of $f'(x)$ is determined by $-x$. The function is increasing when $x < 0$, decreasing when $x > 0$, and has its highest point at $x = 0$, where $f(0) = 1$.

Continue to the second derivative:

$$f''(x) = \frac{2(3x^2 - 1)}{(1+x^2)^3}.$$

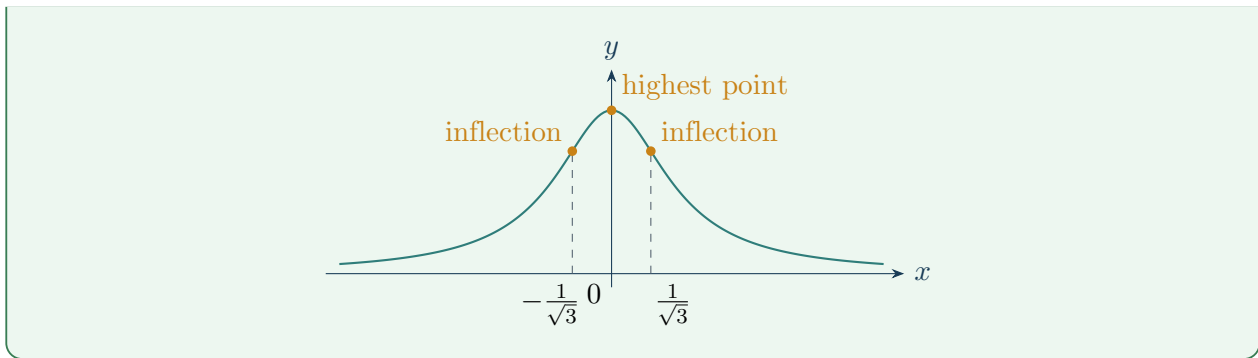
The denominator is always positive, so the sign of $f''(x)$ is determined by $3x^2 - 1$. Solving

$$3x^2 - 1 = 0$$

gives

$$x = \pm \frac{1}{\sqrt{3}}.$$

When $|x| < \frac{1}{\sqrt{3}}$, $f''(x) < 0$, so the graph opens downward. When $|x| > \frac{1}{\sqrt{3}}$, $f''(x) > 0$, so the graph opens upward. Also, when $|x|$ is large, $f(x)$ approaches 0, so both ends of the graph lie close to the x -axis.



9.10 Chapter Summary

Main Thread

- (1) A derivative is the limit of a difference quotient. Geometrically, it is tangent slope; in applications, it is instantaneous rate of change.
- (2) Before differentiating, identify the structure: linear combination, product, quotient, or composition.
- (3) A tangent line must use both the tangent point and the tangent slope.
- (4) The sign of f' determines monotonicity and local extrema; absolute extrema on a closed interval also require endpoint comparison.
- (5) The sign of f'' determines bending direction; an inflection point requires the bending direction to actually change.
- (6) When sketching a function, first find the domain, key points, and asymptotes, then combine monotonicity and concavity/concavity information.

Flowchart for Derivative Applications

When studying a function, use a fixed order: write the domain first, then find $f'(x)$; use $f'(x) = 0$ or undefined points to find critical points; use a sign table to determine monotonic intervals; use changes in increasing/decreasing behavior to determine local extrema; if studying absolute extrema on a closed interval, compare endpoint values with values at interior critical points.

9.11 Chapter Exercises

How to Use These Exercises

Group A checks the minimum mastery for this chapter. If you make mistakes, return to the corresponding subsection and reread the examples. Group B is for integrated quiz preparation. Group C is for extension; you may skip it during a first self-study pass without affecting your ability to enter the next chapter.

Difficulty: Basic

Difficulty: Core

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- (6) Study the monotonic intervals, local extrema, and absolute extrema on $[0, 3]$ of $f(x) = xe^{-x}$.
- (7) Study the monotonic intervals, local extrema, concavity/convexity intervals, and inflection point of $f(x) = \ln x/x$.

Group C: Extension and Thinking

Difficulty: Extension

- (1) Find the 2027th derivative of $f(x) = \sin x$.
- (2) Let $f(x) = x^2 - 2a \ln x$ ($x > 0$). Discuss monotonicity according to the value of a .
- (3) Use derivatives to prove that $\ln x \leq x - 1$ for all $x > 0$.

Self-Check

Skill	Yes	Almost	Not yet
Understand average rate of change and the derivative definition	☐	☐	☐
Use the basic derivative table	☐	☐	☐
Use product, quotient, and chain rules	☐	☐	☐
Find tangent lines to curves	☐	☐	☐
Find monotonic intervals	☐	☐	☐
Find local extrema and absolute extrema on closed intervals	☐	☐	☐
Determine concavity/convexity and find inflection points	☐	☐	☐

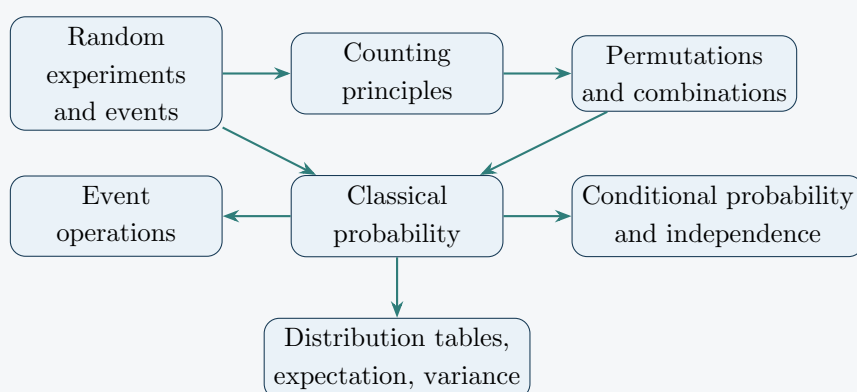
Introductory Probability and Distributions (概率初步与分布)

Learning Goals

After studying this chapter, you should be able to:

- ▶ explain clearly what one trial records in a concrete random situation;
- ▶ understand sample points, sample spaces, and events, and represent events as sets;
- ▶ use the addition principle, multiplication principle, permutations, and combinations for counting;
- ▶ use the classical probability formula in finite equally likely models;
- ▶ understand the Bernoulli, binomial, hypergeometric, and geometric distributions, and write simple distribution tables;
- ▶ use intersections, unions, complements, and mutual exclusiveness of events to simplify probability calculations;
- ▶ compute conditional probabilities and decide whether simple events are independent;
- ▶ understand the intuitive meanings of random variables, distribution tables, expectation, and variance.

Knowledge Map



The main line of this chapter is: first describe the random problem clearly, then count the possible outcomes clearly, and finally translate the counting results into probabilities and distributions.

Learning Route for This Chapter

Probability does not begin with memorizing formulas. It begins with uncertainty in daily life. Tossing coins, drawing cards, drawing balls, and inspecting parts all have the same feature: the result of one trial cannot be known in advance, but all possible results can be listed or counted. When there are few outcomes, list them directly. When there are many outcomes, use counting principles, permutations, and combinations. When the outcomes are equally likely, use "number of favorable outcomes divided by number of total outcomes." On this basis, record a numerical result as a random variable, and then obtain a distribution table, expectation, and variance.

Common Difficulties in This Chapter

The most common mistakes in probability problems are of three kinds: choosing sample points that are too coarse and then treating unequal outcomes as equally likely; double-counting or missing cases in permutation and combination problems; and failing to translate words such as "or," "at least," and "given" into event operations or conditional probability. For every problem, first ask: what exactly is recorded in one trial? Are these outcomes equally likely?

10.1 Random Experiments, Sample Spaces, and Events

We first look at concrete problems, and then introduce terminology. Definitions in probability are essentially a language for describing uncertain events precisely.

10.1.1 Starting from Tossing a Coin

Toss a fair coin once. Before the toss, we do not know whether the result will be heads or tails; but we know that only two results are possible:

heads, tails.

A process that can be repeated, whose single outcome cannot be determined in advance, and whose possible outcomes can be described clearly is called a **random experiment** (随机试验).

If we use H for heads and T for tails, then the most basic outcomes of tossing a coin once are

$$\{H, T\}.$$

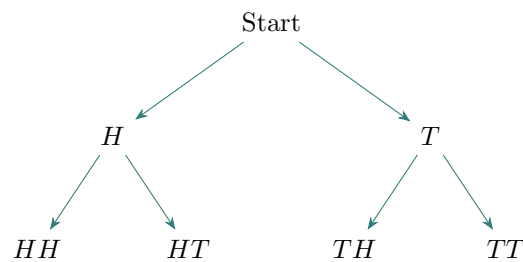
Here H and T are both most basic outcomes. In probability, each most basic outcome is called a **sample point** (样本点); the set of all sample points is called the **sample space** (样本空间), usually denoted by Ω . Thus here

$$\Omega = \{H, T\}.$$

If a coin is tossed twice, one trial records "the results of the two tosses." The sample space is not $\{H, T\}$, but

$$\Omega = \{HH, HT, TH, TT\}.$$

For example, HT means "heads on the first toss and tails on the second toss." Here HT and TH are different, because the order is different.



Each path from "Start" to the bottom of the tree diagram is one sample point. The tree diagram helps us list the sample space without omissions or repetitions.

10.1.2 An Event Is a Part of the Sample Space

Continue with tossing a coin twice. If we care about "exactly one heads," the sample points satisfying the condition are

$$HT, TH.$$

Therefore the event "exactly one heads" can be written as the set

$$A = \{HT, TH\}.$$

In probability, a subset of the sample space is called an **event (事件)**. An event may contain one sample point or many sample points. If the actual outcome of the trial falls in this set, we say that the event occurs.

The sample space Ω itself represents an event that must occur, called the **certain event (必然事件)**. The empty set $\emptyset$ represents an event that can never occur, called the **impossible event (不可能事件)**.

Example 10.1 Writing Sample Spaces and Events from Context

A box contains six cards numbered 1, 2, 3, 4, 5, 6. One card is drawn at random.

- (1) Write the sample space.
- (2) Use a set to represent the event $A =$ "an even number is drawn."
- (3) Use a set to represent the event $B =$ "a number greater than 4 is drawn."
- (4) Explain what determines whether event A occurs.

Solution:

- (1) One trial records only the number on the card, so

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

- (2) The even numbers are 2, 4, 6, so

$$A = \{2, 4, 6\}.$$

(3) The numbers greater than 4 are 5, 6, so

$$B = \{5, 6\}.$$

(4) If the actual card drawn is one of 2, 4, 6, then A occurs. If the actual card drawn is one of 1, 3, 5, then A does not occur.

10.1.3 Sample Points Must Be Fine Enough

In many probability problems, the hard part is not the formula, but choosing the wrong sample points. Consider two dice of different colors. Write the number on the red die first and the number on the blue die second. A most basic outcome should be written as

$$(i, j), \quad i, j = 1, 2, 3, 4, 5, 6.$$

For example, $(1, 2)$ and $(2, 1)$ are different sample points, because the numbers on the red and blue dice have been exchanged. In this way there are $6 \times 6 = 36$ equally likely sample points.

If we only record the "sum of the two dice," the possible values are $2, 3, \dots, 12$, but these values are not equally likely. A sum of 2 has only one way, $(1, 1)$, while a sum of 7 has six ways:

$$(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1).$$

If we directly treat $2, 3, \dots, 12$ as equally likely sample points, the probability will be wrong.

(1, 6)	(2, 6)	(3, 6)	(4, 6)	(5, 6)	(6, 6)
(1, 5)	(2, 5)	(3, 5)	(4, 5)	(5, 5)	(6, 5)
(1, 4)	(2, 4)	(3, 4)	(4, 4)	(5, 4)	(6, 4)
(1, 3)	(2, 3)	(3, 3)	(4, 3)	(5, 3)	(6, 3)
(1, 2)	(2, 2)	(3, 2)	(4, 2)	(5, 2)	(6, 2)
(1, 1)	(2, 1)	(3, 1)	(4, 1)	(5, 1)	(6, 1)

Shaded cells have sum 7

Three Questions for Writing a Sample Space

- (1) What information does one trial record?
- (2) Is this information already fine enough, so that it cannot be split into more basic equally likely outcomes?
- (3) For two outcomes that look similar, should they actually be treated as different outcomes?

In-Class Practice 10.1.1 Random Experiments, Sample Spaces, and Events

- (1) Toss a coin three times. Write the sample space.
- (2) Toss two dice of different colors. Use set language to represent "the two numbers are equal."
- (3) Choose one integer at random from 1 to 10. Write the event "the number is a multiple of 3."
- (4) Explain why the "sum of two dice" cannot be directly used as equally likely sample points.

10.2 Counting Principles (计数原理)

In probability, both the numerator and the denominator are often obtained by counting. First decide whether the task is completed by choosing one category, or by completing several steps.

10.2.1 Addition Principle (分类加法原理)

If a task can be completed by several non-overlapping categories of methods, and choosing any one category completes the task, then the total number is found by addition. More precisely, if there are m non-overlapping categories of methods, and the i -th category has n_i methods, then the total number of methods is

$$n_1 + n_2 + \cdots + n_m.$$

For example, to go to Shanghai, one may take a train or take a plane. There are 3 train services and 2 flights. If only one type of transportation needs to be chosen, then there are

$$3 + 2 = 5$$

choices. We use addition because "taking a train" and "taking a plane" are two categories, and choosing one category completes the task.

10.2.2 Multiplication Principle (分步乘法原理)

If a task must be completed through several consecutive steps, and every step must be completed, then the total number is found by multiplication. More precisely, if completing a task requires m

steps, the i -th step has n_i choices, and the choices in different steps can be combined, then the total number of methods is

$$n_1 n_2 \cdots n_m.$$

For example, to make an outfit, first choose a shirt and then choose a pair of pants. If there are 4 shirts and 3 pairs of pants, then there are

$$4 \times 3 = 12$$

outfits. We use multiplication because both a shirt and a pair of pants must be chosen.

Example 10.2 Categories or Steps

- (1) Choose one representative from Class A with 20 students or Class B with 18 students. How many choices are there?
- (2) Choose one student from Class A and one student from Class B to form a two-person group. How many choices are there?
- (3) How many two-digit numbers with no repeated digits can be formed using the digits 1, 2, 3, 4?
- (4) A password consists of 2 uppercase letters and 3 digits. If repetition is allowed in every position, how many passwords are there?

Solution:

- (1) "Class A or Class B" is a classification, so the number of choices is $20 + 18 = 38$.
- (2) "Choose one from Class A and one from Class B" is a step-by-step task, so the number of choices is $20 \times 18 = 360$.
- (3) The tens digit has 4 choices. The units digit cannot repeat it, so it has 3 choices. Hence there are $4 \times 3 = 12$ numbers.
- (4) Each letter position has 26 choices, and each digit position has 10 choices. Hence there are

$$26^2 \cdot 10^3$$

passwords.

Common Mistake

The word "or" often suggests the addition principle, while phrases such as "first ... then ..." often suggest the multiplication principle. But do not rely only on keywords. The real question is whether one category is enough, or every step must be completed.

In-Class Practice 10.2.1 Counting Principles

- (1) There are 5 shirts and 4 pairs of pants. How many outfits are there?
- (2) From Beijing to Shanghai, one may take 3 trains or 2 flights. How many choices are there?
- (3) How many two-digit numbers with no repeated digits can be formed using 0, 1, 2, 3?
- (4) A die is rolled three times. How many sample points are in the sample space?

10.3 Permutations and Combinations (排列与组合)

Permutations care about order; combinations do not. Permutations and combinations are the most frequently used counting tools in classical probability.

Start with a small example. Choose 2 letters from $\{a, b, c\}$. If the chosen letters must be arranged in a row, then we have

$$ab, ba, ac, ca, bc, cb,$$

so there are 6 results. This type of selection is called a permutation, because the order of the selected elements matters. For example, ab and ba are counted as different results.

If the chosen letters only form a group, then we have

$$\{a, b\}, \quad \{a, c\}, \quad \{b, c\},$$

so there are 3 results. This type of selection is called a combination, because it only cares which elements are chosen and not the order. For example, $\{a, b\}$ and $\{b, a\}$ are the same group.

10.3.1 Permutations

The number of ways to take m elements from n distinct elements and arrange them in order is

$$A_n^m = n(n-1) \cdots (n-m+1) = \frac{n!}{(n-m)!}.$$

This is called the number of permutations. Here

$$n! = n(n-1)(n-2) \cdots 2 \cdot 1, \quad 0! = 1.$$

The formula comes from the multiplication principle. The first position has n choices, the second

position has $n - 1$ choices left, the third position has $n - 2$ choices left, and so on until the m -th position has $n - m + 1$ choices left. Multiplying them gives the formula.

$$\underbrace{\square \square \cdots \square}_{m \text{ positions}} \quad n, n - 1, \dots, n - m + 1$$

10.3.2 Combinations

The number of ways to take m elements from n distinct elements to form a group is

$$C_n^m = \binom{n}{m} = \frac{n!}{m!(n-m)!}.$$

This is called the number of combinations. Because choosing m elements to take is equivalent to choosing $n - m$ elements to leave behind, we have

$$C_n^m = C_n^{n-m}.$$

The combination formula can be obtained from the permutation formula. First take m elements in order; there are A_n^m ways. But for the same group of m elements, the $m!$ internal orders are all counted as the same combination, so we divide by $m!$:

$$C_n^m = \frac{A_n^m}{m!} = \frac{n!}{m!(n-m)!}.$$

Permutation or Combination

If exchanging the positions of selected elements changes the result, it is a permutation. If exchanging positions does not change the result, it is a combination.

Example 10.3 Permutations and Combinations

- (1) From 8 students, choose a monitor and a vice monitor.
- (2) From 8 students, choose 2 representatives.
- (3) Form three-digit numbers with no repeated digits using 1, 2, 3, 4, 5.
- (4) Choose 3 books from 6 different books.
- (5) From 5 boys and 4 girls, choose 2 boys and 1 girl.

Write the counting expression for each situation.

Solution:

- (1) The positions are different. Student A as monitor and student B as vice monitor is different from student B as monitor and student A as vice monitor. So this is a permutation:

$$A_8^2 = 8 \cdot 7.$$

(2) Only representatives are chosen; order has no meaning. So this is a combination:

$$C_8^2 = \frac{8 \cdot 7}{2}.$$

(3) The hundreds, tens, and units digits are chosen in order, so order matters:

$$A_5^3 = 5 \cdot 4 \cdot 3.$$

(4) Only books are chosen; order has no meaning:

$$C_6^3 = \frac{6 \cdot 5 \cdot 4}{3 \cdot 2 \cdot 1}.$$

(5) Choose boys and girls separately, then combine them into one group:

$$C_5^2 C_4^1.$$

Common Mistake

Do not decide only by whether the problem uses the word "arrange." Choosing a monitor and a vice monitor is not lining people up, but the two positions are different, so it is a permutation. Choosing two representatives has no role distinction, so it is a combination.

In-Class Practice 10.3.1 Permutations and Combinations

(1) Compute A_6^2, C_6^2, C_8^3 .

(2) In how many ways can 5 people stand in a row?

(3) In how many ways can 3 people be chosen from 10 people to form a group?

(4) How many three-digit even numbers with no repeated digits can be formed using 0, 1, 2, 3, 4?

10.4 Classical Probability and Distributions in Classical Models

The core of classical probability is "finite and equally likely." If a random result can take different numerical values, we can further write its distribution.

10.4.1 Classical Probability Formula

Suppose the sample space Ω has finitely many sample points, every sample point is equally likely, event A contains m of these sample points, and the sample space contains n sample points in total. Then

$$P(A) = \frac{m}{n} = \frac{|A|}{|\Omega|}.$$

Before using this formula, always check two things:

- (1) Are the sample points finite?
- (2) Are the sample points equally likely?

Four Steps for Classical Probability

- (1) State what the sample points of one trial are.
- (2) Decide whether these sample points are equally likely.
- (3) Count the total number of sample points $|\Omega|$.
- (4) Count the number of sample points in event A , namely $|A|$, and then write $P(A) = |A|/|\Omega|$.

Example 10.4 Classical Probability with Dice

Two fair dice are rolled at the same time. Find:

- (1) the probability that the sum is 7;
- (2) the probability that the two numbers are equal;
- (3) the probability that the sum is at least 10.

Solution: Write the outcome as an ordered pair (i, j) . There are $6 \times 6 = 36$ equally likely sample points.

- (1) The outcomes with sum 7 are

$$(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1),$$

so there are 6 of them. The probability is

$$\frac{6}{36} = \frac{1}{6}.$$

- (2) The outcomes with equal numbers are $(1, 1), (2, 2), \dots, (6, 6)$, so there are 6 of them. The probability is

$$\frac{6}{36} = \frac{1}{6}.$$

- (3) A sum of at least 10 means a sum of 10, 11, 12. The corresponding numbers of outcomes

are 3, 2, 1, respectively, so there are 6 outcomes. The probability is

$$\frac{6}{36} = \frac{1}{6}.$$

10.4.2 Random Variables and Distribution Tables

Sometimes we care not only whether an event occurs, but also a numerical result. For example, after rolling two dice, we may record the sum

$$X = i + j.$$

The value of X changes with the outcome of the trial, so X is called a **random variable** (随机变量) .

A table listing every possible value of a random variable and its corresponding probability is called a **probability distribution table** (分布列) .

In this chapter, we mainly study random variables with finitely many possible values. If the possible values of random variable X are

$$x_1, x_2, \dots, x_n,$$

and the corresponding probabilities are

$$p_1, p_2, \dots, p_n,$$

then the distribution table is written as

X	x_1	x_2	$\cdots$	x_n
P	p_1	p_2	$\cdots$	p_n

and it must satisfy

$$p_i \geq 0, \quad p_1 + p_2 + \cdots + p_n = 1.$$

Example 10.5 Checking Whether a Distribution Table Is Valid

Decide whether the following table can be the distribution table of a random variable:

X	0	1	2
P	0.2	0.5	0.4

Solution: All probabilities are nonnegative, but their sum is

$$0.2 + 0.5 + 0.4 = 1.1 \neq 1.$$

Therefore it cannot be a distribution table. In a distribution table, the probabilities of all possible cases must add up to 1.

Example 10.6 Distribution Table for the Number of Heads in Two Coin Tosses

Toss two fair coins. Let $X =$ "the number of heads." Write the distribution table of X .

Solution: The sample space is

$$\Omega = \{HH, HT, TH, TT\}.$$

These 4 sample points are equally likely. The random variable X can take the values 0, 1, 2.

X	0	1	2
$P(X)$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$

Here $X = 1$ corresponds to the two sample points HT, TH , so the probability is $\frac{2}{4} = \frac{1}{2}$.

10.4.3 Bernoulli Distribution

If one trial has only two outcomes, we often call the outcome we care about "success" and the other outcome "failure." For example,

drawing a red ball = success, not drawing a red ball = failure.

Let

$$X = \begin{cases} 1, & \text{success,} \\ 0, & \text{failure.} \end{cases}$$

If the probability of success is p and the probability of failure is $1 - p$, then X is said to follow a **Bernoulli distribution (伯努利分布)** :

X	0	1
$P(X)$	$1 - p$	p

The role of the Bernoulli distribution is to turn "whether something occurs" into a random variable taking values 0 and 1. It is the starting point of many more complicated distributions.

Example 10.7 Bernoulli Distribution: Drawing a Card

A box contains one card for each number from 1 to 10. One card is drawn at random. Let $X = 1$ mean "an even number is drawn," and $X = 0$ mean "an even number is not drawn." Write the distribution table of X .

Solution: There are 5 even numbers from 1 to 10, so

$$P(X = 1) = \frac{5}{10} = \frac{1}{2}, \quad P(X = 0) = \frac{1}{2}.$$

Therefore

X	0	1
$P(X)$	$\frac{1}{2}$	$\frac{1}{2}$

Example 10.8 Bernoulli Distribution: Quality Inspection

In a batch of parts, the qualified rate is 0.98. One part is randomly inspected. Let $X = 1$ mean "qualified," and $X = 0$ mean "not qualified." Write the distribution table of X .

Solution: The success event is "qualified," and the success probability is $p = 0.98$. Hence

X	0	1
$P(X)$	0.02	0.98

This distribution table tells us that X takes the value 1 most of the time and the value 0 only rarely.

10.4.4 Binomial Distribution

If we repeat Bernoulli trials independently many times, we obtain the binomial distribution. Consider sampling with replacement: suppose a bag contains some red balls and some blue balls, and we draw from the bag n times. After each draw, the ball is put back before the next draw. Therefore the probability of drawing a red ball is the same each time, and the results of different draws can be regarded as independent.

For example, suppose a bag contains 3 red balls and 2 blue balls. Each time, draw one ball at random, record its color, and put it back. If "drawing a red ball" is called success, then the success probability each time is

$$p = \frac{3}{5}.$$

After drawing 4 times, let $X =$ "the number of red balls drawn." Then X can take the values

$$0, 1, 2, 3, 4.$$

Here we care about the total number of successes, not the time at which the first success appears.

In general, suppose each trial has only two outcomes, success and failure; the success probability is p , the failure probability is $1 - p$, and the trial is repeated independently n times. Let $X =$ "the number of successes." Then

$$P(X = k) = C_n^k p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n.$$

This is called the **binomial distribution** (二项分布).

The three parts of the formula have clear meanings:

$$\underbrace{C_n^k}_{\text{choose which } k \text{ trials succeed}} \cdot \underbrace{p^k}_{\text{success } k \text{ times}} \cdot \underbrace{(1 - p)^{n-k}}_{\text{failure } n-k \text{ times}}.$$

Example 10.9 Binomial Distribution: Drawing Balls with Replacement

A bag contains 3 red balls and 2 blue balls. Each time, draw one ball at random, record its color, and put it back. Draw 4 times. Let $X =$ "the number of red balls drawn."

- (1) Find $P(X = 2)$.
- (2) Find the probability of drawing at least one red ball.
- (3) Write the distribution table of X .

Solution: The probability of drawing a red ball each time is $p = \frac{3}{5}$, and the probability of not drawing a red ball is $\frac{2}{5}$. Since the ball is replaced after each draw, this is an independent repeated experiment.

- (1) Exactly 2 draws are red:

$$P(X = 2) = C_4^2 \left(\frac{3}{5}\right)^2 \left(\frac{2}{5}\right)^2.$$

- (2) The complement of "at least one red ball" is "all 4 draws are blue," so

$$P(X \geq 1) = 1 - \left(\frac{2}{5}\right)^4.$$

- (3) The distribution table is

X	0	1	2	3	4
$P(X)$	$\left(\frac{2}{5}\right)^4$	$C_4^1 \left(\frac{3}{5}\right) \left(\frac{2}{5}\right)^3$	$C_4^2 \left(\frac{3}{5}\right)^2 \left(\frac{2}{5}\right)^2$	$C_4^3 \left(\frac{3}{5}\right)^3 \left(\frac{2}{5}\right)$	$\left(\frac{3}{5}\right)^4$

Example 10.10 Binomial Distribution: Independent Shots

A shooter hits the target with probability 0.8 each time, and the shots are independent. If the shooter shoots 5 times, find the probability of hitting exactly 4 times.

Solution: Let $X =$ "the number of hits." Each shot has only two outcomes, hit and miss; the hit probability is the same each time; and the shots are independent. Therefore X follows a binomial distribution. Hence

$$P(X = 4) = C_5^4 (0.8)^4 (0.2).$$

From the expression for binomial probabilities, since the probabilities of all possible cases must add up to 1, we immediately get

$$\sum_{k=0}^n C_n^k p^k (1-p)^{n-k} = 1 = (p + 1 - p)^n.$$

More generally,

$$(a + b)^n = \sum_{k=0}^n C_n^k a^k b^{n-k}.$$

This formula is called the binomial expansion.

10.4.5 Hypergeometric Distribution

The binomial distribution above comes from sampling with replacement or from independent repeated trials. If we sample without replacement, that is, each selected object is not put back before the next selection, then we obtain the hypergeometric distribution.

A typical situation is as follows: there are N objects in total, among which M are success objects and $N - M$ are failure objects. We choose n objects without replacement. Let $X =$ ``the number of success objects chosen.''

If every group of n objects is equally likely, then

$$P(X = k) = \frac{C_M^k C_{N-M}^{n-k}}{C_N^n}.$$

Here k must satisfy

$$0 \leq k \leq M, \quad 0 \leq n - k \leq N - M.$$

This is called the hypergeometric distribution (超几何分布). The source of the formula is direct:

$$\frac{\text{number of ways to choose } k \text{ success objects and } n - k \text{ failure objects}}{\text{number of ways to choose } n \text{ objects from } N \text{ objects}}.$$

Example 10.11 Hypergeometric Distribution: Drawing Balls

A bag contains 5 red balls and 7 blue balls. Four balls are drawn at random all at once, without replacement. Let $X =$ ``the number of red balls drawn.''

- (1) Write the possible values of X .
- (2) Find $P(X = 2)$.
- (3) Write the distribution table of X .

Solution:

- (1) Since 4 balls are drawn, the possible numbers of red balls are 0, 1, 2, 3, 4.
- (2) The total number of ways to draw 4 balls is C_{12}^4 . If exactly 2 red balls are drawn, choose 2 from the 5 red balls and 2 from the 7 blue balls. Thus

$$P(X = 2) = \frac{C_5^2 C_7^2}{C_{12}^4}.$$

- (3) The distribution table is

X	0	1	2	3	4
$P(X)$	$\frac{C_5^0 C_7^4}{C_{12}^4}$	$\frac{C_5^1 C_7^3}{C_{12}^4}$	$\frac{C_5^2 C_7^2}{C_{12}^4}$	$\frac{C_5^3 C_7^1}{C_{12}^4}$	$\frac{C_5^4 C_7^0}{C_{12}^4}$

Example 10.12 Hypergeometric Distribution: Defective Products

A box contains 20 products, among which 3 are defective. Five products are randomly inspected without replacement. Find the probability that exactly 1 defective product is found, and the probability that at least 1 defective product is found.

Solution: Let $X =$ "the number of defective products found." This is a hypergeometric distribution with $N = 20, M = 3, n = 5$.

Exactly 1 defective product:

$$P(X = 1) = \frac{C_3^1 C_{17}^4}{C_{20}^5}.$$

For at least 1 defective product, direct calculation would require adding $X = 1, 2, 3$. It is shorter to use the complement. The complement is "no defective products," so

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{C_3^0 C_{17}^5}{C_{20}^5}.$$

10.4.6 Geometric Distribution

Again consider sampling with replacement. Now we no longer ask, as in the binomial distribution, "if we draw n times, what is the probability of drawing k red balls?" Instead, we ask: "what is the probability that the first red ball appears on the k -th draw?" This leads to the geometric distribution.

For example, suppose a bag contains 3 red balls and 2 blue balls. Each time, draw one ball at random, record its color, and put it back. If "drawing a red ball" is called success, then the success probability each time is

$$p = \frac{3}{5}.$$

Let $X =$ "the number of draws needed until the first red ball appears." Then

$$X = 1, 2, 3, \dots$$

are all possible. Here $X = 4$ means that the first three draws are not red, and the fourth draw is the first red ball.

In general, suppose each trial has success probability p and failure probability $1 - p$, and the trials are independent. Let $X =$ "the trial number on which the first success appears." Then

$$P(X = k) = (1 - p)^{k-1} p, \quad k = 1, 2, 3, \dots$$

This is called the **geometric distribution (几何分布)**.

In the formula, $(1 - p)^{k-1}$ means that the first $k - 1$ trials all fail, and the final factor p means that the k -th trial succeeds:

$$\underbrace{F, F, \dots, F}_{k-1 \text{ failures}}, S.$$

Example 10.13 Geometric Distribution: First Red Ball

A bag contains 3 red balls and 2 blue balls. Each time, draw one ball at random, record its color, and put it back. Let $X =$ ``the number of draws needed until the first red ball appears.''

- (1) Find $P(X = 1)$.
- (2) Find $P(X = 4)$.
- (3) Find the probability of drawing at least one red ball within the first three draws.

Solution: The probability of drawing a red ball each time is

$$p = \frac{3}{5},$$

and the probability of not drawing a red ball, that is, drawing a blue ball, is

$$1 - p = \frac{2}{5}.$$

- (1) A red ball appears on the first draw:

$$P(X = 1) = \frac{3}{5}.$$

- (2) The first red ball appears on the 4-th draw, meaning that the first three draws are blue and the fourth draw is red:

$$P(X = 4) = \left(\frac{2}{5}\right)^3 \cdot \frac{3}{5}.$$

- (3) ``At least one red ball within the first three draws'' is equivalent to ``the first red ball appears on the first, second, or third draw.'' The complement is faster: the first three draws are all blue. Thus

$$P(X \leq 3) = 1 - \left(\frac{2}{5}\right)^3.$$

Example 10.14 Geometric Distribution: First Qualified Product

A machine produces qualified parts with probability 0.98. Each inspection is independent. Let $X =$ ``the number of inspections needed until the first qualified product appears.'' Find $P(X = 3)$.

Solution: The first qualified product appears on the 3-rd inspection, which means that the first two products are not qualified and the third product is qualified. Therefore

$$P(X = 3) = (0.02)^2 \cdot 0.98.$$

We use the geometric distribution because the inspections can be regarded as independent, and the probability of a qualified product is the same each time.

Common Mistake

To distinguish the binomial, hypergeometric, and geometric distributions, look at what the problem is asking. The binomial distribution usually corresponds to "with replacement or independent repeated trials, a fixed number n of trials, and counting successes." The hypergeometric distribution usually corresponds to "without replacement, a fixed number n of objects drawn, and counting successes." The geometric distribution usually corresponds to "independent repeated trials until the first success."

In-Class Practice 10.4.1 Classical Probability and Distributions

- (1) Roll a die. Let $X = 1$ mean "the number is greater than 4," and otherwise let $X = 0$. Write the distribution table of X .
- (2) A bag contains 2 red balls and 3 blue balls. Each draw is followed by replacement, and 4 draws are made. Let $X =$ "the number of red balls drawn." Write $P(X = 0)$, $P(X = 1)$, $P(X = 2)$.
- (3) A bag contains 4 red balls and 6 blue balls. Three balls are drawn without replacement. Let $X =$ "the number of red balls." Write $P(X = 0)$, $P(X = 1)$, $P(X = 2)$, $P(X = 3)$.
- (4) A bag contains 2 red balls and 3 blue balls. Each draw is followed by replacement. Let $Y =$ "the number of draws needed until the first red ball appears." Write $P(Y = 1)$, $P(Y = 2)$, $P(Y = 3)$.
- (5) From a standard deck without jokers, 5 cards are drawn at random. Write an expression for the probability that exactly 2 of them are hearts.

10.5 Intersection, Union, Complement, and Mutual Exclusiveness of Events

Complicated probabilities are often first translated into set operations. "And" corresponds to intersection, "or" corresponds to union, and "not" corresponds to complement.

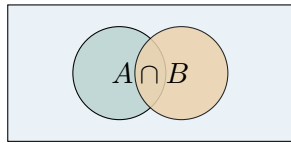
Many earlier problems can be solved by directly counting the outcomes that satisfy a condition. But once the condition becomes slightly more complicated, direct counting can easily become confusing. For example, choose one integer at random from 1 to 30, and find the probability that it is a multiple of 2 or a multiple of 3. If we simply add the number of multiples of 2 and the number of multiples of 3, then 6, 12, 18, 24, 30 are counted twice. As another example, "at least one heads" is troublesome if we split it into exactly one heads, exactly two heads, and so on, but its opposite event "all tails" is simple.

Therefore we need to treat events as sets and operate on them. "And" corresponds to intersection, "or" corresponds to union, and "does not occur" corresponds to complement. The purpose is to split, combine, or transform complicated conditions so that we avoid omissions and repeated counting.

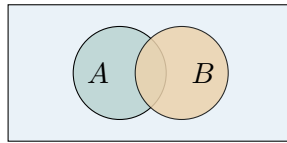
10.5.1 Intersection, Union, and Complement

Let A, B be two events in the same sample space Ω .

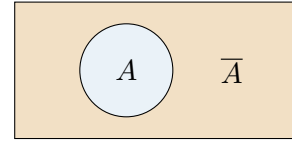
- $A \cap B$ means "A and B occur at the same time."
- $A \cup B$ means "A or B occurs," including only A, only B, or both.
- $\bar{A}$ means "A does not occur," and is called the opposite event or complement of A.



Intersection: both occur



Union: at least one occurs



Complement: does not occur

10.5.2 Mutually Exclusive Events and the Addition Formula

If $A \cap B = \emptyset$, then A and B cannot occur at the same time. They are called **mutually exclusive events** (互斥事件).

In general,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If A, B are mutually exclusive, then $P(A \cap B) = 0$, so

$$P(A \cup B) = P(A) + P(B).$$

For complements,

$$P(\bar{A}) = 1 - P(A).$$

Example 10.15 Using the Complement to Simplify an "At Least" Problem

A fair coin is tossed 5 times. Find the probability that at least one heads appears.

Solution: If we calculate "at least one heads" directly, we need to consider exactly one

heads, exactly two heads, exactly three heads, exactly four heads, and exactly five heads. Its complement is much simpler: no heads at all, that is, all 5 tosses are tails.

$$P(\text{all tails}) = \left(\frac{1}{2}\right)^5 = \frac{1}{32}.$$

Therefore

$$P(\text{at least one heads}) = 1 - \frac{1}{32} = \frac{31}{32}.$$

Example 10.16 Using the Union Formula for "Or"

Choose one integer at random from 1 to 30. Find the probability that it is a multiple of 2 or a multiple of 3.

Solution: Let $A =$ "the number is a multiple of 2," and $B =$ "the number is a multiple of 3." From 1 to 30,

$$|A| = 15, \quad |B| = 10.$$

Being a multiple of both 2 and 3 means being a multiple of 6:

$$|A \cap B| = 5.$$

Thus

$$|A \cup B| = 15 + 10 - 5 = 20, \quad P(A \cup B) = \frac{20}{30} = \frac{2}{3}.$$

Example 10.17 Using the Complement in a Ball-Drawing Problem

A bag contains 4 red balls and 6 blue balls. Three balls are drawn at once without replacement. Find the probability that at least one red ball is drawn.

Solution: Directly calculating "at least one red ball" requires splitting into one red ball, two red balls, and three red balls. Its complement is "no red ball," that is, all three balls are blue. The total number of draws is C_{10}^3 , and the number of all-blue draws is C_6^3 . Therefore

$$P(\text{at least one red ball}) = 1 - P(\text{all blue}) = 1 - \frac{C_6^3}{C_{10}^3}.$$

Common Mistake

Mutually exclusive and independent do not mean the same thing. Mutually exclusive means "cannot occur at the same time." Independent means "whether one event occurs does not change the probability of the other event." Two mutually exclusive events with positive probabilities cannot be independent.

In-Class Practice 10.5.1 Event Operations

- (1) Roll two dice. Find the probability that the sum is not equal to 7.
- (2) Roll a die four times. Find the probability that at least one 6 appears.
- (3) Draw one card from a standard deck. Find the probability of drawing a spade or a face card.
- (4) If $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B) = 0.2$, find $P(A \cup B)$.

10.6 Conditional Probability and Independence

The word “given” changes the range we are looking at. Independence asks whether this change affects the probability of another event.

Probability judgments in real life often come with extra information. For example, choose one student at random from a class. The probability that this student is a boy may be different from the probability that this student is a boy given that the student wears glasses. In the second question, we already know that the selected student lies within the group of students who wear glasses, so students who do not wear glasses are no longer part of the comparison.

For example, suppose a class has 12 boys and 8 girls. Among them, 6 boys wear glasses and 2 girls wear glasses. If one student is chosen at random, the probability that the student is a boy is $\frac{12}{12+8} = \frac{3}{5}$. But if we know that the selected student wears glasses, then the probability that this student is a boy becomes $\frac{6}{6+2} = \frac{3}{4}$. The condition of wearing glasses provides information, so the probability changes.

This is the motivation for conditional probability: whenever a problem says “given that ...” or “under the condition that ...,” the sample space usually shrinks. We no longer take a ratio inside all outcomes; instead, we take a ratio inside the new range determined by the given event.

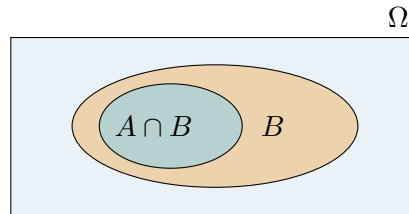
10.6.1 Conditional Probability

After we know that B has occurred, all outcomes not in B are no longer considered. The new range is B , and we look at the proportion of A inside B .

When $P(B) > 0$, define

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

This is read as "the probability that A occurs given that B has occurred."



Given B , recompute the proportion only inside B

Example 10.18 Conditional Probability: Shrinking the Sample Space

Choose one integer at random from 1 to 20. Given that the chosen number is even, find the probability that it is also a multiple of 3.

Solution: Given that the number is even, the new sample space is

$$\{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\},$$

which has 10 outcomes. Among them, the multiples of 3 are

$$\{6, 12, 18\},$$

so there are 3 favorable outcomes. Therefore the conditional probability is

$$\frac{3}{10}.$$

Example 10.19 Conditional Probability: Drawing Balls

A bag contains 3 red balls and 2 blue balls. Two balls are drawn consecutively without replacement. Given that the first ball is red, find the probability that the second ball is also red.

Solution: After the first ball is known to be red, the bag contains 2 red balls and 2 blue balls, for a total of 4 balls. Thus the probability that the second ball is red is

$$\frac{2}{4} = \frac{1}{2}.$$

The key is not to mechanically apply a formula, but to describe clearly the state of the bag after the condition "the first ball is red" is known.

10.6.2 Independent Events

If knowing that B has occurred does not change the probability that A occurs, namely

$$P(A | B) = P(A),$$

then we say that A and B are **independent** (独立). Equivalently, we often use the following multiplication form to judge independence:

$$P(A \cap B) = P(A)P(B).$$

Example 10.20 Independence: Tossing Two Coins

Toss two fair coins. Let $A =$ "the first coin is heads," and $B =$ "the second coin is heads." Decide whether A and B are independent.

Solution: The sample space is

$$\{HH, HT, TH, TT\}.$$

We have

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{2}, \quad P(A \cap B) = P(\{HH\}) = \frac{1}{4}.$$

Since

$$P(A)P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = P(A \cap B),$$

A and B are independent.

Example 10.21 Not Independent: Drawing Balls Without Replacement

A bag contains 3 red balls and 2 blue balls. Two balls are drawn consecutively without replacement. Let $A =$ "the first ball is red," and $B =$ "the second ball is red." Decide whether A and B are independent.

Solution:

$$P(B) = \frac{3}{5},$$

because without any other information, the probability that the second ball is red is the same as the initial proportion of red balls. If we know that the first ball is red, then the bag contains 2 red balls and 2 blue balls, so

$$P(B | A) = \frac{1}{2}.$$

Since

$$P(B | A) \neq P(B),$$

A and B are not independent.

Example 10.22 Mutually Exclusive but Not Independent

Roll one die. Let $A =$ "the number is 1," and $B =$ "the number is 2." Decide whether A and B are mutually exclusive and whether they are independent.

Solution: A and B cannot occur at the same time, so they are mutually exclusive:

$$A \cap B = \emptyset, \quad P(A \cap B) = 0.$$

But

$$P(A)P(B) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36} \neq 0.$$

Therefore A and B are not independent. This example shows that mutual exclusiveness is not

independence.

In-Class Practice 10.6.1 Conditional Probability and Independence

- (1) Given $P(A) = 0.6$, $P(B) = 0.5$, $P(A \cap B) = 0.3$, decide whether A and B are independent.
- (2) Two fair coins are tossed independently. Let $A =$ "the first coin is heads," and $B =$ "exactly one coin is heads." Decide whether A and B are independent.
- (3) A class has 12 boys and 8 girls. Two students are chosen at random without replacement. Given that the first student is a girl, find the probability that the second student is also a girl.
- (4) A part has qualified rate 0.98. Three parts are inspected independently. Find the probability that all are qualified, and the probability that at least one is not qualified.

10.7 Expectation and Variance

Once a distribution table is known, expectation describes the long-term average level, while variance describes how spread out the values are around that average.

Knowing only the probability of one event is sometimes not enough. For example, the score distributions of three shooters in a shooting contest are:

X_A	5	7	10
P	0.3	0.1	0.6

X_B	8	9	10
P	0.2	0.6	0.2

X_C	6	9	10
P	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$

If we only look at the probability of scoring 10, shooter A seems best, shooter C second, and shooter B worst. But this does not fully tell us who is more suitable for a competition, because shooter A also has a fairly large probability of scoring only 5 or 7. Instead of looking only at random outcomes one by one, we want definite numerical quantities to describe the shooters, such as average score and stability.

The average score can be computed by multiplying each score by its probability and adding. This is called expectation:

$$E(X_A) = 5 \cdot 0.3 + 7 \cdot 0.1 + 10 \cdot 0.6 = 8.2,$$

$$E(X_B) = 8 \cdot 0.2 + 9 \cdot 0.6 + 10 \cdot 0.2 = 9,$$

$$E(X_C) = 6 \cdot \frac{1}{6} + 9 \cdot \frac{1}{3} + 10 \cdot \frac{1}{2} = 9.$$

We see that although shooter A has the highest probability of scoring 10, his average score is the lowest. In the long run, A performs worse than B and C on average.

Shooters B and C have the same average score, so how can we compare them? We may also compare stability. From the distribution tables, B's probabilities are concentrated near 9, while C's scores are more spread out. Mathematically, we use variance to describe this observation:

$$D(X_A) = (5 - 8.2)^2 \cdot 0.3 + (7 - 8.2)^2 \cdot 0.1 + (10 - 8.2)^2 \cdot 0.6 = 5.16,$$

$$D(X_B) = (8 - 9)^2 \cdot 0.2 + (9 - 9)^2 \cdot 0.6 + (10 - 9)^2 \cdot 0.2 = 0.4,$$

$$D(X_C) = (6 - 9)^2 \cdot \frac{1}{6} + (9 - 9)^2 \cdot \frac{1}{3} + (10 - 9)^2 \cdot \frac{1}{2} = 2.$$

Therefore, although shooter A has the highest probability of scoring 10, his long-term average score is the lowest and his fluctuation is the largest. Shooters B and C both have expectation 9, so their average levels are the same, but B's variance 0.4 is smaller than C's variance 2, so B is more stable. Among the three shooters, if both average level and stability are considered, shooter B should be chosen.

This analysis shows how to extract definite numerical features, such as expectation and variance, from the distribution table of a random variable. We now introduce these ideas more formally.

10.7.1 Expectation

The **expectation** (期望) of a random variable can be understood as the "average value after a very large number of repeated trials." If

X	x_1	x_2	$\cdots$	x_n
P	p_1	p_2	$\cdots$	p_n

then

$$E(X) = x_1 p_1 + x_2 p_2 + \cdots + x_n p_n.$$

Expectation does not mean that the next trial must take this value. It says that if the same random experiment is repeated many, many times, and the values of X are averaged, that average will stabilize near the expectation.

Example 10.23 Expectation: Number on a Die

Roll a fair die. Let X = "the number shown." Find $E(X)$.

Solution: The distribution table of X is

X	1	2	3	4	5	6
P	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Therefore

$$E(X) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + \cdots + 6 \cdot \frac{1}{6} = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5.$$

A single die roll can never give 3.5, but the long-term average number will be close to 3.5.

Example 10.24 Expectation: Bernoulli Distribution

Suppose X follows a Bernoulli distribution with success probability p . Find $E(X)$.

Solution: The Bernoulli distribution is

X	0	1
P	$1 - p$	p

so

$$E(X) = 0 \cdot (1 - p) + 1 \cdot p = p.$$

This shows that the expectation of a 0-1 random variable is exactly its success probability.

10.7.2 Variance

Two random variables may have the same expectation but different stability. Variance describes how spread out the values are around the expectation.

If the expectation of random variable X is $\mu = E(X)$, then

$$D(X) = E((X - \mu)^2).$$

That is,

$$D(X) = (x_1 - \mu)^2 p_1 + (x_2 - \mu)^2 p_2 + \cdots + (x_n - \mu)^2 p_n.$$

A larger variance means that the values of the random variable are usually farther from the average level. A smaller variance means that the values are more concentrated.

Example 10.25 Computing Expectation and Variance from a Distribution Table

The distribution table of random variable X is

X	0	2	5
P	0.2	0.5	0.3

Find $E(X)$ and $D(X)$.

Solution: First compute the expectation:

$$E(X) = 0 \cdot 0.2 + 2 \cdot 0.5 + 5 \cdot 0.3 = 2.5.$$

Then use 2.5 as the average level and compute the squared distance from each value to the average level:

$$D(X) = (0 - 2.5)^2 \cdot 0.2 + (2 - 2.5)^2 \cdot 0.5 + (5 - 2.5)^2 \cdot 0.3.$$

Thus

$$D(X) = 6.25 \cdot 0.2 + 0.25 \cdot 0.5 + 6.25 \cdot 0.3 = 3.25.$$

Example 10.26 First Write the Distribution Table, Then Compute Expectation and Variance

A bag contains 2 red balls and 1 blue ball. Two balls are drawn at random all at once, without replacement. Let X = "the number of red balls drawn." Write the distribution table of X , and find $E(X), D(X)$.

Solution: The total number of draws is

$$C_3^2 = 3.$$

When 2 balls are drawn, the number of red balls can only be 1 or 2. If $X = 1$, we need to draw 1 red ball and 1 blue ball:

$$P(X = 1) = \frac{C_2^1 C_1^1}{C_3^2} = \frac{2}{3}.$$

If $X = 2$, we need to draw 2 red balls:

$$P(X = 2) = \frac{C_2^2 C_1^0}{C_3^2} = \frac{1}{3}.$$

Therefore the distribution table is

X	1	2
P	$\frac{2}{3}$	$\frac{1}{3}$

The expectation is

$$E(X) = 1 \cdot \frac{2}{3} + 2 \cdot \frac{1}{3} = \frac{4}{3}.$$

The variance is

$$D(X) = \left(1 - \frac{4}{3}\right)^2 \cdot \frac{2}{3} + \left(2 - \frac{4}{3}\right)^2 \cdot \frac{1}{3} = \frac{2}{9}.$$

Example 10.27 Same Expectation, Different Variance

In game A, X takes the values 4 and 6 with equal probability. In game B, Y takes the values 0 and 10 with equal probability. Compare their expectations and spreads.

Solution: The two expectations are

$$E(X) = 4 \cdot \frac{1}{2} + 6 \cdot \frac{1}{2} = 5, \quad E(Y) = 0 \cdot \frac{1}{2} + 10 \cdot \frac{1}{2} = 5.$$

Although both expectations are 5, the results of game B are farther from 5. Compute the variances:

$$D(X) = (4 - 5)^2 \cdot \frac{1}{2} + (6 - 5)^2 \cdot \frac{1}{2} = 1,$$

$$D(Y) = (0 - 5)^2 \cdot \frac{1}{2} + (10 - 5)^2 \cdot \frac{1}{2} = 25.$$

Therefore Y fluctuates more.

Example 10.28 Meaning of the Expectation of a Hypergeometric Distribution

A box contains 20 products, among which 3 are defective. Five products are randomly inspected. Let $X =$ "the number of defective products found." Without computing the full distribution table, first explain the meaning of $E(X)$.

Solution: $E(X)$ means the following: if we repeatedly inspect 5 products from boxes with the same structure, and each time record the number of defective products found, then the long-term average of these recorded numbers is $E(X)$.

This does not mean that one inspection must find exactly $E(X)$ defective products. It means that after many repetitions, the average level will stabilize near this number.

In-Class Practice 10.7.1 Distribution Tables, Expectation, and Variance

- (1) The distribution table of a random variable X is

X	0	1	2
P	0.3	a	0.2

Find a .

- (2) For the distribution table in the previous problem, find $E(X)$.
- (3) Roll a fair die. Let $X = 1$ mean that the number is 6, and otherwise let $X = 0$. Find $E(X)$.
- (4) Random variable A takes the values 2 and 8 with equal probability, while random variable B is always equal to 5. Compare their expectations and variances.

10.8 Chapter Summary

Main Ideas

- (1) The result of a single random experiment cannot be determined in advance, but all possible results should be described clearly.
- (2) A sample point is a most basic outcome; the sample space is the set of all sample points; an event is a subset of the sample space.
- (3) Use addition for categories and multiplication for steps. Permutations care about order; combinations do not.
- (4) Classical probability requires finitely many equally likely outcomes, and is computed as favorable outcomes divided by total outcomes.
- (5) The Bernoulli distribution describes one success-or-failure trial; the binomial distribution describes the number of successes in independent repeated trials; the hypergeometric distribution describes the number of successes in sampling without replacement; the geometric distribution describes the trial number of the first success.
- (6) "And," "or," and "not" correspond to intersection, union, and complement, respectively. "At least" is often simplified by using the complement.
- (7) Conditional probability shrinks the sample space under a given condition. Independence means that one event does not change the probability of another event.
- (8) Expectation describes the long-term average level, and variance describes the spread around that average.

Formula List

$$A_n^m = \frac{n!}{(n-m)!}, \quad C_n^m = \binom{n}{m} = \frac{n!}{m!(n-m)!},$$

$$P(A) = \frac{|A|}{|\Omega|}, \quad P(\bar{A}) = 1 - P(A),$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B), \quad P(A | B) = \frac{P(A \cap B)}{P(B)},$$

$$P(A \cap B) = P(A)P(B) \quad (A, B \text{ independent}),$$

$$P(X = k) = C_n^k p^k (1-p)^{n-k} \quad (k = 0, 1, \dots, n; \text{ binomial}),$$

$$P(X = k) = \frac{C_M^k C_{N-M}^{n-k}}{C_N^n} \quad (\text{hypergeometric}),$$

$$P(X = k) = (1-p)^{k-1} p \quad (k = 1, 2, 3, \dots; \text{ geometric}),$$

$$E(X) = \sum x_i p_i, \quad D(X) = \sum (x_i - E(X))^2 p_i.$$

10.9 Chapter Exercises

How to Use the Exercises

Group A checks the minimum required mastery of this chapter. If you make mistakes, return to the corresponding subsection and reread the examples. Group B is for integrated practice before a quiz. Group C contains extension problems; during a first self-study pass, you may skip them without affecting the main line of the chapter.

Group A: Basic Practice

Difficulty: Basic

- (1) Toss a coin twice. Write the sample space and the event “exactly one heads.”
- (2) Roll two dice. Find the probability that the sum is 8.
- (3) How many three-digit numbers with no repeated digits can be formed using 1, 2, 3, 4, 5?
- (4) In how many ways can 3 people be chosen from 8 people?
- (5) Choose one integer at random from 1 to 50. Find the probability that it is a multiple of 5.
- (6) Draw one card from a standard deck. Find the probability of drawing a red card or a king.
- (7) Toss a coin 4 times. Find the probability that at least one tails appears.
- (8) A bag contains 4 red balls and 6 blue balls. Two balls are drawn without replacement. Find the probability that both are red.
- (9) Roll a die. Let $X = 1$ mean that the number is even, and otherwise let $X = 0$. Write the distribution table of X .
- (10) A bag contains 2 red balls and 3 blue balls. Each draw is followed by replacement, and 4 draws are made. Find the probability that exactly 2 red balls are drawn.
- (11) A bag contains 2 red balls and 3 blue balls. Each draw is followed by replacement. Find the probability that the first red ball appears exactly on the 3-rd draw.
- (12) If $P(A) = 0.7$, $P(B) = 0.4$, $P(A \cap B) = 0.2$, find $P(A \cup B)$.

Group B: Quiz Level

Difficulty: Core

- (1) How many three-digit even numbers with no repeated digits can be formed using 0, 1, 2, 3, 4, 5?
- (2) From 6 boys and 4 girls, choose 3 people. Find the probability that at least one girl is chosen.
- (3) Roll two dice at the same time. Find the probability that the product of the two numbers is even.
- (4) A bag contains 5 qualified products and 2 defective products. Three products are drawn without replacement. Find the probability that exactly one is defective.
- (5) A box contains balls numbered from 1 to 10. Given that the number drawn is greater than 4, find the conditional probability that it is even.
- (6) An event is repeated independently three times, with success probability p each time. Use the binomial distribution to write the probability of exactly two successes and the probability of at least one success.
- (7) If two events are mutually exclusive and both have positive probabilities, can they be independent? Explain why.
- (8) A shooter hits the target with probability 0.8 each time, and the shots are independent. Find the probability that the first hit appears on the 4-th shot, and the probability of at least one hit within the first 4 shots.
- (9) The distribution table of random variable X is

X	-1	0	2
P	0.2	0.5	0.3

Find $E(X)$ and $D(X)$.

- (10) A bag contains 6 red balls and 4 blue balls. Three balls are drawn without replacement. Let $X =$ "the number of red balls." Write the distribution table of X .

Group C: Extension and Thinking

Difficulty: Extension

- (1) Every student chooses at least one of three courses. Use set language to explain how to count the number of students who choose at least two courses.
- (2) Put n different balls randomly into m different boxes, allowing empty boxes. How many outcomes are in the sample space?
- (3) For birthdays, assume a year has 365 days and every day is equally likely. Write an expression for the probability that in a group of n people, at least two people have the same birthday.
- (4) Given the sensitivity and false-positive rate of a test, try to use a tree diagram to represent the four cases "diseased/not diseased" and "positive/negative," and explain why $P(\text{diseased} \mid \text{positive})$ is not the same as $P(\text{positive} \mid \text{diseased})$.
- (5) Compare the binomial, hypergeometric, and geometric distributions. For each one, explain whether it corresponds to sampling with replacement or without replacement, and whether it has a fixed number of trials or continues until the first success.

Full-Book Self-Check

For each item, choose "Yes," "Mostly," or "Not Yet," and use your choices to plan your review order.

Skill	Yes	Mostly	Not Yet
List a sample space and represent events	☐	☐	☐
Use the addition principle and multiplication principle	☐	☐	☐
Distinguish permutations and combinations	☐	☐	☐
Use the classical probability formula	☐	☐	☐
Recognize Bernoulli, binomial, hypergeometric, and geometric distributions	☐	☐	☐
Use intersections, unions, and complements to simplify probability calculations	☐	☐	☐
Compute conditional probabilities and judge independence	☐	☐	☐
Write simple distribution tables and compute expectation and variance	☐	☐	☐
Use algebra, function language, and probability language together	☐	☐	☐

After Finishing the Whole Book

Review the self-check at the end of each chapter. First review the basic items marked ``Not Yet," then complete the corresponding Group A exercises. After the basics are stable, work on Group B, and finally use Group C to train integrated thinking. The key to learning mathematics is not understanding everything once, but being able to complete problems independently without prompts.

Chinese-English Mathematical Vocabulary

This appendix organizes common Chinese and English mathematical vocabulary used in this book by topic. In English textbooks, the same concept may have several names; this table prioritizes the most common expressions and the expressions most suitable for classroom use.

How to Use This Vocabulary List

This vocabulary list is for reading and lookup. It is not recommended to memorize entire pages before studying the main text. After finishing each chapter, return to this appendix and check the core terms that appeared in that chapter. It is enough to match the English term with its Chinese meaning; you do not need to master every English expression at the beginning.

Do Not Only Memorize English Words: Connect Them Back to Concepts

When looking up a word, ask three questions at the same time: in which chapter does this word appear? What mathematical object does it refer to? Which formula or graph is it often used with? For example, domain is not only “定义域,” but also the set of all allowed inputs; slope is not only “斜率,” but also vertical change divided by horizontal change.

A.1 Mathematical Language and Common Notation

Chinese Term	English Term	Note
数学	mathematics	Often abbreviated as math or maths
定义	definition	An exact explanation of a concept
定理	theorem	An important statement that has been proved
性质	property	A rule or feature of an object
公式	formula	A relation expressed with symbols
例子	example	Used to illustrate a concept or method
反例	counterexample	An example used to disprove a false statement
证明	proof	Deriving a conclusion from known conditions
计算	calculation	Numerical or symbolic operation according to rules
化简	simplify	Rewrite an expression in a simpler equivalent form

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Chinese Term	English Term	Note
求值	evaluate	Compute the value of an expression or function
解	solution	A value that satisfies an equation or condition
条件	condition	A restriction given in a problem
结论	conclusion	A result obtained from conditions
等于	equal to	$=$
不等于	not equal to	$\neq$
大于	greater than	$>$
小于	less than	$<$
大于等于	greater than or equal to	$\geq$
小于等于	less than or equal to	$\leq$
约等于	approximately equal to	$\approx$
因此	therefore	$\therefore$
因为	because	$\because$
任意的	arbitrary / any	True for every object in the range
存在	there exists	$\exists$

A.2 Numbers, Operations, and Algebraic Expressions

Chinese Term	English Term	Note
自然数	natural number	$\mathbb{N}$
整数	integer	$\mathbb{Z}$
正整数	positive integer	$1, 2, 3, \dots$
负整数	negative integer	$-1, -2, -3, \dots$
有理数	rational number	$\mathbb{Q}$
无理数	irrational number	Cannot be written as a ratio of two integers
实数	real number	$\mathbb{R}$
复数	complex number	$\mathbb{C}$
正数	positive number	$x > 0$
负数	negative number	$x < 0$
零	zero	0
相反数	opposite / additive inverse	The opposite of a is $-a$
倒数	reciprocal	The reciprocal of a is $1/a$

continued on next page

Chinese Term	English Term	Note
绝对值	absolute value	$ a $
数轴	number line	A line used to represent real numbers
加法	addition	$a + b$
减法	subtraction	$a - b$
乘法	multiplication	ab or $a \times b$
除法	division	a/b or $a \div b$
和	sum	Result of addition
差	difference	Result of subtraction
积	product	Result of multiplication
商	quotient	Result of division
分数	fraction	$\frac{a}{b}$
分子	numerator	Number above the fraction bar
分母	denominator	Number below the fraction bar
小数	decimal	For example, 0.25
百分数	percentage	For example, 25%

A.3 Algebraic Expressions, Equations, and Inequalities

Chinese Term	English Term	Note
变量	variable	Often represented by x, y, t
常数	constant	A quantity with fixed value
未知数	unknown	A quantity to be found in an equation
代数式	algebraic expression	Made of numbers, letters, and operations
项	term	Parts of a polynomial separated by plus or minus signs
系数	coefficient	Numerical factor in front of a letter term
次数	degree	Exponent or highest degree of a polynomial
单项式	monomial	A polynomial with only one term
多项式	polynomial	A sum of monomials
同类项	like terms	Terms with the same letters and exponents
展开	expand	Remove parentheses and write as a polynomial
因式	factor	Each factor in a product
因式分解	factorization	Write a polynomial as a product of factors

continued on next page

Chinese Term	English Term	Note
公因式	common factor	A factor shared by all terms of a polynomial
恒等式	identity	True for all allowed variable values
分式	rational expression	An expression whose denominator contains letters
约分	reduce / cancel	Cancel common factors in numerator and denominator
通分	find a common denominator	Rewrite denominators in the same form
方程	equation	An equality containing unknowns
方程组	system of equations	Several equations that must hold simultaneously
一元一次方程	linear equation in one variable	$ax + b = 0$
一元二次方程	quadratic equation	$ax^2 + bx + c = 0$
根	root	A solution of an equation
重根	repeated root	Two equal roots
判别式	discriminant	$\Delta = b^2 - 4ac$
配方法	completing the square	Rewrite into square form
求根公式	quadratic formula	General formula for quadratic roots
消元法	elimination method	Eliminate an unknown from a system
换元法	substitution method	Replace a complicated structure with a new variable
不等式	inequality	A relation containing $>$, $<$, $\geq$, $\leq$

A.4 Sets, Logic, and Coordinates

Chinese Term	English Term	Note
集合	set	A collection of definite objects
元素	element	$x \in A$
属于	belongs to	$\in$
不属于	does not belong to	$\notin$
空集	empty set	$\emptyset$
有限集	finite set	A set with finitely many elements
无限集	infinite set	A set with infinitely many elements
子集	subset	$A \subseteq B$

continued on next page

Chinese Term	English Term	Note
真子集	proper subset	$A \subsetneq B$
全集	universal set	All elements in the range of discussion
交集	intersection	$A \cap B$
并集	union	$A \cup B$
补集	complement	A^c
区间	interval	A continuous set on the number line
开区间	open interval	(a, b)
闭区间	closed interval	$[a, b]$
命题	proposition / statement	A declarative sentence that is true or false
真命题	true statement	A statement judged true
假命题	false statement	A statement judged false
且	and	$\wedge$
或	or	$\vee$
非	not / negation	$\neg$
充分条件	sufficient condition	$A \text{ in } A \Rightarrow B$
必要条件	necessary condition	$B \text{ in } A \Rightarrow B$
充要条件	necessary and sufficient condition	$\Longleftrightarrow$
坐标系	coordinate system	Determines positions of points by coordinates
横坐标	x -coordinate / abscissa	First coordinate of (x, y)
纵坐标	y -coordinate / ordinate	Second coordinate of (x, y)

A.5 Functions and Graphs

Common Mistake

Function vocabulary is easy to confuse: domain is the set of allowed inputs, range is the set of actual outputs; a zero is an input that makes the function value 0, not the y -coordinate of a point; maximum/minimum refers to function values, while the point where a maximum or minimum is attained should be written as coordinates.

Chinese Term	English Term	Note
函数	function	$y = f(x)$
自变量	independent variable	Input of a function
因变量	dependent variable	Output that changes with the input
定义域	domain	All allowed inputs

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Chinese Term	English Term	Note
值域	range	All possible outputs
函数值	function value	$f(a)$
对应关系	correspondence / mapping	Rule matching inputs with outputs
函数图像	graph of a function	Set of points $(x, f(x))$
描点	plot a point	Mark a coordinate point in a plane
截距	intercept	Corresponding coordinate of an intersection with an axis
横截距	x -intercept	x -coordinate of intersection with the x -axis
纵截距	y -intercept	y -coordinate of intersection with the y -axis
斜率	slope / gradient	$k = \frac{\Delta y}{\Delta x}$
一次函数	linear function	$y = kx + b$
二次函数	quadratic function	$y = ax^2 + bx + c$
抛物线	parabola	Graph of a quadratic function
顶点	vertex	Highest or lowest point of a parabola
对称轴	axis of symmetry	$x = -b/(2a)$ for a quadratic function
零点	zero	An input satisfying $f(x) = 0$
单调递增	increasing	Function values increase as inputs increase
单调递减	decreasing	Function values decrease as inputs increase
单调区间	interval of monotonicity	An interval on which a function keeps increasing or decreasing
最大值	maximum	Largest attainable function value
最小值	minimum	Smallest attainable function value
奇函数	odd function	$f(-x) = -f(x)$
偶函数	even function	$f(-x) = f(x)$
周期函数	periodic function	$f(x + T) = f(x)$
最小正周期	fundamental period	Smallest positive period
复合函数	composite function	$(f \circ g)(x) = f(g(x))$
反函数	inverse function	A function that swaps inputs and outputs

A.6 Power Functions, Exponents, and Logarithms

Chinese Term	English Term	Note
幂	power	a^n
底数	base	The a in a^n
指数	exponent	The n in a^n

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Chinese Term	English Term	Note
平方	square	a^2
立方	cube	a^3
平方根	square root	A solution of $x^2 = a$
算术平方根	principal square root	$\sqrt{a} \geq 0$
根式	radical expression	An expression containing a radical sign
幂函数	power function	$y = x^\alpha$
反比例函数	reciprocal function	$y = k/x$
分式函数	rational function	Ratio of two polynomials
渐近线	asymptote	A line that a graph approaches infinitely closely
水平渐近线	horizontal asymptote	A line of the form $y = c$
竖直渐近线	vertical asymptote	A line of the form $x = c$
对称中心	center of symmetry	Center for point symmetry of a graph
指数函数	exponential function	$y = a^x$
对数	logarithm	$\log_a b$
常用对数	common logarithm	$\lg x = \log_{10} x$
自然对数	natural logarithm	$\ln x = \log_e x$
对数函数	logarithmic function	$y = \log_a x$
换底公式	change-of-base formula	$\log_a b = \frac{\log_c b}{\log_c a}$

A.7 Trigonometric Functions

Chinese Term	English Term	Note
角	angle	Often represented by α, β, θ
锐角	acute angle	$0^\circ < \theta < 90^\circ$
直角	right angle	90°
钝角	obtuse angle	$90^\circ < \theta < 180^\circ$
角度制	degree measure	Measures angles in degrees
弧度制	radian measure	Measures angles by the ratio of arc length to radius
弧度	radian	rad
终边	terminal side	Ray after rotation
单位圆	unit circle	$x^2 + y^2 = 1$
象限	quadrant	Four regions of the coordinate plane
正弦	sine	$\sin \theta$

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Chinese Term	English Term	Note
余弦	cosine	$\cos \theta$
正切	tangent	$\tan \theta$
三角函数	trigonometric function	Sine, cosine, tangent, etc.
勾股定理	Pythagorean theorem	$a^2 + b^2 = c^2$
余弦定理	law of cosines	$c^2 = a^2 + b^2 - 2ab \cos C$
特殊角	special angle	Such as $30^\circ, 45^\circ, 60^\circ$
诱导公式	reduction formula	Change an arbitrary angle into a familiar angle
和角公式	angle-sum formula	For example, $\sin(\alpha + \beta)$
差角公式	angle-difference formula	For example, $\cos(\alpha - \beta)$
二倍角公式	double-angle formula	For example, $\sin 2x$
辅助角公式	auxiliary-angle formula	Combine $a \sin x + b \cos x$
振幅	amplitude	$ A $ in a sinusoidal function
相位	phase	$\omega x + \varphi$
相位移动	phase shift	$-\varphi/\omega$
中线	midline	$y = k$ in a sinusoidal function
轴对称	line symmetry	Symmetric about a line
中心对称	point symmetry	Symmetric about a point
弧长	arc length	$l = r\theta$
扇形面积	area of a sector	$S = \frac{1}{2}r^2\theta$

A.8 Vectors, Complex Numbers, and Analytic Geometry

Chinese Term	English Term	Note
向量	vector	A quantity with magnitude and direction
标量	scalar	A quantity with magnitude only
向量的模	magnitude / norm of a vector	$ \vec{a} $
单位向量	unit vector	A vector with magnitude 1
零向量	zero vector	$\vec{0}$
相等向量	equal vectors	Same magnitude and same direction
相反向量	opposite vectors	Same magnitude and opposite directions
向量加法	vector addition	$\vec{a} + \vec{b}$
向量减法	vector subtraction	$\vec{a} - \vec{b}$
数乘	scalar multiplication	$\lambda \vec{a}$

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Chinese Term	English Term	Note
线性组合	linear combination	$\lambda \vec{a} + \mu \vec{b}$
内积	dot product / inner product	$\vec{a} \cdot \vec{b}$
夹角	included angle	Angle between two vectors
平行	parallel	$\vec{a} \parallel \vec{b}$
垂直	perpendicular	$\vec{a} \perp \vec{b}$
虚数单位	imaginary unit	$i^2 = -1$
实部	real part	$\operatorname{Re} z$
虚部	imaginary part	$\operatorname{Im} z$
共轭复数	complex conjugate	$\bar{z}$
复数的模	modulus of a complex number	$ z $
直线	straight line	A set of points represented by a linear equation
点斜式	point-slope form	$y - y_0 = k(x - x_0)$
斜截式	slope-intercept form	$y = kx + b$
一般式	general form	$Ax + By + C = 0$
两点式	two-point form	A line determined by two points
点到直线距离	distance from a point to a line	Uses the distance formula
圆	circle	Set of points at a fixed distance from a fixed point
圆心	center of a circle	(a, b)
半径	radius	r
切线	tangent line	A line tangent to a circle or curve
圆锥曲线	conic section	Quadratic curves such as ellipse, hyperbola, parabola
椭圆	ellipse	Set of points whose sum of distances to two foci is constant
双曲线	hyperbola	Set of points whose absolute difference of distances to two foci is constant
抛物线	parabola	Set of points equidistant from a focus and a directrix
焦点	focus	Fixed point in a conic section
准线	directrix	Fixed line in the definition of a parabola
渐近线	asymptote	A line that a hyperbola approaches infinitely closely
半焦距	semi-focal distance	Distance c from a focus to the center

A.9 Sequences and Derivatives

Chinese Term	English Term	Note
数列	sequence	An ordered list of numbers
项	term of a sequence	a_n
首项	first term	a_1
末项	last term	Last term of a finite sequence
项数	number of terms	Often represented by n
通项公式	explicit formula / general term	Directly represents a_n
递推公式	recurrence relation	Represents later terms using earlier terms
等差数列	arithmetic sequence	Difference between adjacent terms is constant
公差	common difference	d
等差中项	arithmetic mean	$2b = a + c$
等比数列	geometric sequence	Ratio of adjacent terms is constant
公比	common ratio	q
等比中项	geometric mean	$b^2 = ac$
前 n 项和	sum of the first n terms	S_n
高斯求和法	Gauss summation method	Write forward and backward, then add
裂项相消法	telescoping method	Split terms so neighboring terms cancel
错位相减法	subtraction after shifting	Handles arithmetic terms times geometric terms
平均变化率	average rate of change	Difference quotient
割线	secant line	A line through two points on a curve
极限	limit	$\lim$
导数	derivative	$f'(x)$
可导	differentiable	The derivative exists
瞬时变化率	instantaneous rate of change	Practical meaning of derivative
切线斜率	slope of the tangent	$f'(x_0)$
法线	normal line	A line perpendicular to the tangent
乘积法则	product rule	$(uv)' = u'v + uv'$
商法则	quotient rule	$(u/v)' = (u'v - uv')/v^2$
链式法则	chain rule	Rule for differentiating composite functions
驻点	stationary point	A point where $f'(x) = 0$
极大值	local maximum	Local largest function value
极小值	local minimum	Local smallest function value

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Chinese Term	English Term	Note
极值	local extremum	Local maximum or local minimum
二阶导数	second derivative	$f''(x)$
凹凸性	concavity	Determined by the sign of f''
拐点	inflection point	A point where concavity changes

A.10 Counting and Probability

Common Mistake

Probability vocabulary should be understood together with the sample space. A sample point is the most basic outcome, and an event is a set of sample points. Mutually exclusive means two events cannot occur at the same time, while independent means whether one event occurs does not change the probability of the other; they are not synonyms.

Chinese Term	English Term	Note
随机试验	random experiment	An experiment whose result cannot be determined in advance
样本点	sample point / outcome	One possible result of a trial
样本空间	sample space	Ω
随机事件	random event	A subset of the sample space
必然事件	certain event	Ω
不可能事件	impossible event	$\emptyset$
对立事件	complementary event	A^c
互斥事件	mutually exclusive events	$A \cap B = \emptyset$
等可能	equally likely	Every basic outcome has the same chance
频率	frequency	Proportion of times an event occurs
概率	probability	$P(A)$
古典概率	classical probability	Favorable outcomes divided by total outcomes
列举法	systematic listing	List possible outcomes one by one
树状图	tree diagram	Shows all outcomes step by step
分类加法原理	addition principle	Count by categories and add
分步乘法原理	multiplication principle	Count by steps and multiply
阶乘	factorial	$n!$
排列	permutation	Selection where order matters
排列数	number of permutations	A_n^m
组合	combination	Selection where order does not matter

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Chinese Term	English Term	Note
组合数	binomial coefficient	C_n^m or $\binom{n}{m}$
有利结果	favorable outcome	Outcome that makes the event occur
条件概率	conditional probability	$P(A B)$
独立事件	independent events	$P(A \cap B) = P(A)P(B)$
加法公式	addition rule of probability	Used to compute the probability of a union

Tips for English Mathematical Wording

In English problems, solve usually means ``solve an equation or inequality," simplify means ``make an expression simpler," evaluate means ``find a value," determine means ``decide or find," and show that or prove that means ``give a proof." When reading a problem, identify the verb first, then decide what form of answer is required.

Formula Reference

This appendix collects formulas used repeatedly in the main text. It is for review and lookup. Before using a formula, still check domains, denominators, radicals, logarithm arguments, and parameter restrictions.

The Formula List Is Not the Starting Point

The formula list is suitable only for review and checking gaps. When learning a formula for the first time, return to the corresponding chapter and see which problem it comes from, what conditions it requires, and what kinds of problems it is designed to solve. If you only memorize formulas without checking restrictions, mistakes most often occur in rational expressions, radicals, logarithms, and derivative problems.

Review by Stage

After Chapters 1--2, first consolidate algebraic computation and equations. After Chapters 3--4, focus on function graphs and properties. After Chapters 5--6, review graphs and formulas of exponential, logarithmic, and trigonometric functions. After Chapters 7--10, work on integrated problems involving analytic geometry, sequences, derivatives, and probability.

Four Checks Before Using a Formula

After finding a formula, do not substitute immediately. Check four things in order: whether a denominator may be 0; whether the expression under a radical is nonnegative; whether the logarithm argument is positive and the base is valid; and whether the variables in the problem lie in the range allowed by the formula. If the formula comes from functions or probability, also check the domain, sample space, and equal likelihood.

B.1 Numbers and Algebraic Expressions

Topic	Formula	Conditions or Notes
Absolute value	$ a = \begin{cases} a, & a \geq 0, \\ -a, & a < 0 \end{cases}$	$ a \geq 0$
Distance between two points on a number line	$d(a, b) = a - b $	$a, b \in \mathbb{R}$
Fraction addition and subtraction	$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$	$b, d \neq 0$
Fraction multiplication	$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$	$b, d \neq 0$
Fraction division	$\frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$	$b, c, d \neq 0$

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Topic	Formula	Conditions or Notes
Product of powers with same base	$a^m a^n = a^{m+n}$	Expression is meaningful
Quotient of powers with same base	$\frac{a^m}{a^n} = a^{m-n}$	$a \neq 0$
Power of a power	$(a^m)^n = a^{mn}$	Expression is meaningful
Power of a product	$(ab)^n = a^n b^n$	Expression is meaningful
Zero exponent	$a^0 = 1$	$a \neq 0$
Negative integer exponent	$a^{-n} = \frac{1}{a^n}$	$a \neq 0$
Fractional exponent	$a^{m/n} = \sqrt[n]{a^m}$	Must be meaningful over the reals
Product rule for square roots	$\sqrt{a} \sqrt{b} = \sqrt{ab}$	$a, b \geq 0$
Quotient rule for square roots	$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$	$a \geq 0, b > 0$
Square of a sum	$(a + b)^2 = a^2 + 2ab + b^2$	Identity
Square of a difference	$(a - b)^2 = a^2 - 2ab + b^2$	Identity
Difference of squares	$(a + b)(a - b) = a^2 - b^2$	Identity
Sum of cubes	$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$	Identity
Difference of cubes	$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$	Identity
Completing the square	$x^2 + px = \left(x + \frac{p}{2}\right)^2 - \frac{p^2}{4}$	Identity
Proportion	$\frac{a}{b} = \frac{c}{d} \implies ad = bc$	$b, d \neq 0$
Percent change	new amount = original amount $(1 \pm r)$	Write r as a decimal
Scientific notation	$N = a \times 10^n$	$1 \leq a < 10$

B.2 Equations and Inequalities

Common Mistake

After an equation formula gives candidate solutions, substitute them back into the original equation when dealing with rational equations, radical equations, and substitution equations. Clearing denominators, squaring both sides, and reverting a substitution may introduce or lose solutions.

Topic	Formula	Conditions or Notes
Linear equation in one variable	$ax + b = 0 \implies x = -\frac{b}{a}$	$a \neq 0$

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Topic	Formula	Conditions or Notes
Zero-product principle	$AB = 0 \iff A = 0 \text{ or } B = 0$	Over real or complex numbers
Quadratic discriminant	$\Delta = b^2 - 4ac$	$a \neq 0$
Quadratic formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	Real roots when $\Delta \geq 0$
Relations between roots and coefficients	$x_1 + x_2 = -\frac{b}{a}, \quad x_1 x_2 = \frac{c}{a}$	$ax^2 + bx + c = 0$
Completing the square for a quadratic	$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 - \frac{\Delta}{4a}$	$a \neq 0$
Adding or subtracting the same amount in an equation	$a = b \implies a + c = b + c$	Any c
Multiplying or dividing by the same amount in an equation	$a = b \implies ac = bc$	For division, $c \neq 0$
Adding or subtracting the same amount in an inequality	$a < b \implies a + c < b + c$	Any c
Multiplying an inequality by a positive number	$a < b, \ c > 0 \implies ac < bc$	Direction unchanged
Multiplying an inequality by a negative number	$a < b, \ c < 0 \implies ac > bc$	Direction reversed
Sign of a quadratic expression	$a(x - x_1)(x - x_2)$	Judge by a and the zero intervals
Sign of a rational inequality	$\frac{P(x)}{Q(x)} \geq 0$	Zeros and undefined points split intervals together

B.3 Coordinates, Lines, and Basic Functions

Topic	Formula	Conditions or Notes
Distance between two points	$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	$A(x_1, y_1), B(x_2, y_2)$
Midpoint coordinates	$M \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$	Midpoint of segment AB
Slope of a line	$k = \frac{y_2 - y_1}{x_2 - x_1}$	$x_1 \neq x_2$
Linear function	$y = kx + b$	A linear function when $k \neq 0$
Point-slope form	$y - y_0 = k(x - x_0)$	Given a point and a slope

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Topic	Formula	Conditions or Notes
Parallel lines	$k_1 = k_2$	Non-vertical lines; they may also coincide
Perpendicular lines	$k_1 k_2 = -1$	Both slopes exist
General form of a quadratic function	$y = ax^2 + bx + c$	$a \neq 0$
Vertex form of a quadratic function	$y = a(x - h)^2 + k$	Vertex (h, k)
Axis of symmetry of a quadratic function	$x = -\frac{b}{2a}$	$a \neq 0$
Vertex of a quadratic function	$\left(-\frac{b}{2a}, -\frac{\Delta}{4a}\right)$	$\Delta = b^2 - 4ac$
Odd function	$f(-x) = -f(x)$	Domain symmetric about the origin
Even function	$f(-x) = f(x)$	Domain symmetric about the origin
Periodic function	$f(x + T) = f(x)$	$T \neq 0$
Composite function	$(f \circ g)(x) = f(g(x))$	Both layers of domains must be satisfied
Reciprocal function	$y = \frac{k}{x}$	$k \neq 0, x \neq 0$
Linear rational function	$\frac{ax+b}{cx+d} = \frac{a}{c} + \frac{bc-ad}{c(cx+d)}$	$c \neq 0, cx+d \neq 0$
Asymptotes of a rational function	$x = -\frac{d}{c}, \quad y = \frac{a}{c}$	$ad - bc \neq 0$
Center of symmetry of a rational function	$\left(-\frac{d}{c}, \frac{a}{c}\right)$	Intersection of the two asymptotes

B.4 Exponents and Logarithms

Common Mistake

Restrictions for exponential and logarithmic formulas must be written in the first step. For logarithm problems, first require the argument to be positive. When comparing exponential or logarithmic inequalities, if the base lies in $(0, 1)$, the monotonic direction is opposite to the case $a > 1$.

Topic	Formula	Conditions or Notes
Exponential function	$y = a^x$	$a > 0, a \neq 1$
Range of an exponential function	$a^x > 0$	Range $(0, +\infty)$
Definition of logarithm	$a^x = b \iff \log_a b = x$	$a > 0, a \neq 1, b > 0$

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Topic	Formula	Conditions or Notes
Basic logarithm values	$\log_a 1 = 0, \quad \log_a a = 1$	$a > 0, a \neq 1$
Logarithm of a product	$\log_a(MN) = \log_a M + \log_a N$	$M, N > 0$
Logarithm of a quotient	$\log_a \frac{M}{N} = \log_a M - \log_a N$	$M, N > 0$
Logarithm of a power	$\log_a(M^r) = r \log_a M$	$M > 0$
Change-of-base formula	$\log_a b = \frac{\log_c b}{\log_c a}$	$a, b, c > 0; a, c \neq 1$
Common change of base	$\log_a b = \frac{\ln b}{\ln a}$	$a, b > 0, a \neq 1$
Exponential-logarithmic cancellation	$a^{\log_a x} = x, \quad \log_a(a^x) = x$	Within corresponding domains
Same-base exponential equation	$a^{f(x)} = a^{g(x)} \iff f(x) = g(x)$	$a > 0, a \neq 1$
Same-base logarithmic equation	$\log_a f(x) = \log_a g(x) \iff f(x) = g(x)$	$f(x), g(x) > 0$
Monotonicity of exponential functions	$a > 1$ increasing; $0 < a < 1$ decreasing	Pay attention to the base when comparing
Monotonicity of logarithmic functions	$a > 1$ increasing; $0 < a < 1$ decreasing	The argument must be positive

B.5 Trigonometric Functions and Trigonometric Transformations

Topic	Formula	Conditions or Notes
Sine of an acute angle	$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$	Right triangle
Cosine of an acute angle	$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$	Right triangle
Tangent of an acute angle	$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$	Adjacent side nonzero
Pythagorean identity	$\sin^2 x + \cos^2 x = 1$	Any allowed x
Quotient identity	$\tan x = \frac{\sin x}{\cos x}$	$\cos x \neq 0$
Law of Cosines	$c^2 = a^2 + b^2 - 2ab \cos C$	Any triangle
Degrees and radians	$180^\circ = \pi \text{ rad}$	Unit conversion
Arc length	$l = r\theta$	θ in radians
Area of a sector	$S = \frac{1}{2}r^2\theta$	θ in radians
Coterminal angles	$\theta + 2k\pi$	$k \in \mathbb{Z}$
Sine period	$\sin(x + 2k\pi) = \sin x$	$k \in \mathbb{Z}$
Cosine period	$\cos(x + 2k\pi) = \cos x$	$k \in \mathbb{Z}$
Tangent period	$\tan(x + k\pi) = \tan x$	Expression is meaningful
Negative-angle formulas	$\sin(-x) = -\sin x, \quad \cos(-x) = \cos x$	Parity

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Topic	Formula	Conditions or Notes
Negative-angle formula for tangent	$\tan(-x) = -\tan x$	Expression is meaningful
π minus angle	$\sin(\pi - x) = \sin x, \quad \cos(\pi - x) = -\cos x$	Reduction formula
π plus angle	$\sin(\pi + x) = -\sin x, \quad \cos(\pi + x) = -\cos x$	Reduction formula
Cofunction relation	$\sin\left(\frac{\pi}{2} - x\right) = \cos x$	Reduction formula
Cofunction relation	$\cos\left(\frac{\pi}{2} - x\right) = \sin x$	Reduction formula
Sine sum formula	$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$	Angle-sum formula
Sine difference formula	$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$	Angle-difference formula
Cosine sum formula	$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$	Angle-sum formula
Cosine difference formula	$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$	Angle-difference formula
Tangent sum formula	$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$	Denominator nonzero
Tangent difference formula	$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$	Denominator nonzero
Sine double-angle formula	$\sin 2x = 2 \sin x \cos x$	Double-angle formula
Cosine double-angle formula	$\cos 2x = \cos^2 x - \sin^2 x$	Double-angle formula
Cosine double-angle variants	$\cos 2x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$	Double-angle formula
Tangent double-angle formula	$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$	Denominator nonzero
Auxiliary-angle formula	$a \sin x + b \cos x = R \sin(x + \varphi)$	$R = \sqrt{a^2 + b^2}$
Auxiliary-angle parameters	$\cos \varphi = \frac{a}{R}, \quad \sin \varphi = \frac{b}{R}$	$R > 0$
Sinusoidal function	$y = A \sin(\omega x + \varphi) + k$	$A\omega \neq 0$
Amplitude and period	$ A , \quad T = \frac{2\pi}{ \omega }$	Sinusoidal function
Phase shift	$h = -\frac{\varphi}{\omega}$	Write as $\omega(x - h)$
Range	$[k - A , k + A]$	Sinusoidal function

B.6 Vectors, Complex Numbers, and Plane Analytic Geometry

Topic	Formula	Conditions or Notes
Vector coordinates	$\overrightarrow{AB} = (x_2 - x_1, y_2 - y_1)$	$A(x_1, y_1), B(x_2, y_2)$
Vector addition	$(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 + b_2)$	Coordinate operation
Vector subtraction	$(a_1, a_2) - (b_1, b_2) = (a_1 - b_1, a_2 - b_2)$	Coordinate operation
Scalar multiplication of a vector	$\lambda(a_1, a_2) = (\lambda a_1, \lambda a_2)$	$\lambda \in \mathbb{R}$

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Topic	Formula	Conditions or Notes
Magnitude of a vector	$ \vec{a} = \sqrt{a_1^2 + a_2^2}$	$\vec{a} = (a_1, a_2)$
Dot product of vectors	$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2$	Coordinate form
Dot product and included angle	$\vec{a} \cdot \vec{b} = \vec{a} \vec{b} \cos\theta$	Two nonzero vectors
Perpendicular vectors	$\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$	Two nonzero vectors
Parallel vectors	$a_1b_2 - a_2b_1 = 0$	Two-dimensional coordinate criterion
Complex addition and subtraction	$(a + bi) \pm (c + di) = (a \pm c) + (b \pm d)i$	$i^2 = -1$
Complex multiplication	$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$	$i^2 = -1$
Complex conjugate	$\overline{a + bi} = a - bi$	Change the sign of the imaginary part
Modulus of a complex number	$ a + bi = \sqrt{a^2 + b^2}$	Nonnegative real number
Conjugate product	$z\bar{z} = z ^2$	$z \in \mathbb{C}$
Complex division	$\frac{a + bi}{c + di} = \frac{(a + bi)(c - di)}{c^2 + d^2}$	$c + di \neq 0$
General form of a line	$Ax + By + C = 0$	A, B not both zero
Distance from a point to a line	$d = \frac{ Ax_0 + By_0 + C }{\sqrt{A^2 + B^2}}$	Point (x_0, y_0)
Distance between parallel lines	$d = \frac{ C_1 - C_2 }{\sqrt{A^2 + B^2}}$	$Ax + By + C_i = 0$
Standard equation of a circle	$(x - a)^2 + (y - b)^2 = r^2$	Center (a, b) , radius r
General equation of a circle	$x^2 + y^2 + Dx + Ey + F = 0$	$D^2 + E^2 - 4F > 0$
Center of a general circle	$\left(-\frac{D}{2}, -\frac{E}{2}\right)$	Obtained by completing the square
Radius of a general circle	$r = \frac{1}{2}\sqrt{D^2 + E^2 - 4F}$	The expression under the radical is positive
Standard equation of an ellipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$a > b > 0, c^2 = a^2 - b^2$
Standard equation of a hyperbola	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$a, b > 0, c^2 = a^2 + b^2$
Asymptotes of a hyperbola	$y = \pm \frac{b}{a}x$	Foci on the x -axis
Standard equation of a parabola	$y^2 = 2px$	$p > 0$
Focus and directrix of a parabola	$F\left(\frac{p}{2}, 0\right), x = -\frac{p}{2}$	$y^2 = 2px$

continued on next page

Topic	Formula	Conditions or Notes
Line and conic section	$\Delta = B^2 - 4AC$	Substitute and inspect the discriminant of the quadratic equation

B.7 Sequences

Topic	Formula	Conditions or Notes
General term of an arithmetic sequence	$a_n = a_1 + (n - 1)d$	Common difference d
Arithmetic mean	$2b = a + c$	a, b, c form an arithmetic sequence
Sum of an arithmetic sequence	$S_n = \frac{n(a_1 + a_n)}{2}$	Sum of the first n terms
Sum of an arithmetic sequence	$S_n = na_1 + \frac{n(n-1)}{2}d$	Sum of the first n terms
General term of a geometric sequence	$a_n = a_1q^{n-1}$	Common ratio q
Geometric mean	$b^2 = ac$	a, b, c form a geometric sequence
Sum of a geometric sequence	$S_n = \frac{a_1(1 - q^n)}{1 - q}$	$q \neq 1$
Sum of a geometric sequence	$S_n = na_1$	$q = 1$
Finding terms from partial sums	$a_1 = S_1, \quad a_n = S_n - S_{n-1}$	$n \geq 2$
Common telescoping form	$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$	Telescoping cancellation
General telescoping form	$\frac{1}{(ak+b)(ak+a+b)} = \frac{1}{a} \left(\frac{1}{ak+b} - \frac{1}{ak+a+b} \right)$	$a \neq 0$
Shift for first-order recurrence	$a_{n+1} = pa_n + q, \quad b_n = a_n - \frac{q}{1-p}$	If $p \neq 1$, then $b_{n+1} = pb_n$
Summing a difference recurrence	$a_n = a_1 + \sum_{k=1}^{n-1} f(k)$	$a_{n+1} - a_n = f(n)$

B.8 Derivatives and Applications

Order for Using Derivative Formulas

First identify the outermost structure, then choose a formula: use the linear rule for addition and subtraction, the product rule for multiplication, the quotient rule for division, and the chain rule for functions inside functions. When finding monotonicity, local extrema, or absolute extrema, the formula list only provides the derivative; afterward you still need to write the domain, critical points, and sign table.

Topic	Formula	Conditions or Notes
Average rate of change	$\frac{f(x_2) - f(x_1)}{x_2 - x_1}$	$x_1 \neq x_2$
Definition of derivative	$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$	Limit exists
Constant function	$(C)' = 0$	C is a constant
Power function	$(x^n)' = nx^{n-1}$	Within the domain
Exponential function	$(e^x)' = e^x$	Any real x
General exponential function	$(a^x)' = a^x \ln a$	$a > 0$
Natural logarithm	$(\ln x)' = \frac{1}{x}$	$x > 0$
General logarithm	$(\log_a x)' = \frac{1}{x \ln a}$	$x > 0, a > 0, a \neq 1$
Sine function	$(\sin x)' = \cos x$	x in radians
Cosine function	$(\cos x)' = -\sin x$	x in radians
Tangent function	$(\tan x)' = \frac{1}{\cos^2 x}$	$\cos x \neq 0$
Linear rule	$(au + bv)' = au' + bv'$	a, b constants
Product rule	$(uv)' = u'v + uv'$	u, v differentiable
Quotient rule	$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$	$v \neq 0$
Chain rule	$[f(g(x))]' = f'(g(x))g'(x)$	Composite function differentiable
Tangent line equation	$y - f(x_0) = f'(x_0)(x - x_0)$	Point of tangency $(x_0, f(x_0))$
Normal line slope	$k_n = -\frac{1}{f'(x_0)}$	$f'(x_0) \neq 0$
Monotonicity test	$f'(x) > 0 \Rightarrow f$ increasing; $f'(x) < 0 \Rightarrow f$ decreasing	On the corresponding interval
Local extremum test	$f' : + \rightarrow -$ gives a local maximum; $- \rightarrow +$ gives a local minimum	Check signs on both sides of a stationary point
Concavity test	$f''(x) > 0$ and $f''(x) < 0$	Judge concavity direction according to this book's convention
Implicit differentiation	$F_x + F_y y' = 0, \quad y' = -\frac{F_x}{F_y}$	$F_y \neq 0$

B.9 Permutations, Combinations, and Probability

Common Mistake

Probability formulas cannot be separated from the sample space. Before using classical probability, confirm that the sample points are finite and equally likely. Before using combinations, confirm that order does not create a new result. Before using independent repeated-trial formulas, confirm that trials are independent and that the success probability is the same each time.

Topic	Formula	Conditions or Notes
Factorial	$n! = n(n-1) \cdots 2 \cdot 1$	$0! = 1$
Number of permutations	$A_n^m = \frac{n!}{(n-m)!}$	$0 \leq m \leq n$
Full permutation	$A_n^n = n!$	n distinct elements
Number of combinations	$C_n^m = \binom{n}{m} = \frac{n!}{m!(n-m)!}$	$0 \leq m \leq n$
Symmetry of combinations	$C_n^m = C_n^{n-m}$	$0 \leq m \leq n$
Recursion for combinations	$C_n^m = C_{n-1}^{m-1} + C_{n-1}^m$	$1 \leq m \leq n-1$
Classical probability	$P(A) = \frac{ A }{ \Omega }$	Finite equally likely sample space
Range of probability	$0 \leq P(A) \leq 1$	Any event
Certain and impossible events	$P(\Omega) = 1, \quad P(\emptyset) = 0$	Basic properties
Complement	$P(A^c) = 1 - P(A)$	Any event
Addition formula	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	Any events
Addition for mutually exclusive events	$P(A \cup B) = P(A) + P(B)$	$A \cap B = \emptyset$
Conditional probability	$P(A B) = \frac{P(A \cap B)}{P(B)}$	$P(B) > 0$
Multiplication formula	$P(A \cap B) = P(B)P(A B)$	$P(B) > 0$
Independent events	$P(A \cap B) = P(A)P(B)$	A, B independent
Independent repeated trials	$P(\text{exactly } k \text{ successes}) = C_n^k p^k (1-p)^{n-k}$	n trials, success probability p each time
At least one success	$1 - (1-p)^n$	n independent repeated trials